

Biochemistry AIIMS 2017

Question 1:

Identify the anaplerotic reaction among the following?

- a) Conversion of pyruvate to Acetyl CoA
- b) Conversion of pyruvate to Oxaloacetate
- c) Conversion of pyruvate to Acetaldehyde
- d) Conversion of pyruvate to Lactic acid

Question 2:

True regarding mitochondrial DNA is:

- a) One set from each parents
- b) It has a higher mutation rate than nuclear DNA
- c) It codes for more than 25% of oxidative phosphorylation proteins
- d) It has 3×10^9 base pairs

Question 3:

Which of the following is Thermogenic food

- a) High fat diet
- b) It does not depend on the macronutrients
- c) High protein diet
- d) High carbohydrate diet

Question 4:

Klenow fragment lacks the activity of

- a) $3' \rightarrow 5'$ Exonuclease
- b) $5' \rightarrow 3'$ Exonuclease
- c) $5' \rightarrow 3'$ DNA polymerase

d) 3'→5' DNA polymerase

Question 5:

In SCID, what type of DNA repair is defective:

- a) Base excision repair
- b) Nucleotide excision repair
- c) Mismatch Repair
- d) Non-homologous end joining repair

Question 6:

A 10-year-old boy rapidly develops hypoglycemia after moderate activity. Blood examination reveals raised levels of ketone bodies, lactic acid, and triglycerides. On examination, the liver and kidneys are found to be enlarged. Histopathology of the liver shows deposits of glycogen in an excess amount. What is the diagnosis?

- a) von Gierke's disease
- b) Pompe's disease
- c) McArdle's disease
- d) Cori's disease

Question 7:

Which one of the following is not involved in iron metabolism:

- a) Ceruloplasmin
- b) Heparin
- c) Transferrin
- d) Transferrin

Question 8:

Which of the following amino acids is not formed after post-translational modification?

- a) Selenocysteine
- b) Triiodothyronine

- c) Hydroxy proline
- d) Hydroxylysine

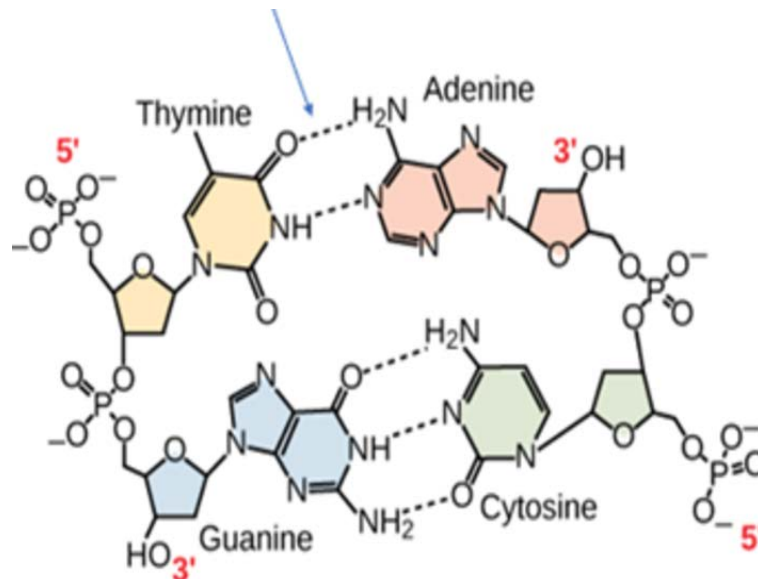
Question 9:

Which test will be positive in starvation state in urine?

- a) Benedict test and heat coagulation test
- b) Heat coagulation test
- c) Rothera's test
- d) Heat coagulation test and Rothera's test

Question 10:

Identify the marked bond:



- a) Ionic bond
- b) Covalent bond
- c) Hydrogen bond
- d) Phosphodiester Bond

Question 11:

Which among the following serves as cofactor for Glycogen phosphorylase:

- a) Thiamine
- b) Niacin
- c) Pyridoxal Phosphate
- d) Biotin

Question 12:

Glucosamines are used in which of the following conditions:

- a) Arthritis
- b) Niemann Pick disease
- c) Alzheimer's disease
- d) Cancer

Question 13:

Which of the following site does the restriction enzyme act when RFLP is used in-order to identify the five different species of staphylococci in a surgical ICU?

- a) ATGGAC-TACGTG
- b) GATATC- CTATAG
- c) TAGATA-ATCTCT
- d) AATATA-TATAAT

Question 14:

Thiamine deficiency results in decreased energy production, because TPP _____

- a) Interferes with alcohol metabolism
- b) Interferes with transketolase activity
- c) Is coenzyme for pyruvate and alpha ketoglutarate dehydrogenase
- d) Interferes with energy production from amino acids

Question 15:

A young patient came with complaints of acute abdominal pain on and off and tingling sensation of limbs and at times weakness of limbs. She had a history of eating paint from the

wall of a newly built house. Which of the following enzyme defect will be the cause of her condition?

- a) ALA synthase
- b) ALA dehydratase
- c) Heme synthase
- d) Coproporphyrinogen oxidase

Question 16:

Under anaerobic metabolism which of the following substrates yields 3 molecules of ATP?

- a) Glycogen
- b) Amino acid
- c) Glucose
- d) Galactose

Question 17:

Which of the following is the reason for using phenylbutyrate in urea cycle disorders?

- a) It maintains renal output
- b) It scavenges nitrogen
- c) It maintains energy production
- d) It activates enzymes

Question 18:

Which of the following can alter the gene expression by methylation and acetylation but not change the gene sequence?

- a) Mutation
- b) Epigenetics
- c) Translocation
- d) Inversion

Question 19:

Which of the following is not seen in low insulin-glucagon ratio?

- a) Glycogen breakdown
- b) Ketogenesis
- c) Glycogen storage
- d) Gluconeogenesis

Answer Key

Question No.	Correct Option
1	b
2	b
3	c
4	b
5	d
6	a
7	c
8	a
9	c
10	c
11	c
12	a
13	b
14	c
15	b
16	a
17	b
18	b
19	c

Detailed Explanations

Solution to Question 1:

The conversion of pyruvate to oxaloacetate is an anaplerotic reaction. It is catalyzed by pyruvate carboxylase.

Anaplerotic reactions are chemical reactions that form intermediates of a metabolic pathway. The above reaction is important in maintaining an adequate concentration of oxaloacetate for the condensation reaction with acetyl-CoA. If acetyl-CoA accumulates, it acts as an activator of pyruvate carboxylase and an inhibitor of pyruvate dehydrogenase, thereby ensuring an adequate supply of oxaloacetate.

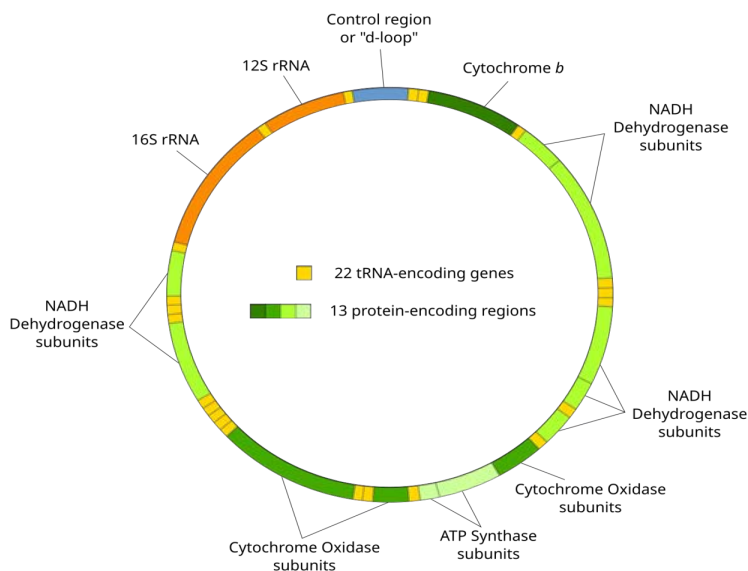
Other anaplerotic reactions are:

- Compounds that are metabolized to yield propionyl CoA, which can be then carboxylated and isomerized to succinyl CoA.
- Glutamate and glutamine are important anaplerotic substrates because they yield α -ketoglutarate as a result of the reactions catalyzed by glutaminase and glutamate dehydrogenase.
- Transamination of aspartate leads directly to the formation of oxaloacetate.

Solution to Question 2:

The true statement is that mitochondrial DNA has a high mutation rate.

Mitochondrial DNA is circular and double-stranded. It has 16,569 bp (that makes up approximately 1% of total cellular DNA) while nuclear DNA has 3.3 billion bp. It has 37 genes that encode 13 protein subunits of the respiratory chain (out of a total of 67 proteins) i.e. about 20%. It mutates at a rate 5-10 times that of nuclear DNA. Mitochondrial DNA is transmitted by maternal non-mendelian inheritance.



Solution to Question 3:

A high intake of protein results in a higher rate of protein turnover, which explains the increased diet-induced thermogenesis seen in people consuming a high protein diet.

The state of protein nutrition can be determined by measuring the difference between intake and output of nitrogenous compounds which are known as nitrogen balance.

- In a healthy adult, nitrogen balance is in equilibrium, when intake equals output, and there is no change in the total body content of protein.
- In a growing child, a pregnant woman, the excretion of nitrogenous compounds is less than the dietary intake of nitrogen in the body as protein— positive nitrogen balance.
- In response to trauma or infection, or if the intake of protein is inadequate to meet requirements, there is the net loss of protein nitrogen from the body—negative nitrogen balance.

A high intake of protein does not lead to positive nitrogen balance; although it increases the rate of protein synthesis, it also increases the rate of protein catabolism, so that nitrogen equilibrium is maintained with a higher rate of protein turnover.

Both protein synthesis and catabolism are ATP expensive, hence the increased diet-induced thermogenesis seen in people consuming a high protein diet.

Solution to Question 4:

Klenow fragment lacks 5' → 3' exonuclease activity.

DNA polymerase I is an enzyme involved in DNA replication. It has polymerase activity along with both 3' → 5' and 5' → 3' exonuclease activity. When the 5' → 3' exonuclease domain is removed, the remaining fragment is called Klenow fragment. It retains the polymerization and proofreading activities of DNA polymerase I.

Solution to Question 5:

SCID is due to defective non-homologous end joining repair.

Defects in DNA Repair:

Non-Homologous End Joining Repair	Severe combined immunodeficiency disease (SCID)
Homologous Repair	Bloom syndrome Werner syndrome Breast cancer susceptibility
Nucleotide Excision Repair	Xeroderma pigmentosum
Mismatch Repair	Hereditary nonpolyposis colorectal cancer (HNPCC)
Base Excision Repair	MUTYH-associated polyposis (MAP)

Solution to Question 6:

The given features are consistent with von Gierke's disease. It is the most common glycogen storage disease.

Enzyme deficient in this condition is glucose-6-phosphatase (common to both glycogenolysis and gluconeogenesis). Inheritance is autosomal recessive.

Von Gierke disease (type I glycogen storage disorder) occurs due to the deficiency of glucose-6-phosphatase enzyme in the liver, kidney, and intestinal mucosa. Patients present commonly by 3-4 months of age with hypoglycemia (leading to seizures), lactic acidosis, and hepatomegaly. Affected children often have a doll-like face with fat cheeks, relatively thin extremities, short stature, and a protuberant abdomen due to massive hepatomegaly. The kidneys are also enlarged, whereas the spleen and heart are not involved. Laboratory findings include hypoglycemia, lactic acidosis, hyperuricemia, and hyperlipidemia.

A definitive diagnosis is made using a liver biopsy. Fructose and galactose restriction in the diet is advised.

Solution to Question 7:

- Hepcidin, ceruloplasmin, and transferrin are involved in iron metabolism.
- Transthyretin transports thyroxine and retinol-binding protein and has no role in iron metabolism

Proteins involved in iron metabolism:

- Ferritin and hemosiderin: Iron storage
- Transferrin: Iron transport
- Ceruloplasmin: a copper-containing enzyme with ferroxidase activity
- Hepcidin: Chief regulator of systemic iron homeostasis

Solution to Question 8:

Selenocysteine, the 21st amino acid, is formed by co-translational modification. It is coded by the UGA codon, which is normally a stop codon. The protein synthetic apparatus can identify a selenocysteine-encoding UGA by the presence of the selenocysteine insertion sequence (SECIS) in the untranslated region of mRNA.

Selenocysteine is an analog of cysteine and is synthesized from serine.

Selenocysteine is present at the active site of several human enzymes that catalyze redox reactions such as thioredoxin reductase, glutathione peroxidase, and deiodinase. Since Glutathione peroxidase contains selenocysteine and hence it is an antioxidant.

Impairments in human selenoproteins have been implicated in tumorigenesis and atherosclerosis and are associated with selenium deficiency cardiomyopathy (Keshan disease).

Solution to Question 9:

Detectable levels of ketone bodies in urine (ketonuria) are seen in conditions characterized by increased ketogenesis such as starvation. The urine test done for ketone bodies is the Rothera's test in which formation of a purple ring indicates presence of ketone bodies. Gerhard's test is also done to detect ketone bodies in urine.

Benedict's test done for reducing sugars.

Heat coagulation test done for proteins.

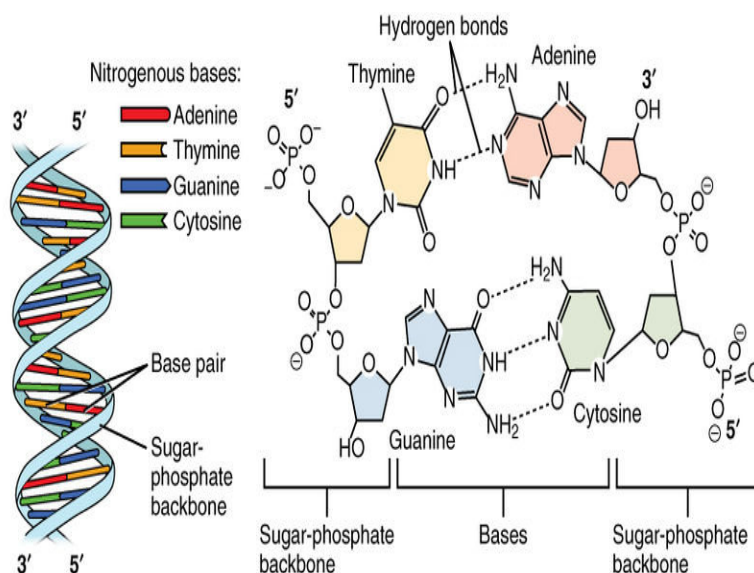
Solution to Question 10:

The marked bond represents a hydrogen bond between thymine and adenine.

The complementary nitrogenous bases of the 2 strands of the DNA double helix are linked by hydrogen bonds. Adenine and thymine are linked by two hydrogen bonds (A=T) whereas three hydrogen bonds are present between cytosine and guanine (C≡G). Van der Waals and hydrophobic interactions are present between the stacked adjacent base pairs.

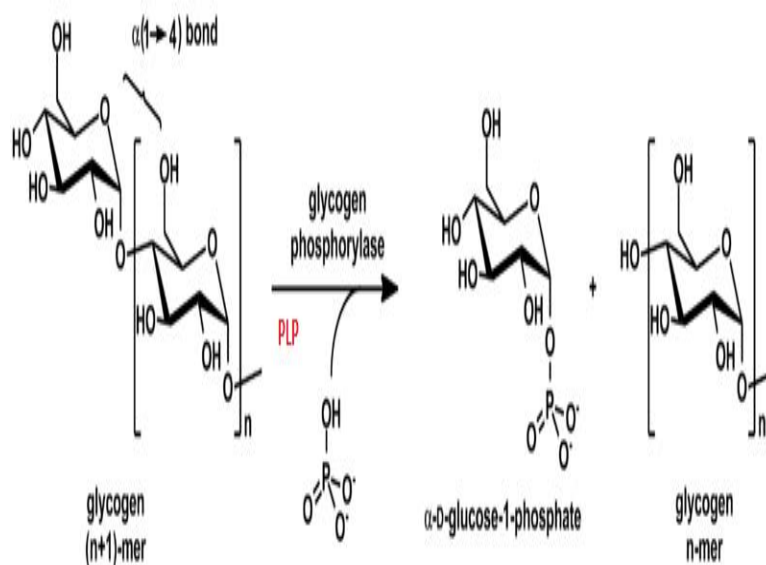
A β -N-glycosidic bond is present between the nitrogenous base and deoxyribose sugar. C1 of deoxyribose/ribose sugar is joined covalently to N1 of pyrimidines and N9 of purines by β -N-glycosidic bond. An ester bond is present between the sugar and phosphate group of the same nucleotide. An acid anhydride bond is present between two phosphate groups.

A polymer of nucleotides is joined by 3'-5' phosphodiester bond. The bond is present between the sugar of one nucleotide and the phosphate group of another.



Solution to Question 11:

- Pyridoxal phosphate is a cofactor of glycogen phosphorylase, where the phosphate group is catalytically important.
- PLP also acts as the coenzyme for many enzymes involved in amino acid metabolism, especially transamination and decarboxylation.
- Glycogen phosphorylase is a debranching enzyme which cleaves alpha-1,4 bonds of the glycogen chain to produce glucose-1-phosphate in glycogenolysis.



Solution to Question 12:

The combined use of glucosamine sulfate and chondroitin sulfates in the treatment of degenerative joint disease has become an extremely popular supplementation protocol in arthritic conditions of the joints.

- Glucosamines are amino sugars which form the major component of glycosaminoglycans. (GAG's are tandem repeats of amino sugar and uronic acid)
- GAG's combine with proteins to form proteoglycans
- One such proteoglycan is aggrecan containing several GAGs (hyaluronic acid, chondroitin sulphate, and keratan sulphate) and both link and core proteins
- Aggrecan is the major proteoglycan in cartilage
- Hence, from this we can deduce that glucosamines being a major part of GAG's and proteoglycans such as aggrecan are protective against arthritis.

Solution to Question 13:

Restriction endonuclease enzymes act at specific palindromic sequences that create DNA fragments of different lengths. By analyzing the fragments, specific species of bacteria can be identified. Option B is the only palindromic sequence here, hence it is the answer.

RFLP is a restriction fragment length polymorphism (RFLP) analysis technique that involves cutting a particular region of DNA with known variability, with restriction enzymes, then separating the DNA fragments by agarose gel electrophoresis and determining the number of fragments and relative sizes.

Cleavage of a sample of DNA with a restriction enzyme produces a characteristic set of smaller DNA fragments. Deviations in the normal product pattern, produce RFLPs that occur if a mutation renders a restriction site unrecognizable and alternatively, generates a new recognition site.

Solution to Question 14:

Thiamine deficiency results in decreased energy production because thiamine pyrophosphate (TPP) is a coenzyme of pyruvate and α -ketoglutarate dehydrogenase, which are required for the production of ATP using NADH.

Thiamine pyrophosphate is an important coenzyme in three multienzyme complexes that catalyze oxidative decarboxylation reactions:

- Pyruvate dehydrogenase in carbohydrate metabolism.
- α -ketoglutarate dehydrogenase in the citric acid cycle.
- Branched-chain ketoacid dehydrogenase involved in the metabolism of leucine, isoleucine and valine.

NAD is a co-factor in pyruvate and α -ketoglutarate dehydrogenase. Hence in thiamine deficiency, there is decreased ATP production owing to reduced NADH entering into the electron transport chain.

Option A: Thiamine does not interfere in alcohol metabolism but alcohol interferes directly with the absorption of thiamine and with the synthesis of thiamine pyrophosphate.

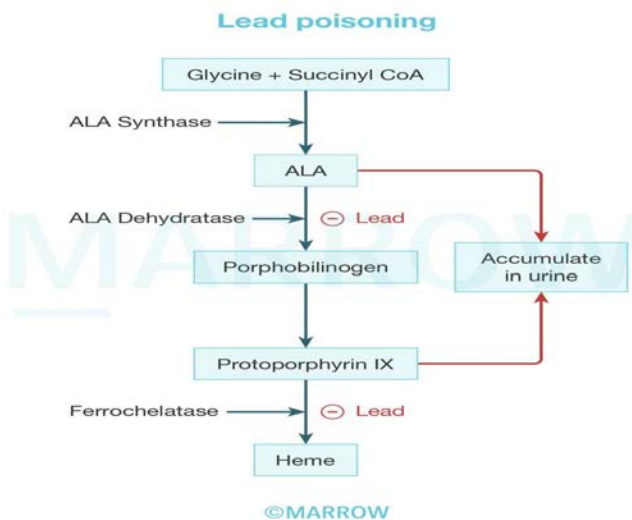
Option B: Thiamine is also the coenzyme for transketolase in the pentose phosphate pathway (HMP shunt) but there is no ATP synthesized in this pathway.

Solution to Question 15:

A child with acute abdominal pain, tingling sensation and weakness of limbs with history of eating paint likely has lead poisoning. It is characterized by impaired ALA dehydratase enzyme activity.

Lead is a potent inhibitor of both ALA dehydratase and ferrochelatase/ heme synthase (option C). However, it has the greatest inhibitory action on ALA dehydratase. Lead poisoning thus leads to the buildup of protoporphyrin and ALA. Lead also substitutes for calcium in many fundamental

cellular processes. It can cross red blood cell membranes as well as the blood-brain barrier and enter the neuroglial cells which support brain function. The fatal dose of lead is 20g of lead acetate and the fatal period is 1 to 2 days.



Other options:

Option A: ALA synthase 2 defect causes X-linked protoporphyria.

Option C: Heme synthase, or ferrochelatase, defect causes erythropoietic protoporphyria.

Option D: Coproporphyrinogen oxidase defect causes hereditary corproporphyria.

Solution to Question 16:

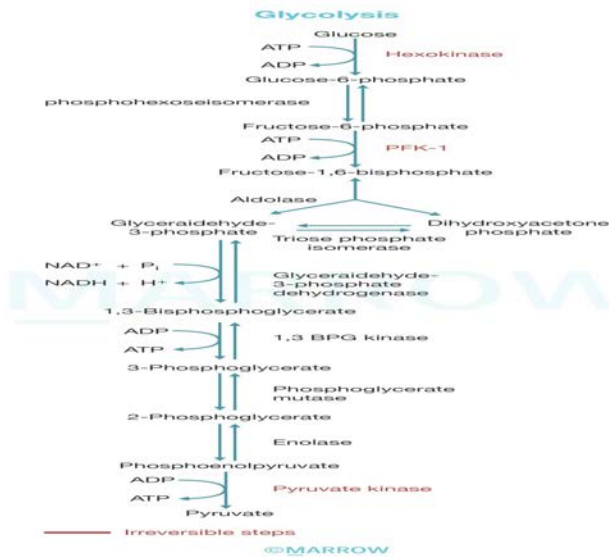
Breakdown of glycogen under anaerobic conditions yields 3 ATPs.

Breakdown of glycogen yields glucose-6-phosphate which can enter the glycolysis pathway. Since it bypasses the first step, one ATP is saved and anaerobic glycolysis yields 3 ATPs.

The first step in glycolysis involves the conversion of glucose to glucose-6-phosphate which requires one ATP. Glycogen breakdown leads to formation of glucose-6-phosphate which can enter glycolysis hence bypassing the first step and saving one ATP.

One ATP is used for conversion of fructose-6-phosphate to fructose 1,6 biphosphate hence net ATP formed from anaerobic glycolysis of glycogen is 3ATP

REACTION	ENZYME	METHOD OF ATP FORMAT ION	NET ATP PRODUCED
1,3-bisphosphoglycerate to 3 phosphoglycerate	Phosphoglycerate kinase	Substrate level phosphorylation	2 ATP
Phosphoenol pyruvate to Pyruvate	Pyruvate kinase	Substrate level phosphorylation	2 ATP
		TOTAL PRODUCED	4 ATP



Solution to Question 17:

Phenylbutyrate scavenges nitrogen by combining with amino acids to form a more water-soluble product, which can be easily excreted by the kidneys.

Phenylbutyrate combines with glutamate to form phenylbutylglutamine.

Other substances that can be used as nitrogen scavengers are:

- Combination of benzoate (forms hippurate on combining with endogenous glycine)
- Phenylacetate (forms phenylacetylglutamine on combining with glutamate) available as ammonia.
- Arginine is also useful.

Solution to Question 18:

Epigenetics is defined as heritable changes in gene expression that are not caused by alteration in gene sequence.

It means "above genetics" and refers to the fact that gene regulatory mechanisms do not change the underlying regulated DNA sequence, but rather simply the expression patterns of the DNA.

Epigenetic mechanisms play a key role in establishment, maintenance, and reversibility of transcriptional states. A key feature is that the controlled transcriptional on/off states can be maintained through multiple rounds of cell division. This observation indicates there must be robust mechanisms to maintain and stably propagate these epigenetic states.

Histone Methylation Mostly Makes DNA Mute.

Histone Acetylation: Relaxes DNA coiling, allowing for transcription. Histone Acetylation makes DNA Active.

Solution to Question 19:

A low insulin/glucagon ratio indicates low insulin and high glucagon levels, which are observed in the fasting state. In the fasting state, all of the above processes (i.e., glycogen breakdown, ketogenesis, and gluconeogenesis) occur, but not glycogen storage.

Metabolic processes	Fasting state/ low insulin-glucagon ratio	Fed state/ high insulin-glucagon ratio
Gluconeogenesis	Increased	Decreased
Glycogenolysis	Increased	Decreased
Ketogenesis	Increased	Decreased
Glycogen storage	Decreased	Increased

Biochemistry AIIMS 2018

Question 1:

An adolescent male patient presented with pain in the calf muscles on exercise. On biopsy, an excessive amount of glycogen was found to be present in the muscle. What is the most likely enzyme deficiency?

- a) Muscle-debranching enzyme
- b) Phosphofructokinase 1
- c) Glucose 6 phosphatase
- d) Muscle glycogen phosphorylase

Question 2:

Alkali disease of livestock is caused by an excess of which of the following minerals?

- a) Selenium
- b) Manganese
- c) Molybdenum
- d) Iron

Question 3:

Which of the following types of collagens is involved in wound healing?

- a) Type II
- b) Type III
- c) Type IV
- d) Type V

Question 4:

An infant is brought with a history of vomiting and poor feeding. A musty odor is noted in the baby. Guthrie test was done and it was found to be positive. All are true regarding this disease except

- a) Phenylalanine hydroxylase enzyme defect
- b) White patch of hair due to tryptophan deficiency
- c) Phenylacetate positive in urine
- d) Mental retardation is present

Question 5:

In a reaction, the substrate is available in a concentration that is 1000 times the K_m value of the enzyme. After 9 minutes of reaction, 1% of substrate is converted to product (12 microgram/ml). If the concentration of the enzyme is changed to $1/3$ and concentration of substrate is doubled, what is the time taken to convert the substrate into the same amount of product that i.e. 12 microgram/ml?

- a) 9 minutes
- b) 4.5 minutes
- c) 27 minutes
- d) 13.5 minutes

Question 6:

The best method to measure HbA_{1c} is _____

- a) Isoelectric focusing
- b) Affinity chromatography
- c) Ion exchange chromatography
- d) Electrophoresis

Question 7:

Which of the following is the best screening investigation for metabolic disorders?

- a) Western blot
- b) Tandem mass spectrometry
- c) PCR
- d) Gel electrophoresis

Question 8:

Which of the following enzyme activity decreases in fasting?

- a) Hormone-sensitive lipase
- b) Glycogen phosphorylase
- c) Pyruvate carboxylase
- d) Phosphofructokinase 1

Question 9:

Fluoroacetate is a component of pesticides. Which of the following metabolic pathways is inhibited by this compound?

- a) TCA cycle
- b) Glycolytic pathway
- c) Oxidative phosphorylation
- d) ETC

Question 10:

Which of the following does not require 5' capping?

- a) tRNA of Alanine
- b) U6 snRNA
- c) mRNA for Histone
- d) siRNA

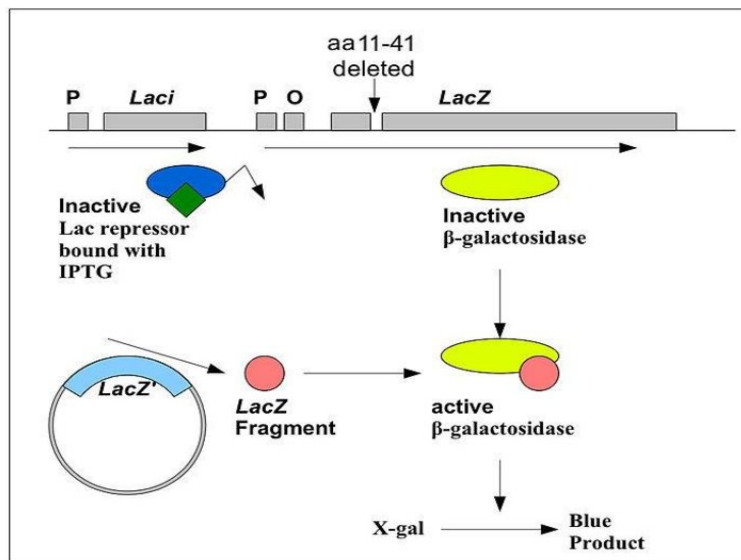
Question 11:

Vitamin K in its coenzyme form is regenerated by the enzyme ____

- a) Glutathione reductase
- b) Quinone reductase
- c) Vitamin K reductase
- d) Epoxide reductase

Question 12:

Identify the phenomenon shown below



- a) Translation
- b) Transformation
- c) Complementation
- d) Conjugation

Question 13:

A middle-aged woman presents with fissured tongue and peripheral neuropathy. Investigations showed reduced glutathione reductase activity. Which vitamin deficiency is the likely cause of this?

- a) Vitamin B1
- b) Vitamin B2
- c) Vitamin B6
- d) Vitamin B12

Question 14:

Which of the following enhances iron absorption?

- a) Vitamin A
- b) Vitamin C
- c) Thiamine

d) Riboflavin

Question 15:

Which of the following does not favor euchromatin formation due to changes occurring at cytosine residues at CpG islands in DNA?

- a) Methylation**
- b) Phosphorylation**
- c) Alkylation**
- d) Sumoylation**

Question 16:

Which of the following techniques is based on RNA?

- a) RT PCR**
- b) Sanger's technique**
- c) Next generation sequencing**
- d) Western blot**

Question 17:

Vitamin C cannot be produced in humans due to lack of

- a) L- gulonolactone oxidase**
- b) Xylitol reductase**
- c) Pyruvate dehydrogenase**
- d) UDP glucose dehydrogenase**

Question 18:

Klenow fragment lacks the activity of

- a) 3'→5' Exonuclease**
- b) 5'→3' Exonuclease**
- c) 5'→3' DNA polymerase**

d) 3'→5' DNA polymerase

Question 19:

Child presents with hypotonia and seizures. It was confirmed to be Zellweger syndrome. Which among the following accumulates in brain in this condition?

- a) Glucose
- b) Lactic acid
- c) Long chain fatty acid
- d) Triglycerides

Question 20:

A 12-year-old girl suffers from an inadequately treated type 1 diabetes mellitus. She is currently hyperglycemic. Which of the following is most likely to occur?

- a) Glycogenesis in muscle
- b) Increased protein synthesis
- c) Increased conversion of fatty acids to acetyl-CoA
- d) Decreased cholesterol synthesis

Question 21:

What would be the respiratory quotient if someone is on an exclusive carbohydrate diet?

- a) 0.7
- b) 0.8
- c) 1
- d) 1.2

Answer Key

Question No.	Correct Option
1	d

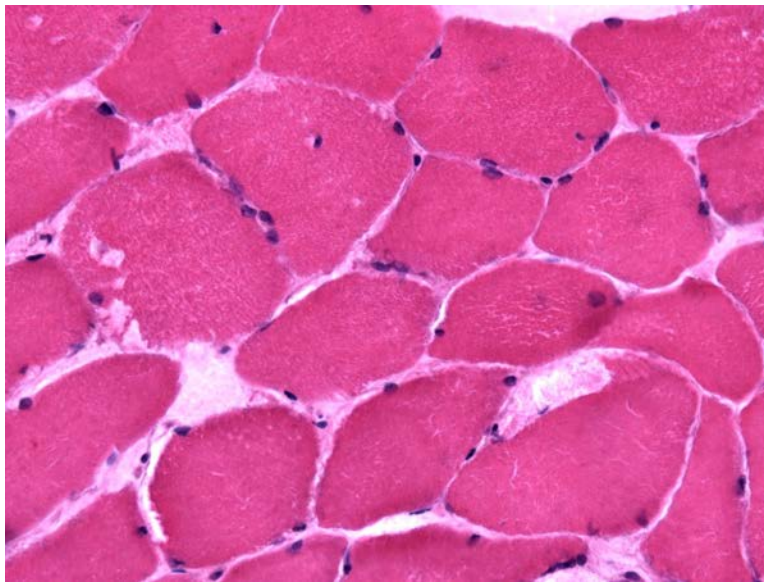
2	a
3	b
4	b
5	c
6	c
7	b
8	d
9	a
10	a
11	d
12	c
13	b
14	b
15	a
16	a
17	a
18	b
19	c
20	c
21	c

Detailed Explanations

Solution to Question 1:

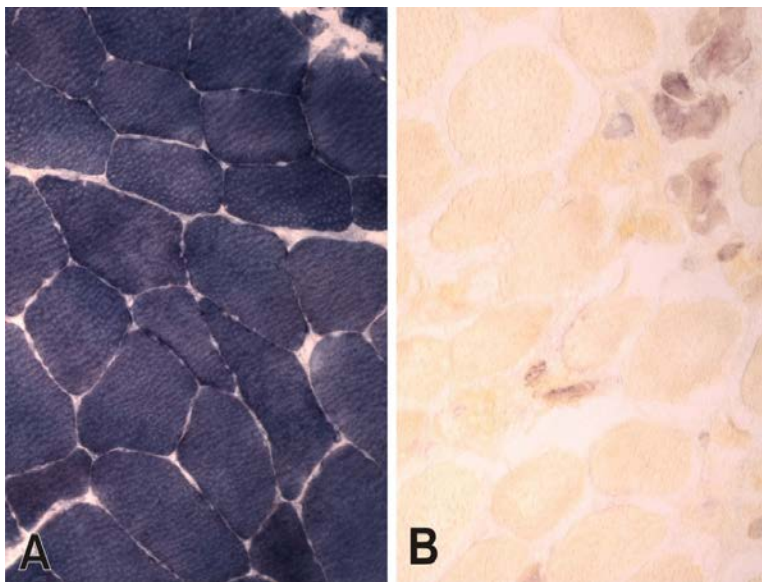
The given clinical features are seen in McArdle's disease. This patient has muscle glycogen phosphorylase deficiency.

The lack of muscle glycogen phosphorylase results in excess glycogen build-up in muscles. The image given below shows the vacuolar myopathy seen in McArdle's disease.



Clinical features include poor exercise tolerance, abnormally high muscle glycogen (2.5-4%), and a very low blood lactate after exercise

The images given below show myophosphorylase enzyme histochemistry showing (A) Normal muscle and (B) McArdle's disease.



Solution to Question 2:

Excess of selenium causes alkali disease in livestock.

Clinical types of selenium intoxication

- Acute selenosis
- Subacute selenosis (i.e., blind staggers type)

- Chronic selenosis (i.e., alkali disease type)

Alkali disease occurs when animals consume moderate levels of selenium for a period of weeks to months.

The common clinical signs of chronic selenosis in horses, cattle, and swine are loss of hair (horses and cattle lose long hair from the mane and tails), emaciation, hoof lesions, weight reduction, and lameness.

Solution to Question 3:

Type III collagen is involved in wound healing.

Collagen types I and III predominate in the skin and aponeurotic layers. In the early stages of wound healing, type III collagen is synthesized. It is replaced by type I collagen as wound healing progresses.

The tensile strength of the healing wound is directly proportional to the amount and type of collagen present. During healing, as disorganized collagen is degraded and reformed, covalent cross-links are formed that enhance tensile strength. The maximum strength of the healed wound depends upon the interconnection of collagen subunits. Approximately 80 percent of the original strength of the tissue is obtained by six weeks after surgery. Wounds slowly continue to get stronger but may never achieve 100 percent of their previous strength.

Solution to Question 4:

The given scenario of an infant with a history of vomiting and poor feeding with a musty odour is suggestive of phenylketonuria (PKU). Tryptophan deficiency is not a feature of this disorder. The deficiency of the enzyme phenylalanine hydroxylase causes PKU.

Characteristic clinical findings in Phenylketonuria:

- Mental retardation
- Failure to walk or talk
- Seizures
- Hyperactivity
- Tremor
- Microcephaly
- Failure to grow
- Deficiency of pigmentation (fair hair, light skin color, and blue eyes)

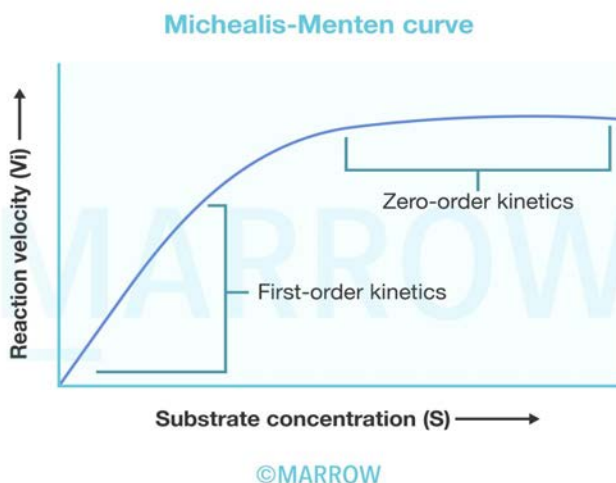
The Guthrie bacterial inhibition test is a screening test for phenylketonuria. This test uses a phenylalanine requiring strain of *Bacillus subtilis*. The ferric chloride test is also used as a screening test for PKU. Tandem mass spectrometry is the gold standard investigation.

Solution to Question 5:

The time taken for the formation of the same amount of product is 27 minutes.

In this reaction, the substrate concentration becomes 1000 times i.e much greater than the K_m value. Thus, the rate of this reaction is equal to V_{max} and is thus independent of the substrate concentration i.e. it follows zero-order kinetics. Thus, doubling the substrate concentration does not change the velocity of the reaction i.e. it remains as V_{max} .

No matter the substrate concentration, the rate of the reaction is directly proportional to the concentration of the enzyme. In this situation the enzyme concentration becomes $1/3$ rd. Thus, the velocity of the reaction (V_{max}) also becomes $1/3$ rd. This means that the time taken for the formation of the same amount of product will become tripled i.e $9 \times 3 = 27$ minutes.



Solution to Question 6:

The best method to measure HBA1C is ion-exchange chromatography.

In ion-exchange chromatography:

- Proteins interact with the stationary phase by charge-charge interactions.
- Proteins with a net positive charge at a given pH will tightly adhere to beads with negatively charged functional groups such as carboxylates or sulfates (cation exchangers).
- Proteins with a net negative charge adhere to beads with positively charged functional groups, typically tertiary or quaternary amines (anion exchangers).
- Non-adherent proteins flow through the matrix and are washed away.
- Proteins elute in inverse order of the strength of their interactions with the stationary phase.

Solution to Question 7:

Tandem mass spectrometry (MS) is the technique of choice for screening neonates for inherited metabolic diseases.

Tandem MS employs the equivalent of two mass spectrometers linked in series (MS-MS or MS²). It can be used to screen blood samples from newborns for the presence and concentrations of amino acids, fatty acids, and other metabolites.

Abnormalities in metabolite levels indicate inherited metabolic diseases, such as phenylketonuria, ethylmalonic encephalopathy, and glutaric acidemia type 1.

Solution to Question 8:

During fasting, there is low insulin, and hence, the activity of phosphofructokinase-1 gets inhibited.

Phosphofructokinase 1 is an enzyme of glycolysis. It is induced by insulin and a well-fed state.

Other options:

Option A: Hormone-sensitive lipase mobilizes stored fats in a fasting state.

Option B: Glycogen phosphorylase is active in fasting state so as to produce glucose from glycogen.

Option C: Pyruvate carboxylase is an enzyme of gluconeogenesis. During the fasting state, the liver is flooded with fatty acids mobilized from adipose tissue. The resulting elevated hepatic acetyl CoA activates pyruvate carboxylase. Other enzymes of gluconeogenesis that are elevated in fasting include Phosphoenolpyruvate carboxykinase, Fructose-1,6-bisphosphatase, and Glucose-6-phosphatase.

Solution to Question 9:

Fluoroacetate inhibits the TCA cycle enzyme aconitase.

Fluoroacetyl CoA condenses with oxaloacetate to form fluorocitrate which inhibits aconitase, causing citrate to accumulate. This way it inhibits TCA cycle enzyme non-competitively.

Sources of fluoroacetate exposure:

- Fluoroacetate in some plants
- Fluorinated compounds (anticancer agents, industrial chemicals, and pesticides) are metabolized to fluoroacetate.

Solution to Question 10:

tRNA of Alanine does not require 5' capping.

Capping process protects the 5' end of mRNA from the action of exonucleases and polyadenylation protects the 3' end of the mRNA. All mRNA, snRNA, and siRNA undergo this

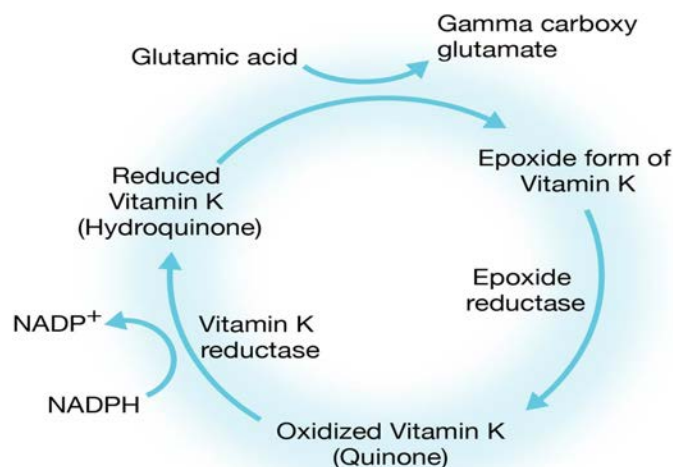
process, but tRNA are comparatively stable and do not require a capping at the 5' end.

Note: rRNAs also do not require 5' capping.

Solution to Question 11:

The coenzyme form of Vitamin K is regenerated by the enzyme epoxide reductase.

Epoxide reductase reduces Vitamin K epoxide to its coenzyme form hydroquinone.



Note: In addition to vitamin K reductase, epoxide reductase can also catalyse the reduction of quinone to hydroquinone. Thus, it can regenerate hydroquinone by itself.

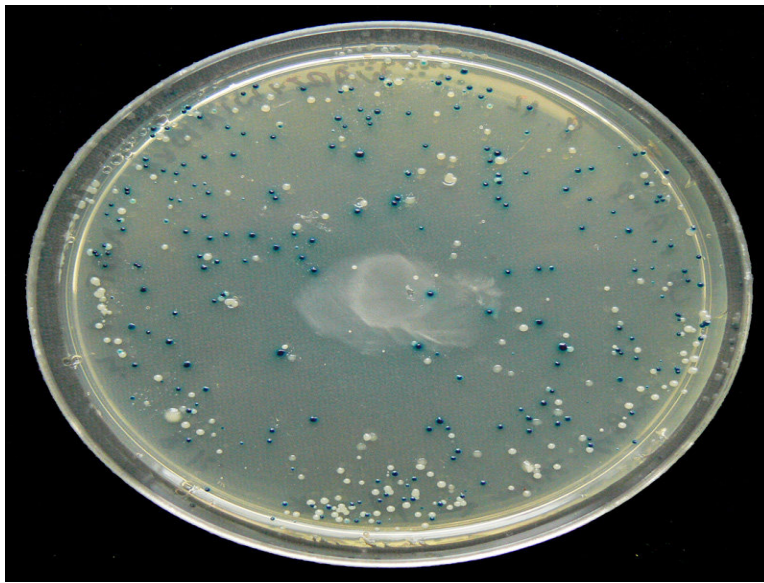
Solution to Question 12:

The image shows a schematic of the blue-white assay used in the screening of recombinant vectors for inserted DNA fragments. It uses the phenomenon of complementation (α -complementation of the β -galactosidase gene).

Blue-white assay:

The DNA of interest is ligated into a vector. The vector is then inserted into a competent host cell viable for transformation, which is then grown in the presence of X-gal (X-gal detects active β -galactosidase). Cells transformed with vectors containing recombinant DNA will produce white colonies; cells transformed with non-recombinant plasmids (i.e., only the vector) grow into blue colonies.

The image below shows the result of a blue-white assay.



Solution to Question 13:

This patient is most likely suffering from riboflavin/vitamin B2 deficiency.

Deficiency of riboflavin is characterized by cheilosis, desquamation, and inflammation of the tongue, seborrheic dermatitis, corneal vascularization, anemia, and personality changes.

Riboflavin nutritional status is assessed by measurement of the activation of erythrocyte glutathione reductase by FAD added in vitro.

Solution to Question 14:

Iron absorption is enhanced by vitamin C.

Vitamin C reduces the ferric iron to its ferrous form, thereby helping in the process of absorption of dietary iron from the intestine.

Vitamin C deficiency causes

- Decreased absorption of iron leading to microcytic-hypochromic anemia
- Decreased hydroxylation of proline and lysine, which results in defective collagen synthesis and bleeding manifestations.

Conversely, vitamin C toxicity (>2000 mg/ day) can lead to iron overloaded states.

Note: Alcohol and fructose also enhance iron absorption. However, the absorption of both inorganic and heme iron is impaired by calcium. So, a glass of milk with a meal significantly reduces iron availability.

Solution to Question 15:

Methylation does not favour euchromatin formation due to changes cytosine residues at CpG islands in DNA.

The transcriptional regulatory regions of active genes often contain a high frequency of CpG dinucleotides (referred to as CpG islands) and they are normally unmethylated. Therefore, methylation is an epigenetic modification that inactivates DNA, and hinders the formation of euchromatin (transcriptionally active chromatin). Epigenetic changes include alterations of chromatin structure by:

- Methylation of cytosine residues in CpG dinucleotides
- Modification of histones by acetylation or methylation
- Changes in higher-order chromosome structure

Sumoylation of histones (SUMO - Small Ubiquitin-related MOdifier) is associated with transcription repression. Other possible roles of posttranslationally modified histones:

- Acetylation of core histones is associated with chromosomal assembly during DNA replication.
- Phosphorylation of histone H1 is associated with the condensation of chromosomes during the replication cycle.
- ADP-ribosylation of histones is associated with DNA repair.
- Methylation of histones is correlated with activation and repression of gene transcription.
- Acetylation of histones H3 and H4 is associated with the activation or inactivation of gene transcription.
- Monoubiquitylation is associated with gene activation, repression, and heterochromatic gene silencing.

Solution to Question 16:

RT PCR is a RNA-based technique.

RT PCR is a method used to quantitate mRNA levels that relies upon the first step of cDNA copying of mRNAs catalyzed by reverse transcriptase prior to PCR amplification and quantitation.

Other techniques:

- Sanger's technique and Southern blot: DNA
- Western blot: protein
- Northern blot: RNA
- Next-generation DNA sequencing technologies include Illumina (Solexa) sequencing, Roche 454 sequencing, Ion torrent: Proton/PGM sequencing, SOLID sequencing.

Solution to Question 17:

Ascorbic acid (vitamin C) cannot be synthesized in human beings because of the absence of L-gulonolactone oxidase.

Uronic acid pathway catalyzes the conversion of glucose to glucuronic acid, ascorbic acid, and pentoses. Glucuronate is reduced to L-gulonate, the direct precursor of ascorbate in the animals capable of synthesizing this vitamin.

Solution to Question 18:

Klenow fragment lacks 5' → 3' exonuclease activity.

DNA polymerase I is an enzyme involved in DNA replication. It has polymerase activity along with both 3' → 5' and 5' → 3' exonuclease activity. When the 5' → 3' exonuclease domain is removed, the remaining fragment is called Klenow fragment. It retains the polymerization and proofreading activities of DNA polymerase I.

Solution to Question 19:

Elevated levels of very long chain polyenoic acids have been found in the brains of patients with Zellweger syndrome.

Zellweger syndrome:

- Autosomal recessive
- Cerebro-hepato-renal syndrome: Impaired neuronal migration and myelination, hepatomegaly, renal cysts
- Reduction or absence of functional peroxisomes in tissues.
- Accumulation of very long chain polyenoic acids in the brain.
- Severe neurological symptoms.
- Most patients die within the 1st year of life

Solution to Question 20:

With the background of inadequately treated type I diabetes mellitus, an insulin-deficient state in the body is expected which will lead to increased conversion of fatty acids to acetyl-CoA.

Type 1 diabetes mellitus is commonly found in the adolescent population. In response to certain precipitating factors, such patients may develop a severe insulin-deficient state. Since insulin is an inhibitor of hormone-sensitive lipase (HSL), an insulin-deficient state will cause HSL overactivity in adipocytes and the breakdown of lipid into fatty acids (lipolysis). Free fatty acids will undergo beta-oxidation and produce a large amount of acetyl CoA, which gets converted to ketone bodies (acetoacetate, acetone, beta-hydroxybutyrate, etc.)

Solution to Question 21:

The respiratory quotient, if someone is on an exclusive carbohydrate diet, is 1.

The respiratory quotient is the amount of CO₂ produced /O₂ consumed.

The respiratory quotient for various substances:

Substance	RQ
Carbohydrate	1.00
Protein	0.81
Fat	0.71
Alcohol	0.66

Biochemistry AIIMS 2019 (May & Nov)

Question 1:

Child presents with hypotonia and seizures. It was confirmed to be Zellweger syndrome. Which among the following accumulates in brain in this condition?

- a) Glucose
- b) Lactic acid
- c) Long chain fatty acid
- d) Triglycerides

Question 2:

Which of the following does not happen in 5' to 3' direction?

- a) Transcription
- b) DNA replication
- c) DNA repair
- d) RNA editing

Question 3:

Chimeric DNA is associated with:

- a) Paternity test
- b) Maternity test
- c) Personal identification
- d) Organ transplantation

Question 4:

A bodybuilder starts eating raw eggs for protein. He starts to develop fatigue on moderate exercise. The doctor prescribes a vitamin. Which enzyme is deficient in him?

- a) Glucose 6 Phosphate
- b) Pyruvate Carboxylase

- c) PECK
- d) Glycogen Phosphorylase

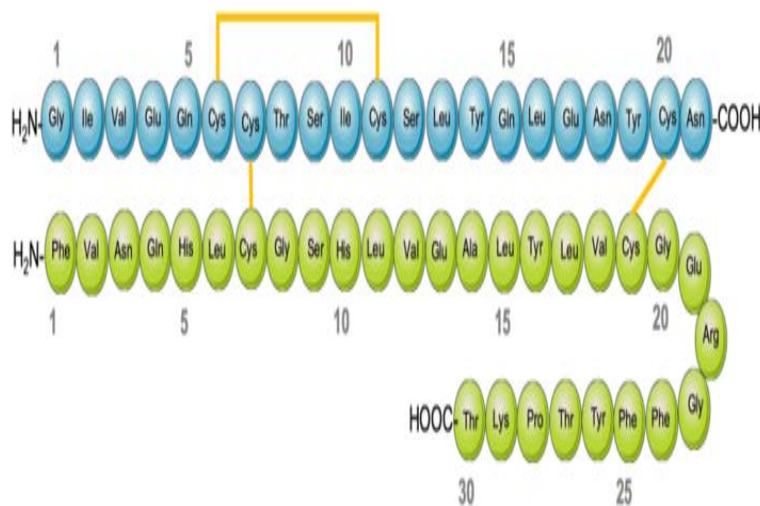
Question 5:

Which of the following shows functional assessment of B1 deficiency?

- a) RBC Transketolase
- b) RBC Glutathione reductase
- c) Serum Thiamine levels
- d) RBC Glutathione Peroxidase

Question 6:

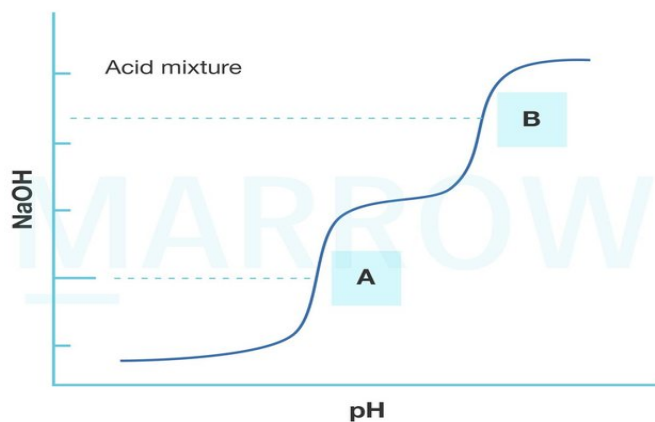
Which modification will lead to a change in the biological activity of the hormone given in the picture?



- a) Hydrolysis of disulphide bonds
- b) Change in 5-6 in A chain
- c) Change in 29-30 in B chain
- d) Change in 14-16 in A chain

Question 7:

The graph shown below is the titration curve of a biochemical compound. Which of the following statement is true?



©MARROW

- a) The maximum buffering capacity of the compound is represented by points A and B
- b) The points A and B represent the range of maximum ionization of the amine and carboxyl group
- c) The compound has three ionisable side chains
- d) The compound has one ionisable group

Question 8:

Which of the following TCA cycle intermediate is a part of heme metabolism?

- a) Alpha ketoglutarate
- b) Fumarate
- c) Succinyl CoA
- d) Malate

Question 9:

The catalytic efficiency of enzyme is best expressed by which of the following kinetic constants:

- a) K_{cat}/K_m
- b) K_m/K_{cat}

- c) Km/Ka
- d) Ka/Km

Question 10:

Type of mutation seen in sickle cell anemia:

- a) Insertion
- b) Deletion
- c) Point mutation
- d) Frameshift mutations

Question 11:

By which mechanism does gene repair occur when mediated by CRISPR-Cas9?

- a) Base excision
- b) Non-homologous end joining
- c) Homologous end joining
- d) Mismatch repair

Question 12:

Which of the following is deficient in a patient with features as shown below:



- a) Phenylalanine
- b) Tryptophan
- c) Glutamine
- d) Arginine

Question 13:

The attachment of Eukaryotic mRNA to the ribosome is mediated through _____

- a) Poly A tail
- b) tRNA
- c) Guanyl Cap
- d) Shine-Dalgarno Sequence

Question 14:

Which of the following would have no effect on the function of the protein product?

- a) Glutamine replaced by asparagine
- b) Glutamine replaced by alanine
- c) Glutamine replaced by glutamate
- d) Glutamine replaced by arginine

Question 15:

Which of the following is not true about proteoglycans?

- a) Carries negative charge
- b) Chondroitin sulfate is a proteoglycan
- c) Made of amino acids and sugars
- d) Holds less water

Question 16:

Which of the following has protective as well as lubricating property?

- a) Collagen

- b) Mucin
- c) Albumin
- d) Immunoglobulin

Question 17:

In urea cycle disorders, which of the following can be used to reduce the levels of ammonia?

- a) Glutamate
- b) Phenylbutyrate
- c) Isoleucine
- d) L-carnitine

Question 18:

Restriction endonuclease will act on which of the following:

- a) AAGAAG
- b) GAGAGG
- c) AAGCTT
- d) TACGAG

Answer Key

Question No.	Correct Option
1	c
2	d
3	d
4	b
5	a
6	c
7	a
8	c
9	a
10	c

11	b
12	b
13	c
14	a
15	d
16	b
17	b
18	c

Detailed Explanations

Solution to Question 1:

Elevated levels of very long chain polyenoic acids have been found in the brains of patients with Zellweger syndrome.

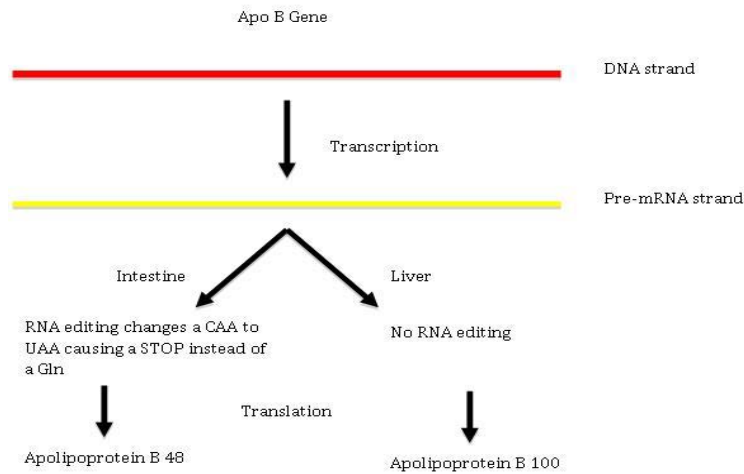
Zellweger syndrome:

- Autosomal recessive
- Cerebro-hepato-renal syndrome: Impaired neuronal migration and myelination, hepatomegaly, renal cysts
- Reduction or absence of functional peroxisomes in tissues.
- Accumulation of very long chain polyenoic acids in the brain.
- Severe neurological symptoms.
- Most patients die within the 1st year of life

Solution to Question 2:

RNA editing occurs in 3' to 5' direction. It is the process by which the coding sequence is changed at the mRNA level before translation.

RNA editing occurs by insertion of additional Us at the 3' end of the mRNA and proceeds in a 3' to 5' direction. The guide RNA (gRNA) acts as a template for the editing process. An example of RNA editing is apoB gene and mRNA shown in the image below:



Solution to Question 3:

Chimeric DNA is associated with organ transplantation, especially bone-marrow transplantation.

In organ transplantation, chimeric DNA refers to the mixture of donor and recipient genetic material that occurs when donor cells begin functioning within the recipient's body. It is used for long-term monitoring of bone marrow transplant recipients. Detecting changes in the levels of donor-derived cells can provide early warning signs of graft failure, rejection, or relapse of underlying disease.

Note: The practical applications of recombinant DNA technology include:

- Gene mapping using techniques such as fluorescence in situ hybridization (FISH).
- Producing proteins such as:
 - Interferons, tPA (tissue plasminogen activator)
 - Insulin, growth hormone
 - Hepatitis B vaccine
- Molecular analysis of disease, to detect gene variations and mutations.
- Restriction Fragment Length polymorphism (RFLP) and single nucleotide polymorphism (SNP)
- Microsatellite DNA polymorphisms
- Gene therapy and bone marrow transplantation

Solution to Question 4:

The patient developed pyruvate carboxylase enzyme deficiency.

- Biotin serves as co-enzyme for the synthesis of several carboxylase enzymes such as:

- Acetyl-CoA carboxylase
- Pyruvate carboxylase
- Propionyl CoA carboxylase
- Methylcrotonyl-CoA carboxylase
- Individuals who eat an excess of uncooked egg-white, develop biotin deficiency and as a result deficiency of the carboxylase enzymes. This is because, a protein called, avidin present in the egg-white binds to biotin in the diet, and makes biotin unavailable for intestinal absorption.

Solution to Question 5:

Erythrocyte (RBC) transketolase test is the most widely used method for the functional assessment of thiamine deficiency.

Thiaminediphosphate is a co-enzyme for transketolase. Therefore, in thiamine (Vitamin B1) deficiency, the RBC transketolase activity is reduced. When thiamine diphosphate is added in-vitro to erythrocyte lysate, activation of apo-transketolase is suggestive of B1 deficiency.

Solution to Question 6:

The hormone shown in the above image is insulin. Among the given options, changes in 29-30 amino acids in the B chain cause the generation of short and long-acting insulins depending on the modification.

Structure:

The primary structure of insulin consists of 2 polypeptide chains: A-chain(21 Amino acids) and B-chain(30 amino acids).

It has two interchain disulfide bonds,

- Between 7th amino acid of A chain and 7th amino acid of B chain
- Between 20th amino acid of A chain and 19th amino acid of B chain

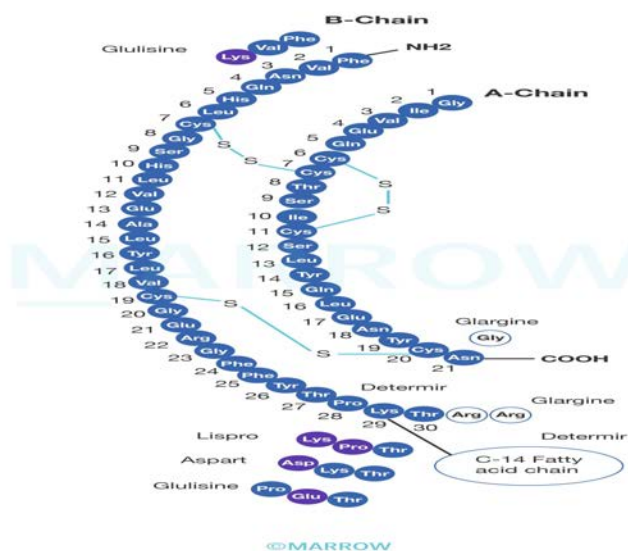
One intrachain disulfide bond is present between the 6th amino acid of A-chain and 11th amino acid in A-chain.

Modification:

Various short and long-acting insulin analogs have been developed by modifying these amino acid residues.

- Insulin lispro: Proline at position 28 and lysine at position 29 in the B-chain are interchanged.
- Insulin Aspart: Proline at position 28 in the B-chain is replaced by aspartic acid.
- Insulin glulisine: Asparagine at position 3 and lysine at position 29 in the B-chain are replaced by lysine and glutamic acid.

- Insulin glargine: Asparagine at position 21 in the A-chain is replaced by glycine and the B-chain contains two extra amino acids (two arginine residues) at positions 31 and 32.
- Insulin detemir: Threonine at position 30 of the B-chain is removed and a 14-carbon fatty acid chain is added to position 29 of the B-chain.



Solution to Question 7:

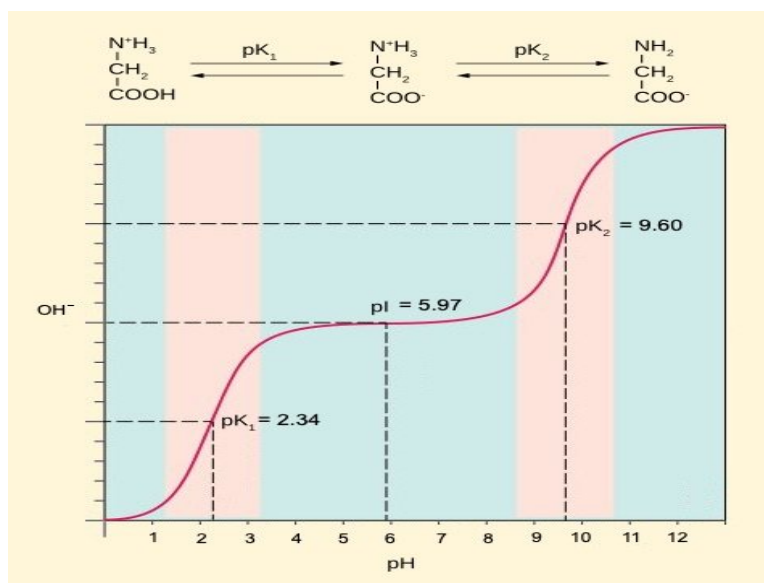
The maximum buffering capacity of the compound is represented by points A and B, which corresponds to the pK₁ and pK₂ of the compound.

Other options:

- The points A and B represent the pK_a of carboxyl and amino groups respectively.
- The amino acid has no ionizable side chains as the graph shows only two pK_as.
- The amino acid has two ionizable groups: an amino group and a carboxyl group.

The above graph resembles the titration curve of the amino acid glycine.

Titration curve of Glycine:



The titration is completed at a pH of 12, where the glycine predominantly exists in the form of $\text{H}_2\text{N} - \text{CH}_2 - \text{COO}^-$.

pK_1 and pK_2 correspond to the point where the buffering capacity is maximum.

Characteristics of the titration curve:

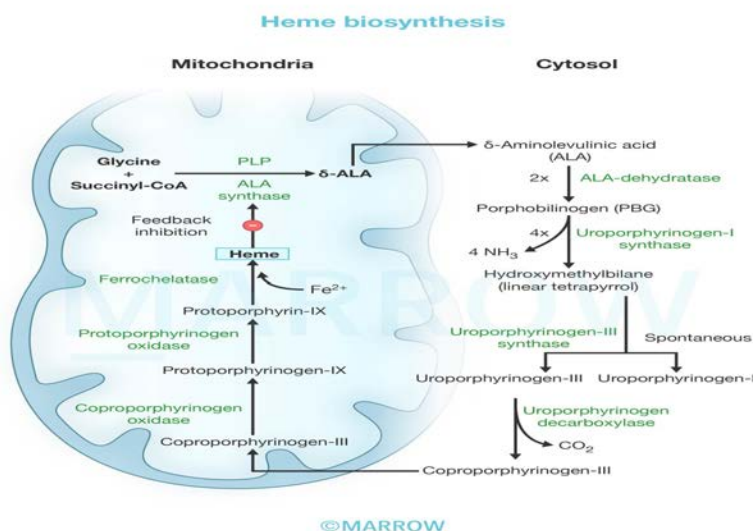
- Buffering capacity is referred to as the ability of the amino acid to resist the change in pH due to the addition of acid (H^+) or base (OH^- equivalents), by accepting or losing the protons respectively.
- An ionizable group refers to the group which can either lose or accept a proton and becomes negatively charged or positively charged ion respectively.
- During the titration, the buffering capacity of the compound reaches its maximum, at points where the pK_a of the ionizable groups is reached.
- Each of these pK_a are labeled as pK_1 , pK_2 , pK_3 , etc.
- The presence of two pK_a s indicates that the compound has two ionizable groups. Ex. Glycine.
- The presence of three or more pK_a s indicates the presence of side chains on the amino acid, with additional ionizable groups.
- Ex. histidine has three ionizable groups, an amino group, an imidazole group (on the side chain), and a carboxyl group. The pK_a s of these groups respectively correspond to the pK_1 , pK_2 , pK_3 of the titration curve of histidine.

Solution to Question 8:

Succinyl Co-A, an essential part for heme biosynthesis, is an intermediate of the TCA cycle.

The synthesis of heme begins by condensation of succinyl Co-A and glycine to form α -amino- β -ketoadipate, which is then converted to δ -aminolevulinate. These steps are mediated by the enzyme ALA synthase in the mitochondria.

Various steps involved in the synthesis of heme is illustrated in the image given below:



Solution to Question 9:

The catalytic efficiency of an enzyme is expressed as K_{cat}/K_m .

K_{cat} is the rate constant, which is the limiting rate of an enzyme-catalyzed reaction at saturation. It is given by V_{max} divided by number of active sites of the enzyme. K_m is the Michaelis-Menten constant, which is defined as the substrate concentration at which the initial velocity (V_i) is half the maximum velocity ($V_{max}/2$).

For some enzymes, when a substrate (S) binds to the active site of the enzyme (E), and forms an enzyme-substrate complex (ES), the product is instantaneously formed. For such enzymes, the reaction of $E+S \rightarrow ES$ becomes the rate-limiting step. Its rate is determined on the rate of diffusion of the molecules in a solution. Hence, these enzymes are called diffusion-limited or catalytically perfect.

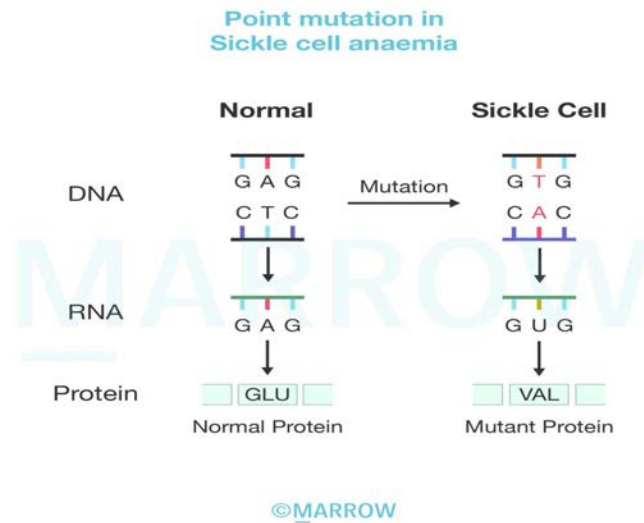
Examples of the catalytically perfect enzymes, where the catalytic efficiency reaches the diffusion limit of 10^8 - $10^9 \text{ M}^{-1}\text{s}^{-1}$:

- Triosephosphate isomerase
- Carbonic anhydrase
- Acetylcholinesterase
- Adenosine deaminase

Solution to Question 10:

Point mutation is seen in sickle cell anemia.

- Sickle cell anemia is caused by the mutation of a single base in the β -globin gene, where adenine is replaced with thymine in the sixth codon of the DNA sequence.
- This leads to change from GAG in the normal gene to GTG in the sickle cell gene, leading to a change in mRNA sequence from GAG to GUG.
- As a result, the mutant gene now codes for valine instead of glutamic acid.

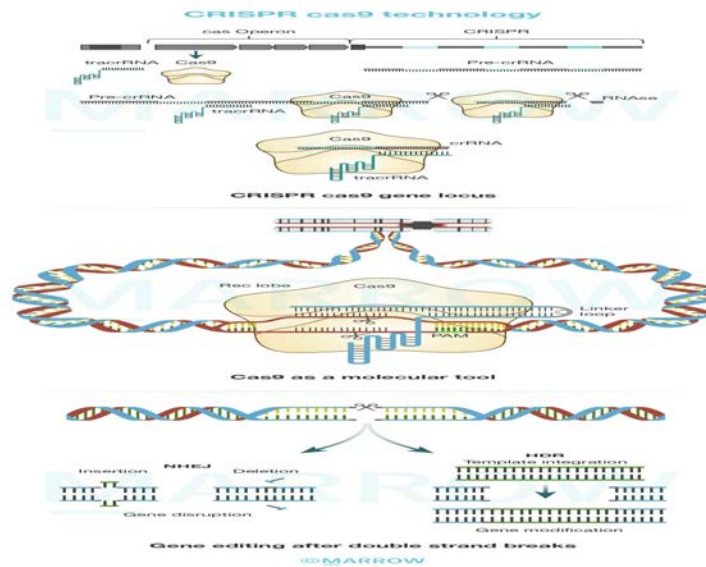


Solution to Question 11:

Non-homologous end joining (NHEJ) is the major mechanism used by CRISPR-Cas9 for gene repair.

The CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) are clustered palindromic sequences found in many bacteria. It is a form of acquired immunity to infection by bacteriophages which complements the system of restriction endonucleases and methylases. CRISPR uses RNA-based targeting to bring the Cas9 endonuclease to foreign (or any complementary) DNA. Within bacteria this CRISPR-RNA-Cas9 complex then degrades and inactivates the targeted DNA.

The CRISPR system has been adapted for use in eukaryotic cells for gene deletion, gene editing and even modulation of gene transcription.



Solution to Question 12:

The image provided is characteristic of Casal's necklace, seen in pellagra, and indicates a deficiency of Tryptophan.

Pellagra can occur due to:

- Vitamin B3 (niacin)
- Niacin is derived from tryptophan and needs B2 and B6 for its synthesis.
- Hartnup's disease (tryptophan malabsorption)
- Malignant carcinoid syndrome ($\uparrow$ tryptophan metabolism)
- Isoniazid ($\downarrow$ vitamin B6)

In Hartnup's disease, defective membrane transport of tryptophan across PCT and enterocytes leads to malabsorption of tryptophan. This causes a deficiency of tryptophan.

As a result, the production of niacin from tryptophan is reduced and this manifests as pellagra with dermatitis, diarrhea and dementia (3D's) and glossitis. A characteristic skin rash called the Casal's necklace may also be seen.

Solution to Question 13:

The attachment of mRNA to the ribosome is mediated through the guanyl cap at the 5' end. Eukaryotic mRNAs have 7-methylguanosine caps at the 5' end and poly-A tails at the 3' end.

The 7-methylguanosine caps (Guanyl caps) are necessary for:

- Processing of primary transcripts to mRNA.
- Mediating the attachment of mRNA to the ribosome for translation.

- Protection of mRNA against nucleolytic attack by 5'exonucleases.

Other options:

Option A: AAUAAA is present at the 3' end, which is the polyadenylation sequence (poly-A tail). They help stabilize the mRNA, facilitate its exit from the nucleus and aid in translation. After the mRNA enters the cytosol, the poly-A tail is gradually shortened.

Option B: tRNA molecules serve as adapters for the translation of the information in the sequence of nucleotides of the mRNA into specific amino acids.

Option D: Shine-Dalgarno sequence initiates translation in prokaryotes.

Note: The guanyl cap and poly-A tail aid in attaching the mRNA to the ribosome for translation, however, it is initiated and mediated first through the Guanyl cap. The poly(A) tail stimulates recruitment of the 40S ribosomal subunit to the mRNA, after which the ribosomal subunit binds to the guanyl cap at 5'end of mRNA.

Solution to Question 14:

When glutamine is replaced by asparagine, there is no change in the function of the protein product.

This is due to the concept of conservative or homologous replacement of amino acids. Asparagine and glutamine are both uncharged polar (hydrophilic) amino acids. Thus, when glutamine is replaced with asparagine there is no change in the polarity or the charge of the peptide/protein.

Solution to Question 15:

Proteoglycans can hold large amounts of water, and not less.

Proteoglycans are made of a central protein core with O-linked glycosaminoglycan side-chains. They hold large quantities of water and occupy space.

They provide the ground substance of connective tissue.

Chondroitin sulfate is a glycosaminoglycan that is bound to a protein to form chondroitin sulfate proteoglycan. It is present in the extracellular matrix. Hyaluronic acid and heparin are examples of glycosaminoglycans.

All of the other statements regarding proteoglycans are true.

Solution to Question 16:

Among the given options, mucin has both a protective as well as a lubricating properties.

Solution to Question 17:

Phenylbutyrate can be used to reduce ammonia levels in patients with urea cycle disorders.

Phenylbutyrate diverts urea into an alternate pathway and reduces the amount of ammonia. Hence, acting as an ammonia scavenger. Other drugs that can be used to achieve the same outcome (ammonia scavengers) include phenylacetate and benzoate.

Note: Arginine is used as first-line treatment as it provides ornithine; a positive regulator of the urea cycle. Arginine also activates NAG synthase which increases NAG; a positive regulator of the urea cycle

Solution to Question 18:

Restriction endonucleases can act on AAGCTT.

Restriction endonucleases act and cleave at the site of a palindromic sequence. A palindromic sequence is a sequence that reads the same backward and forward. An example is listed below.

5' --AAGCTT-- 3'

3' --TTCGAA-- 5'

Biochemistry AIIMS 2020 (May and Nov INI-CET)

Question 1:

Dicoumarol inhibits the action of vitamin K by?

- a) Suicide inhibition
- b) Competitive inhibition
- c) Non competitive inhibition
- d) Uncompetitive inhibition

Question 2:

A TPN bag consists of 100g glucose, 30g amino acids, and 40g lipids. What is the amount of calories delivered?

- a) 840 KCal
- b) 880 KCal
- c) 640 KCal
- d) 680 KCal

Question 3:

What is true about the Hayflick limit?

- a) Human cells can divide a finite number of times only
- b) Free radical oxidation level is unrestricted
- c) Telomerase activity is unrestricted
- d) RNA polymerase can function for. a finite number of times only

Question 4:

Ammonia in the brain is detoxified to form _____

- a) Urea
- b) Glutamine

- c) Ammonium ions
- d) Alanine

Question 5:

The sequence of vitamin D synthesis is?

- a) Liver- Skin- Kidney
- b) Skin- Liver- Kidney
- c) Skin- Kidney- Liver
- d) Liver- Kidney- Skin

Question 6:

Glucose is stored in the form of glycogen because _____

- a) It is compact
- b) It has multiple reducing ends
- c) It can be stored at multiple organs
- d) It can provide glucose for about 1 week when needed

Question 7:

DNA methylation is not related to?

- a) Mismatch repair
- b) Chromatin Remodelling
- c) DNA replication
- d) Gene silencing

Question 8:

The enzyme that is not used in cholesterol synthesis?

- a) Thiolase
- b) HMG CoA reductase
- c) HMG CoA synthase

d) HMG CoA lyase

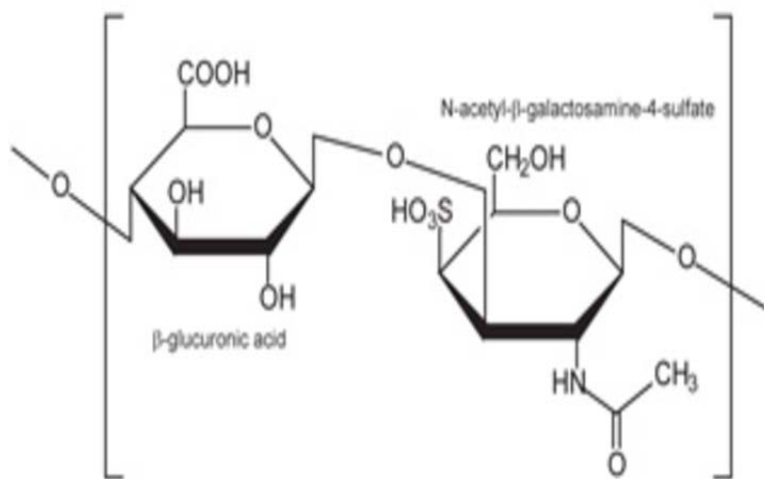
Question 9:

The enzyme that is common for both glycolysis and gluconeogenesis is?

- a) Pyruvate kinase
- b) Phosphofructokinase
- c) Phosphoenolpyruvate carboxykinase
- d) Phosphoglycerate kinase

Question 10:

The given chemical structure is of:



- a) Keratan sulfate
- b) Chondroitin sulfate
- c) Heparin
- d) Dermatan sulfate

Question 11:

The correct order of steps in Polymerase Chain Reaction (PCR) is:

- a) Hybridisation, Annealing, Elongation, Denaturation

- b) Elongation, Denaturation, Hybridisation, Annealing
- c) Denaturation, Annealing, Extension
- d) Annealing, Extension, Denaturation

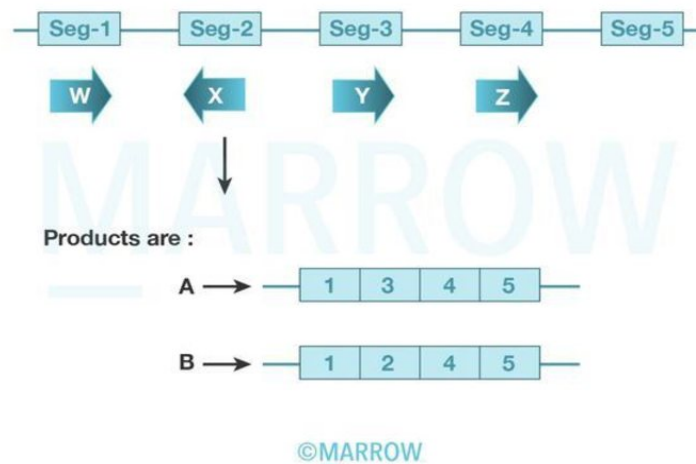
Question 12:

Which of the following is not a true statement regarding telomerase?

- a) Has reverse transcriptase activity
- b) Present only in eukaryotes
- c) Maintain the chromosome length
- d) Involved in DNA repair

Question 13:

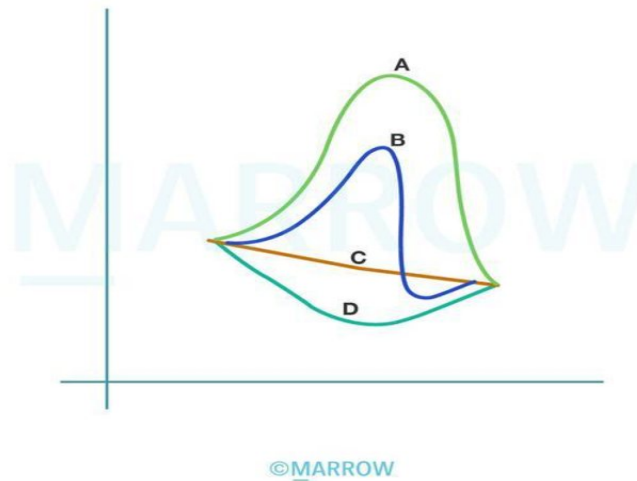
The illustration below shows alternative splicing of a gene which has intron segments 1, 2, 3, 4, and 5. It yields two products A and B, by the action of different promoters. You are provided with the promoters w, x, y, and z, each of which binds to a different segment as shown below and acts in the marked direction. Which of the following promoters will you use to get product A?



- a) W,X,Y
- b) W,Z
- c) W,X
- d) W, Y

Question 14:

Gibbs free energy curve of an enzyme-catalyzed reaction is given by



- a) B
- b) A
- c) C
- d) D

Question 15:

The maximum thermic effect is seen with a meal which has

- a) Protein
- b) Fat
- c) Carbohydrates
- d) Mixed meal having all 3 in equal amount

Question 16:

DNA methylation is not involved in?

- a) 1 and 2
- b) 2 and 3

- c) 1 and 4
- d) 2 and 4

Question 17:

Which of the following enzymes has the highest catalytic efficiency?

- a) K_m of an enzyme = 10 micromole and K_{cat} = 20 per sec
- b) K_m of an enzyme = 2000 nanomole and K_{cat} = 50 per sec
- c) K_m of an enzyme = 2 micromole and K_{cat} = 200 per sec
- d) K_m of an enzyme = 4 micromole and K_{cat} = 200 per sec

Question 18:

Which of the following are vitamin-B12 dependent enzymes?

- a) 1, 2, 3 and 4
- b) 1 and 4 only
- c) 1 and 3 only
- d) 1, 2, and 3

Question 19:

Which of the following enzymes are involved in gluconeogenesis? A. Pyruvate carboxylase B. Phosphoenol pyruvate kinase C. Hexokinase D. Pyruvate kinase E. Glucose 6 phosphatase

- a) A,B,and E only
- b) B and C only
- c) A and E only
- d) A, B and C only

Question 20:

Ketosis is common in type 1 DM due to:

- a) 2,3 and 4
- b) 1,2,3 and 4

- c) 4 only
- d) 1 and 2 only

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	b
5	b
6	a
7	c
8	d
9	d
10	b
11	c
12	d
13	d
14	b
15	a
16	d
17	c
18	b
19	c
20	a

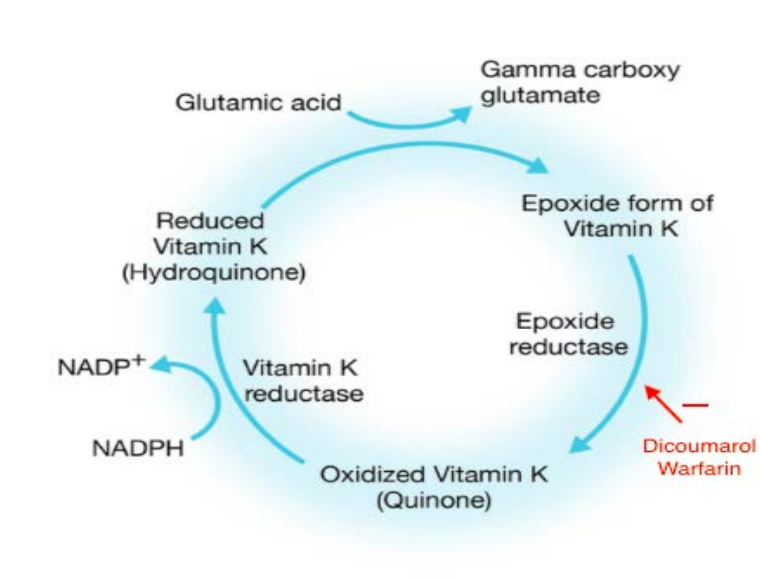
Detailed Explanations

Solution to Question 1:

Dicoumarol inhibits the action of vitamin K by competitive inhibition.

Dicoumarol and warfarin competitively inhibit the gamma-carboxylation system due to structural similarity with vitamin K. Therefore, they can act as anticoagulants used for therapeutic purposes.

Vitamin K epoxide is reduced to the quinone form by epoxide reductase, which is sensitive to inhibition by warfarin and dicoumarol.



Solution to Question 2:

The amount of calories delivered by the given TPN (Total Parenteral Nutrition) bag is 880 KCal.

The caloric value of different food components are:

- Carbohydrate (glucose) - 4KCal/g
- Proteins (amino acids) - 4.2KCal/g = ~4KCal/g
- Fat (lipids) - 9KCal/g

Calculation of the number of calories:

- 100 gm of glucose = 100 x 4KCal=400 KCal
- 30 gm of amino acids = 30 x 4 KCal= 120 KCal
- 40 gm of lipids = 40 x 9 KCal= 360 KCal

Total amount of calories delivered= 400+120+360=880 Kcal

Note: The caloric value of alcohol is 7KCal/g.

Solution to Question 3:

The Hayflick limit says that human cells can divide for a finite number of times.

The Hayflick limit is the number of divisions a cell population can go through in culture before division stops and enters a state known as senescence. With the exception of stem cells, all the other human cells have only a limited capacity for cell division. This biological clock is of great help in the study of the normal ageing process.

The number of cell divisions of the cultured cell is roughly related to:

- The age of the cell donors
- The longevity of the species

For example, fibroblasts from the human embryos, when sustained in tissue culture, divide about 50 times before they die. Those taken from person after birth divide only 20 to 30 times.

Solution to Question 4:

Ammonia in the brain is detoxified to form glutamine.

Glutamine synthetase combines ammonia with glutamate to form glutamine. It requires the hydrolysis of ATP.

Deficiency of glutamine synthetase in neonates results in severe brain damage, multiorgan failure, and death.

Note: Glutamine is the transport form of ammonia from the brain and intestine to the liver; while alanine is the transport form from the muscle.

Solution to Question 5:

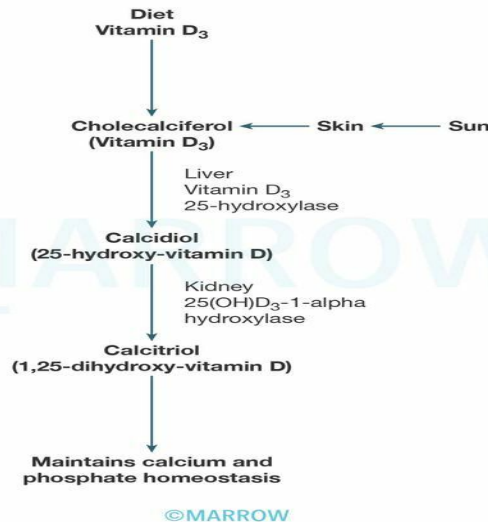
The sequence of vitamin D synthesis is skin-liver-kidney.

The sequence of vitamin D synthesis:

Step 1: In the skin: 7-Dehydrocholesterol (an intermediate in cholesterol synthesis that accumulates in the skin) undergoes a nonenzymic reaction on exposure to ultraviolet light, yielding cholecalciferol, which is absorbed into the bloodstream.

Step 2: In the liver: Cholecalciferol is hydroxylated to 25-hydroxycholecalciferol (calcidiol). This is released into the circulation bound to a vitamin D-binding globulin, which is the main storage form of the vitamin.

Step 3: In the kidney: 25-hydroxycholecalciferol is converted to either 1,25 dihydroxycholecalciferol (calcitriol), the biologically active form by the enzyme 1α -hydroxylase.



Solution to Question 6:

Among the given options, the most appropriate answer would be option A i.e Glucose is stored in the form of glycogen because it is compact.

Glycogen is the major storage carbohydrate in animals, corresponding to starch in plants. It is a branched polymer of α -d-glucose.

Other options:

The major sites of storage are liver and muscle, that is, it can't be stored at multiple sites.

It does not have multiple reducing ends. It instead has multiple non-reducing ends.

Liver glycogen functions as a reserve to maintain the blood glucose concentration in the fasting state. After about 18 hours of fasting, most of the liver glycogen is depleted and hence it can not provide glucose for up to 1 week.

Solution to Question 7:

DNA methylation is not related to DNA replication as DNA methylation is stable during replication and cell cycle arrest.

Chromatin remodeling by methylation/histone acetylation is an important aspect of transcription regulation.

Methylation of genes also plays an important part in gene silencing. Ex:- Promoter hypermethylation in tumor suppressor genes contributes to the development of cancers.

The process of mismatch repair relies upon a DNA-methylation excision signal, especially in *E. coli*.

Solution to Question 8:

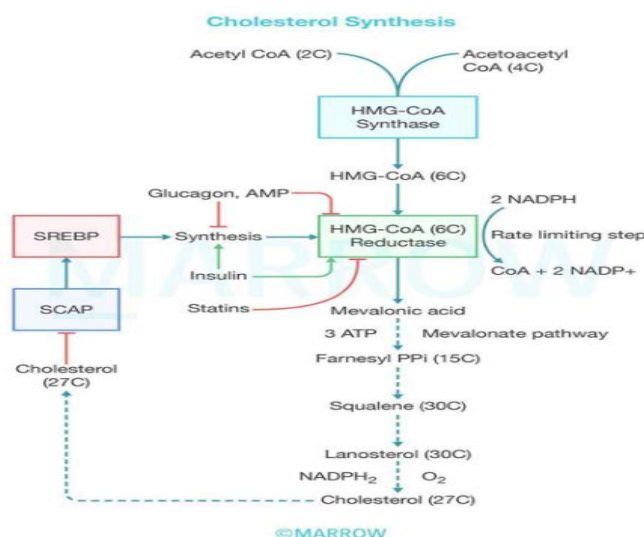
Among the given options, the enzyme HMG CoA lyase is not used in cholesterol synthesis. It is a mitochondrial enzyme involved in the breakdown of proteins and fats.

Cholesterol is a 27-carbon compound consisting of four rings and a side chain. It is synthesized from acetyl-CoA.

The major sites of synthesis of cholesterol are the liver, adrenal cortex, testes, ovaries, and intestine. All nucleated cells can synthesize cholesterol, including arterial walls. The enzymes involved in the synthesis of cholesterol are partly located in the endoplasmic reticulum and partly in the cytoplasm.

Initially, two molecules of acetyl-CoA condense to form acetoacetyl-CoA catalyzed by cytosolic thiolase. This acetoacetyl-CoA enters the cholesterol synthesis pathway along with a molecule of acetyl-CoA.

The image below shows the enzymes involved in cholesterol synthesis:



Solution to Question 9:

The enzyme phosphoglycerate kinase is common for both glycolysis and gluconeogenesis. Phosphoglycerate kinase catalyzes the conversion of 1,3-bisphosphoglycerate to 3-phosphoglycerate, which is a reversible step involved in both the processes.

Glycolysis and gluconeogenesis are processes that are reverse of each other. All the reversible steps of these two processes are catalyzed by the same enzymes, therefore are common to both.

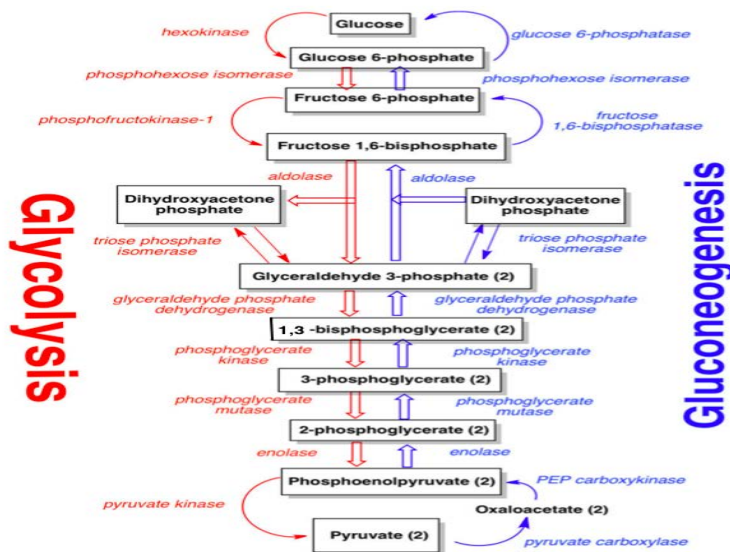
The irreversible steps of glycolysis are catalyzed by the following enzymes:

- Hexokinase

- Phosphofructokinase
- Pyruvate kinase

The irreversible steps of gluconeogenesis are catalyzed by the following enzymes:

- Pyruvate carboxylase
- Phosphoenolpyruvate carboxykinase
- Fructose-1,6-bisphosphatase
- Glucose-6-phosphatase



Solution to Question 10:

The given chemical structure is of chondroitin sulfate.

Chondroitin sulfate is composed of repeating units of N-acetyl galactosamine with sulfate and glucuronic acid. Some important features of chondroitin sulfate are as follows:

- The most abundant GAG in the body.
- Location: Cartilage, CNS, bone
- They are located at sites of calcification in endochondral bone and are a major component of cartilage.
- They are found in high amounts in the ECM of the central nervous system.

Solution to Question 11:

The correct sequence of events for PCR among the given options is denaturation, annealing, and extension.

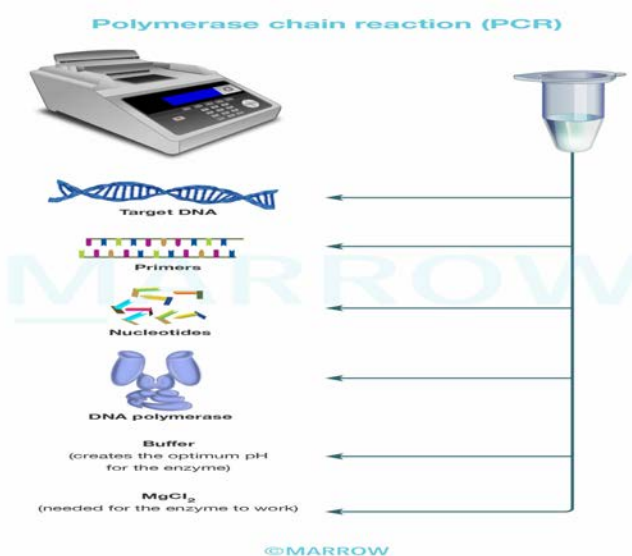
The process of PCR is as follows:

The DNA sample is first heat-denatured at $\approx 90^{\circ}\text{C}$ (90-96 degrees Celsius for 3 minutes) to separate the two strands of the template DNA containing the target sequence; the primers (which attach to the 3' end of the flanking sequence) are added in excess, are allowed to anneal to the DNA (typically at $50-75^{\circ}\text{C}$) in order to generate the required template-primer hybridization complex. Under suitable conditions and presence of deoxynucleotides, elongation of the DNA strand occurs.

The number of copies produced in the presence of an unlimited supply of deoxynucleotides and enzymes is given by 2^n (where n is the number of PCR cycles)

- 20 cycles provide an amplification of $2^{20} = 1,048,576$ or 1 million copies
- 30 cycles provides an amplification of $2^{30} = 1,073,741,824$ or 1 billion copies

Below is an image showing the various ingredients for carrying out PCR.



Solution to Question 12:

The false statement is that telomerase is involved in DNA repair.

Telomeres are the end of chromosomal DNA. Since prokaryotic DNA is circular, it does not have free ends and hence no telomeres. Thus, telomeres are present only in eukaryotes. Telomeres have TG-rich repeats and usually do not code for any proteins. With every DNA replication during cell division, telomere shortening occurs. Once telomeres are almost negligible or non-existent in length, the DNA is unable to replicate and the cell dies. Hence, there is a limit to the number of times a cell can divide.

In stem cells and cancer cells, an enzyme telomerase having reverse transcriptase activity (RNA dependent DNA polymerase) adds telomeres to the chromosome ends, thereby preserving chromosomal length. It allows the stem cells and cancer cells to multiply a lot more number of times than a normal adult cell.

Solution to Question 13:

Among the given options, using primers W and Y will yield the desired product A.

Promoters or DNA regulatory elements usually precedes the desired exon to be transcribed. It helps redirect transcription of only the desired exons of a gene to yield a product. Alternative splicing utilizes different promoters to code slightly different products from the same gene.

In this question, the desired product A has segments 1,3,4,5 whilst the other product B has 1,2,4,5. The difference between both is the presence of seg 3 in product A and the presence of seg 2 in product B. As the product in question is A, only seg 3 is required while excluding seg 2 and ensuring seg 1,4 and 5 intact.

- Seg 1- Needs to be transcribed and so W is essential
- Seg 2- Does not need to be transcribed.
- Seg 3- Needs to be transcribed and so Y is essential
- Seg 4- Needs to be transcribed but Z is not essential as Y continues the transcription in the direction shown by the arrow.

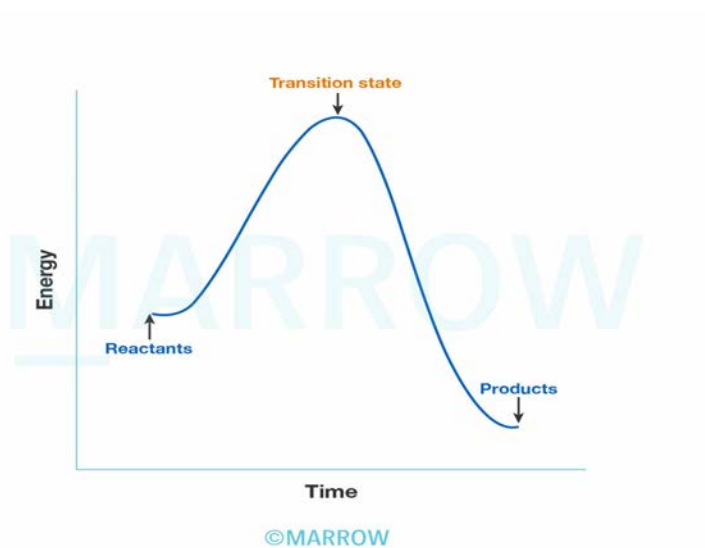
The answer is hence promoter W and Y.

Solution to Question 14:

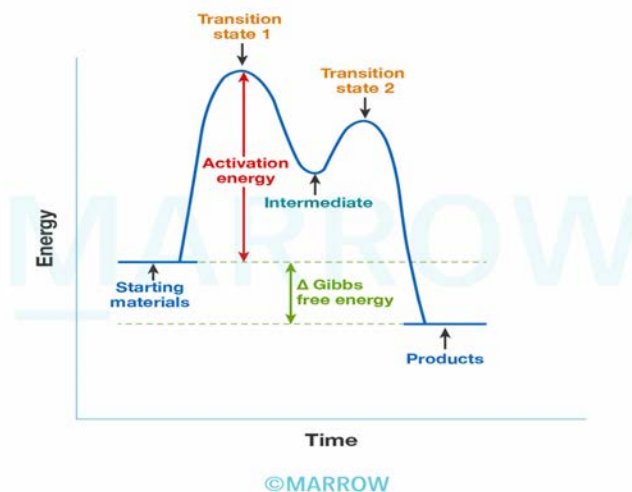
Among the given graphs, the Gibbs free energy change of an enzyme-catalyzed reaction is shown by graph A (Option B).

Graph B (Option A) cannot represent the Gibbs free energy of an enzyme-catalyzed reaction, as the energy state of an intermediate cannot be lower than the product. A lower energy state of the intermediate implies that the enzyme-intermediate complex is stable and such a complex will usually not further form a product.

During any chemical reaction, the product is formed by the breaking of substrate bonds and the conformational change of the substrate into a highly unstable transition state. This transition state is denoted by the peak of the graph. The Gibbs free energy curve of an enzyme-catalyzed reaction (with no intermediate state) is shown in the image given below:

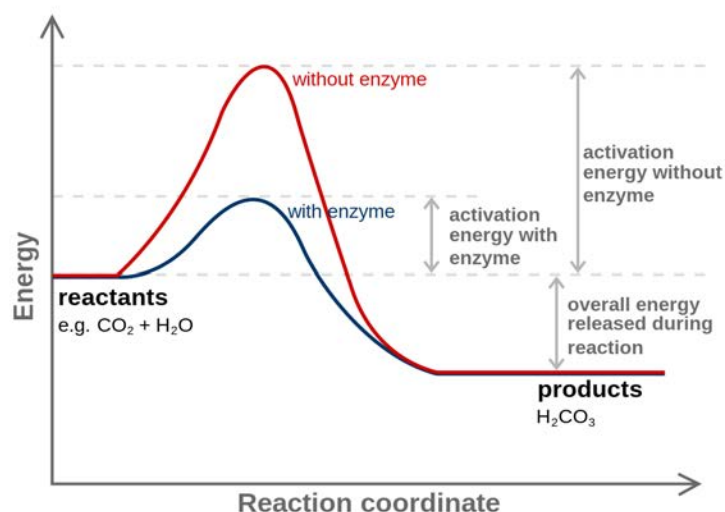


The below graph shows the normal intermediate formation with an energy state still higher than that of the product:



All the reactions have a transition state, but not all reactions have an intermediate state. Every intermediate state is again always followed by another transition state.

Activation energy is the energy difference between the free energy of the substrates and that of the transition state. Because of the high activation energy, the rates of uncatalyzed chemical reactions are often slow.



In an enzyme-catalyzed reaction, the activation energy is reduced by forming a relatively stable intermediate-enzyme complex. Enzymes provide attachment sites for the substrate and speed up the rate of reaction. They don't change the energy state of either the substrate or the product. The Gibbs free energy change remains the same.

Graph C: Free energy change cannot be a straight line. Every reaction passes through a high-energy intermediate transition state.

Graph D: The activation energy of any reaction is never lower than the free energy of the product.

Solution to Question 15:

Among the given options, maximum thermic effect is seen with proteins

Specific dynamic allowance (previously called as thermic effect of food): This refers to the increased heat production or increased metabolic rate following the intake of food (diet-induced thermogenesis).

The values of SDA are:

- 30% for proteins: This means that out of every 100 grams of proteins consumed, the energy available for doing useful work is 30% less than the calculated value.
- 15% for lipids, and
- 5% for carbohydrates.

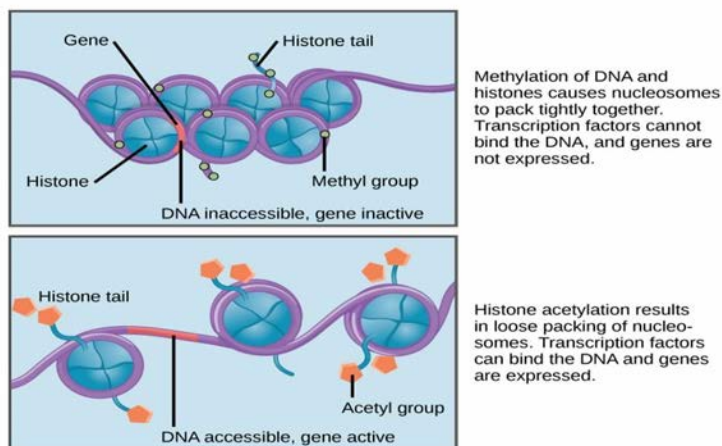
Solution to Question 16:

DNA methylation is not involved in the base pair excision repair mechanism (statement 4) and RNA splicing (statement 2).

DNA methylation is an epigenetic modification used as a gene expression regulation mechanism. It causes gene silencing by inactivating DNA and hindering the formation of euchromatin (transcriptionally active chromatin). Thus, it also aids in chromatin remodelling.

It however is not involved in the base pair repair mechanism or RNA splicing.

Methylation and acetylation of histones



Solution to Question 17:

The highest catalytic efficiency is seen when $K_m = 2$ micromole and $K_{cat} = 200$ per sec.

Catalytic efficiency is calculated as K_{cat}/K_m , where K_m is inversely proportional to the binding affinity of the enzyme and K_{cat} is the catalytic constant (V_{max} divided by the total number of active sites of the enzyme present). Higher values of K_{cat} and lower values of K_m gives better catalytic efficiency. Catalytic efficiency can be used to compare the performance of 2 different enzymes in catalyzing a reaction.

Step 1: K_m of each option after converting respective units to moles will be 10, 2, 2 and 4 (each multiplied by 10^{-6}) moles.

Step 2: The catalytic efficiency is calculated as K_{cat}/K_m . On substituting the values in various options after converting to moles, the values come out to be 2, 25, 100 and 50 (each multiplied by 10^6) $M^{-1} s^{-1}$.

Some enzymes can be so efficient that they convert the substrate to product instantaneously. They are limited only by the rate at which substrate diffuses towards the enzyme in a solution. These are called diffusion-limited enzymes and their catalytic efficiency is as high as $10^8 M^{-1} s^{-1}$.

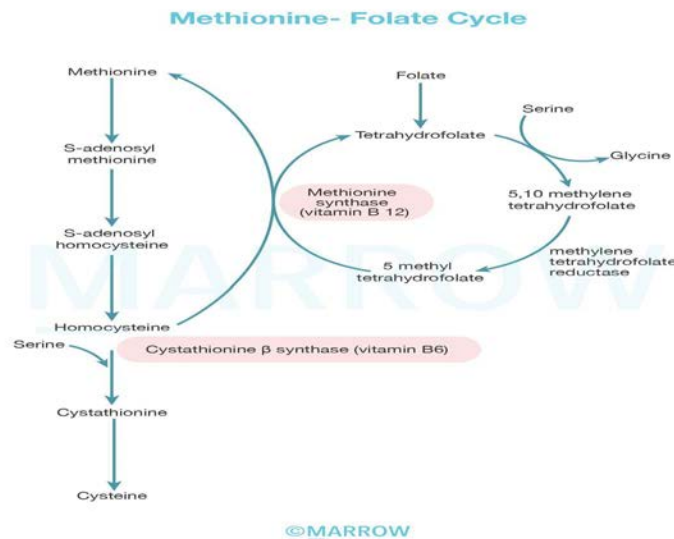
Solution to Question 18:

Among the given enzymes, methyl malonyl CoA isomerase and methionine synthase are vitamin B12 dependent enzymes.

There are two vitamin B12-dependent enzymes:

- Methionine synthase (homocysteine methyltransferase): Catalyzes the conversion of homocysteine to methionine
- Methylmalonyl CoA mutase: methylmalonyl CoA, formed as an intermediate in the catabolism of valine, isoleucine, and propionyl CoA is converted to succinyl CoA

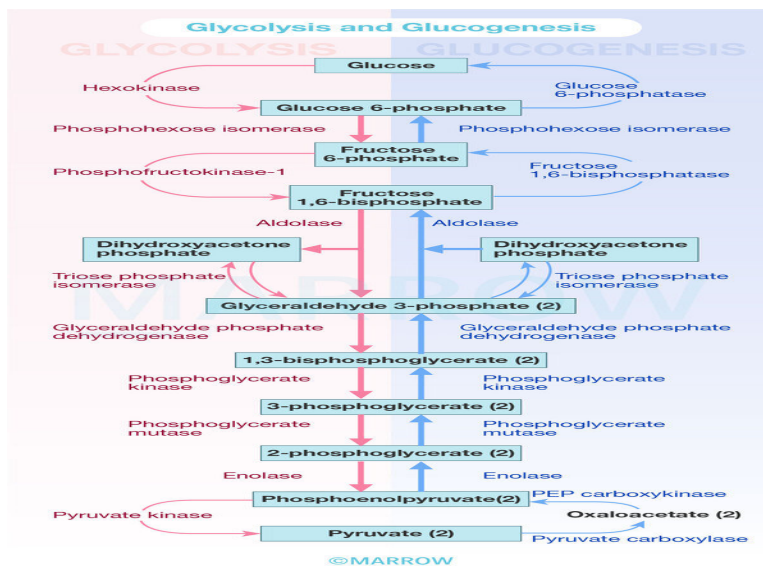
The below figure shows B12 serving as a cofactor of methionine synthase. Its deficiency may cause a folate trap.



Solution to Question 19:

Among the given options, A and E are correct. Pyruvate carboxylase and Glucose-6-phosphatase are enzymes used in gluconeogenesis.

Glycolysis and gluconeogenesis are 2 metabolic pathways in opposite directions (glucose being the substrate and product respectively). But a few reactions are not reversible in glycolysis. So a different pathway is used or a different enzyme is used by the body to make the reactions feasible.



Solution to Question 20:

2, 3 and 4 statements correctly explain the reason behind increased ketogenesis as seen in type 1 DM.

Type 1 DM is characterized by a deficiency in insulin production. Since insulin is deficient, GLUT4 (insulin-dependent uptake) mediated glucose uptake is not possible in the heart or skeletal muscles. So adipose tissue mobilizes fat to be used as an energy source. These FFA (free fatty acids) are oxidized to acetoacetyl CoA which serves as a precursor for ketone body synthesis.

As the level of serum FFA is raised, proportionately more of the acetyl-CoA produced from their breakdown is converted to ketone bodies and less is oxidized via the citric acid cycle to CO₂. A fall in the concentration of oxaloacetate, particularly within the mitochondria, can impair the ability of the citric acid cycle to metabolize acetyl-CoA and divert fatty acid oxidation toward ketogenesis. Such a fall may occur because of an increase in the (NADH)/(NAD⁺) ratio caused when increased β-oxidation alters the equilibrium between oxaloacetate and malate so that the concentration of oxaloacetate is decreased.

Beta-hydroxy butyrate dehydrogenase is one of the enzymes used in ketone body synthesis. Increased rate of ketone body synthesis in insulin-deficient state implies an increased activity of beta-hydroxy butyrate dehydrogenase enzyme.

Biochemistry NEET 2018

Question 1:

Which of the following is responsible for the negative charge in fibrinopeptide A?

- a) Glutamate and Valine
- b) Histidine and Lysine
- c) Aspartate and glutamate
- d) Serine and Threonine

Question 2:

Serotonin is also known as?

- a) 5 hydroxytryptamine
- b) 5 hydroxytryptophan
- c) 5 carboxytryptamine
- d) 5 carboxytryptophan

Question 3:

Which of the following is true about DNA polymerase I?

- a) It is involved in the creation of okazaki fragments.
- b) It is found in prokaryotes.
- c) It synthesises RNA primers.
- d) It is the primary enzyme of DNA synthesis.

Question 4:

A patient presents with recurrent kidney stones. Microscopic examination finding of the urine specimen is shown below. Which of the following is not seen in the urine of this patient?



- a) Ornithine
- b) Lysine
- c) Arginine
- d) Histidine

Question 5:

In which condition is excessive urinary excretion of 5- hydroxy indole acetic acid seen?

- a) Alkaptonuria
- b) Albinism
- c) Carcinoid syndrome
- d) Phenylketonuria

Question 6:

Which of the following should be avoided by a patient of fish odour syndrome?

- a) Biotin
- b) Choline
- c) Niacin
- d) Pantothenic acid

Question 7:

Which of the following is true regarding an allosteric modifier?

- a) Desaturates the enzyme
- b) Causes the enzyme to work faster only.
- c) Binds to the catalytic site
- d) Participates in feedback regulation

Question 8:

Which of the following is the most common end-product of fatty acid synthesis pathway?

- a) Arachidonic acid
- b) Oleic acid
- c) Palmitic acid
- d) Acetyl CoA

Question 9:

Vanillyl mandelic acid is excreted in urine in which of the following conditions?

- a) Pheochromocytoma
- b) Alkaptonuria
- c) Carcinoid syndrome
- d) Diabetic ketoacidosis

Question 10:

Which of the following statements is true about vitamin K?

- a) Deficiency of vitamin K leads to hypercoagulability.
- b) It is a water soluble vitamin.
- c) Prolonged use of antibiotics leads to vitamin K deficiency.
- d) Menaquinone is the dietary source of vitamin K found in green vegetables.

Question 11:

What do chaperones assist in?

- a) Ubiquitination of proteins
- b) Protein modification
- c) Protein folding
- d) Protein cleavage

Question 12:

Defective fumarylacetoacetate hydrolase enzyme is associated with _____.

- a) Type 1 Tyrosinemia
- b) Type 2 Tyrosinemia
- c) Type 3 Tyrosinemia
- d) Type 4 Tyrosinemia

Question 13:

A newborn baby refuses breast milk since the 2nd day of birth but accepts glucose water. He develops vomiting and severe jaundice by the 5th day. Benedict's test was positive for urine and blood glucose was low. The most likely cause is due to the deficiency of

- a) Aldose reductase
- b) Galactokinase
- c) Galactose-1-phosphate uridyl transferase
- d) UDP galactose-4-epimerase

Question 14:

A 45-year-old woman with hypertriglyceridemia is considered for niacin therapy. She is a known diabetic managed adequately with sugars well under control. Why should niacin be used cautiously in this patient?

- a) It can cause hyperglycemia.
- b) It can cause scleroderma which makes injecting insulin difficult.
- c) It can increase the metabolism of oral hypoglycemic drugs.
- d) It can cause hypoglycemia.

Question 15:

Which of the following can alter the gene expression by methylation and acetylation but not change the gene sequence?

- a) Mutation
- b) Epigenetics
- c) Translocation
- d) Inversion

Question 16:

The flipped LDH ratio implies _____.

- a) LDH 1 > LDH 2
- b) LDH2 > LDH1
- c) LDH 2 > LDH 3
- d) LDH 3 > LDH 2

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	d
5	c
6	b
7	d
8	c
9	a
10	c
11	c
12	a
13	c
14	a

15	b
16	a

Detailed Explanations

Solution to Question 1:

Aspartate and glutamate are responsible for the negative charge in fibrinopeptide A and B of fibrinogen.

The negative charges contribute to the solubility of fibrinogen in plasma and importantly also serve to prevent aggregation by causing electrostatic repulsion between fibrinogen molecules.

Solution to Question 2:

Serotonin is also known as 5 hydroxy-tryptamine, and it is a monoamine neurotransmitter. Serotonin is produced by hydroxylation of tryptophan and subsequent decarboxylation. It is a potent vasoconstrictor and stimulator of smooth muscle contraction.

Clinical correlation:

In carcinoid syndrome, excessive levels of serotonin is produced which causes flushing, diarrhea, wheezing, damage to the tricuspid valve, etc. The levels of 5-OH indole acetic acid (serotonin metabolite) is elevated in the urine.

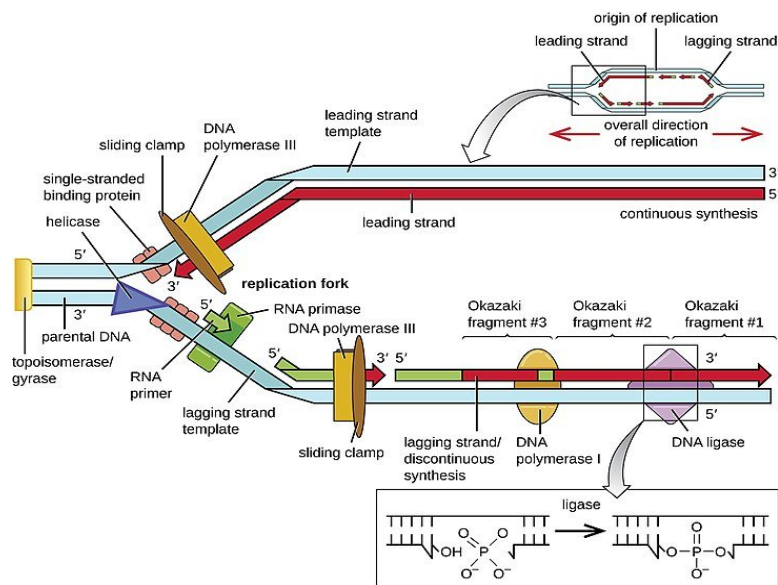
Solution to Question 3:

The true statement is that DNA polymerase I (pol I) is found in prokaryotes.

The prokaryotic polymerases are DNA polymerases I, II, and III. DNA polymerase I recognize a “nick” or break in the phosphate backbone, and then removes each RNA primer and fills the gaps with DNA. It does this through its 5'→3' exonuclease activity. It has DNA repair activity through 3'→5' exonuclease activity. DNA polymerase II carries out DNA proofreading and repair. DNA polymerase III is the primary enzyme for the synthesis of the leading strand of DNA.

Option A: DNA polymerase δ (eukaryotic polymerase) synthesizes Okazaki fragments which are short, newly synthesized DNA fragments that are formed on the lagging template strand during DNA replication. They are complementary to the lagging template strand.

Option C: Primase synthesizes RNA primers to start the replication.



Solution to Question 4:

The image shows hexagonal, translucent and white crystals suggestive of cystinuria. Histidine is not seen in the urine of this patient.

Cystinuria is an autosomal-recessive disease due to a defect in renal PCT & intestinal amino acid transporter that is essential for reabsorptive transport of Cystine and the dibasic amino acids Ornithine, Lysine, and Arginine (COLA).

Diagnosis of cystinuria is by the cyanide–nitroprusside test. Initial treatment is with adequate hydration, alkalization of the urine with citrate supplementation or acetazolamide, and dietary modification to reduce salt and protein intake. If this fails then patients are usually started on chelation therapy with penicillamine.

Solution to Question 5:

Excessive urinary excretion of 5-hydroxy indole acetic acid (5-HIAA) is seen in carcinoid syndrome.

Carcinoid syndrome is marked by symptoms of bronchospasm, cutaneous flushing, abdominal cramps, and diarrhea. It occurs due to excessive production of serotonin, a specialized product of tryptophan metabolism by tumors commonly in the small intestine, liver and lungs. It is then metabolized and excreted in urine as 5-HIAA.

Solution to Question 6:

A fishy odor is seen in trimethylaminuria. Consumption of food products with choline should be avoided in fish odor syndrome.

Fish odor syndrome (trimethylaminuria) is an autosomal recessive disease where TMA(trimethylamine) is not converted to the non-odorous TMAO (TMA-N-oxide) in the liver. This is due to the deficiency of flavin-containing mono oxygenase 3 (FMO3). TMA then builds up in the body and is released in sweat, urine, and breath, giving off a strong odor of rotting fish from the body.

Choline, lecithin, and TMAO are precursors of trimethylamine which are found in food products. These food products are eggs, liver, peanuts, meat, poultry, fish, dairy foods, pasta, and rice.

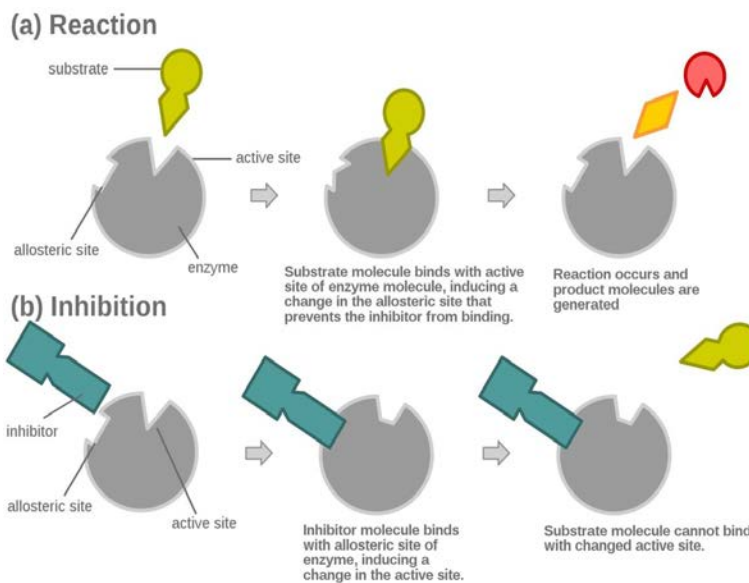
Solution to Question 7:

An allosteric modifier of an enzyme usually participates in feedback regulation.

Allosteric regulations are a natural example of control loops, such as feedback from downstream products or feedforward from upstream substrates. Allosteric regulation is also particularly important in the cell's ability to adjust enzyme activity.

An allosteric modifier or effector binds to the allosteric site to induce a conformational change in the catalytic or active site. Those modifiers or effectors that enhance the protein's activity are referred to as allosteric activators, whereas those that decrease the protein's activity are called allosteric inhibitors.

Some examples of allosteric modifiers		
Enzyme	Allosteric inhibitor	Allosteric activator
1. ALA synthase	Heme	-
2. Aspartate transcarbamoylase	CTP	ATP
3. HMG CoA reductase	Cholesterol	-
4. Phosphofructokinase	Citrate, ATP	AMP, F2,6P
5. Acetyl CoA carboxylase	Acyl CoA	Citrate
6. Citrate synthase	ATP	-
7. Carbamoyl phosphate synthase-I	-	NAG
8. Carbamoyl phosphate synthase-II	ATP	-



Solution to Question 8:

The most common end product of fatty acid synthesis is palmitic acid.

Palmitic acid is a 16 carbon saturated long chain fatty acid (LCFA). The substrate for its synthesis is acetyl CoA. Acetyl CoA is also the end product of fatty acid beta-oxidation.

Solution to Question 9:

- Vanillyl Mandelic Acid is the end product of catabolism of catecholamines and is excreted in urine in Pheochromocytoma.
- VMA is the urinary product of both epinephrine and norepinephrine.
- In pheochromocytoma and neuroblastoma, there is an excessive synthesis of catecholamines which causes enhanced synthesis of VMA and its excretion in the urine.
- It is a good screening test for pheochromocytoma and is also used to diagnose and follow up neuroblastoma and ganglioneuroma.

Solution to Question 10:

Prolonged use of antibiotics leads to vitamin K deficiency. This is due to the loss of normal bowel flora that synthesizes menaquinones (vitamin K₂).

Vitamin K is a fat-soluble vitamin. The following are the vitamers of vitamin K:

- Phylloquinone (Vitamin K₁) is the normal dietary source of Vitamin K that is found in green vegetables.

- Menaquinones (Vitamin K₂) are synthesized by intestinal bacteria.
- Menadione (Vitamin K₃) is a synthetic compound that is metabolized to phyloquinone.

Solution to Question 11:

Chaperones assist in protein folding.

Chaperones are heat shock proteins. They bind to the short sequences of hydrophobic amino acids of an extended polypeptide. Thus, they shield the new polypeptide being synthesized from aggregating. This provides the polypeptide an opportunity to form an appropriate secondary structure.

When a protein is thermodynamically trapped in a misfolded dead-end form, chaperones can rescue the protein by unfolding their hydrophobic ends. Thus, they provide the protein a second chance to fold productively.

Solution to Question 12:

Defective fumarylacetoacetate hydrolase enzyme is associated with type I tyrosinemia (tyrosinosis). It is the most common type of tyrosinemia.

There are three types of tyrosinemias:

- Type 1 (Hepatorenal tyrosinemia): Fumaryl acetoacetate hydrolase deficiency
- Type 2 (Oculocutaneous tyrosinemia/Richner-Hanhart syndrome): Tyrosine transaminase deficiency
- Type 3 (Neonatal tyrosinemia)

Solution to Question 13:

The above clinical features on ingestion of breastmilk at birth suggest classical galactosemia, a disorder of galactose metabolism. Galactose-1-phosphate uridyl transferase deficiency leads to classical galactosemia.

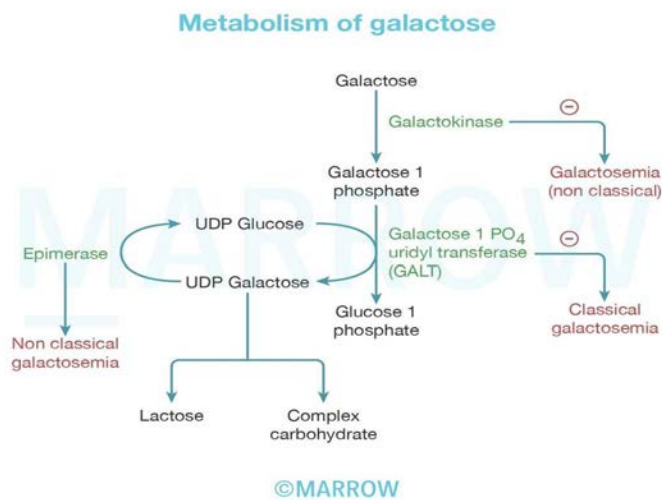
It is an autosomal recessive disorder. Clinical features include:

- Accumulation of galactose-1-phosphate in the liver causes liver damage & leads to hepatomegaly, vomiting, convulsions.
- Accumulation of galactose-1-phosphate depletes the liver of inorganic phosphate leading to diminished ATP synthesis which affects gluconeogenesis resulting in hypoglycemia.
- Accumulation of galactose-1-phosphate in the nerves causes intellectual disability.
- Accumulation of galactitol in the lens causes an oil-drop cataract.

Urinary reducing sugar (Benedict's test) and the Mucic acid test are positive.

Treatment is a galactose-free diet.

The flowchart given below depicts galactose metabolism.



Solution to Question 14:

Niacin should be used cautiously in diabetics as it can cause hyperglycemia.

Niacin induces insulin resistance and also decreases glucose-stimulated insulin secretion, which may result in severe hyperglycemia. When niacin is prescribed for patients with known or suspected diabetes, blood glucose levels should be monitored at least once a week.

Solution to Question 15:

Epigenetics is defined as heritable changes in gene expression that are not caused by alteration in gene sequence.

It means “above genetics” and refers to the fact that gene regulatory mechanisms do not change the underlying regulated DNA sequence, but rather simply the expression patterns of the DNA.

Epigenetic mechanisms play a key role in establishment, maintenance, and reversibility of transcriptional states. A key feature is that the controlled transcriptional on/off states can be maintained through multiple rounds of cell division. This observation indicates there must be robust mechanisms to maintain and stably propagate these epigenetic states.

Histone Methylation Mostly Makes DNA Mute.

Histone Acetylation: Relaxes DNA coiling, allowing for transcription. Histone Acetylation makes DNA Active.

Solution to Question 16:

The flipped lactate dehydrogenase (LDH) ratio is LDH 1 > LDH 2.

Under normal conditions, LDH 2 is predominantly present in the blood and LDH 1 is predominantly present in the heart. When cardiac tissue is damaged LDH 1 is released into the bloodstream.

LDH 1 > LDH 2 (flipped pattern) in the blood is suggestive of myocardial infarction.

LDH is a tetrameric enzyme with 4 subunits. It has four subunits with two isoforms: H isoform (heart) and M isoform (muscle)

LDH Type	Composition	Location
LDH-1	HHHH	Heart and RBC
LDH-2	HHHM	Heart and RBC
LDH-3	HHMM	Lungs, brain, lymphatic tissues, and platelets
LDH-4	HMMM	Liver and skeletal muscle
LDH-5	MMMM	Liver and skeletal muscle

Biochemistry NEET 2019

Question 1:

Which of the following diseases have autosomal recessive inheritance?

- a) Albinism
- b) G6PD deficiency
- c) Marfan's syndrome
- d) Lesch-Nyhan syndrome

Question 2:

Deficiency of which of the following causes increased glutamine in blood?

- a) Arginosuccinate lyase
- b) Alpha-galactosidase-A
- c) Ornithine transcarbamoylase
- d) Arginase

Question 3:

Which of the following amino acid is necessary for the conversion of nor-epinephrine to epinephrine

- a) Tyrosine
- b) Tryptophan
- c) Phenyl alanine
- d) Methionine

Question 4:

An infant presents with hypotonia and seizures. It was confirmed to be cerebrohepatorenal syndrome. Which of the following is accumulated in the brain in cerebrohepatorenal syndrome?

- a) Glucose

- b) Long chain fatty acid
- c) Lactic acid
- d) Triglycerides

Question 5:

Excess ammonia reduces which of the following substrates of the Krebs cycle?

- a) Malate
- b) Oxaloacetate
- c) Fumarate
- d) Alpha ketoglutarate

Question 6:

Which of the following is autosomal dominant?

- a) Cystic fibrosis
- b) Duchene muscular dystrophy
- c) Sickle cell anemia
- d) Achondroplasia

Question 7:

In Menke's disease, which of the following enzymes would be affected?

- a) Lysyl hydroxylase
- b) Lysyl oxidase
- c) Prolyl oxidase
- d) Prolyl hydroxylase

Question 8:

Common precursor amino acid for urea, creatine and nitric oxide is ____.

- a) Glycine
- b) Arginine

- c) Aspartate
- d) Alanine

Question 9:

True regarding mitochondrial genome:

- a) Maternal inheritance
- b) Low rate of mutation
- c) Single stranded DNA
- d) Paternal inheritance

Question 10:

In a patient with lipoprotein lipase deficiency, which of the following is increased following a fatty meal?

- a) Chylomicron
- b) LDL
- c) HDL
- d) Apo A

Question 11:

In a patient with Wilson's disease which of the following is decreased in urine?

- a) Tyrosine
- b) Phosphorous
- c) Serine
- d) 3-methylhistidine

Question 12:

NADPH is used in _____.

- a) Fatty acid synthesis
- b) Ketone synthesis

- c) Gluconeogenesis
- d) Glycolysis

Answer Key

Question No.	Correct Option
1	a
2	c
3	d
4	b
5	d
6	d
7	b
8	b
9	a
10	a
11	d
12	a

Detailed Explanations

Solution to Question 1:

Albinism is inherited in an autosomal recessive pattern.

Solution to Question 2:

Deficiency ornithine transcarbamoylase causes increased glutamine in blood

Ornithine transcarbamoylase catalyzes the condensation of carbamoyl phosphate with ornithine to form citrulline in the urea cycle. Its deficiency leads to the accumulation of carbamoyl phosphate. The carbamoyl phosphate is then shunted to pyrimidine biosynthesis, resulting in high ammonia and orotic acid levels in the blood. This condition is thus called hyperammonemia type 2.

As a compensatory mechanism, ammonia is then combined with glutamate to form glutamine (by the enzyme glutamine synthetase). The levels of glutamine are elevated in the blood, cerebrospinal

fluid, and urine, probably as a result of this dysregulated glutamine synthesis.

Other options:

Option A: Deficiency of argininosuccinate lyase leads to argininosuccinic aciduria

Option C: Deficiency of alpha-galactosidase-A leads to Fabry's disease and an accumulation of globotriaosylceramide.

Option D: Deficiency of arginase leads to hyperargininemia

Solution to Question 3:

Methionine is necessary for the conversion of nor-epinephrine to epinephrine.

Solution to Question 4:

Cerebrohepatorenal syndrome or Zellweger's syndrome is characterized by the accumulation of very-long-chain fatty acid, abnormalities of synthesis of bile acids and a marked reduction of plasmalogens.

It occurs in individuals with a rare inherited absence of peroxisomes. It is due to mutations in genes encoding certain proteins called peroxins (involved in various steps of peroxisome biogenesis) in all tissues. They accumulate C26-C38 polyenoic acids in brain tissues and also exhibit a generalized loss of peroxisomal functions. It causes severe neurological symptoms and most patients die in the first year of life.

Solution to Question 5:

Excess ammonia results in depletion of levels of alpha-ketoglutarate.

This happens because excess ammonia reacts with alpha-ketoglutarate to form glutamate, mainly in the CNS.

Solution to Question 6:

Achondroplasia is an autosomal dominant disorder.

It is caused by a mutation of the fibroblast growth factor receptor-3 (FGFR-3) gene.

The primary defect is abnormal chondrocyte proliferation at the growth plate that causes development of short and thick bones.

Clinical features:

- Short limbs

- Normal trunk
- Large head
- Saddle nose
- Exaggerated lumbar lordosis
- Severe spinal deformity may lead to cord compression



Solution to Question 7:

Menke's disease is due to defective lysyl oxidase.

It is an X-linked recessive disorder caused by a mutation in ATP7A, which encodes for a copper-transporting ATPase α polypeptide.

Solution to Question 8:

Arginine is the common precursor for urea, creatine and nitric oxide.

Solution to Question 9:

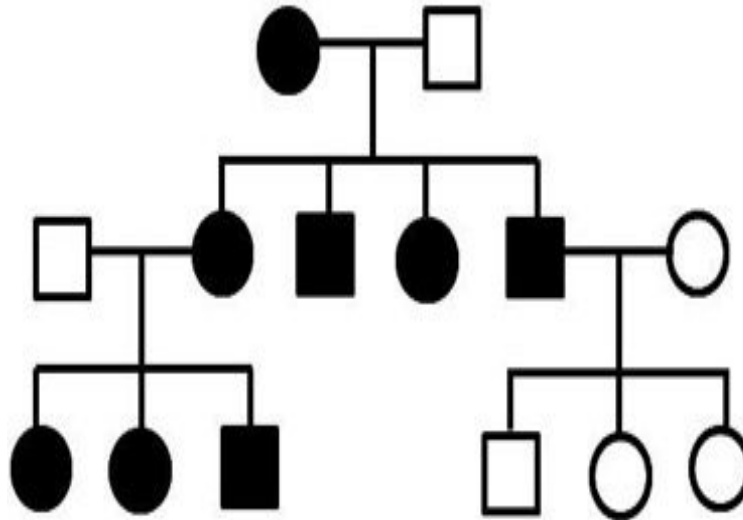
Mitochondrial genome is maternally inherited.

Mitochondrial inheritance:

- Inherited exclusively from the affected mother (ovum) as an individual's mitochondrial genome is entirely derived from the mother because sperms contain few mitochondria, which are typically shed upon fertilization.

- Both male and female children born to an affected mother will inherit the disease.

Pedigree shows mitochondrial inheritance.



Solution to Question 10:

After a fatty meal, chylomicrons are elevated in lipoprotein lipase deficiency.

Chylomicrons transport dietary lipids from the intestine to the peripheral tissues. They are synthesized in the intestine by assembling apo B-48 with lipids (mainly triacylglycerols). This is called a nascent chylomicron, which upon receiving apo C-II and apo E from HDL is remodelled into mature chylomicron.

Apo C-II activates lipoprotein lipase present on the endothelium. Lipoprotein lipase hydrolyzes the triacylglycerols into fatty acids and glycerol which are either used for storage or energy.

Therefore, patients who are deficient in lipoprotein lipase or apo C-II (type I hyperlipoproteinemia) show increased levels of chylomicrons in the plasma after a fatty meal.

Solution to Question 11:

Urinary levels of 3-methylhistidine are decreased in patients with Wilson disease.

Amino acids in urine exist either in conjugated form (most) or in free form. In Wilson's disease the amount of conjugated amino acids far exceeds the amount of free amino acids. Methyl Histidine largely exists in free form and does not increase in amount on hydrolysis of acids in urine. Hence low amounts is seen in Wilson's disease.

Wilson disease is an autosomal-recessive disorder caused by mutation of the ATP7B gene located on chromosome 13. The ATP7B gene, encodes a transmembrane copper-transporting ATPase, expressed on the hepatocyte canalicular membrane.

Deficiency in the ATP7B protein causes a decrease in copper excretion into the bile, impairs its incorporation into ceruloplasmin, and inhibits ceruloplasmin secretion into the blood, leading to the accumulation of toxic levels of copper in the liver, brain, and eye.

Solution to Question 12:

NADPH is the source of reducing equivalent for fatty acid synthesis.

This NADPH is derived from 3 sources:

- HMP shunt pathway (main source)
- Extramitochondrial isocitrate dehydrogenase (minor source)
- Malic enzyme

Biochemistry NEET 2020

Question 1:

What is the function of the proteasome?

- a) Protein folding
- b) Post-translational modification
- c) Protein degradation
- d) Protein sorting

Question 2:

Replacing alanine by which amino acid, will increase UV absorbance of protein at 280nm wavelength?

- a) Leucine
- b) Proline
- c) Arginine
- d) Tryptophan

Question 3:

Vitamin B12 is found maximum in:

- a) Animal products
- b) Green leafy vegetables
- c) Roots and tubers
- d) Vegetarian diet

Question 4:

The process of apolipoprotein B48 synthesis involves which of the following?

- a) RNA alternate splicing
- b) RNA editing

- c) DNA editing
- d) RNA interference

Question 5:

A patient presents with a clinical scenario of obstructive jaundice. Which of the following results can be seen on Van den bergh's reaction?

- a) Direct positive
- b) Indirect positive
- c) Both positive
- d) Coproporphyrin

Question 6:

If the percentage of thymine residues in DNA is 28%. What is the percentage of cytosine?

- a) 28%
- b) 44%
- c) 36%
- d) 22%

Question 7:

Enzyme activated by a decrease in insulin: glucagon ratio is:

- a) Glucokinase
- b) Hexokinase
- c) Phosphofructokinase
- d) Glucose 6-phosphatase

Question 8:

A patient presented with symptoms of peripheral neuropathy and macrocytic anemia. She was given folate 5mg and the blood picture improved. However, the neurologic manifestations were aggravated. What is the most likely cause?

- a) Malabsorption of folate

- b) Treatment with folate unmasking pyridoxine deficiency
- c) Deficiency of folate reductase in CNS
- d) Folate therapy using the B12 stores

Question 9:

Which of the following is true regarding non-competitive inhibition?

- a) Constant K_m and increased V_{max}
- b) Constant K_m and decreased V_{max}
- c) Decreased K_m and decreased V_{max}
- d) Decreased K_m and increased V_{max}

Question 10:

A patient came to the hospital with complaint of abdominal pain. Routine investigations were unremarkable. On observation, the urine turned black on standing. Which of the following enzymes is likely to be defective in this patient?

- a) Homogentisate oxidase
- b) Xanthine oxidase
- c) Phenylalanine hydroxylase
- d) Dihydro orotate dehydrogenase

Question 11:

A 34-year-old male was on maize as a staple diet for a prolonged period. He presented with diarrhoea and lesion over his neck as shown in the image below. The condition occurs due to deficiency of:



- a) Niacin
- b) Zinc
- c) Thiamine
- d) Iron

Question 12:

Which vitamin, when infused in supraphysiological doses, causes macular edema and macular cysts?

- a) Vitamin A
- b) Vitamin D
- c) Vitamin E
- d) Vitamin B3

Question 13:

Which of the following is true about the structural organization of proteins?

- a) Tertiary structure is three dimensional
- b) Primary, secondary and tertiary structures are destroyed by denaturation
- c) Secondary structure is stabilised by disulphide bonds
- d) Secondary and tertiary structures depend upon amino acid sequence

Question 14:

Nitric oxide is synthesized from:

- a) L - arginine
- b) L - citrulline
- c) Glycine
- d) Lysine

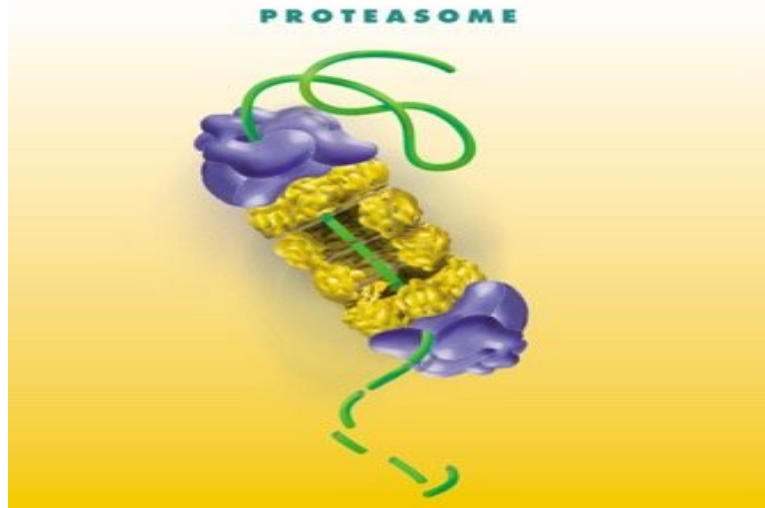
Answer Key

Question No.	Correct Option
1	c
2	d
3	a
4	b
5	a
6	d
7	d
8	d
9	b
10	a
11	a
12	d
13	a
14	a

Detailed Explanations**Solution to Question 1:**

The proteasome takes part in protein degradation.

The proteasome is a cylindrical structure in the cytosol that degrades polyubiquitinated proteins. It has a central core with proteolytic enzymes. Proteins that are tagged by ubiquitin to be degraded enter the core through regulatory rings on either side of the cylinder.



Clinical relevance: It plays an important role in antigen presentation to T lymphocytes as it degrades viruses and other molecules to small peptides.

Solution to Question 2:

Replacing alanine with tryptophan will increase UV absorbance of protein at 280nm wavelength. The amino acids which absorb 250-290nm (maximum at 280nm) UV light are Tryptophan (Maximum Absorption), Tyrosine, Phenylalanine.

Solution to Question 3:

Vitamin B12 is found maximum in animal products.

Vitamin B12 or cobalamin is synthesized only by microorganisms. Thus, the only source of cobalamin for humans is animal products. It is not found in plant sources. Therefore, individuals who are strict vegetarians are usually at the risk of developing vitamin B12 deficiency.

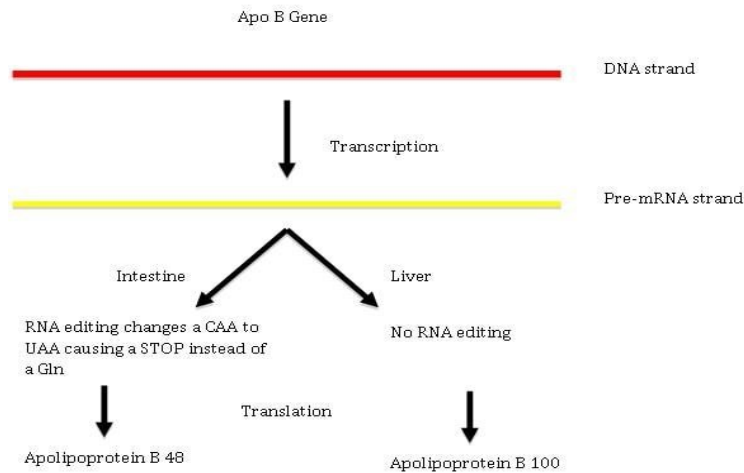
Solution to Question 4:

The process of apolipoprotein B48 synthesis involves RNA editing.

RNA editing is the process by which the coding sequence of the mRNA is changed before translation. As a result, the mRNA template differs from the DNA from which it was transcribed. This leads to the formation of a different protein. This occurs by insertion of additional uracils at the 3' end of the mRNA and proceeds in 3' to 5' direction.

RNA editing in apoB synthesis:

- In the liver, the apoB gene is transcribed to mRNA which is then translated to apoB 100.
- In the intestine, the same apoB gene is transcribed to the same mRNA. But now, the mRNA undergoes editing by an enzyme cytidine deaminase which converts the CAA codon in the mRNA to UAA at a single specific site. This edited mRNA translates to apoB 48.



Solution to Question 5:

In a patient with obstructive jaundice, van der Bergh reaction gives direct positive result due to conjugated hyperbilirubinemia.

Van den Bergh reaction is used to determine the serum bilirubin level.

Van den Bergh reaction is a colorimetric method to measure serum bilirubin. It is based on the formation of red azodipyrroles when diazotised sulfanilic acid reacts with bilirubin.

Interpretation:

- In the absence of methanol, the water-soluble conjugated bilirubin reacts rapidly with the reagent and is called direct bilirubin.
- In the presence of methanol, both conjugated and unconjugated bilirubin react, providing the total bilirubin.
- Unconjugated (Indirect) bilirubin = Total-direct bilirubin

Solution to Question 6:

The percentage of cytosine in the DNA is 22%.

As per the Chargaff's rule, in the DNA molecules, the concentration of deoxyadenosine (A) equals that of thymidine (T), and the concentration of deoxyguanosine (G) equals that of deoxycytidine (C). Therefore, A=T and G=C.

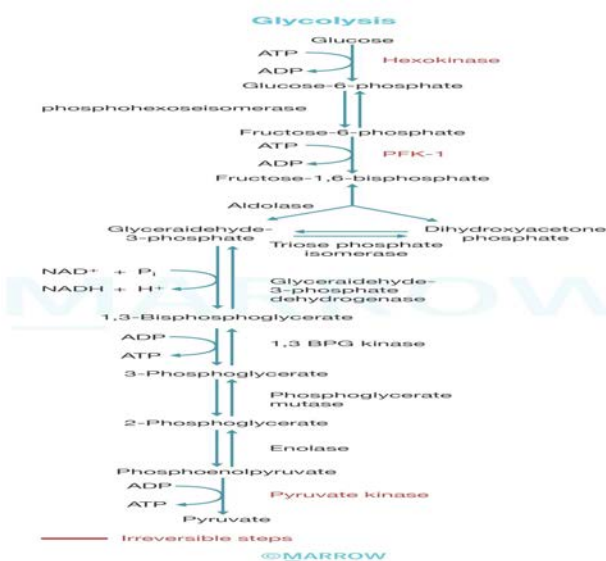
In the question, the concentration of thymine is given as 28%, therefore the concentration of adenine is also 28%, both of which add up to 56%. The remaining 44% (100-56) of the DNA consists of G-C base pairs. Of which the guanine constitutes 22% and the cytosine constitutes another 22%.

Solution to Question 7:

Enzyme activated by a decrease in insulin: glucagon ratio is glucose 6-phosphatase.

A decrease in insulin: glucagon ratio is seen in fasting state, in which there is the activation of gluconeogenesis and inhibition of glycolysis.

Among the given options, only glucose 6-phosphatase is an enzyme of gluconeogenesis and therefore is activated. The enzymes mentioned as other options are irreversible enzymes of glycolysis and are inhibited in the fasting state.

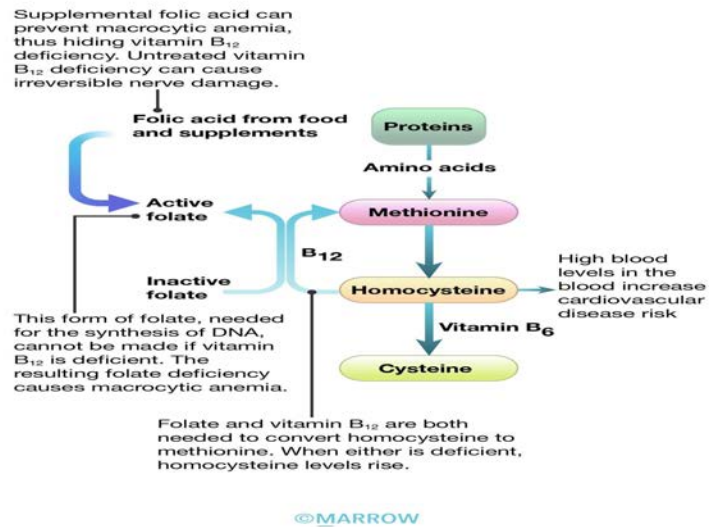


Solution to Question 8:

The most likely cause is folate therapy which used up the B12 stores, aggravating the symptoms.

Administration of folic acid for patients with macrocytic anemia can partially reverse the hematologic abnormalities associated with vitamin B12 deficiency. However, the neurologic manifestations of vitamin B12 deficiency are not treated by folic acid.

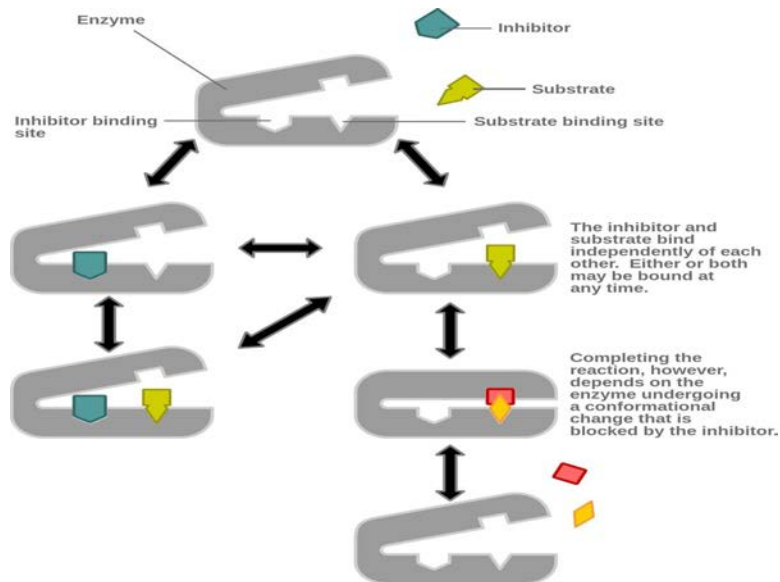
Administration of folic acid to an individual with vitamin B12 deficiency can potentially mask untreated vitamin B12 deficiency or even worsen the neurologic complications. Hence, testing for and/or treatment of vitamin B12 deficiency is required in all patients being treated with folic acid.



Solution to Question 9:

Noncompetitive inhibition has constant K_m and decreased V_{max} .

Noncompetitive inhibition occurs when the inhibitor binds at a site different from the substrate-binding site. The structure of the inhibitor does not resemble that of the substrate and the inhibitor does not affect the binding of the substrate.



Solution to Question 10:

Urine turning black on exposure to air is suggestive of alkaptonuria which occurs due to the defect in homogentisate oxidase enzyme.

Alkaptonuria is a disorder of tyrosine metabolism, that develops due to the defect in homogentisate oxidase, which oxidizes homogentisate to maleylacetoacetate. The defect in this enzyme leads to the accumulation of homogentisate, which is then excreted in the urine. The excreted homogentisate gets oxidized when exposed to the air and causes the urine to turn black.

In some cases, the excessive homogentisate may get oxidized to benzoquinone acetate, which binds to the connective tissue. This leads to arthritis and pigmentation of the connective tissue, a condition called ochronosis.

Other options:

Option B: Xanthine oxidase deficiency results in hypouricemia, xanthinuria, and xanthine lithiasis.

Option C: Defects in phenylalanine hydroxylase can lead to phenylketonuria characterised by hyperreflexia, seizures, hypopigmentation, low IQ, hyperactivity, and tremors.

Option D: Defects in dihydroorotate dehydrogenase are rarely seen in clinical practice. Miller syndrome is associated with this defect.

Solution to Question 11:

The above history and clinical features suggest niacin/vitamin B3 deficiency.

Excessive consumption of maize-based diets can cause pellagra. Corn/Maize has low levels of niacin and tryptophan. Jowar has high leucine content which inhibits QPRTase, an enzyme in the synthesis of Niacin.

Niacin deficiency leads to a disease called pellagra which is characterized by

- Diarrhea
- Dermatitis
- Dementia

The lesion in the neck is characteristic Casal's necklace pattern, seen in pellagra. The Casal collar or Casal necklace is a clinical sign in which there is an erythematous pigmented skin rash in the distribution of a broad collar (dermatomes C3 and C4).



Solution to Question 12:

Macular cysts and edema may develop in patients due to the infusion of supraphysiological doses of vitamin B₃ (niacin toxicity).

The toxic effects of niacin include:

- Flushing associated with skin dryness, itching, paresthesia, and headache is seen. These effects may be seen with even a low dose of 30mg/day. Pre-medication with aspirin alleviates these symptoms as these symptoms are mediated by prostaglandins.
- Macular edema and macular cysts
- Nausea, vomiting, and abdominal pain.
- Hepatic toxicity presents as jaundice with elevated AST and ALT levels.
- Fulminant hepatitis requiring liver transplantation.
- Glucose intolerance
- Hyperuricemia

Solution to Question 13:

The tertiary structure is three dimensional is the right answer among the given options.

Proteins are structurally organised into four levels, as follows:

- Primary: The linear sequence of amino acids in the polypeptide chain
- Secondary: Folded and geometrically ordered units of short segments of polypeptides. Eg: alpha helix, beta-pleated sheets, loops, and bends
- Tertiary: Secondary structures arranged into larger functional units called domains

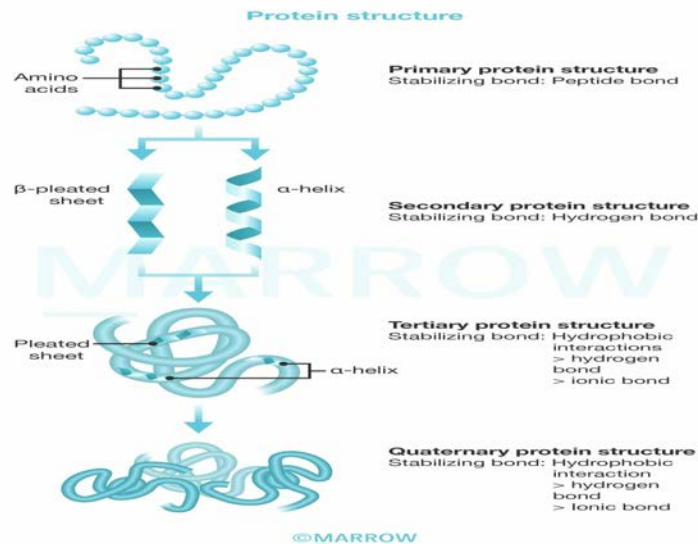
- Quaternary: Different types of polypeptides arranged in a spatial manner

Other options:

Option B: Protein denaturation results in the unfolding and disorganization of the secondary and tertiary structures without the hydrolysis of peptide bonds, which form the primary structure.

Option C: Hydrogen bond and Van der Waal interactions are involved in the stabilization of secondary structure(helix). Disulfide bonds stabilize the tertiary structure of proteins.

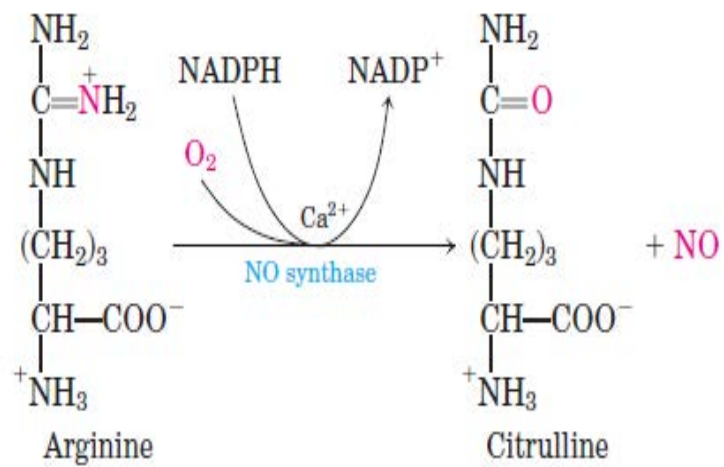
Option D: Primary structure depends upon the amino acid sequence, not secondary or tertiary.



Solution to Question 14:

Nitric oxide is synthesized from arginine by a calcium-dependent reaction catalyzed by NO synthase.

The co-enzymes of NO synthase include FMN, FAD, Heme, and Tetrahydrobiopterin (BH₄).



Biochemistry INI-CET 2021

Question 1:

What is the effect of RNAi on gene expression?

- a) Knock out
- b) Knock down
- c) Knock up
- d) Knock in

Question 2:

Methyl-tetrahydro-folic acid gets accumulated in deficiency of which of the following?

- a) B2
- b) B12
- c) B1
- d) B6

Question 3:

Cytochrome oxidase is inhibited in all of the following poisonings except:

- a) Hydrogen sulphide
- b) Carbon monoxide
- c) Methane
- d) Cyanide

Question 4:

Which of the following molecular analysis methods is used to detect RNA ?

- a) Sanger's method
- b) G banding
- c) Western blot

d) RT-PCR

Question 5:

Insulin inhibits which of the following lipase enzymes?

- a) Lipoprotein lipase**
- b) Pancreatic lipase**
- c) Gastric lipase**
- d) Hormone sensitive lipase**

Question 6:

Glycated hemoglobin (HbA1c) is best measured using?

- a) Ion exchange chromatography**
- b) Affinity chromatography**
- c) Isoelectric focussing**
- d) Electrophoresis**

Question 7:

What is the main substrate and metabolic fuel used by RBCs during fasting?

- a) Ketone bodies**
- b) Glucose**
- c) Free fatty acids**
- d) Amino acids**

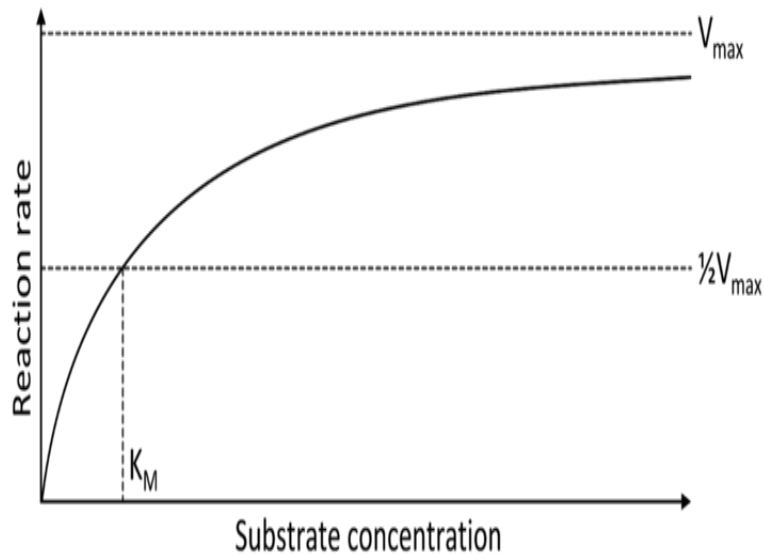
Question 8:

miRNA binds to which part of the mRNA to inhibit translation?

- a) Gene body**
- b) Gene promoter**
- c) 3' UTR**
- d) 5' UTR**

Question 9:

The enzyme kinetics depicted in the Graph is suggestive of-



- a) Allosteric Activation
- b) Cooperative Binding
- c) Michaelis Menten kinetics
- d) Sigmoid Curve

Question 10:

Which of the following methods cannot be used to precipitate proteins?

- a) Add alcohol and acetone
- b) Using heavy metal ions
- c) Moving pH away from isoelectric pH
- d) Adding trichloroacetic acid

Question 11:

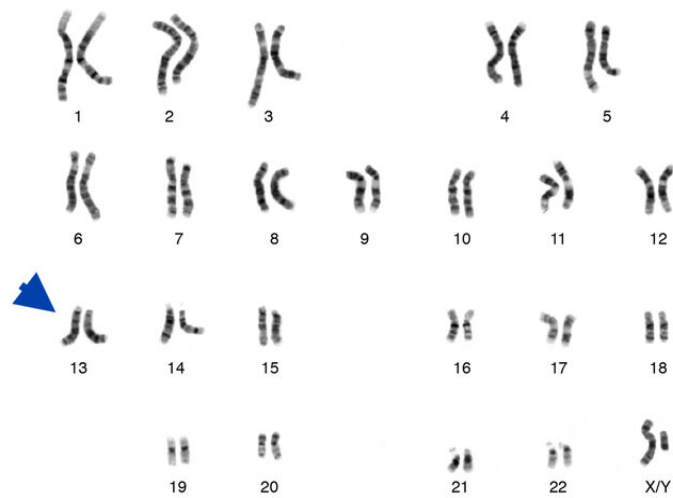
Which of the following diagnostic markers is/are incorrectly paired?

- a) 1 only
- b) 1 and 2

- c) 1, 2 and 4
- d) 2, 3 and 4

Question 12:

Identify the type of marked chromosome in the given karyotype.



- a) Acrocentric
- b) Metacentric
- c) Submetacentric
- d) Telocentric

Question 13:

Which of the following consumes the highest energy during assimilation?

- a) Proteins
- b) Carbohydrates
- c) Mono-unsaturated fatty acids
- d) Poly-unsaturated fatty acids

Question 14:

Which of the following is the best method to detect HbA1c?

- a) Ion-exchange chromatography
- b) Electrophoresis
- c) Immunoassay
- d) Affinity chromatography

Question 15:

Which of the following options is false in a patient with advanced liver disease?

- a) Hyperammonemia
- b) Decreased ATP synthesis
- c) Oxidative decarboxylation of alpha-keto glutarate
- d) Increased synthesis of glutamine and glutamate from alpha-ketoglutarate

Question 16:

Which of the following is associated with a defect in collagen maturity?

- a) Scurvy
- b) Rickets
- c) Beri-beri
- d) Night blindness

Question 17:

Which of the following is not a substrate for glucose formation?

- a) Acetyl coenzyme A
- b) Glycerol
- c) Alanine
- d) Lactate

Question 18:

Which of the following is the best test for quantitative analysis of an RNA sample?

- a) Real time PCR

- b) Reverse transcriptase PCR
- c) PCR - Restriction Fragment Length Polymorphism
- d) RTPCR followed by Real time PCR

Question 19:

Wobble hypothesis explains which of the following concepts of the genetic code?

- a) Universal
- b) Unambiguous
- c) Degeneracy
- d) Non-punctuated

Question 20:

Which of the following statements are true regarding mitochondrial DNA?

- a) 1 and 3
- b) 1 and 2
- c) 2 and 4
- d) 2 and 3

Question 21:

What is the repair mechanism associated with CRISPR-Cas9?

- a) Non-homologous end joining
- b) Base excision repair
- c) Nucleotide excision repair
- d) Mismatch repair

Question 22:

A patient with thiamine deficiency develops lactic acidosis. Which of the following enzymes is affected?

- a) Pyruvate carboxylase

- b) Pyruvate dehydrogenase
- c) Phosphoenolpyruvate carboxykinase
- d) Transketolase

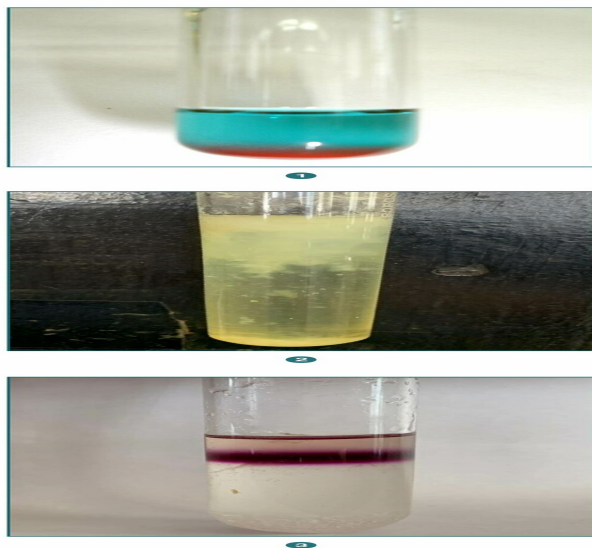
Question 23:

Which of the following is not amphipathic?

- a) Sphingolipids
- b) Phosphoglycerol
- c) Triglycerides
- d) Glycolipids

Question 24:

Which of the following tests will be positive in the urine of a patient who has undergone prolonged starvation?



- a) 1 and 3
- b) Only 3
- c) 1, 2 and 3
- d) 1 and 2

Question 25:

The action of inhibition of acetylcholinesterase enzyme by organophosphorus compounds is through which of the following mechanisms?

- a) Non-competitive inhibition
- b) Reversible competitive inhibition
- c) Irreversible competitive inhibition
- d) Suicidal inhibition

Answer Key

Question No.	Correct Option
1	b
2	b
3	c
4	d
5	d
6	a
7	b
8	c
9	c
10	c
11	a
12	a
13	a
14	a
15	c
16	a
17	a
18	d
19	c
20	b
21	a
22	b
23	c

24	b
25	c

Detailed Explanations

Solution to Question 1:

RNA interference(RNAi) is a biological process in which miRNA/siRNA molecules inhibit gene expression ("knockdown") by neutralizing the targeted mRNAs.

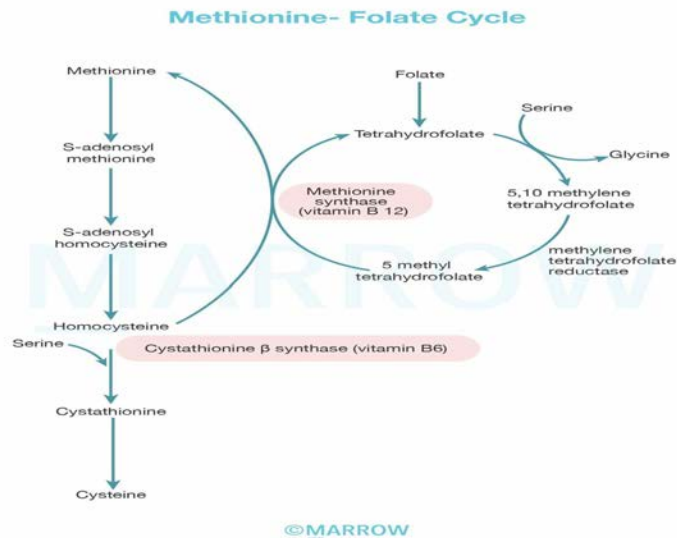
- miRNA regulates the expression of the target gene by post-transcriptional silencing.
- Small interfering RNAs (siRNAs) are short RNA sequences that can be introduced experimentally into cells and then serve as substrates for the Dicer enzyme and interact with the RNA-induced silencing complex (RISC) in a manner similar to that observed for endogenous miRNAs.
- Synthetic siRNAs targeted against specific mRNA species are useful in the study of gene function and this technique is referred to as knockdown technology.
- siRNAs are also being developed as possible therapeutic agents to silence pathogenic genes like oncogenes involved in neoplastic transformation.

Knock out: Mice generated through genetic manipulation of embryonic stem (ES) cells such that a specific genetic locus is targeted and rendered non-functional ("knocked-out"), either by insertion of an irrelevant DNA sequence or by deletion of the endogenous DNA sequence.

Knock-in: Mice generated through genetic manipulation of embryonic stem (ES) cells such that a specific genetic locus is altered either by one-for-one substitution of DNA sequence or by addition of sequence not found in the endogenous genetic locus.

Solution to Question 2:

Accumulation of methyl tetrahydro folic acid is seen in vitamin B12 deficiency. Vitamin B12 acts as a cofactor of methionine synthase, an enzyme involved in the regeneration of tetrahydrofolate from methyl-tetrahydrofolate. Hence, in vitamin B12 deficiency methyl-tetrahydrofolate accumulates leading to functional folate deficiency known as the folate trap.



The conversion of 5,10-methylene-THF into 5-methyl-THF (methyltetrahydrofolate) is an irreversible process. The major source of tetrahydrofolate for tissues is 5-methyl-THF. The only way to make further use of 5-methyl-THF and to maintain the folate cycle is by the vitamin B12-dependent remethylation of homocysteine to methionine by the methionine synthase enzyme.

Decreased activity of methionine synthase in vitamin B12 deficiency results in the accumulation of methyltetrahydrofolate, which cannot be used, resulting in a folate trap. The folate cycle acts as the link between the functions of folate and vitamin B12, and deficiency of vitamin B12 can cause a functional deficiency of folate.

Note: Since vitamin B12 is also linked to the regeneration of methionine from homocysteine, its deficiency can result in hyperhomocysteinemia as well. This in turn increases the risk of atherosclerosis, hypertension, and thrombosis.

Solution to Question 3:

Cytochrome oxidase is a hemoprotein, which is the terminal component of the respiratory chain carriers (complex IV) in mitochondria. The action of cytochrome oxidase is inhibited in carbon monoxide, cyanide, and hydrogen sulfide poisoning, resulting in prevention of cellular respiration.

Hemoproteins are metalloproteins that have a typical heme prosthetic group. They are present in myoglobin, hemoglobin, and other cytochromes. Cytochrome oxidase contains two molecules of heme, each having one Fe atom that oscillates between Fe^{3+} and Fe^{2+} during oxidation and reduction. It also contains two atoms of copper, each associated with a heme unit.

Complexes of ETC :

Complex I: NADH-CoQ oxidoreductase

Contains - FeS, FMN, Coenzyme Q

Inhibitors – Rotenone

Complex II: Succinate-CoQ reductase

Contains - FeS, Coenzyme Q

Inhibitor - Carboxin

Complex III: CoQH₂-cytochrome c oxidoreductase

Contains - cytochrome c, b, C₁, and FeS

Inhibitor - Antimycin A

Complex IV: cytochrome oxidase

Contains - Cytochrome a, a₃, and Copper

Inhibitor - Cyanide, Azide, Carbon monoxide

Complex V (informally): ATP Synthase

Inhibitor - Oligomycin

Note: MELAS (mitochondrial encephalopathy, lactic acidosis, and stroke) is an inherited condition due to NADH-Q oxidoreductase (Complex I) or cytochrome oxidase (Complex IV) deficiency.

Solution to Question 4:

RT PCR(Reverse Transcriptase PCR) is a molecular analysis method used for the detection of RNA. It can be used to quantitate mRNA levels by generating complementary DNA (cDNA) copies of the mRNA using the reverse transcriptase enzyme.

The northern blot technique can also be used to detect RNAs. Here, the mixture of mRNA molecules is separated by electrophoresis and then transferred to a nitrocellulose membrane and hybridized with radiolabeled probes.

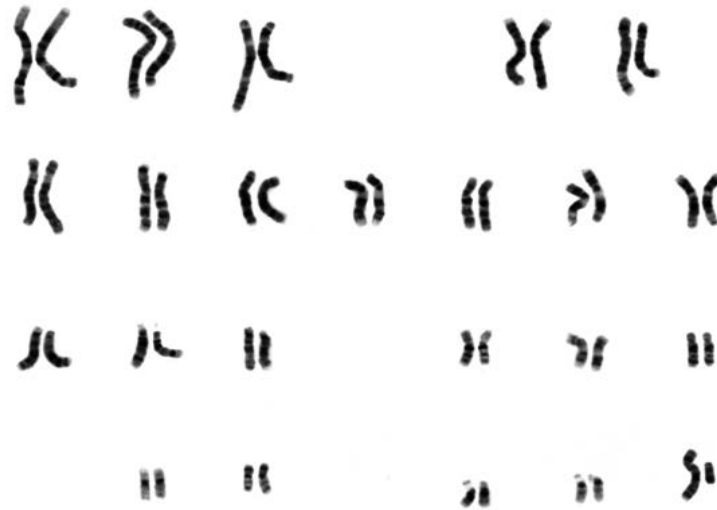
Sanger's technique:

- This was the first technique used to determine the sequence of proteins.
- Sanger's reagent is 1- Fluoro 2,4 Dinitro benzene.
- The first protein to be sequenced by this method was insulin.
- Only dipeptides and tripeptides can be sequenced by this method.

G banding (Giemsa banding):

It is the most common technique used in cytogenetics to produce a visible karyotype by staining condensed chromosomes. It is useful for identifying genetic diseases through the photographic representation of the entire chromosome complement. The metaphase chromosomes are treated with trypsin (to partially digest the chromosome) and stained with Giemsa stain.

Heterochromatic regions, which tend to be rich with adenine and thymine (AT-rich) DNA and relatively gene-poor, stain darker and appear as dark bands in G-banding. In contrast, less condensed chromatin, which tends to be rich with guanine and cytosine (GC-rich) and more transcriptionally active incorporates less Giemsa stain, and these regions appear as light bands in G-banding. The image below shows a G-banding karyogram of a normal human male.



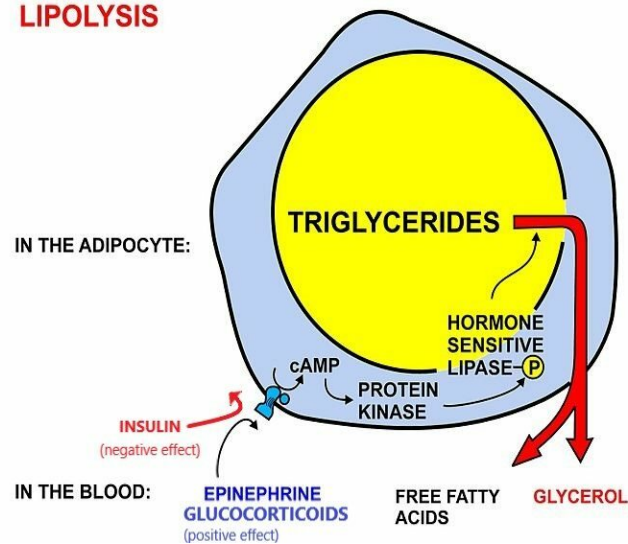
Western blot, also called immunoblots, is a technique used to study protein molecules. The proteins undergo electrophoresis and then are transferred to a membrane. These proteins are then tagged with the specific antibodies and visualized.

Solution to Question 5:

Insulin has an inhibitory effect on lipolytic hormones like hormone sensitive lipase (HSL). In adipose tissues, the stored triglycerides are broken down (lipolysis) by hormone sensitive lipase (HSL) into free fatty acids (FFAs). Insulin inhibits HSL and thus reduces the release of FFAs and also glycerol.

In diabetic patients, low insulin level corresponds to an increase in HSL activity, promoting the breakdown of lipids and release of FFAs from adipose tissue. Hormone-sensitive lipase activity is accelerated by hormones like epinephrine, norepinephrine, glucagon, ACTH, TSH, GH, ADH, and Melanocyte stimulating hormones (MSH).

LIPOLYSIS



Insulin induces the activity of lipoprotein lipase (LPL) which is present on the walls of capillaries. LPL promotes hydrolysis of TGs and degradation of VLDL and chylomicrons. In diabetes, low insulin leads to decreased LPL activity resulting in increased VLDL and TGs levels.

Effects of insulin on various tissues:

Well-fed state (Insulin)	During fasting (Glucagon)
Lipogenesis increased	Lipogenesis inhibited
Lipolysis inhibited	Lipolysis increased
Insulin inhibits HS-lipase	Glucagon activates HS-lipase
Lipoprotein lipase active	Lipoprotein lipase inactive

Tissue	Increases	Decreases
Adipose tissue	Increased glucose entry Increased fatty acid synthesis Increased glycerol phosphate synthesis Increased triglycerides deposition Activation of lipoprotein lipase Increased K ⁺ uptake	Inhibition of hormone-sensitive lipase
Muscle	Increased glucose entry Increased glycogen synthesis Increased amino acid uptake Increased protein synthesis in ribosome Increase ketone uptake Increase K ⁺ uptake	Decrease protein catabolism Decrease release of glucogenic amino acids
Liver	Increased protein synthesis Increased lipid synthesis Increased glycogen synthesis & glycolysis	Decrease ketogenesis Decreased glucose output due to decreased gluconeogenesis

Tissue	Increases	Decreases
General	Increased cell growth	

Solution to Question 6:

Glycated hemoglobin (HbA1c) is best measured using ion-exchange chromatography.

HbA1c is formed by non-enzymatic glycation of hemoglobin A present in RBCs. Normally this occurs to a small extent but is increased in diabetic patients with poor glycemic control. Since the half-life of RBCs is typically 60 days, HbA1c can be used to monitor mean blood glucose concentration over the past 6 to 8 weeks.

In ion-exchange chromatography:

- Proteins interact with the stationary phase by charge-charge interactions.
- Proteins with a net positive charge at a given pH will tightly adhere to beads with negatively charged functional groups such as carboxylates or sulfates (cation exchangers).
- Proteins with a net negative charge adhere to beads with positively charged functional groups, typically tertiary or quaternary amines (anion exchangers).
- Non-adherent proteins flow through the matrix and are washed away.
- Proteins elute in inverse order of the strength of their interactions with the stationary phase.

Solution to Question 7:

Glucose is the main substrate and metabolic fuel used by RBCs during fasting. RBCs lack mitochondria and are dependent fully on glycolysis (anaerobic) and pentose phosphate pathway during fasting state.

Glucose is always required by brain and RBCs. Brain can meet only about 20 percent of energy requirements by using ketone bodies and the remaining is supplied by glucose. Metabolic changes occurring during fasting state and starvation therefore serve to preserve glucose and glycogen reserves for brain and RBCs.

Solution to Question 8:

Micro-RNA (miRNA) is an endogenous non-coding RNA that typically binds to the 3'-untranslated regions (UTRs) of target messenger-RNAs (mRNAs). The primary function of miRNA is to modulate the translation of target mRNAs into their corresponding proteins. After the target RNA is transcribed from its gene (post-transcriptional), the miRNA may either induce mRNA cleavage or repress its translation. This is referred to as post-transcriptional gene silencing or RNA interference.

RNA interference (RNAi) is a biological process in which miRNA or siRNA molecules inhibit gene expression by neutralizing targeted mRNAs. The microRNA (miRNA) belongs to a group of non-coding genes. The miRNA genes are transcribed but not translated into proteins. They play a role in the regulation of the expression of the target gene by post-transcriptional silencing.

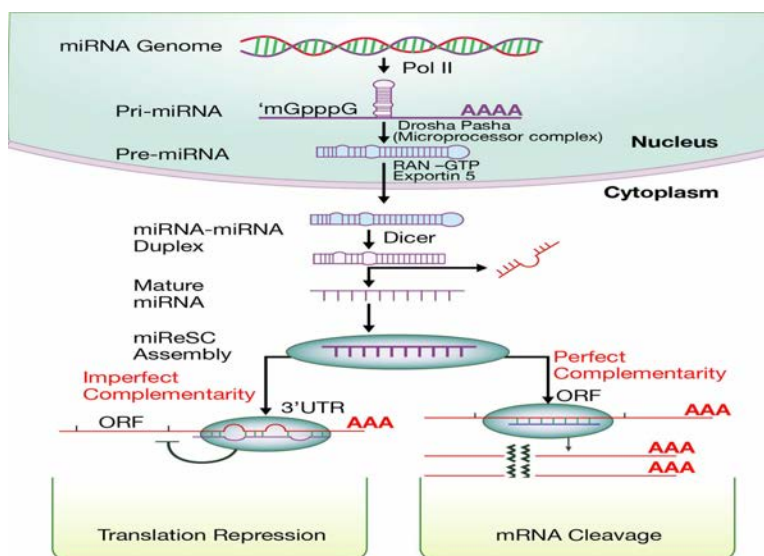
Mechanism:

- The miRNA gene present in the nucleus is transcribed to primary miRNA, which is further converted to pre-miRNA by an enzyme DROSHA intranuclearly and exported to the cytoplasm by an export protein called Exportin 5.
- The pre-miRNA undergoes trimming, by an enzyme called DICER, into mature single-stranded miRNA.
- The mature miRNA is associated with a multiprotein aggregate called RNA-induced silencing complex (RISC).
- The target mRNAs contain seed sequences in their 3' untranslated region (UTR) that determines the specificity of miRNA binding and gene silencing.

Depending on the base pairing between the miRNA strand and its target mRNA, the RISC causes gene silencing either by mRNA cleavage (degradation or deadenylation) or by repressing its translation.

siRNA is similar in function to miRNA and is used in RNA interference.

miRNA expression has been found to be dysregulated in many cancers. Such misregulation is thought to have a causal role in the pathogenesis of such cancers. Some types of miRNAs are oncogenic (referred to as oncomiRs) and are overexpressed in cancerous tissue. On the other hand, tumor-suppressive miRNAs that counteract oncogenic characteristics are found to be underexpressed in cancers.



Solution to Question 9:

The graph shows a hyperbola which is suggestive of Michaelis - Menten Kinetics of the enzyme.

For enzymes showing Michaelis-Menten kinetics, there is an increase in Rate of reaction, V , with increase in substrate concentration till a point V_{max} is reached beyond which rate of reaction shows saturation.

Cooperative Binding would result in a sigmoid curve and is suggestive of Non Michaelis-Menten Kinetics.

Solution to Question 10:

Precipitation of proteins is maximum at isoelectric pH and decreases as pH moves away from it. Hence, moving pH away from isoelectric pH cannot be used for precipitating proteins.

The isoelectric point or isoelectric pH (pI) is the pH at which the molecule carries no net charge.

Significance of Iso-electric pH (pI):

- The number of anions and cations present on the protein molecule will be equal and the net charge is zero.
- The proteins will not migrate in an electrical field
- Buffering capacity is minimum (note: maximum buffering action of an amino acid is seen when $pH=pK_a$, and not at isoelectric pH)
- Viscosity will be minimum
- Precipitability will be maximum and solubility is minimum

Various precipitation reactions of proteins are:

- Salting out: higher the molecular weight of protein, the lesser the salt required. Globulins would require only half-saturated salt whereas albumin would need full saturation.
- Isoelectric precipitation: proteins are least soluble at their isoelectric pH and get precipitated immediately.
- Precipitation by organic solvents like alcohol and acetone.
- Precipitation by heavy metal ions: positively charged metal ions precipitate the negatively charged proteins in an alkaline medium. Salts of heavy metals can precipitate normal proteins of the GI tract in poisoning.
- Precipitation by alkaloidal reagents like tungstic acid, trichloroacetic acid, and picric acid.
- Heat coagulation is an irreversible precipitation process.

Solution to Question 11:

Beta glucocerebrosidase deficiency is seen in Gaucher's disease. It is an inborn error of metabolism due to the deficiency of the enzyme β -glucocerebrosidase (β -glucosidase) that leads to the accumulation of glucocerebroside in the lysosomes of cells.

- It is the most common lysosomal storage disorder. It is inherited in an autosomal recessive manner.

- The clinical features result from the accumulation of lipid-laden macrophages in the spleen, liver, bone marrow, bone, and other tissues.
- There are 3 clinical subtypes, delineated by the difference of neurologic manifestations: type 1, or the adult, non-neuronopathic form; type 2, the infantile or acute neuronopathic form; and type 3, the juvenile or subacute neuronopathic form
- Typical clinical features include delayed puberty, painless hepatomegaly and splenomegaly, liver cirrhosis, hypersplenism and pancytopenia, bleeding tendencies, bone pain due to avascular necrosis, osteoporosis, and pathological fractures.
- Neurological symptoms such as developmental delay, hypertonia, strabismus, and supranuclear gaze palsy (seen in type 3).
- Bone marrow examination usually reveals the presence of Gaucher cells having a characteristic wrinkled-paper appearance.
- Enzyme replacement therapy for Gaucher's disease includes imiglucerase, velaglucerase alfa, taliglucerase alfa, and eliglustat.

Von Gierke disease occurs due to deficiency of glucose-6-phosphatase enzyme in the liver, kidney, and intestinal mucosa. Patients present commonly by 3-4 months of age with hypoglycemia (leading to seizures), lactic acidosis, and hepatomegaly. Affected children often have a doll-like face with fat cheeks, relatively thin extremities, short stature, and a protuberant abdomen due to massive hepatomegaly. The kidneys are also enlarged, whereas spleen and heart are not involved. Laboratory findings include hypoglycemia, lactic acidosis, hyperuricemia, and hyperlipidemia.

Elevated serum lipase and serum amylase levels are associated with acute pancreatitis. Serum amylase levels do not correlate with the severity of pancreatitis and lack specificity whereas serum lipase is a more specific marker.

Low serum ceruloplasmin level is suggestive of hepatolenticular degeneration seen in Wilson's disease. However, it should not be used for definitive diagnosis as levels are normal in 10% of the affected patients and reduced in 20% of carriers. The diagnosis of Wilson's disease is based on the following:

- Increased 24-hour urinary excretion of copper (screening test)
- Decrease in serum ceruloplasmin
- Kayser–Fleischer rings under slit lamp
- Genetic testing for mutations in ATP7B gene
- Liver biopsy with quantitative copper assay (>200 $\mu\text{g/g}$ dry weight of liver) is the gold standard investigation.

A higher level of LDH1 than LDH 2 in the blood is suggestive of myocardial infarction. Under normal conditions, LDH 2 is predominantly present in the blood and LDH 1 is predominantly present in the heart. When cardiac tissue is damaged LDH 1 is released into the bloodstream resulting in a high LDH1:LDH2 ratio known as the flipped LDH ratio ($\text{LDH1} > \text{LDH2}$). LDH is a tetrameric enzyme with 4 subunits. It has four subunits with two isoforms: H isoform (Heart) and M isoform (Muscle)

- LDH-1: Heart

- LDH-2: Blood
- LDH-3: Brain, kidney, lungs, spleen, pancreas
- LDH-4: Liver and skeletal muscle
- LDH-5: Liver and skeletal muscle

Solution to Question 12:

The marked chromosome is chromosome 13 and the type is acrocentric.

Each chromatid has a long arm q and a short arm p. The two arms are separated by a centromere. In this image, the centromere is not midway between the two ends of the chromatids. It is seen towards one end with a marked difference in length between the p and q arms, hence it is an acrocentric chromosome.

Chromosomes are classified based on the position of the centromere as follows:

-

Metacentric: Centromere is centrally placed and the two arms are equal in length

-

Submetacentric: Centromere is slightly away from the center and the two arms are only slightly different in length

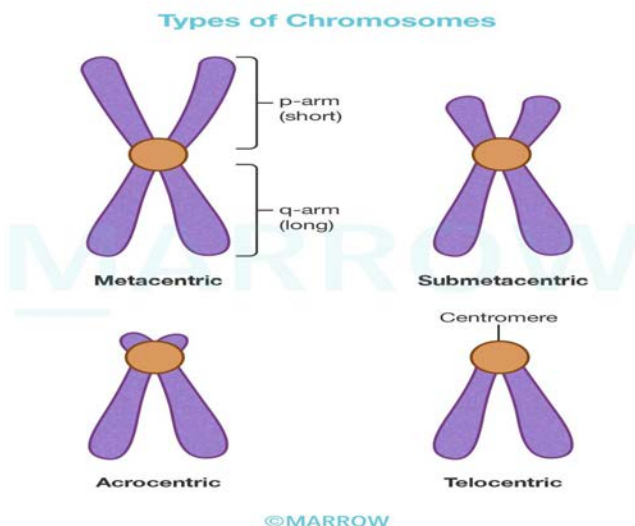
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Acrocentric: Centromere is nearer to one end and the difference in length of arms is marked

-

Telocentric: Centromere lies at one end.

Group	Pairs of chromosomes	Features
A	1–3	Long, metacentric
B	4, 5	Long, submetacentric
C	6–12, X-chromosome	Medium, Submetacentric
D	13–15	Medium, acrocentric Satellite bodies +
E	16–18	Short, metacentric (16) or submetacentric (17 and 18)
F	19, 20	Short, metacentric
G	21, 22 + Y-chromosome	Short, acrocentric Satellite bodies +



Solution to Question 13:

Proteins consume the highest energy during assimilation.

The energy wasted in the assimilation of food is represented as diet-induced thermogenesis or specific dynamic action (SDA). Diet-induced thermogenesis (DIT) can be defined as the increase in energy expenditure above the basal metabolic rate and is expressed as a percentage.

The values for diet-induced thermogenesis are relatively higher protein and alcohol consumption and lower at a high-fat consumption, the hierarchy for it, being in the sequence of alcohol > protein > fat > carbohydrate. The value of DIT for protein is 30%, the lipid is 15% and carbohydrate is 5%. Hence, taking into account the loss of energy as DIT, additional 10% calories should be provided in a mixed diet.

Solution to Question 14:

Ion exchange chromatography is the best method to detect HbA1c.

Ion-exchange chromatography allows the separation of ionizable molecules according to differences in charge. It involves the use of a column which contains a matrix consisting of resin-bound charged groups.

If the charged group is anionic or negatively charged, it is called a cation exchanger and is used in cation exchange chromatography where proteins with a net positive charge migrate through the matrix more slowly than those with a net negative charge due to retardation of the migration of the former which is attracted to the negative charges bound to the resin. Cation exchange chromatography is used for negatively charged HbA1c estimation and is used as a diagnostic tool for diabetes mellitus.

Glycated hemoglobin, reported as HbA1c, has been used to monitor glycemic control in patients with diabetes for many years. HbA1c results from the attachment of glucose to the N-terminal

valine of the hemoglobin β -chain.

The concentration of HbA_{1c} depends on both the concentration of glucose in the blood and the lifespan of the erythrocyte. Since the half-life of an RBC is 60 days, the level of glycated haemoglobin reflects the mean blood glucose concentration over the preceding 6 to 8 weeks. Normal glucose tolerance is characterized by an HbA_{1c} $\leq$ 5.7%; any value of HbA_{1c} equal to or exceeding 6.5 % is diagnostic of the patient being diabetic. Also, ion exchange chromatography has the advantage of separating hemoglobin variants into peaks with the least interference by other hemoglobin variants; its disadvantage is that it is temperature dependant.

Other options:

Option B: Electrophoresis is an electrochemical process where macromolecules or colloidal particles depending on the electric charge migrate in a solution under the influence of an electric current. It can also be used for HbA_{1c} estimation, its advantage being able to identify different variants of haemoglobin and disadvantage being interference of the process by fetal hemoglobin(HbF).

Option C: Immunoassays are used to quantify molecules based on the specificity and selectivity of antibody reagents generated. It can provide quantitative estimates of HbA_{1c} but HbF produces falsely low results and hemoglobin variants such as HbS and HbC may cause a false increase in HbA_{1c} measurements.

Option D: Affinity chromatography is a type of liquid chromatography. It involves the passing of a matrix through a solution that consists of the molecules of interest. These molecules would have a specific affinity for ligands present in the matrix and bind with them and end up getting separated from the solution. It can be used to measure HbA_{1c} without being affected by temperature. High levels of HbF seen in thalassemia may interfere with its result.

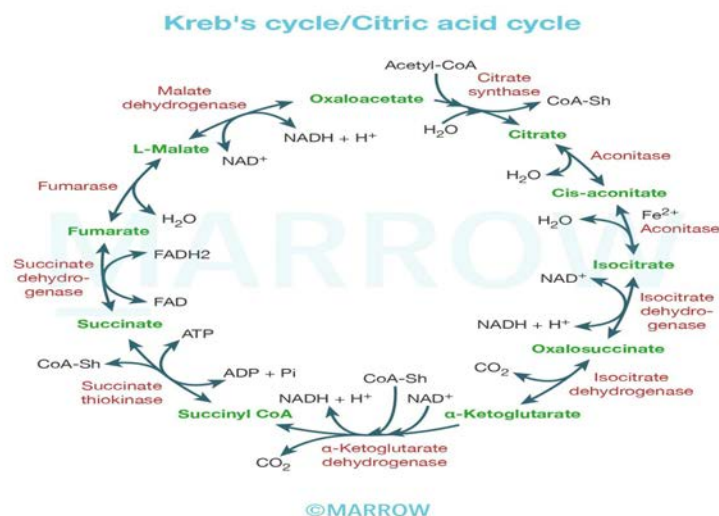
Solution to Question 15:

Oxidative decarboxylation of alpha-ketoglutarate is not seen in a patient with advanced liver disease. Option C is false.

Normally, ammonia, a toxic by-product of protein catabolism, is excreted as urea upon being metabolised in the urea cycle, which takes place in the liver. In advanced liver disease, impaired ammonia metabolism leads to hyperammonemia. Ammonia withdraws α -ketoglutarate to form glutamate and glutamine. This leads to lowered concentrations of all citric acid cycle intermediates, and hence reduced generation of ATP.

The Krebs cycle involves a sequence of reactions occurring in mitochondria that oxidize the acetyl moiety of acetyl-CoA to CO₂ and reduces the coenzymes that are reoxidized through the electron transport chain. It is the final common pathway for the oxidation of carbohydrates, lipids, and protein also has a central role in gluconeogenesis, lipogenesis, and interconversion of amino acids. Although processes occur in many tissues, the liver is the only tissue in which all occur to a significant extent.

The reactions that occur in Kreb's cycle are shown in the image below.



One of the steps of the Krebs cycle involves α -ketoglutarate undergoing oxidative decarboxylation in a reaction catalyzed by α -ketoglutarate dehydrogenase, resulting in the formation of succinyl-CoA. High concentrations of ammonia inhibit α -ketoglutarate dehydrogenase, thereby preventing its oxidative decarboxylation.

Solution to Question 16:

Scurvy is associated with a defect in collagen maturity.

Scurvy results from ascorbic acid or vitamin C deficiency. Vitamin C is an essential cofactor for the enzyme prolyl hydroxylase, required for the hydroxylation of prolyl residues in collagen. This enables each collagen fibril to form a uniform and flexible triple helix configuration in the body's connective tissues.

Its deficiency results in a defective prolyl hydroxylase activity, thereby resulting in the impaired formation of mature connective tissue; the clinical manifestations of this deficiency include petechiae, ecchymosis, coiled hairs, inflamed and bleeding gums, joint effusion, tooth loss, bone fracture, poor wound healing, and fatigue which can be attributed to impaired collagen synthesis all of which are typical of scurvy.

Risk factors include chronic alcoholism, diseases such as celiac disease and Crohn's disease, associated with malabsorption and poor nutrition.

It is treated with vitamin C supplementation.

Other options:

Option B- Rickets is caused due to deficiency of vitamin D in children which is characterized by the continued formation of the collagen matrix of bone but its incomplete mineralization or demineralization. Children have soft, pliable bones.

Option C- Beriberi is caused due to deficiency of thiamine. It can either be dry beriberi characterized by peripheral neuropathy or wet beriberi characterized by edema due to dilated cardiomyopathy.

Option D- Night blindness is caused by a deficiency of vitamin A. Deficiency can also lead to pancreatic insufficiency, repeated gastrointestinal, pulmonary, and urinary infections, and xerophthalmia.

Solution to Question 17:

Acetyl coenzyme A (CoA) is not a substrate for glucose formation.

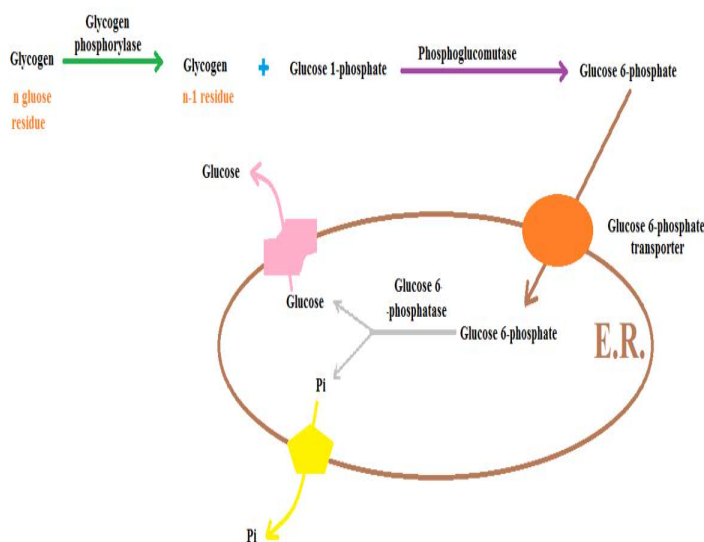
Acetyl CoA and compounds that give rise to acetyl CoA alone like acetoacetate, lysine, and leucine cannot give rise to a net synthesis of glucose. This is because of the irreversible nature of the pyruvate dehydrogenase complex, which converts pyruvate to acetyl CoA. These compounds give rise to ketone bodies instead.

The synthesis of glucose in the body, in the absence of dietary carbohydrates, occurs in two ways:

- From liver glycogen, by a process known as glycogenolysis in the absence of dietary carbohydrate, for up to 24 hours.
- De novo, from non-carbohydrate sources, by a process known as gluconeogenesis, when glycogen stores have depleted. This occurs primarily in the liver and to a lesser extent in the kidneys while fasting overnight or primarily in the kidneys during prolonged fasting of 48 hours or longer.

The substrate for glycogenolysis is glycogen and during this process, glycogen is first converted into glucose-1-phosphate by the enzyme glycogen phosphorylase. This is a rate-limiting step in the process. Glucose-1-phosphate is then converted to glucose-6-phosphate by the enzyme phosphoglucomutase and eventually, glucose is obtained from glucose-6-phosphate by the enzyme phosphatase. There is no involvement of acetyl CoA.

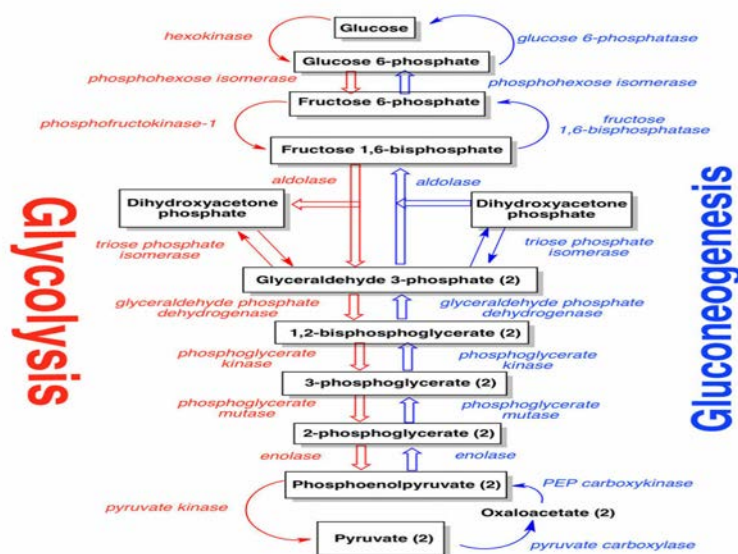
The steps with enzymes involved in glycogenolysis are shown in the image below.



In gluconeogenesis, the substrates can be:

- Oxaloacetate (from its conjugate acid, oxaloacetic acid) is a direct precursor of phosphoenolpyruvate
- Pyruvate, an α -keto acid that can be converted to glucose. Alanine is a glucogenic amino acid that is transaminated to form pyruvate by the enzyme alanine amino-transferase in the liver; the pyruvate formed, can be used for the synthesis of glucose.
- Lactate, obtained from anaerobic glycolysis, released from skeletal muscles and RBCs into the blood, to be taken up by the liver in the Cori cycle to be oxidised to pyruvate, to eventually synthesize glucose
- Glycerol from adipose tissue is converted to dihydroxyacetone phosphate in the liver, which itself is an intermediate of glycolysis and gluconeogenesis.

The steps and enzymes involved in gluconeogenesis are shown in blue in the flowchart below.



Solution to Question 18:

RTPCR followed by real-time PCR is the best test for quantitative analysis of an RNA sample.

RTPCR (reverse transcriptase PCR) involves the generation of complementary DNA (cDNA) copies of the mRNA using the reverse transcriptase enzyme followed by amplification of the cDNA copies by PCR technique.

Quantification of DNA involves the assessment of the amount/concentration of DNA that is present in a sample when carrying out a technique like PCR. Real-time PCR or quantitative PCR/qPCR involves the quantification of DNA after each cycle of amplification, or, in "real-time" than at the end of the procedure; therefore real-time PCR helps in quantitative analysis of DNA.

When reverse transcriptase PCR and qPCR are combined, they can be used to determine the relative concentrations of a particular mRNA molecule in a cell.

Other options:

Option A: Real-time PCR alone can only be used for the quantification of DNA only.

Option B: RTPCR can convert the RNA into cDNA by reverse transcriptase enzyme and then amplify the cDNA copies by means of PCR but cannot quantify the DNA/RNA.

Option C: PCR-Restriction fragment length polymorphism (PCR-RFLP) is also called cleaved amplified polymorphic sequence (CAPS) assay. Polymorphisms are inherited differences found among individuals in more than 1% of the normal population. PCR-RFLP is a technique in which organisms may be differentiated by variations in their DNA sequence by restriction endonucleases. Typically, gel electrophoresis is used to visualize RFLPs. Thus, DNA but not RNA molecules can be quantified in this process.

Solution to Question 19:

The wobble hypothesis explains the degeneracy of the genetic code.

The wobble hypothesis states that codon-anticodon pairing follows the traditional Watson–Crick rules (G pairs with C and A pairs with U) for the first two bases of the codon but can be less stringent for the third/last base. It has been observed that a tRNA can recognize more than one codon for a specific amino acid, especially in the last nucleotide of the codon triplet. This is referred to as degeneracy and its mechanism can be explained by the wobble hypothesis.

The first base of the anticodon moves to permit nontraditional pairing with the last base of the codon. This movement is called wobble and is possible because the first base of the anticodon is not as spatially defined as the other two bases. Due to the wobbling, 61 tRNA species are not required to read the 61 codons.

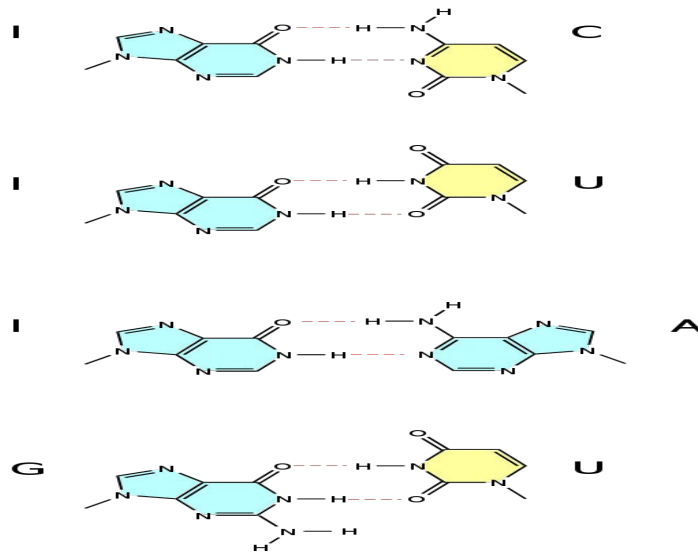
For example, three codons for glycine—GGU, GGC, and GGA—can form a base pair from one anticodon, 3' CCI 5' ie, I, inosine, can base pair with U, C, and A, as shown in the diagram below.

Other options:

Option A: The genetic code is universal because it is the same for all living organisms.

Option B: The genetic code is unambiguous because each codon is specific for only one amino acid. Example: GGA only codes for Glycine.

Option D: The genetic code is non-punctuated. This means that once the reading commences at a specific start codon, it is continuous and is read continuously as triplets until a stop codon is reached without stopping.



Solution to Question 20:

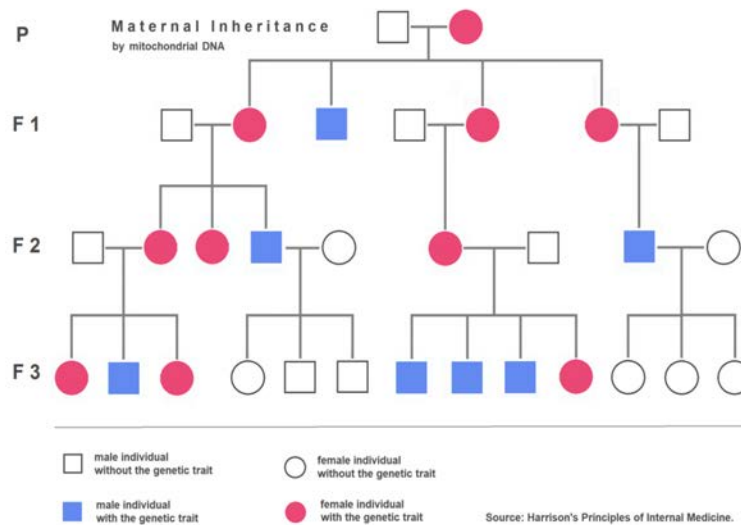
Statements 1 and 2 are true regarding mitochondrial DNA; it has a higher rate of mutations as compared to the nuclear genome and is a circular double-stranded DNA with about 16,000 base pairs.

Mitochondrial DNA is circular, double-stranded DNA composed of heavy and light chains and comprises about 1% of total cellular DNA. It contains about 16,569 base pairs and encodes about 13 protein subunits of the electron transport chain, of a total of about 67 protein subunits.

Its genetic code/ nucleotide sequence largely differs from that of the standard cellular code. For example, the standard stop codon UGA is read as tryptophan (Trp); in addition, few untranslated sequences exist and a very high mutation rate, about 5-10 times compared to that of nuclear DNA of the cell is observed.

As all mitochondria are contributed by the ovum during zygote formation, human mitochondrial DNA is transmitted by maternal non-mendelian inheritance. This is clinically significant because in diseases resulting from mutations of mitochondrial DNA, except in a few cases, an affected mother would pass the disease to all of her children but only her daughters would transmit the trait.

The below image shows the mitochondrial inheritance



Solution to Question 21:

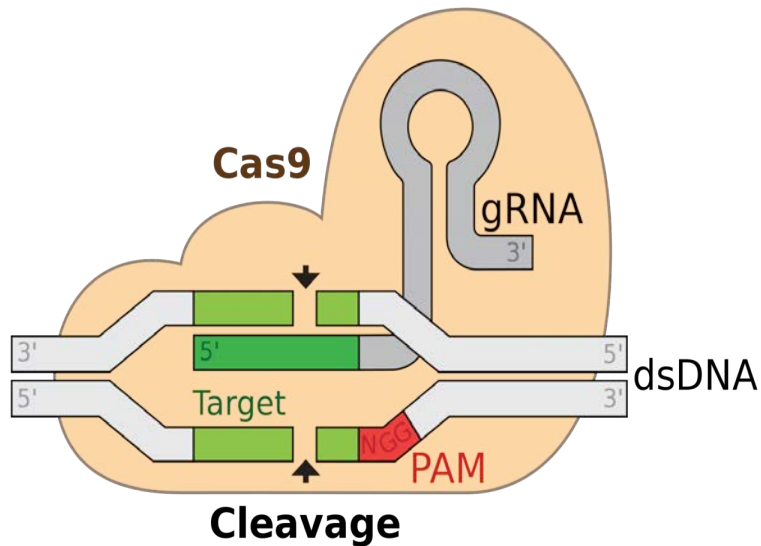
The repair mechanism associated with CRISPR-Cas9 is non-homologous end-joining.

Non-homologous end-joining (NHEJ) is a mechanism by which a group of proteins mediates the recognition, processing, and direct ligation of the ends of two DNA fragments where a double-strand break in DNA has occurred. This is in contrast to homologous recombination (HR), another mechanism to repair a double-strand break, where DNA repair needs an undamaged homologous DNA template to replace an adjacent damaged one.

Clustered regularly interspaced short palindromic repeats-associated gene 9 or CRISPR-Cas9 is a DNA editing/gene regulatory system. It involves two essential components, Cas9, an endonuclease that causes a double-stranded DNA break, allowing modifications to the genomic DNA to be made; and a synthetic single guide RNA or sgRNA responsible for guiding Cas9 to its target.

Once the sgRNA has directed the Cas9 to the section of the genomic DNA to be edited, the 3' end of the sgRNA comes to lie across a specific sequence of DNA adjacent to the editing site called the protospacer adjacent motif or PAM. The Cas9 now causes a double-stranded break in the DNA at the PAM site which is then repaired by a non-homologous end-joining.

The following image shows CRISPR-Cas9 system



Solution to Question 22:

Pyruvate dehydrogenase is the enzyme affected in a patient with thiamine deficiency who develops lactic acidosis.

Pyruvate is transported from the cytosol into the mitochondria, where undergoes oxidative decarboxylation to form acetyl CoA in the presence of the multi-enzyme pyruvate dehydrogenase complex (PDHC) in the mitochondrial matrix.

The PDHC primarily consists of three enzymes: Pyruvate decarboxylase (also called Pyruvate dehydrogenase), dihydrolipoyl transacetylase, and dihydrolipoyl dehydrogenase. It also requires the following co-enzymes:

- Thiamine pyrophosphate, essential for pyruvate decarboxylase and is tightly bound to it
- Lipoic acid and coenzyme A, essential for dihydrolipoyl transacetylase
- FAD and NAD⁺, essential for dihydrolipoyl dehydrogenase.

In a patient with thiamine deficiency, due to the inactive thiamine pyrophosphate, the pyruvate dehydrogenase complex is unable to act on pyruvate, which begins to accumulate on the body and is then shunted towards lactic acid formation by the enzyme lactate dehydrogenase.

This results in lactic acid accumulation and subsequent lactic acidosis in the patient.

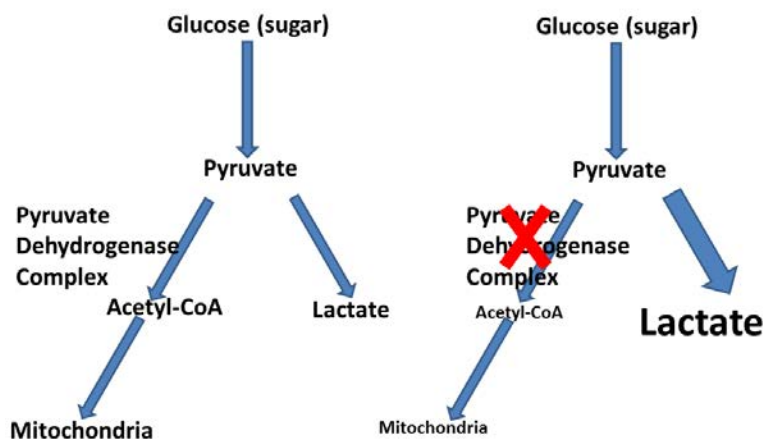
Wernicke–Korsakoff, an encephalopathy-psychosis syndrome due to thiamine deficiency, may be seen in chronic alcoholics. It occurs due to insufficient production of ATP by the brain cells via the citric acid cycle as the PDHC is inactive.

Other options:

Option A: Pyruvate carboxylase facilitates the conversion of pyruvate to oxaloacetate by carboxylation.

Option C: Phosphoenolpyruvate carboxykinase is involved in the decarboxylation and phosphorylation of oxaloacetate to form phosphoenolpyruvate seen in gluconeogenesis.

Option D: Transketolase with coenzyme thiamine pyrophosphate is an enzyme involved in the non-oxidative phase of the pentose phosphate pathway.



Solution to Question 23:

Triglycerides are not amphipathic.

Amphipathic substances are those which have both a hydrophilic or polar region and a hydrophobic or nonpolar region.

Phosphoglycerols or glycerol-based phospholipids have a hydrophilic head which is a phosphate group with alcohol attached to it and a hydrophobic tail containing fatty acids or fatty acid-based hydrocarbons. Therefore, they are amphipathic.

Sphingolipids, like all membrane-based lipids, are amphipathic molecules having hydrophobic and hydrophilic properties.

Glycolipids are amphipathic molecules consisting of a hydrophilic oligosaccharide chain attached to a hydrophobic lipid that anchors the glycolipid in cell membranes.

But lipids like triacylglycerol and cholesteryl esters are termed neutral lipids and thus are not amphipathic.

Cholesterol is amphipathic.

Solution to Question 24:

Only test 3, which is Rothera's test for ketone bodies in the urine will be positive in the urine of a patient with prolonged starving.

Urine tests

Test 1 - Benedict's test for reducing sugars

Test 2 - Heat and acetic acid test for presence of albumin

Test 3 - Rothera's test for ketone bodies

In case of prolonged starvation,

- After a meal till about 2 hours later, maintenance of blood glucose is primarily from the diet. During this time, hepatic glycogen stores are getting replenished.
- As dietary blood glucose levels start to decline, glucagon secretion increases and insulin secretion decreases, resulting in rapid mobilisation of hepatic glycogen stores.
- If there is fasting for a longer period of time, hepatic glycogen stores cannot maintain blood glucose levels. 12 hours after the last meal, gluconeogenesis becomes the primary source of blood glucose, since most of the hepatic glycogen would have depleted. If an individual continues fasting, gluconeogenesis continues to be the main source.
- However, over the first few days of fasting, ketogenesis happens. This is when the concentration of acetyl CoA from fatty acid oxidation exceeds the oxidative capacity of the TCA cycle. Also, ketogenesis releases CoA, ensuring its availability for continued fatty acid oxidation. Water-soluble ketone bodies are used as fuel by most tissues, including the brain during fasting. It also reduces the need for gluconeogenesis from amino acid carbon skeletons, thus preserving essential protein in the body.

Therefore, during prolonged fasting, the levels of ketone bodies in the body rise and subsequently begin to be eliminated in urine. Hence, in starvation ketosis, Rothera's test will be positive, but Benedict's test will be negative.

Both Benedict's and Rothera's tests will be positive in diabetic ketoacidosis. (Option 1)

- Benedict's test (Test 1) for reducing sugars is a semiquantitative method for the estimation of sugar in the urine.
- Rothera's test (Test 3) is performed by saturating urine with solid ammonium sulfate. On adding sodium nitroprusside and liquor ammonia along the sides of the test tube, a purple ring is formed at the junction of the two liquids if there is acetone or acetoacetic acid in urine. Beta-hydroxybutyrate does not give a purple ring.
- Heat and acetic acid test (Test 2) for proteinuria is based on the fact that when heated to an isoelectric point, proteins get coagulated.

Color	Concentration
Green	0.5 g%
Yellow	1 g%
Orange	1.5 g%
Red	2 g%

Solution to Question 25:

The action of inhibition on acetylcholinesterase enzyme by organophosphorus compounds is through irreversible competitive inhibition.

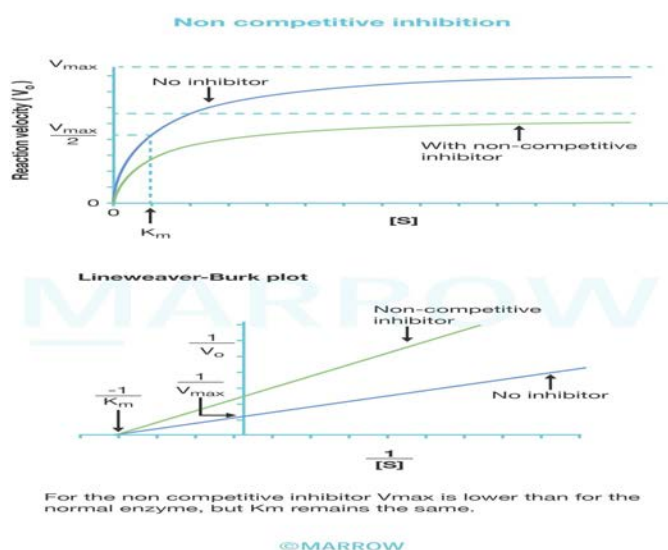
Competitive inhibition is a type of inhibition that occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy. It, therefore, competes with the substrate for binding to the enzyme active site. The effect of competitive inhibition can be overcome by increasing the concentration of the substrate. K_m is increased, but V_{max} is unaltered in this type of inhibition.

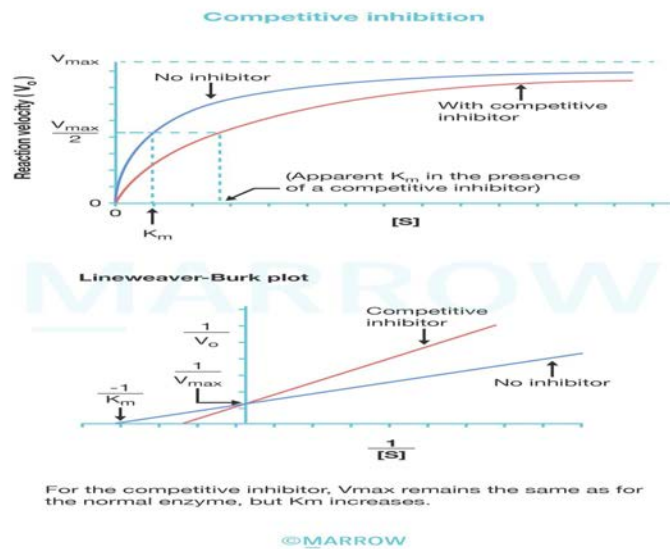
Competitive inhibition is of two types:

- Irreversible (non-equilibrium type) competitive inhibition: The inhibitor forms strong covalent bonds or has such a high affinity for the enzyme that the normal substrate is not able to displace the inhibitor. For example, organophosphorus poisoning inhibits acetylcholinesterase (AChE). Once an organophosphate binds to AChE, endogenous hydrolysis of the phosphorylated enzyme by esterases further leads to irreversible binding and permanent enzyme inactivation which is termed aging.
- Reversible (equilibrium type) competitive inhibition: Substrate concentration is sufficiently increased, it can displace the inhibitor. Statin drug inhibits HMG CoA reductase.

Noncompetitive inhibitors bind to enzymes at remote sites from the active site. Therefore, noncompetitive inhibition cannot be overcome by increasing the concentration of the substrate. Thus, noncompetitive inhibitors decrease the V_{max} of the reaction. They do not interfere with the binding of the substrate to the enzyme. Thus, the enzyme shows the same K_m in the presence or absence of noncompetitive inhibitors. For eg, acetazolamide inhibits carbonic anhydrase, digoxin inhibits $Na^+ K^+$ ATPase

Graph showing the changes in V_{max} and K_m noncompetitive and competitive inhibition.





Suicidal inhibition is a special type of irreversible inhibition. It is also known as mechanism-based inactivation. The inhibitor is a specialised substrate analog that can be transformed by the enzyme it is inhibiting. After binding to the active site of the enzyme, a highly reactive group is generated. This covalently binds to and inhibits the enzyme. An example is allopurinol, which is oxidised by xanthine oxidase to alloxanthine. Alloxanthine in turn is a strong inhibitor of xanthine oxidase.

Biochemistry NEET 2021

Question 1:

An alcoholic patient presented with itchy lesions as shown in the image, He also has a red-colored beefy tongue. Which of the following histories would you like to additionally ask/ elicit in this patient?



- a) Ask for diarrhea and history of memory loss
- b) Ask for diarrhea, thiamine deficiency features, and detailed dietary history
- c) History of allergy
- d) Ask for diarrhea and conjunctivitis

Question 2:

A patient presented with symptoms of delayed adaptation to vision in dark places, gritty sensation in the eyes, dry eyes, and softening of the cornea. What is the likely deficiency in this case?

- a) Retinoic acid
- b) Riboflavin
- c) Niacin
- d) Pantothenic acid

Question 3:

You are conducting an experiment on mitochondrial respiration. You add malate/pyruvate and respiration is normal. You add succinate and respiration is normal. When you add another substance in the presence of pyruvate/succinate and malate, respiration is blocked. Which of the following substances is most likely added?

- a) Rotenone
- b) Antimycin A
- c) Oligomycin
- d) 2,4-dinitrophenol

Question 4:

A patient presented with tendon xanthomas. His lipid profile showed serum cholesterol levels to be 283 mg/dL and serum LDL to be 202 mg/dL. He was started on lovastatin. What is the likely diagnosis?

- a) Familial hypercholesterolemia
- b) Tangier's disease
- c) Lipoprotein lipase deficiency
- d) Familial hyperchylomicronemia

Question 5:

Which of the following glycosaminoglycans (GAGs) are responsible for corneal transparency?

- a) Keratan sulfate
- b) Chondroitin sulfate
- c) Heparan sulfate
- d) Hyaluronic acid

Question 6:

A person who had dinner at 8 pm the previous night, tested his blood glucose levels at 7 am the next morning. What is the source of blood glucose in this case?

- a) Hepatic glycogenolysis
- b) Hepatic gluconeogenesis

- c) Muscle glycogenolysis
- d) Dietary glucose

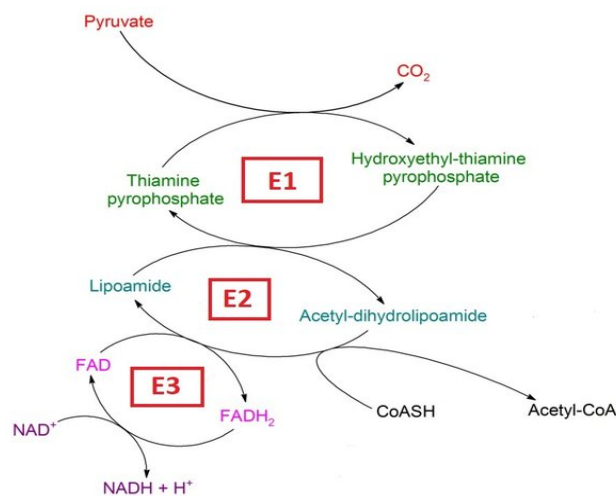
Question 7:

A patient who has been diagnosed with multiple gastrointestinal polyps and carcinoma, with a positive family history/history of hereditary non-polyposis colon carcinoma has a defect in which of the following?

- a) Mismatch repair
- b) Nucleotide excision repair
- c) Base excision repair
- d) Point mutation

Question 8:

The enzymes that convert pyruvate to acetyl CoA are given in the diagram. What are the enzymes marked as E1 and E2 respectively?



- a) Pyruvate dehydrogenase and dihydrolipoyl transacetylase
- b) Pyruvate carboxylase and dihydrolipoyl transacetylase
- c) Pyruvate dehydrogenase and dihydrolipoyl dehydrogenase
- d) Pyruvate carboxylase and dihydrolipoyl dehydrogenase

Question 9:

A young girl presented to the OPD for examination. She is tall and thin with slender fingers, and dislocated lenses. Workup revealed cystathionine beta-synthase deficiency. Which amino acid supplement is required in this case?

- a) Cysteine
- b) Methionine
- c) Serine
- d) Tyrosine

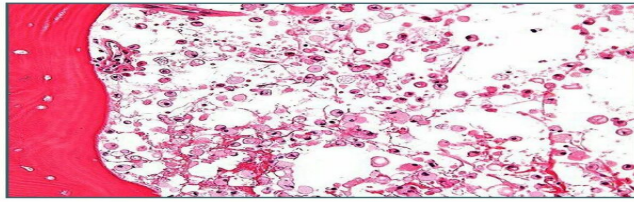
Question 10:

A patient presented with joint pain and on evaluation, urine turns black on standing. What is the likely enzyme deficiency?

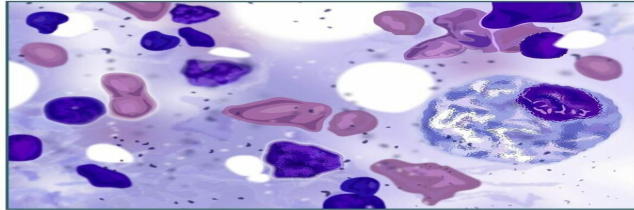
- a) Homogentisic acid 1, 2 - dioxygenase
- b) Tyrosine transaminase
- c) Fumarylacetoacetate hydrolase
- d) 4-hydroxy-phenyl pyruvate dioxygenase

Question 11:

A 5-year-old boy presented with easy fatigability, irritability and inability to concentrate. Labs revealed the following findings a and b (light and oil immersion respectively) on the bone marrow aspiration. Which of the following is the most likely enzyme deficient in this condition?



a



b

©Marrow

- a) Hexosaminidase
- b) Glucocerebrosidase
- c) Sphingomyelinase
- d) N-acetylglucosaminidase

Question 12:

A farmer presented with loose stools and rashes over the sun-exposed areas. Careful diet history reveals a diet rich in maize. Which of the following deficiencies would you suspect?

- a) Niacin
- b) Pyridoxine
- c) Thiamine
- d) Biotin

Answer Key

Question No.	Correct Option
1	a
2	a
3	c
4	a

5	a
6	a
7	a
8	a
9	a
10	a
11	b
12	a

Detailed Explanations

Solution to Question 1:

An alcoholic patient with a red-colored beefy tongue, itching and the characteristic Casal's necklace, shown in the image, is suggestive of pellagra. It is caused by niacin/vitamin B₃ deficiency. Hence, one should ask for a history of diarrhea and memory loss in this case.

Pellagra presents with 4 Ds: Diarrhea, Dermatitis, Dementia, and if untreated, Death. Early symptoms of pellagra are loss of appetite, generalized weakness and irritability, abdominal pain, and vomiting. Photosensitive dermatitis is seen over sun-exposed areas. Diarrhea occurs due to proctitis or malabsorption. Bright red glossitis (beefy tongue) and Casal's necklace (a pigmented, scaly skin rash that forms a ring around the neck) follow. Depressive psychosis and dementia also commonly occur.

Corn/ maize/ jowar based diets are the main causes of niacin deficiency. Corn/maize lacks niacin and tryptophan. The excess leucine in jowar and maize disrupts the conversion of tryptophan to niacin. Niacin deficiency is also seen in alcoholism. Hence, additional histories of diarrhea and memory loss will aid in supporting the diagnosis of pellagra (along with dietary history).

Treatment includes oral supplementation with 100–200 mg of nicotinamide or nicotinic acid three times daily for 5 days.

Option B: Thiamine deficiency (vitamin B₁) features include CVS manifestations like heart failure and edema (wet beriberi), peripheral neuritis (dry beriberi), and Wernicke encephalopathy (in alcoholics).

Option C: Dermatitis seen in pellagra is due to dietary deficiency of niacin and so allergy history is not relevant.

Option D: Conjunctivitis is not a feature of niacin deficiency.

Solution to Question 2:

The clinical picture of delayed adaptation to vision in dark places, gritty sensation in the eyes, dry eyes, and softening of the cornea is suggestive of vitamin A deficiency in the patient. Hence

retinoic acid (the active form of vitamin A) is likely to be deficient.

Vitamin A is a fat-soluble vitamin that can occur as carotenoids (most common - beta carotene from plant sources) or retinoids (from animal sources).

3 forms of retinoids are:

- Retinal is required mainly for normal vision (present in the visual pigment rhodopsin).
- Retinol is important for the normal functioning of the reproductive system.
- Retinoic acid has a role in the regulation of gene expression and tissue differentiation.

Ocular manifestations of vitamin A deficiency include loss of sensitivity to green light (earliest to appear), nyctalopia (night blindness), xerophthalmia.

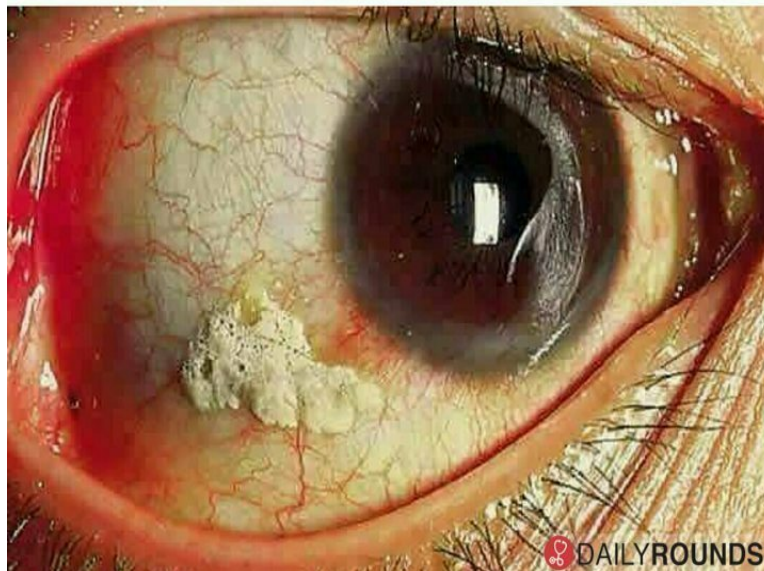
Xerophthalmia includes conjunctival xerosis (dryness) with Bitot's spots (white patches of keratinized epithelium appearing on the sclera), corneal ulcers, and keratomalacia (softening of the cornea) that leads to corneal scarring and ultimately, blindness.

Other manifestations: General impairment of immunity, keratinization of skin (phrynoderma/toad skin).

Treatment: The WHO recommended schedule for all patients above the age of 1 year (except women of reproductive age): 200,000 IU of vitamin A orally or 100,000 IU by intramuscular injection should be given immediately on diagnosis (day 0) and repeated the following day (day 1) and 14 days later.

Dietary sources of vitamin A are liver, fish, eggs, dark green, and deeply colored fruits and vegetables.

Bitot's spots



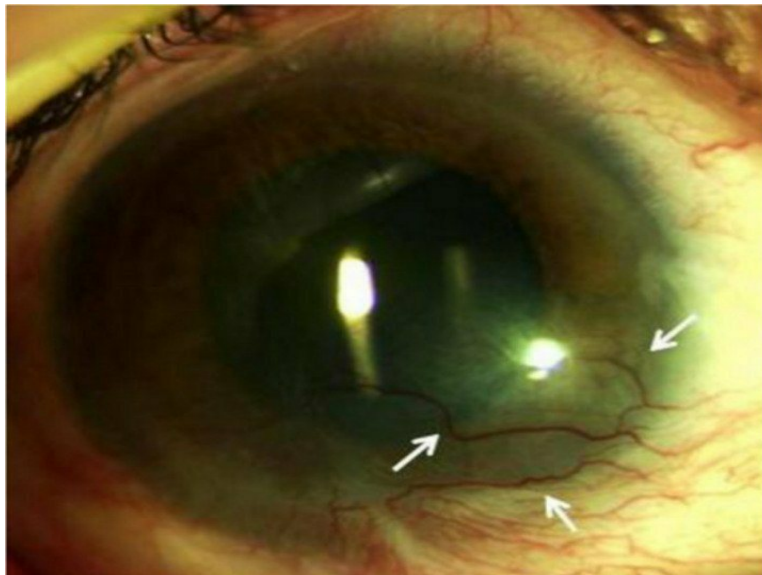
The below image shows phrynoderma with multiple keratotic papules



Other options:

Option B: Ocular manifestations of riboflavin deficiency (vitamin B2) are photophobia, burning sensation, and vascularization of the cornea. Other features include cheilosis, magenta-colored tongue, glossitis, and seborrheic dermatitis.

Corneal vascularization. Seen in riboflavin (Vitamin B2) deficiency.



Option C: Niacin deficiency (vitamin B3) leads to pellagra. Dermatitis, diarrhea, and dementia are its common features.

Casal's necklace finding in niacin deficiency



Option D: Features of pantothenic acid deficiency (vitamin B5) are burning foot syndrome ("nutritional melalgia"), dermatitis, enteritis, alopecia, ataxia.

Solution to Question 3:

In the experiment conducted on mitochondrial respiration, respiration will be blocked by the addition of oligomycin, an antibiotic, when added in the presence of pyruvate/ succinate and malate. Oligomycin inhibits oxidative phosphorylation in the mitochondria and hence blocks respiration.

Electron transport chain (ETC)/respiratory chain is the final pathway by which all the electrons derived from the oxidation of carbohydrates, fatty acids, and amino acids are directed towards the final reaction with oxygen, to form water and adenosine triphosphate (ATP).

The chemiosmotic theory states that the flow of electrons through the respiratory chain generates a proton motive force. This force drives the formation of ATP by the process of oxidative phosphorylation.

In the given experiment, malate and pyruvate are electron donors (NADH) at complex I of the ETC, and succinate is the electron donor (FADH₂) at complex II. Normal respiration is observed on the addition of substances at both complexes I and II i.e., both oxygen consumption and ATP synthesis from ADP (adenosine diphosphate) take place.

Oligomycin is a competitive inhibitor of the F_o (oligomycin-sensitive) complex of ATP synthase (complex V) and blocks oxidation and phosphorylation by stopping the flow of protons through the ATP synthase enzyme complex. Hence the addition of oligomycin will inhibit complex V and stop respiration.

Other options:

Options A and B: Rotenone (option A) if added, will inhibit complex I. Electron transfer before complex I will be blocked but not after. Hence, normal respiration will occur. Similarly, antimycin A (option B) inhibits complex III, and electron transfer prior to complex III will be blocked and

not after. Hence, respiration will not be completely hindered.

Option C: 2,4-dinitrophenol (DNP) is an uncoupler that dissociates oxidation in the respiratory chain from phosphorylation. Hence, respiration continues without ATP synthesis.

Complexes of ETC	Inhibitors
Complex I: NADH-CoQ oxidoreductase	Piericidin, amobarbital, rotenone
Complex II: Succinate-CoQ reductase	TTFA (thenoyltrifluoroacetone), carboxin, malonate
Complex III: CoQH ₂ -cytochrome c oxidoreductase	BAL(British anti-lewisite), antimycin A
Complex IV: cytochrome oxidase	Hydrogen sulfide, carbon monoxide, cyanide
Complex V (informally): ATP Synthase	Atractyloside, oligomycin

Solution to Question 4:

The clinical presentation with tendon xanthomas and elevated serum LDL (low-density lipoprotein) and elevated serum cholesterol points to a diagnosis of familial hypercholesterolemia (FH).

Familial hypercholesterolemia (type IIa hyperlipoproteinemia) is an inherited disorder due to defects in LDL receptors. This defect causes a decrease in the uptake of LDL and cholesterol. As a result, the blood levels of both LDL and cholesterol are elevated. Such patients are at increased risk to develop atherosclerosis and coronary artery disease.

Features of FH:

- Painful polyarthritis and tendinitis are common.
- Physical findings seen are corneal arcus and tendon xanthomas over Achilles tendon and the dorsum of hands. Xanthomas are tumorous masses produced by foamy cell clusters (lipid-laden macrophages) found in the connective tissue of the skin and in tendons.

Diagnosis is done clinically based on hypercholesterolemia with

- LDL $\geq$ 190 mg/dl,
- Family history of hypercholesterolemia and/or
- Premature coronary artery disease.

Treatment includes lifestyle changes, restriction of a fatty diet, and pharmacotherapy with statins in heterozygous FH patients.

Recent drugs for specific treatment of homozygous FH are:

- Lomitapide, a microsomal triglyceride transfer protein (MTP) inhibitor, and
- Mipomersen, an antisense oligonucleotide against apoB.

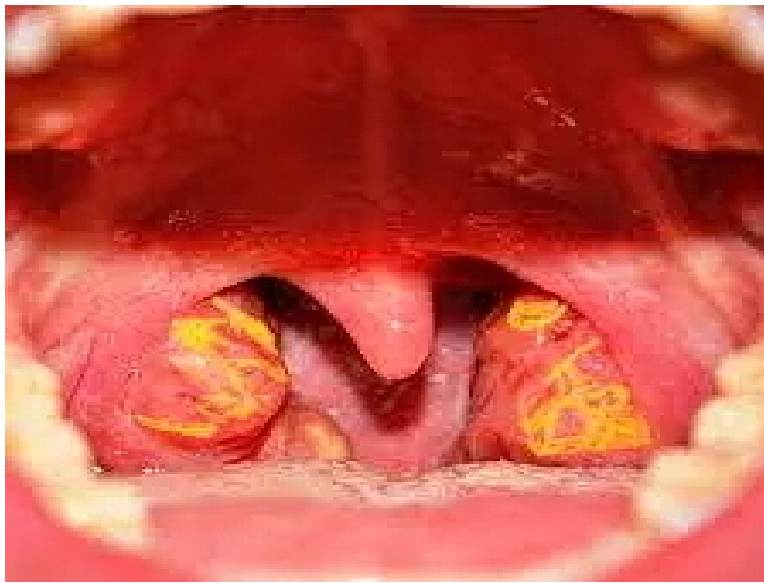
The image below shows Achilles tendon xanthoma and a plain lateral radiograph of both ankles showing bilateral symmetrical swellings (arrows) superior to the calcaneum respectively.



Other options

Option B: Tangier's disease is a hypolipoproteinemia caused by a mutation of the ABCA1 gene. Enlarged orange tonsils are pathognomonic. Low HDL levels are seen. There is an increased risk for atherosclerosis.

The image below shows enlarged yellow-orange colored tonsils in Tangier's disease.



Option C: Lipoprotein lipase (LPL) deficiency causes type I hyperlipoproteinemia due to defects in LPL or apo C-II. Eruptive xanthomas are seen. Chylomicrons and VLDL are elevated. LDL and HDL are low. There is no risk for coronary disease.

Option D: Familial hyperchylomicronemia syndrome is caused due to either LPL deficiency, ApoC-II deficiency, ApoA-V deficiency, or GPIHBP1 mutation. Clinically, patients present with recurrent episodes of severe abdominal pain due to acute pancreatitis. The retinal blood vessels may be opalescent and are called lipemia retinalis. Increased levels of triglycerides (mostly >1000 mg/dL) is seen. The risk of premature coronary heart disease is low.

Solution to Question 5:

The glycosaminoglycan (GAG) responsible for corneal transparency is keratan sulfate.

Glycosaminoglycans or mucopolysaccharides are complex carbohydrates made up of amino sugars and uronic acids. They may be attached to a protein molecule to form a proteoglycan.

Keratan sulfate is a mucopolysaccharide that does not contain uronic acid. It is constituted by repeating disaccharide units of N-acetyl glucosamine and galactose.

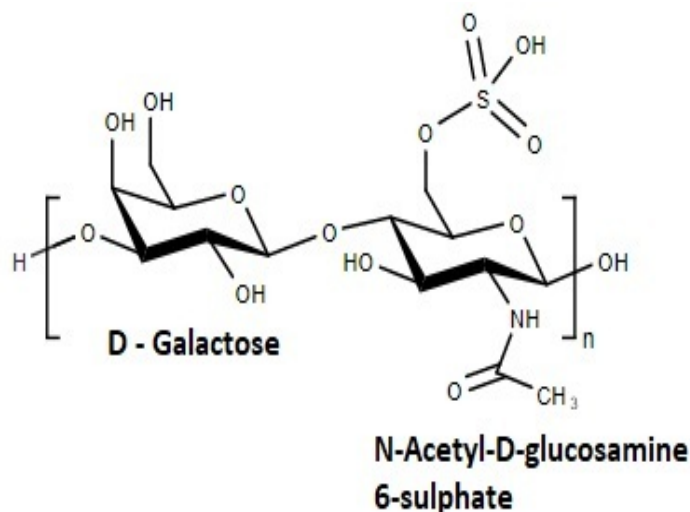
Keratan sulfate is present between the collagen fibrils in the cornea and plays a major role in corneal transparency.

2 types of keratan sulfate are seen:

- Keratan sulfate I is present in the cornea.
- Keratan sulfate II is mainly present in the cartilage.

Keratan sulfate I and II differ in their linkage to the core proteins. Keratan sulfate I links to a core protein L-asparagine via N-acetylglucosamine and Keratan sulfate II to L-Serine (or L-Threonine) via N-acetylgalactosamine.

The chemical structure of Keratan sulfate is shown in the image below.



Other options:

Option B: In chondroitin sulfates (chondroitin 4-Sulfate & chondroitin 6-Sulfate), the repeating disaccharide units are glucuronic acid and N-acetylgalactosamine. They maintain the structure of the extracellular matrix (ECM) and are located in endochondral bone and cartilage. They are also found in the ECM of the central nervous system. They act as signaling molecules in the prevention of the repair of nerve endings after injury.

Option C: Heparan sulfate is found on many extracellular cell surfaces and contains glucosamine with fewer N-sulfates than heparin. The predominant uronic acid is d-glucuronic acid. In the plasma membrane of cells, they act as receptors and mediate cell growth and cell-cell communication. It is also found in the basement membrane of the kidney along with type IV collagen and laminin, where it determines the charge selectiveness of glomerular filtration.

Option D: Hyaluronic acid is present in synovial fluid. It is also found in skin, bone, cartilage, vitreous humor, and embryonic tissues. It contains repeating units of N Acetyl Glucosamine and glucuronic acid. It plays a role in morphogenesis and wound repair. The high concentrations of hyaluronic acid along with chondroitin sulfates in cartilage contribute to its compressibility.

Solution to Question 6:

The source of blood glucose in a person fasting overnight for 11 hours (8 pm to 7 am) is hepatic glycogenolysis.

First stage of fasting: The fasting state begins two to four hours after the absorptive/well-fed state. In early fasting i.e; 4-18 hours after the well-fed state, the blood glucose level is maintained by hepatic glycogenolysis. The glycogen stores are sufficient for around 18 hours after a meal following which they start to deplete. So the source of blood glucose in a person fasting for 11 hours will be liver glycogenolysis.

Second stage of fasting: Gluconeogenesis begins 4–6 hours after the last meal but becomes fully active only after stores of liver glycogen are depleted (after 18 hours). They can serve as a source

of glucose for up to 2 days.

Substrates for gluconeogenesis are:

- Glucogenic amino acids (mainly alanine) from the muscle and
- Glycerol from adipose tissue

During fasting, the glucogenic amino acids are transported to the liver, transaminated into pyruvate, and converted to glucose.

Third stage of fasting:

- Lipolysis occurs in the adipose tissue due to the activity of hormone-sensitive lipase, stimulated by increased glucagon:insulin ratio.
- Free fatty acids are the source of energy in this stage and not glucose. Triacylglycerols will be hydrolyzed to fatty acids and glycerol.
- Beta oxidation of fatty acids takes place which provides acetyl CoA.
- In prolonged starvation (>3 days), acetyl CoA is converted to ketone bodies in the liver which will act as the main fuel source for skeletal and heart muscle and can meet up to 20% of the brain's energy needs.

Final stage of fasting: After several weeks of starvation, ketone body synthesis will not take place due to depleted triacylglycerol stores. Proteins remain the only source of fuel. Protein degradation increases, and death results from a loss of heart, liver, or kidney function.

Role of insulin and glucagon:

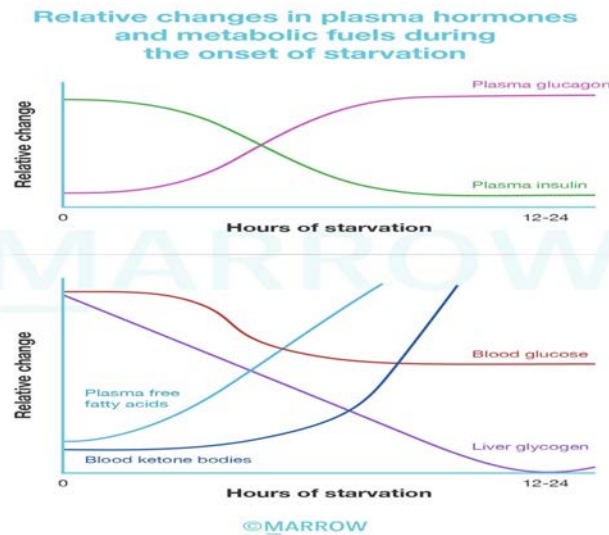
When blood glucose falls during fasting, there is increased secretion of glucagon and decreased release of insulin. Hence glucagon:insulin ratio is increased.

Effects of increased glucagon:insulin ratio on enzymes of glycogenolysis and gluconeogenesis:

- It activates glycogen phosphorylase, the key enzyme of glycogenolysis, and deactivates glycogen synthase. This causes a rapid mobilization of liver glycogen stores to be utilized as fuel.
- It favors the synthesis of glucose-6-phosphatase which converts glucose-6-phosphate to glucose. The resulting glucose is used by red blood cells (lack mitochondria) and the brain as they are largely dependant on glucose in the fasting state.
- Phosphoenol pyruvate carboxykinase and fructose-1, 6-bisphosphatase are induced, leading to increased gluconeogenesis.

Metabolic processes	Fasting state	Fed state
Glucagon: insulin	Increased	Decreased
Gluconeogenesis	Increased	Decreased
Glycogenolysis	Increased	Decreased
Ketogenesis	Increased	Decreased

Metabolic process	Fasting state	Fed state
Glycogen storage	Decreased	Increased

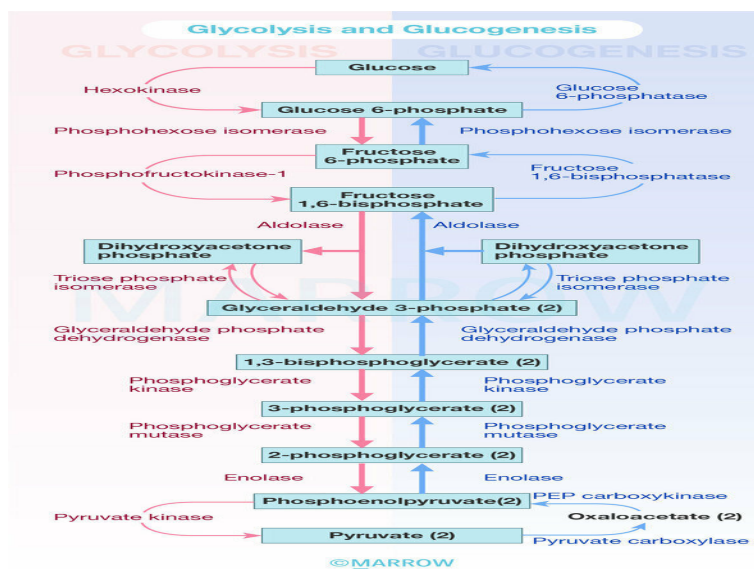


Other options:

Option C: Muscle glycogen cannot contribute directly to plasma glucose, since muscle lacks glucose-6-phosphatase. So glycogen cannot be converted to glucose to be used as fuel during starvation.

Option D: In the well-fed state/ absorptive state of two to four hours after a meal, dietary glucose is the main source of fuel.

The below image shows steps of glycolysis and gluconeogenesis.



Solution to Question 7:

Mismatch repair will be defective in a patient with multiple gastrointestinal polyps and carcinoma, with a positive family history/history of hereditary non-polyposis colon carcinoma (HNPCC)

Mismatch repair is a DNA repair mechanism that corrects single-nucleotide mismatches, insertions, and deletions introduced by the DNA polymerase during DNA replication.

HNPCC or Lynch syndrome is an autosomal dominant genetic condition. It occurs due to defects in DNA mismatch repair genes like MLH1 and MSH2. The genomes of these patients show microsatellite instability. HNPCC is associated with a high risk of colon cancer. It may also be associated with a risk of developing carcinomas of the endometrium, stomach, small bowel, ovary, pancreas, and transitional cell carcinoma of the ureter and renal pelvis.

The diagnostic criteria for HNPCC are referred to as the Amsterdam criteria or the 3-2-1-0 rule. It includes:

- Three or more relatives with HNPCC-associated cancer (colorectal cancer, endometrial cancer, cancer of the small bowel, ureter, or renal pelvis), one of whom is a first-degree relative of the other two.
- At least two successive generations affected
- At least one case diagnosed before age 50 years
- Familial adenomatous polyposis excluded
- Tumors verified by pathologic examination.

Other options:

Option B: Patients with xeroderma pigmentosum have a defect in the nucleotide excision repair pathway.

Option C: Defect in base excision repair pathway is seen in patients with MUTYH-associated polyposis.

Option D: Sickle cell disease and beta-thalassemia are examples of diseases seen in patients with point mutations.

Solution to Question 8:

E1 is pyruvate dehydrogenase and E2 is dihydrolipoyl transacetylase in the image showing oxidative decarboxylation of pyruvate. This is catalyzed by the pyruvate dehydrogenase (PDH) complex where E1, E2, and E3 are the three enzyme components of the complex.

The pyruvate dehydrogenase (PDH) complex is a multienzyme complex that is associated with the inner mitochondrial membrane. It catalyzes the oxidative decarboxylation of pyruvate to acetyl-CoA which enters the citric acid/Kreb's cycle.

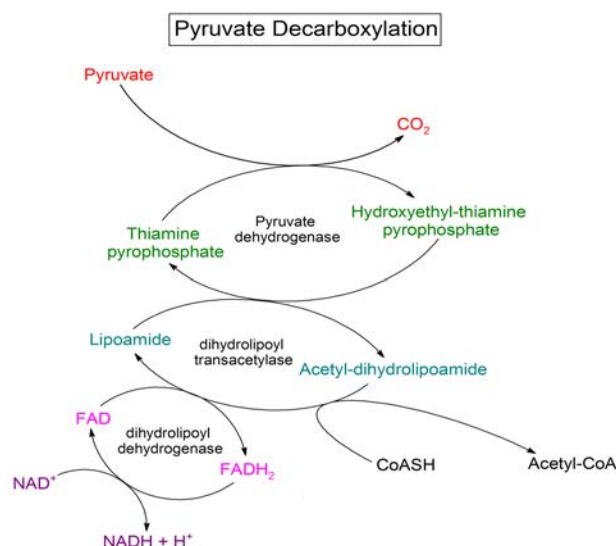
Steps of pyruvate decarboxylation:

- Pyruvate is transported into the mitochondria by a proton symporter.

- Pyruvate is decarboxylated by pyruvate dehydrogenase (E1). TPP is required for this step. The decarboxylated pyruvate remains bound to the enzyme, as hydroxyethyl thiamine pyrophosphate.
- The hydroxyethyl group is oxidized to form an acetyl group. It is then transferred from TPP to lipoamide to form acetyl lipoamide. This step is catalyzed by dihydrolipoyl transacetylase (E2).
- Acetyl lipoamide reacts with coenzyme A to form acetyl-CoA and reduced lipoamide.
- The reduced lipoamide is reoxidized by FAD-dependent dihydrolipoyl dehydrogenase (E3).

The overall reaction is : $\text{Pyruvate} + \text{NAD}^+ + \text{CoA} \rightarrow \text{Acetyl-CoA} + \text{NADH} + \text{H}^+ + \text{CO}_2$

Enzyme components of PDH complex	Co-factors
Pyruvate dehydrogenase (E1)	Thiamine pyrophosphate (TPP)
Dihydrolipoyl transacetylase (E2)	Lipoic acid Coenzyme A
Dihydrolipoyl dehydrogenase (E3)	Flavin adenine dinucleotide (FAD) Nicotinamide adenine dinucleotide (NAD ⁺)



Clinical significance:

- Dietary deficiency of thiamine inhibits PDH. Chronic alcoholics may develop potentially fatal pyruvic and lactic acidosis. Lactic acidosis is a common presentation in patients with inherited pyruvate dehydrogenase deficiency too.
- In arsenic and mercury poisoning, the heavy metal ions react with -SH groups of lipoic acid and inhibit pyruvate dehydrogenase.

Other multienzyme complexes include alpha-ketoglutarate dehydrogenase and branched-chain ketoacid dehydrogenase. All the five coenzymes of PDH are common to these two complexes too.

Solution to Question 9:

The clinical picture of a tall and thin young girl with slender fingers, dislocated lenses, and cystathionine beta-synthase deficiency is suggestive of classic homocystinuria. The amino acid supplement required in this scenario is cysteine.

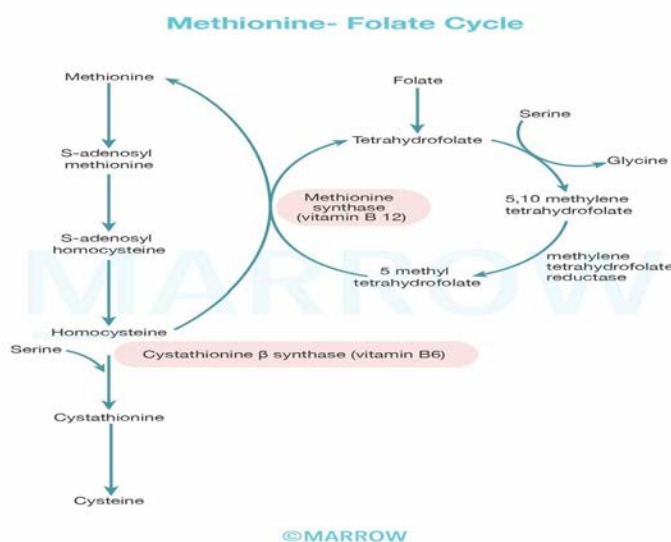
Classic homocystinuria is a metabolic disorder of methionine metabolism, due to cystathionine beta-synthase deficiency. It is an autosomal recessive condition.

Cystathionine beta-synthase is a pyridoxal phosphate-dependent enzyme that condenses homocysteine with serine to form cystathionine. The absence of this enzyme will cause cysteine deficiency, hence cysteine needs to be supplemented in the diet.

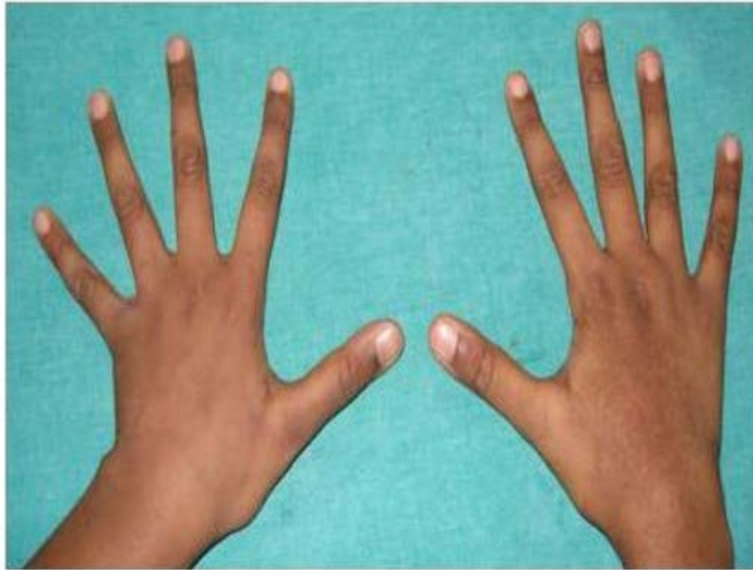
Clinical features:

- Affected children are initially asymptomatic.
- Ectopia lentis (subluxation of the lens) is seen first, after 3 years of age.
- Intellectual disability and psychiatric and behavioral disorders.
- Thromboembolic episodes.
- Skeletal abnormalities resembling those of Marfan syndrome: They are tall and thin with elongated limbs and arachnodactyly (slender fingers).

Elevations of both methionine and homocysteine in body fluids are the diagnostic laboratory findings. The diagnosis is established by molecular analysis of cystathionine β -synthase (CBS) or by assay of the enzyme in cultured fibroblasts, phytohemagglutinin-stimulated lymphocytes, or liver biopsy specimens. Treatment includes restriction of methionine intake and supplementation with cysteine, betaine, vitamins B6, B12, and folate.



The image below shows arachnodactyly.



Methionine supplementation is required in non-classical homocystinuria. It is due to impaired re-methylation of homocysteine to methionine. This can be caused by defective methionine synthase or reduced availability of two essential cofactors, 5-methyltetrahydrofolate, and methylcobalamin.

Elevated homocysteine levels with low levels of methionine will be seen. Treatment includes methionine supplementation and administration of methyl folate, hydroxocobalamin, and betaine.

Solution to Question 10:

The urine turning black in color, on exposure to air, along with joint pain is suggestive of alkaptonuria. It occurs due to the defect in the homogentisic acid 1, 2 - dioxygenase enzyme.

Alkaptonuria is an autosomal recessive disorder of tyrosine metabolism, that develops due to the defect in homogentisate oxidase/homogentisic acid 1, 2 - dioxygenase, which oxidizes homogentisate to maleylacetoacetate. The defect in this enzyme leads to the accumulation of homogentisate, which is then excreted in the urine.

Alkaptonuria is characterized by homogentisic aciduria, ochronosis, and arthropathy (joint pain).

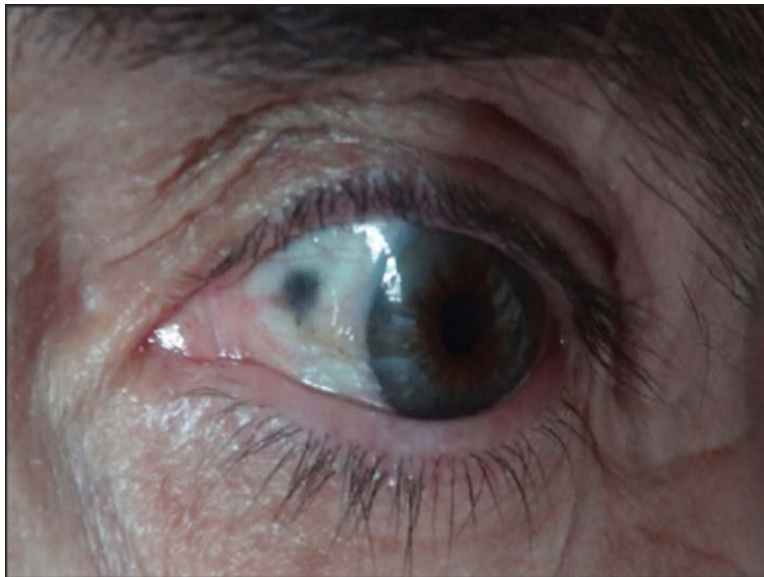
Homogentisic aciduria is the black discoloration of urine on prolonged exposure to air caused due to oxidation of the excreted homogentisate.

Ochronosis is the accumulation of oxidized homogentisic acid pigment (alkapton bodies) in connective tissues leading to bluish-black discoloration of tissues mostly involving ear cartilage, ocular tissue, the skin of the eyelids, forehead, cheeks, and hands, nail beds, and intervertebral discs. The compound that accumulates in the tissues in ochronosis is benzoquinone acetate.

Painful arthritis (arthropathy) may be seen from the fourth or fifth decade.

Diagnosis: Blackish discoloration of urine on standing is diagnostic of alkaptonuria (shown below). The ferric chloride test will be positive.

The image further below shows Ochronosis.



Treatment: No specific treatment is required. Symptomatic treatment with analgesics, spinal surgery, and arthroplasty along with ascorbic acid and protein restriction can be considered. Nitisinone (p-hydroxyphenylpyruvic acid (pHPPA) oxidase inhibitor) reduces urinary excretion of homogentisic acid.

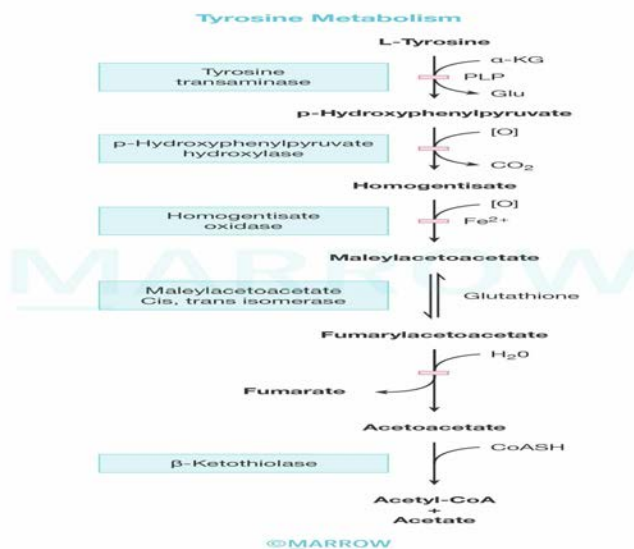
Other options:

Option B: Tyrosine transaminase deficiency causes type II tyrosinemia (Richner-Hanhart syndrome). Patients present with painful corneal erosions with photophobia, learning disability, and palmoplantar keratosis.

Option C: Fumarylacetoacetate hydrolase deficiency causes type I tyrosinemia (Tyrosinosis). Patients generally present with failure to thrive, neuropathy, boiled-cabbage urine, and liver failure secondary to cirrhosis.

Option D: 4-hydroxy-phenyl pyruvate dioxygenase deficiency causes type III tyrosinemia (neonatal tyrosinemia). Patients might rarely have a mental disability. The liver function is generally normal.

The steps of tyrosine metabolism are shown below.



Solution to Question 11:

The given clinical scenario of a child with easy fatiguability, irritability, and inability to concentrate with the images showing a crumpled tissue paper appearance (Gaucher's cell) is suggestive of Gaucher's disease. The deficient enzyme is glucocerebrosidase (also known as beta-glucosidase). I

Gaucher's disease is an inborn error of metabolism caused due to a deficiency of beta glucocerebrosidase. It is the most common lysosomal storage disorder and the mode of inheritance is autosomal recessive. There is an accumulation of lipid-laden macrophages in the spleen, liver, and other tissues.

There are three clinical subtypes - type I (chronic non-neuronopathic), type II (acute neuronopathic), and type III (intermediate). Clinical features include anemia, thrombocytopenia, hepatosplenomegaly, and neurological features depending on the subtype.

Gaucher's cells - These are formed as a result of the accumulation of massive amounts of glucocerebrosides within the phagocytes. The cells have a fibrillary cytoplasm and give a crumpled tissue paper appearance.

Option A: Deficiency of beta-hexosaminidase A is seen in Tay-Sachs disease. The deficiency of beta-hexosaminidase A and B causes Sandhoff's disease. Clinical features are seizures, macrocephaly, and hyperacusis.

Option C: Deficiency of sphingomyelinase leads to Niemann-Pick disease. It is characterized by the presence of foam cells in the bone marrow. There may be pulmonary infiltrates leading to lung failure.

Option D: Deficiency of N-acetylglucosaminidase leads to Sanfillipo's disease. It is a type of mucopolysaccharidosis (MPS - III). There is an accumulation of heparan sulfate leading to severe cognitive degeneration.

Solution to Question 12:

History of a maize-rich diet along with loose stools and rashes over the sun-exposed areas is suggestive of niacin/vitamin B3 deficiency.

Niacin deficiency causes pellagra, found mostly among people eating corn/maize-based diets, which is low in niacin and tryptophan. Excess leucine in jowar and maize disrupts the conversion of tryptophan to niacin.

Niacin can be synthesized in the body from tryptophan. Two compounds- nicotinic acid, and nicotinamide have the biological activity of niacin. The nicotinamide ring of the coenzymes NAD (nicotinamide adenine dinucleotide) and NADP (nicotinamide adenine dinucleotide phosphate) participate in oxidation/reduction reactions.

The initial symptoms of pellagra include loss of appetite, generalized weakness, irritability, abdominal pain, and vomiting. This is followed by bright red glossitis and a characteristic pigmented, scaly skin rash in sun-exposed areas. This rash is known as Casal's necklace when it forms a ring around the neck.

The 4Ds of pellagra:

- Diarrhea
- Dermatitis
- Dementia, leading to
- Death

It is diagnosed by measuring urinary levels of 2-methyl nicotinamide and 2-pyridone.

It is treated by oral supplementation of 100–200 mg of nicotinamide or nicotinic acid three times daily for 5 days.

Casal's necklace finding in niacin deficiency



Other options:

Option B: Dietary pyridoxine deficiency (vitamin B6) is rare but has been observed in alcoholics, newborn infants formulas low in vitamin B6, and women taking oral contraceptives. Isoniazid, an anti-TB drug, leads to pyridoxine deficiency resulting in peripheral neuropathy. Hence pyridoxine supplementation is included in ATT regimens.

Option C: Thiamine deficiency (vitamin B1) can manifest as wet beriberi (CVS features like heart failure and edema) or Shoshin beriberi, an acute and fulminant form of wet beriberi, dry beriberi (peripheral neuritis), or as Wernicke encephalopathy with Korsakoff psychosis (in alcoholics). Dietary deficiency of thiamine inhibits pyruvate dehydrogenase. Chronic alcoholics may develop potentially fatal pyruvic and lactic acidosis.

Option D: Dietary biotin deficiency is unknown but rarely, dermatitis, glossitis, loss of appetite, and nausea may occur after consumption of large amounts of uncooked egg white, which contains the protein avidin. It binds biotin and makes it unavailable for absorption.

Note: A similar image-based question on vitamin B3 deficiency was previously asked in NEET PG 2020. Vitamin deficiencies are a high-yield topic and are repeatedly tested every year.

Biochemistry INI-CET 2022

Question 1:

Which is the correct sequence of steps in the isolation of desirable protein using recombinant DNA technology?

- a) 2, 1, 3, 5, 4
- b) 1, 2, 4, 3, 5
- c) 1, 5, 2, 4, 3
- d) 2, 4, 5, 3, 1

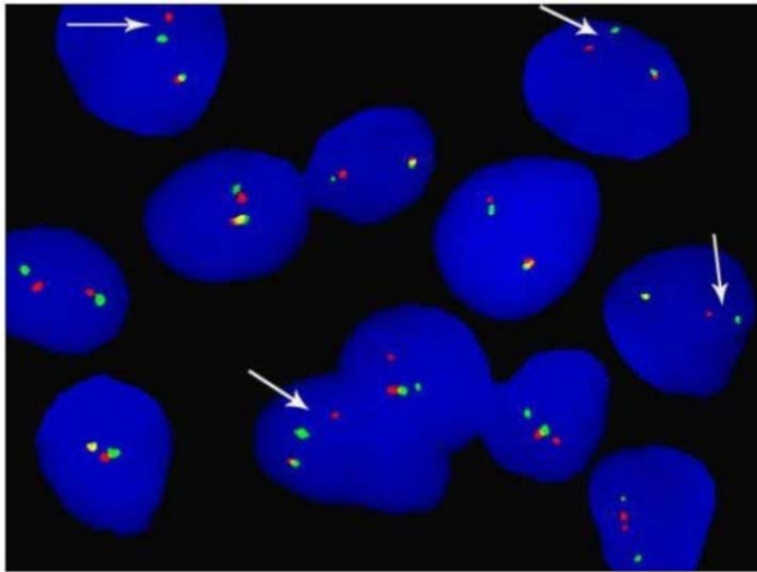
Question 2:

Which of the following techniques is used to detect DNA-protein interaction?

- a) DNA footprinting
- b) DNA fingerprinting
- c) ELISA
- d) Northern blotting

Question 3:

A patient was diagnosed with Ewing sarcoma with EWSR1 gene mutation. The technique shown in the image was performed to confirm the diagnosis. Identify the technique shown in the image.



- a) Fluorescent in situ hybridisation
- b) Karyotyping
- c) Immunohistochemistry
- d) Sanger's technique

Question 4:

In a patient with maple syrup urine disease, all of the following amino acids should be restricted in diet except?

- a) Methionine
- b) Valine
- c) Leucine
- d) Isoleucine

Question 5:

Which of the following statements about the human genome project is false?

- a) Total 20,500 genes were identified
- b) The project spanned over 13 years
- c) It required international collaboration
- d) Next generation sequencing was used

Question 6:

Banding technique used for dicentric chromosomes is:

- a) G banding
- b) C banding
- c) NOR
- d) R banding

Question 7:

Affinity maturation of antibodies is because of?

- a) Somatic hypermutation
- b) Differential mRNA processing
- c) CD40
- d) Gene rearrangements

Question 8:

Best method for detection of mutations with low allele frequency is

- a) Droplet digital PCR
- b) FISH
- c) Nested PCR
- d) Sanger Sequencing

Question 9:

In Sanger sequencing, ddNTP is used because:

- a) It is used to increase DNA polymerase activity
- b) It can be made fluorescent
- c) They exhibit higher affinity for DNA polymerases
- d) It doesn't allow further elongation of DNA

Question 10:

Misincorporation of uracil into DNA is seen in deficiency of:

- a) Vitamin B12
- b) Folate
- c) Vitamin B6
- d) Vitamin E

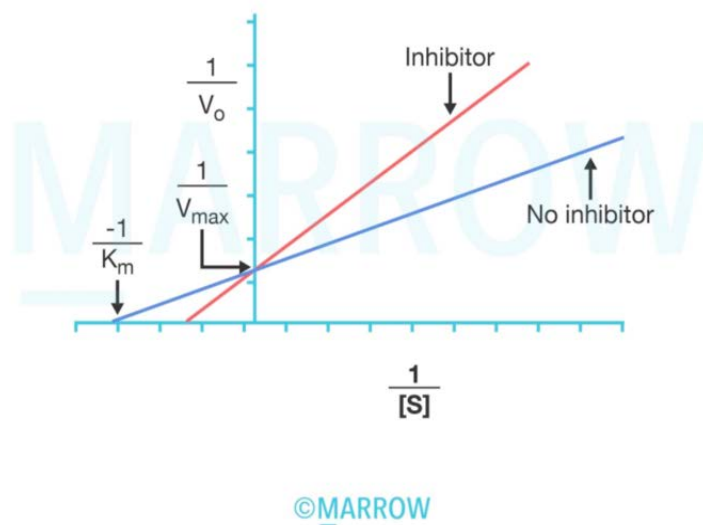
Question 11:

Deamination of methylated cytosine forms which of the following?

- a) Uracil
- b) Cytosine
- c) Guanine
- d) Thymine

Question 12:

Which of the following types of inhibition is depicted in the given Lineweaver-Burk plot?



- a) Competitive inhibition
- b) Non-competitive inhibition
- c) Uncompetitive inhibition
- d) Suicidal inhibition

Question 13:

Glycogen phosphorylase is activated by:

- a) Calcium
- b) Glucose-6-phosphate
- c) Insulin
- d) Protein phosphatase

Question 14:

Mutation of the gene of cystic fibrosis occurs with:

- a) Tryptophan at 708 position, chloride channel
- b) Phenylalanine at 708 position, calcium channel
- c) Phenylalanine at 508 position, chloride channel
- d) Tryptophan at 508 position, calcium channel

Question 15:

A baby is brought to the OPD with complaints of lethargy and poor feeding. On examination, it is noted that the urine had a smell of burnt sugar. A defect in which of the following enzymes causes this disease?

- a) Branched-chain ketoacid dehydrogenase
- b) Branched-chain amino acid dehydrogenase
- c) Branched-chain ketoacid oxidase
- d) Branched-chain amino acid oxidase

Question 16:

All are true about the hexose monophosphate shunt, except:

- a) Source of ribose-5-phosphate
- b) Source of Acetyl CoA
- c) Source of NADPH
- d) Utilises ribose

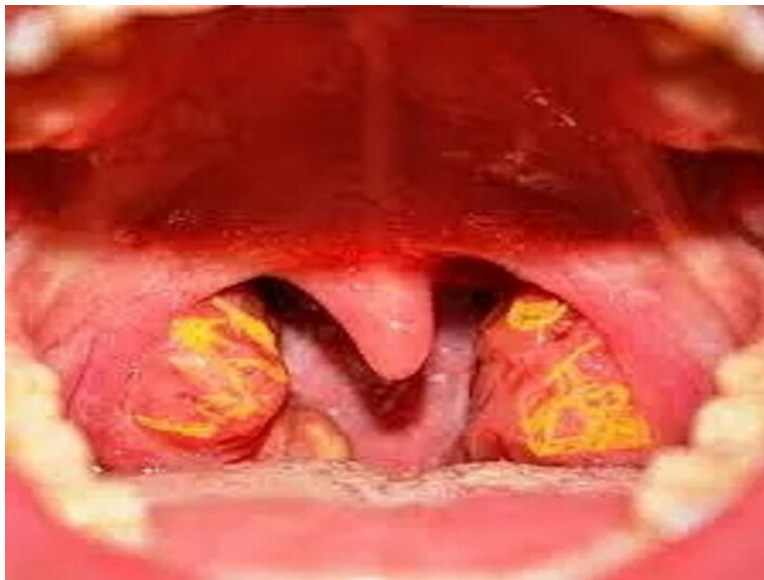
Question 17:

Which of the following enzymes use selenocysteine?

- a) 1, 3, 4
- b) 1, 2, 3
- c) 2, 3, 4
- d) 1, 2, 4

Question 18:

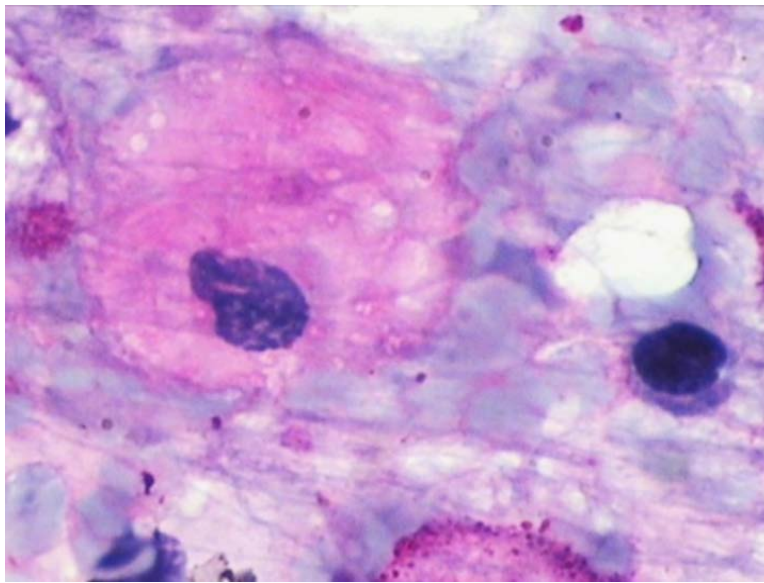
A patient presented with reduced levels of high-density lipoprotein, and ABCA1 mutation. On examination, tonsils appeared as shown in the image. What is the diagnosis?



- a) Tangier's disease
- b) Abetalipoproteinemia
- c) Hyperlipoproteinemia
- d) Wolman's disease

Question 19:

A baby is being evaluated for delayed developmental milestones. On examination, the child has hepatosplenomegaly. A microscopic image of the bone marrow evaluation is shown below. What is the diagnosis?



- a) Gaucher disease
- b) Niemann-Pick disease
- c) Hunter disease
- d) Hurler disease

Question 20:

Refsum disease is due to:

- a) Absence of alpha oxidation in peroxisomes
- b) Absence of beta oxidation in peroxisomes
- c) Absence of alpha oxidation in lysosomes
- d) Absence of beta oxidation in lysosomes

Question 21:

Choose the right sequence of steps in the conversion of cholesterol to testosterone.

- a) b, d, a, c
- b) a, c, d, b
- c) b, a, c, d
- d) d, a, b, c

Question 22:

Which of the following describes the relationship between the melting temperature (T_m) of DNA and the concentration of G-C and A-T?

- a) No relation of GC with T_m
- b) T_m increases with more A-T
- c) T_m decreases with more G-C
- d) T_m increases with more G-C

Question 23:

An elderly patient on a vegetarian diet presenting with easy fatigability and pallor was investigated. Peripheral smear showed findings of macrocytic anemia. What is the probable cause of anemia in this patient? Other lab findings are as follows:

- a) Folate deficiency
- b) Alcohol-induced macrocytosis
- c) Vitamin B12 deficiency
- d) Thiamine deficiency

Question 24:

Zellweger syndrome is due to a defect in the biogenesis of:

- a) Peroxisomes
- b) Lysosomes
- c) Mitochondria
- d) Nucleus

Question 25:

True statement about the action of enhancers of gene expression?

- a) Work only when in close conjugation with promoter region
- b) Work only when in upstream of promoter region
- c) They are very specific to the gene
- d) Recruit chromatin and modify regulatory mechanisms

Question 26:

Increased H⁺ ions in the intermembrane space of mitochondria are due to?

- a) Decreased oxidative phosphorylation
- b) Increased ATP production
- c) Reduced chemiosmosis
- d) Water retention in inner mitochondrial space

Question 27:

Fructosamine is obtained by glycation of:

- a) Fructose
- b) Albumin
- c) Urea
- d) Glycosaminoglycans

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	a
5	d
6	b
7	a
8	a
9	d
10	b
11	d
12	a
13	a

14	c
15	a
16	d
17	a
18	a
19	a
20	a
21	a
22	d
23	a
24	a
25	d
26	a
27	b

Detailed Explanations

Solution to Question 1:

The correct sequence of isolation of desirable protein in recombinant DNA technology is 2, 1, 3, 5, 4.

I) Incorporation of genes into bacteria: The gene needed to produce the desired proteins is to be incorporated into the bacteria. This is done using cloning vectors like bacterial plasmids, phages, or cosmids. The DNA of interest is ligated into the vector DNA and is called recombinant DNA. This vector is then inserted into a competent host cell (eg: E.coli bacteria) where it is replicated.

II) Expression of protein and lysis of bacterial cell: After incorporation of recombinant DNA, bacteria express the inserted gene and produce the desired protein on a large scale along with other indigenous bacterial proteins. To release these proteins bacterial cell is lysed.

III) SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis): It is the most widely used method for determining the purity of a protein. After the proteins are released, to separate the desirable protein, we subject it to SDS-PAGE. It separates proteins based on molecular weight.

IV) Hybridization and blotting: It is done to confirm that the isolated protein is the desired protein. It is done using western blotting techniques on nitrocellulose paper.

V) Protein elution - extraction of the produced protein.

The practical applications of recombinant DNA technology include:

- Gene mapping using techniques such as fluorescence in situ hybridization (FISH).
- Producing proteins such as:

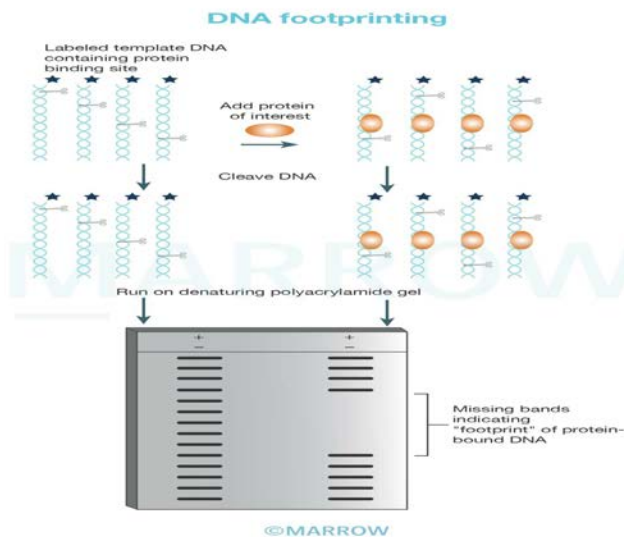
- Interferons, tPA (tissue plasminogen activator)
- Insulin, growth hormone
- Hepatitis B vaccine
- Molecular analysis of disease, to detect gene variations and mutations.
- Restriction Fragment Length polymorphism (RFLP) and single nucleotide polymorphism (SNP)
- Microsatellite DNA polymorphisms
- Gene therapy and bone marrow transplantation

Solution to Question 2:

The technique used to detect DNA-protein interaction is DNA footprinting.

The parts of DNA that are protein-bound (Eg: RNA polymerase bound to the promoter region of DNA) are resistant to digestion by nuclease enzymes. Since the nucleases are unable to cleave the protein-bound regions of DNA, when the cleaved DNA fragments are run on polyacrylamide gel electrophoresis, there are missing bands indicating, protected sites of protein-bound DNA. These missing areas represent the footprint of the protected DNA-protein complexes. Hence DNA footprinting technique helps to identify the specific binding site of protein on a DNA sequence.

The image below illustrates the DNA footprinting technique.



Other options:

Option B: DNA fingerprinting uses RFLPs (restriction fragment length polymorphism) or repeat sequence DNA to identify a person's unique pattern of DNA. Polymorphisms are inherited differences found among individuals. With this technique, individuals can be differentiated by variations in their DNA sequence, using restriction endonucleases followed by gel electrophoresis, to visualize RFLPs. DNA fingerprinting is used for forensic analysis of crime scenes, and for paternity testing.

Option C: ELISA uses antibodies linked to certain enzymes to detect certain antigens present in the sample.

Option D: Northern blotting technique can be used to detect RNA. Here, the mixture of mRNA molecules is separated by electrophoresis and then transferred to a nitrocellulose membrane and hybridized with radiolabeled probes.

Solution to Question 3:

The technique shown in the image is FISH demonstrating a positive EWSR1 gene rearrangement, characteristic of Ewing sarcoma. The fused region denoted by Yellow corresponds to a normal intact gene, while a split signal pattern showing separate red and green signals is indicative of EWSR1 gene rearrangement.

Fluorescent In Situ Hybridisation (FISH) is a simple technique for the detection of specific genetic information on morphologically intact tissue using fluorescent probes. FISH can be performed on prenatal samples, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections. It is used to detect

- numeric abnormalities of chromosomes (aneuploidy)
- subtle microdeletions and rearrangements.
- complex translocations that are not demonstrable by routine karyotyping
- gene amplification (e.g., HER2 in breast cancer or NMYC amplification in neuroblastomas)

Other options:

Option B: Karyotyping is the process by which chromosomes are isolated, stained, and examined to find chromosomal anomalies. It is used in prenatal diagnosis, investigation of repeat pregnancy loss, investigation of children and adults suspected of having a chromosomal disorder, and subtyping of lymphomas and leukemias for the purpose of exact diagnosis, giving a prognosis, and treatment.

Option C: IHC (Immunohistochemistry) is a staining method for histopathological specimens. It helps in the diagnosis and classification of tumors based on their distinct protein expression patterns. Eg: The expression of estrogen receptor (ER), progesterone receptor (PR), and HER2/neu in breast cancer cells is mainly assessed by immunohistochemistry.

Option D: Sanger's technique: This was the first technique used to determine the sequence of proteins. The first protein to be sequenced by this method was insulin. Sanger's reagent is 1-Fluoro 2,4 Dinitro benzene. Only dipeptides and tripeptides can be sequenced by this method.

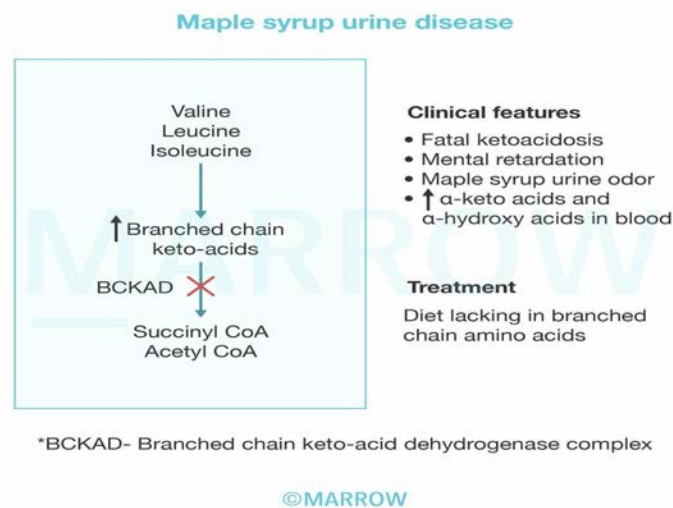
Solution to Question 4:

In maple syrup urine disease (MSUD), only branched-chain amino acids are restricted, hence, methionine does not need any restriction in diet.

Branched-chain amino acids are:

- Leucine (L)
- Isoleucine (I)
- Valine (V)

MSUD is an autosomal recessive disorder in which there is a partial or complete deficiency of branched-chain ketoacid dehydrogenase complex (BCKD). This impairs the metabolism of branched-chain amino acids. These amino acids, particularly leucine and their metabolite accumulate in the blood, causing toxic effects on the brain. Features include feeding problems, vomiting, ketosis, and a characteristic burnt sugar odor /maple syrup-like odor found in urine, sweat, and cerumen.



Laboratory investigations will show increased levels of branched-chain amino acids in the plasma and urine. Leucine levels are usually higher than isoleucine, and valine.

Treatment involves restricted intake of branched-chain amino acids using commercially available formulas and judicious supplementation of isoleucine and valine. Very high levels of leucine are removed by hemodialysis.

Solution to Question 5:

The human genome project did not use next-generation sequencing.

The Human Genome Project (HGP) was an international, collaborative research program, which involved agencies such as UNESCO, HUGO, the National Institute of Health/Department of Energy (USA), the Medical Research Council (UK), Genethon (France), and the European Union. The primary goal of the Human Genome Project was to determine the sequence of all nucleotide base pairs, that make up human DNA and to create a single linear map of all genes from the human genome. The project was active from 1990 to 2003, taking around 13 years to complete. A total of 20500 genes were identified and mapped. Scientists used various methods of first-generation sequencing technologies including sanger's technique, SGS technique, etc. The HGP has produced a curated and accurate reference sequence for the majority of human

chromosomes. The reference sequence has provided the foundation for the use of second-generation sequencing technologies such as array techniques.

Next-generation sequencing (NGS) is a new technology used for sequencing DNA and RNA and mutation detection. NGS can sequence the genome faster and in a shorter period of time than first-generation sequencing technologies. NGS allows for the detection of a spectrum of mutations, including single nucleotide substitutions, small insertions and deletions, large genomic duplications and deletions, and rare variations. Next-generation DNA sequencing technologies include

- Illumina sequencing
- Roche 454 sequencing
- Ion torrent: Proton/PGM sequencing
- SOLID sequencing.

Solution to Question 6:

The banding technique used for dicentric chromosomes is C banding.

Heterochromatin includes non-coding regions of DNA, such as centromeric regions. The staining of these constitutive heterochromatins, is done by C-banding using silver salt staining. A dicentric chromosome having two centromeres can hence be identified by C banding.

Various techniques are used in karyotyping to study chromosomes. Staining with various compounds produces various results and helps us to understand the chromosomes better.

Common techniques used are:

- Q banding - Fluorescence staining with quinacrine mustard.
- G banding - Giemsa staining after partial denaturation of chromatin. It is the most common technique used in cytogenetics to produce a visible karyotype by staining condensed chromosomes.
- R banding - Reverse Giemsa staining. In this, the light and dark areas are reversed.
- C banding - staining of heterochromatin with silver salts.
- T banding - staining the telomeres of chromosomes.

Solution to Question 7:

Affinity maturation of antibodies is because of somatic hypermutation.

Affinity maturation is the process by which antibodies gain increased affinity, avidity, and anti-pathogen activity. It is a result of somatic hypermutation of immunoglobulin genes in B cells, coupled with selection for antigen binding.

Somatic hypermutation is a cellular response by which the immune system adapts to the new foreign elements that confront it. They do so by adding nucleotides in V gene regions of heavy and

light chains, resulting in a better fit of antigen with the antibodies. This hence increases the affinity of antibodies to antigens. Due to this, the subset of plasma cells with these improved hypervariable regions are more strongly

selected by the antigen and therefore constitute a larger part of the population of antibody-producing cells. This process is called affinity maturation.

This process occurs in the germinal centers and proceeds for weeks after acute infection or vaccination, or for many cycles during chronic infection. The resulting antibodies can be highly mutated from their germline-encoded counterparts, with an increase in several orders of magnitude in affinity for antigen compared to the corresponding naïve B cell receptors (BCRs).

Solution to Question 8:

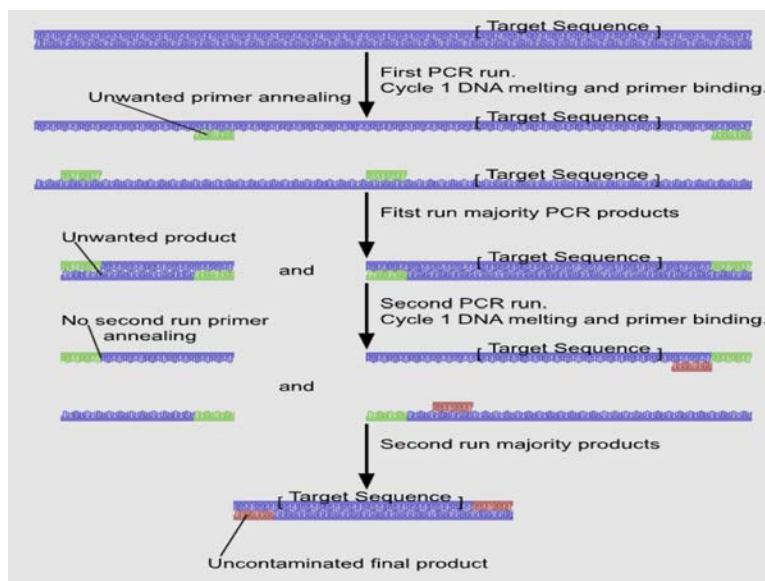
Best method for detection of mutation with low allele frequency is Droplet digital PCR (ddPCR)

It is an advanced and more precise type of quantitative PCR which gives more consistent and reproducible results. It can accurately quantify the target nucleic acid sequence within a sample. It is useful for determining somatic mutation with low allele frequency. Mutant alleles of frequency as low as 0.25% can also be detected using this technique.

Polymerase chain reaction (PCR) is a rapid automated method for the amplification of specific DNA sequences (or genes). Reverse transcriptase PCR- The enzyme reverse transcriptase (RT) is used to synthesize a complimentary copy of DNA (cDNA) from RNA and this cDNA is then used as a template for PCR.

Other special PCR types:

Nested PCR- It involves the sequential use of two primer sets. The first primer set is used for amplification and the target sequence for the second primer is present only on the amplicon with the target sequence. Nested PCR is intended to reduce non-specific binding due to the amplification of unexpected primer binding sites.



Multiplex PCR- Multiple primers are used to amplify several DNA sequences simultaneously. As this process targets multiple genes at once, a single test run gives additional information that would otherwise require several times the reagents and technician time to perform.

The Sanger sequencing technique is used for DNA sequencing. It is based on controlled termination of synthesis of DNA by using chain terminating agents like dideoxynucleotides.

Fluorescent In Situ Hybridisation (FISH) is a simple technique for the detection of specific genetic information on morphologically intact tissue using fluorescent probes. FISH can be performed on prenatal samples, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections.

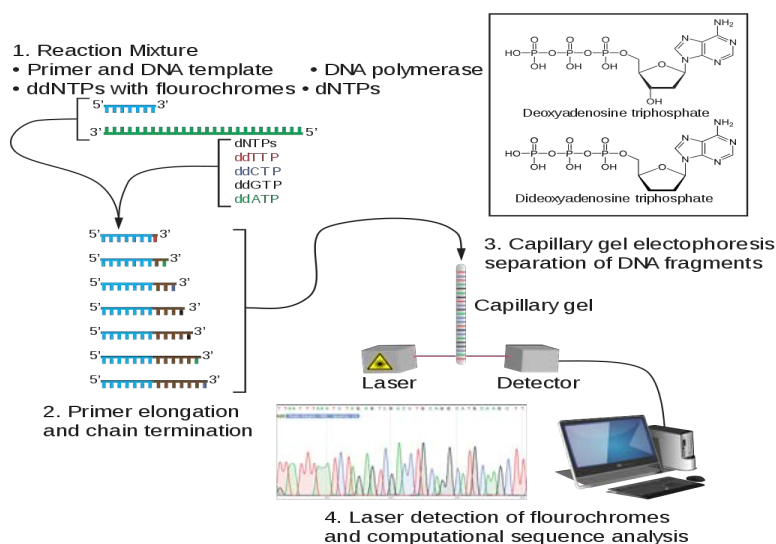
Solution to Question 9:

In Sanger sequencing, dideoxynucleotides triphosphates (ddNTP) are used because it doesn't allow further elongation of DNA.

The Sanger method is used for DNA sequencing. This method uses the mechanism of synthesis of DNA by DNA polymerases. In the process of synthesis of DNA, the 3'-hydroxyl group of primer reacts with an incoming dNTP to form a new phosphodiester bond. This reaction is catalyzed by DNA polymerase.

In the Sanger sequencing method, ddNTP is used because it halts the DNA synthesis. ddNTPs can bind to the template strand but lack the 3'-hydroxyl group which is required to add the next nucleotide. The products of this reaction are a mixture of DNA strands of varying lengths. This mixture is then separated using polyacrylamide gel electrophoresis and autoradiography. This produces a pattern of bands from which the DNA sequence can be read.

The image given below shows the steps involved in Sanger's method.



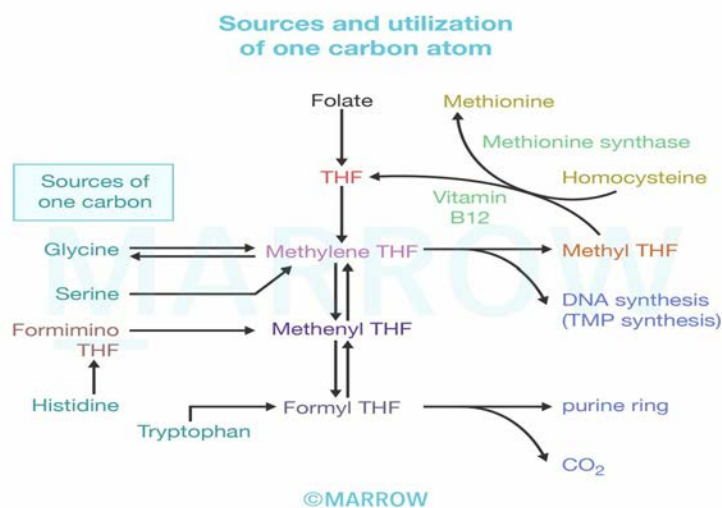
Sanger's technique was the first technique used to determine the sequence of proteins. The first protein to be sequenced by this method was insulin. Sanger's reagent is 1-fluoro 2,4-dinitrobenzene.

Solution to Question 10:

Misincorporation of uracil into DNA is seen in the deficiency of folate.

Folate deficiency leads to decreased thymidylate synthesis from deoxyuridylate due to deficient methylation of deoxyuridine monophosphate (dUMP) to thymidine monophosphate (TMP). The methylation of dUMP to TMP is catalyzed by thymidylate synthase and requires tetrahydrofolate (THFA) as a methyl group donor. This decreased TMP synthesis can result in an imbalance of deoxyribonucleotides and hence leading to excessive uracil misincorporation into DNA during replication and repair.

Folic acid (vitamin B9) is a water-soluble vitamin, and its active form is tetrahydrofolate (THFA). It acts as a carrier of one-carbon groups. Tetrahydrofolate receives one-carbon fragments from donors such as serine, glycine, and histidine and transfers them to intermediates in the synthesis of amino acids, purines, and pyrimidines.



Folate deficiency results in:

- Megaloblastic anemia
- Homocystenemia
- Neural tube defects (spina bifida, Anencephaly) in the fetus.

Solution to Question 11:

Deamination of methylated cytosine forms thymine.

Deamination is the removal of an amino group(NH₂) from a chemical compound. Enzymes that catalyze this reaction are called deaminases. Deamination of 5-methylcytosine forms thymine.

Other significant deaminations are given below:

- Deamination of adenosine forms inosine. This is a metabolic step involved in uric acid formation by purine catabolism. It is mediated by the enzyme adenosine deaminase (ADA). ADA deficiency leads to sparse or defective T cells and B cells and patients suffer from Severe combined immunodeficiency (SCID).
- Deamination of cytosine leads to uracil formation - The production of apolipoprotein B-48 (apoB-48) in the intestine is the result of RNA editing in the intestine. This involves post-transcriptional deamination of cytosine to uracil by enzyme cytidine deaminase. This converts sense codon (CAA) in the mRNA to nonsense codon (UAA) at a single specific site. This edited mRNA translates to apoB-48.
- Deamination of guanine forms xanthine. This is a metabolic step involved in uric acid formation by purine catabolism.

Solution to Question 12:

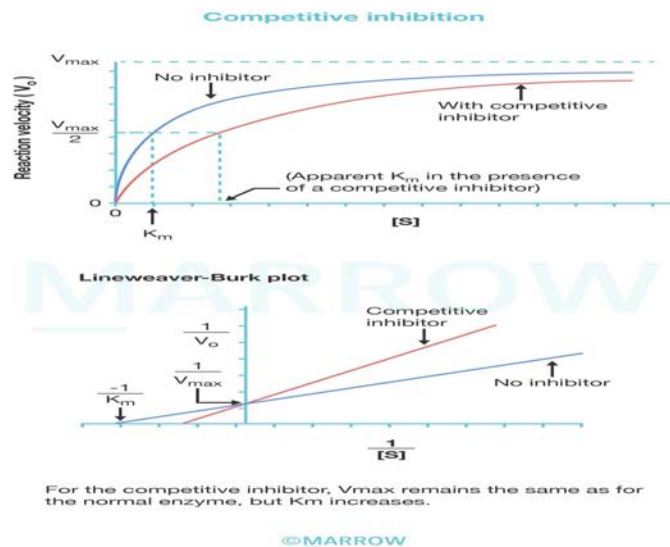
The given graph shows competitive inhibition, in which the value of V_{max} remains constant and the K_m value increases.

- K_m is the substrate concentration at which velocity is half maximum.
- V_{max} is the maximum velocity of a reaction.

Competitive inhibition is when the inhibitor binds reversibly to the substrate-binding site and competes with the substrate for the active enzyme-binding site. This effect of competitive inhibition can be overcome by increasing the substrate concentration by which maximum velocity can still be achieved. Therefore, V_{max} is unaltered in this type of inhibition. However, the concentration of the substrate required to achieve half of the maximum velocity is increased. Therefore, K_m is increased in this condition.

The graph shows the changes in V_{max} and K_m in competitive inhibition.

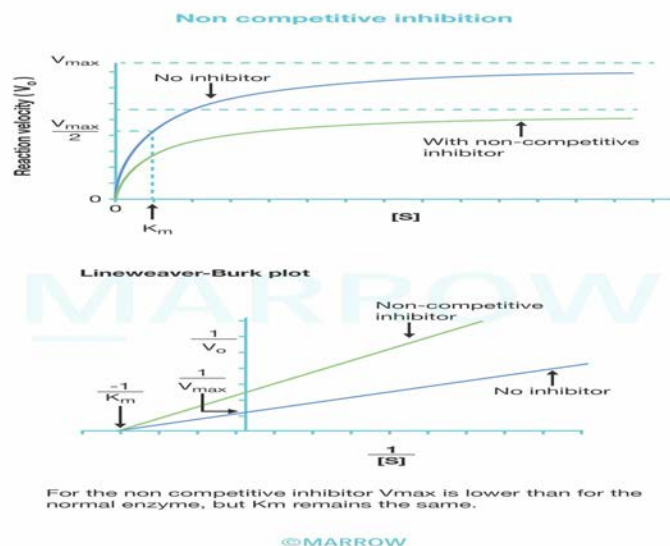
Inhibition	K_m	V_{max}
Competitive	Increases	Same
Non Competitive	Same	Decreases
Uncompetitive	Decreases	Decreases



Other options:

Option B: Non-competitive inhibition is by binding inhibitors to enzymes in locations remote from the active site. Increasing the concentration of the substrate does not overcome non-competitive inhibition. Thus, noncompetitive inhibitors decrease the V_{max} of the reaction. They do not interfere with the binding of the substrate to the enzyme. Thus, the enzyme shows the same K_m in the presence or absence of noncompetitive inhibitors.

The graph shows the changes in V_{max} and K_m in non-competitive inhibition.



Option C: In uncompetitive inhibition, inhibitors bind with the enzyme-substrate intermediate complex, and inhibit it. This leads to decrease in both V_{max} and K_m . An example is the inhibition of placental alkaline phosphatase (ALP) by phenylalanine.

Option D: Suicidal inhibition is a special type of irreversible inhibition. The inhibitor here makes use of the enzyme's own reaction mechanism to inactivate it and is hence known as mechanism-based inactivation. An example is allopurinol, which is oxidized by xanthine oxidase

to alloxanthine. Alloxanthine in turn is a more potent inhibitor of xanthine oxidase.

Solution to Question 13:

Glycogen phosphorylase is activated by calcium in the muscle.

Glycogen phosphorylase is the rate-limiting enzyme of glycogenolysis and is decreased in the fed state.

In the muscle, the increased cytosolic calcium ion concentration initiates both contraction and glycogenolysis. Muscle phosphorylase kinase has four different subunits, α , β , γ , and δ . The α and β subunits are phosphorylated by cAMP-dependent protein kinase. The δ subunit is identical to the calcium-binding protein calmodulin and binds four calcium ions. Calcium coordinates the activation of Glycogen Phosphorylase with the contraction of the muscle.

Glycogen phosphorylase catalyzes the cleavage of $\alpha 1 \rightarrow 4$ linkages of glycogen to single units of glucose 1-phosphate. It uses pyridoxal phosphate as the coenzyme. Different isoenzymes of glycogen phosphorylase are seen in the liver, muscles, and brain.

Other options:

Option B: During glycogenolysis, glucose 6-phosphate allosterically inhibits glycogen phosphorylase.

Option C: Insulin entirely suppresses glycogenolysis in the skeletal muscle and liver.

Option D: In glycogenolysis protein phosphatase is inactivated by inhibitor proteins in response to cAMP, to maintain the glycogen phosphorylase in an active state.

Solution to Question 14:

The mutant gene of cystic fibrosis (CF gene) produces a mutant CFTR protein with the deletion of a phenylalanine residue at 508 (F508del), characterised by impaired chloride secretion.

Cystic fibrosis is a condition with an autosomal recessive mode of inheritance. It results due to a defective mutation in the gene encoding for cystic fibrosis transmembrane conductance regulator (CF gene). It plays an important role in mucociliary transport along the respiratory mucosa. It also helps to regulate the amount of exocrine gland secretion.

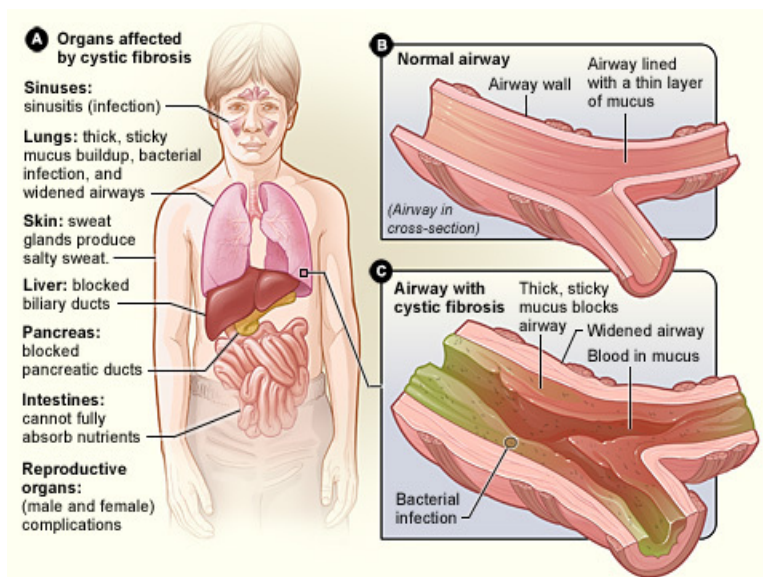
The defect in the ion channel results in the production of hyper-viscous, adherent secretions which can obstruct the airways. Organisms such as *Pseudomonas aeruginosa* and *Staphylococcus aureus* can grow in the sputum. The exocrine pancreas can undergo fibrotic scarring, cyst formation, and acinar tissue loss. The viscid secretions can lead to pancreatic duct obstruction. The flow of digestive enzymes to the duodenum is hence impaired.

Symptoms include sinusitis, steatorrhea, and abdominal pain. Distal intestinal obstruction is a complication due to difficulty in excreting intestinal contents. In neonates, it can present as meconium ileus.

Detection of high levels of trypsinogen is used in newborn screening. Chloride level $>60\text{mEq/L}$ in the sweat of affected patients is diagnostic. Hyponatremic hypochloremic alkalosis is seen.

Postural drainage, short-term antibiotics, and inhaled bronchodilators can be used for medical management. The definitive treatment is lung transplantation.

The image below shows the features of the disease.



Solution to Question 15:

The diagnosis, in this case, is that of maple syrup urine disease (MSUD). This disease occurs due to a defect of the enzyme branched-chain keto acid dehydrogenase.

MSUD is an autosomal recessive disorder that occurs due to a deficiency of branched-chain ketoacid dehydrogenase complex. This impairs the metabolism of branched-chain amino acids valine, leucine, isoleucine, and alloisoleucine. These amino acids, particularly leucine and its metabolite accumulate in the blood, causing toxic effects on the brain.

The affected child develops poor feeding and vomiting a few days after birth. There is a period of hypertonicity with a period of flaccidity in between. These appear as repetitive boxing and bicycling movements. The babies may also have hypoglycemia, but this is not corrected by the administration of glucose. The urine, sweat, and cerumen of the baby have a sweetish maple syrup-like odor.

Laboratory investigations will show increased levels of branched-chain amino acids in the plasma and urine. Leucine levels are usually higher than isoleucine, valine, and alloisoleucine. Since the patient excretes ketone bodies in the urine, the urine will be positive for the dinitrophenylhydrazine test.

Treatment involves controlled intake of branched-chain amino acids using commercially available formulas and judicious supplementation of isoleucine and valine. Very high levels of leucine are removed by hemodialysis.

Solution to Question 16:

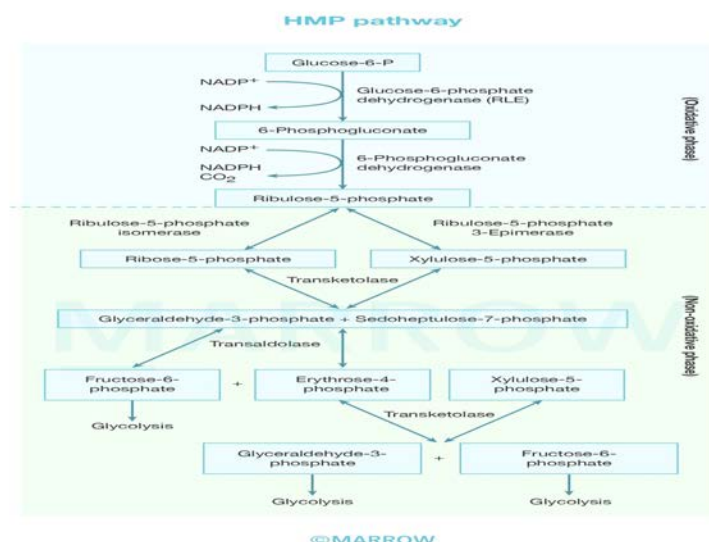
The hexose monophosphate (HMP) shunt produces ribose and does not utilize it.

The pentose phosphate pathway, also known as the HMP shunt pathway, is an alternative route for the metabolism of glucose. It contributes in two major ways:

- Formation of NADPH for the synthesis of fatty acids and steroids and maintenance of reduced glutathione for antioxidant activity.
- Formation of ribose for the synthesis of nucleic acid and nucleotides

The enzymes for this pathway are found in the cytosol. There are two phases:

- Irreversible oxidative phase: Glucose-6-phosphate is dehydrogenated and decarboxylated to form ribulose-5-phosphate which in turn gets converted to ribose-5-phosphate.
- Reversible non-oxidative phase: Ribulose-5-phosphate is converted back to glucose-6-phosphate.



Solution to Question 17:

The enzymes that use selenocysteine are glutathione peroxidase, deiodinase, and thioredoxin reductase.

Selenocysteine is the 21st amino acid coded by the UGA codon, which is normally a stop codon. It is formed by co-translational modification. The protein synthetic apparatus can identify a selenocysteine-encoding UGA by the presence of the selenocysteine insertion sequence (SECIS) in the untranslated region of mRNA.

Enzymes such as thioredoxin reductase, glutathione peroxidase, and deiodinase use selenocysteine to catalyze redox reactions. Defects in human selenoproteins are associated with tumorigenesis and atherosclerosis. It can cause cardiomyopathy due to selenium deficiency, which is known as Keshan disease.

Glutathione synthase is the second enzyme in the synthesis of glutathione. Glutathione plays an important role in oxidative stress, redox signalling, and regulates cell proliferation and apoptosis.

Solution to Question 18:

The diagnosis in this case with reduced levels of high-density lipoprotein (HDL), ABCA1 mutation, and orange tonsils on examination is Tangier's disease. It is caused by a defect in ABCA1 receptors.

ABCA1 plays a role in the efflux of cholesterol from peripheral cells of HDL. LCAT (lecithin cholesterol acyltransferase) converts cholesterol to cholesterol esters, which accumulate in the core of HDL. Thus, defective ABCA1 will lead to low levels of HDL in the blood.

Other options:

Option B: Abetalipoproteinemia is a hypolipoproteinemia caused by a defect in the MTP gene. Here, loading apolipoprotein B with lipids is defective. Hence, few very low density-lipoproteins (VLDLs) or chylomicrons are formed, leading to the accumulation of triacylglycerols in the liver and intestine.

Option C: Hyperlipoproteinemia is a disorder that causes the body to metabolize lipids incorrectly. This results in hyperlipidemia and can lead to the development of multiple small xanthomas in the skin.

Option D: Acid lipase deficiency can lead to Wolman's disease which is a lysosomal storage disease. The pathognomic feature of this condition is adrenal calcification. So patients generally present with adrenal insufficiency and are neonates.

Solution to Question 19:

The microscopic image of the bone marrow biopsy shows Gaucher cells and the diagnosis is Gaucher disease.

Gaucher disease is a lysosomal storage disease with autosomal recessive inheritance. The condition results due to the deficiency of acid β -glucosidase. This results in the accumulation of glucosylceramide (a glycolipid substrate) in the reticuloendothelial cells. There is progressive infiltration of the liver, spleen, and bone marrow.

There are three clinical types:

- Type 1: Adult (non-neuronopathic form)
- Type 2: Infantile (acute neuronopathic form)
- Type 3: Juvenile (subacute neuronopathic form)

Type 1 is the most common. It presents with a myriad of symptoms, such as bruising secondary to thrombocytopenia, fatigue due to anemia, hepatosplenomegaly, and bone pain. Lytic lesions and osteosclerosis of the femur, ribs, and pelvis may also be seen. A few patients may have pulmonary involvement. Growth retardation may be observed in cases presenting in the first decade.

Type 2 is rare, and presents with hypertonia, strabismus, and organomegaly. Neurodegeneration is the characteristic feature. Stridor and failure to thrive can also be the presenting features.

Ultimately, death can occur due to respiratory arrest.

Type 3 presents with intermediate symptoms, as seen in the other two types. By the age of 10-15 years, the child succumbs to death.

Gaucher cells are reticuloendothelial cells with glucosylceramide, and have a wrinkled-paper appearance due to intracytoplasmic substrate inclusions. These cells stain strongly with periodic acid-Schiff (PAS) stain. This is characteristic of Gaucher's disease.

Other options:

Option B - Acid sphingomyelinase deficiency results in Niemann-Pick disease types A and B. Both are autosomal recessive conditions. Type A has a neurodegenerative course, while type B is non-neuronopathic. The former presents with hepatosplenomegaly, lymphadenopathy, and psychomotor retardation. Type B is associated with hepatosplenomegaly and mild pulmonary involvement.

Option-C - Hunter disease is an X-linked inherited disorder due to a deficiency of the enzyme iduronate sulfatase. It leads to the accumulation of dermatan and heparan sulfate. Clinical features involve coarse facies, cardiovascular involvement, and distinctive skin lesions.

Option-D - Hurler/Scheie disease is a mucopolysaccharidosis that has childhood and adult onset. It presents with cognitive degeneration.

Solution to Question 20:

Refsum disease is due to the absence of alpha oxidation in peroxisomes.

Refsum's disease is a rare autosomal recessive disorder. The defective enzyme is phytanoyl CoA hydroxylase, which causes alpha oxidation of fatty acids and leads to elevated levels of phytanic acid in the blood.

Clinical features are dysmorphic facies and severe neurological symptoms such as polyneuropathy, retinitis pigmentosa, nerve deafness, ichthyosis, and cerebellar ataxia.

Restricted dietary intake of phytanic acid reduces the symptoms. Management includes avoiding sources of phytanic acid such as dairy products, meat, nuts, and green leafy vegetables.

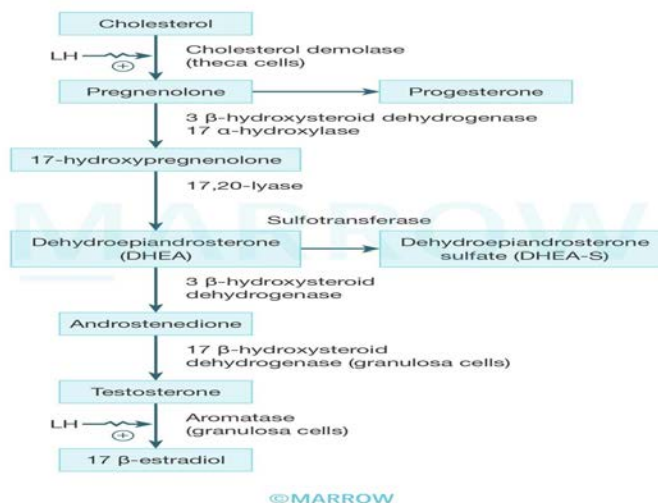
Solution to Question 21:

The correct sequence of steps in the conversion of cholesterol to testosterone is:

- b. 20, 22 desmolase
- d. 3 beta-hydroxy steroid dehydrogenase
- a. 17 alpha-hydroxylase
- c. 17, 20 lyase

The step of synthesis of sex steroids is illustrated in the diagram below.

Synthesis of Sex Steroids



Solution to Question 22:

The correct statement about the melting temperature of DNA (T_m) is that it increases with more G-C concentration.

On heating the DNA, the temperature at which one-half of the helical structure loses its structure defines the melting temperature T_m . The two strands of the double helix structure of the DNA get separated when the hydrogen bonds between the base pairs are disrupted. Thus, the T_m is influenced by the concentration base composition of the DNA.

Since G-C pairing has 3 hydrogen bonds and A-T pairing has 2 hydrogen bonds, DNA rich in G-C base pairs requires higher temperatures to denature in comparison to DNA rich in A-T base pairs.

Solution to Question 23:

The probable cause of anemia in this patient of folate deficiency.

The given clinical scenario of macrocytic anemia in a vegetarian with elevated homocysteine levels is due to megaloblastic anemia, which can be caused by the deficiency of vitamin B12 or folic acid. Vitamin B12 deficiency is associated with raised serum methylmalonic acid (MMA) levels, which are normal in this patient; hence, folic acid deficiency is the cause of megaloblastic anemia in this patient.

Folic acid deficiency is associated with the following clinical and laboratory findings:

- Raised MCV (normal range: 80–100 fL) - macrocytosis
- Normal MCHC (normal range: 33–36 g/dL)
- Increased MCH
- Elevated RDW

- Decreased reticulocyte count - due to destruction of abnormal erythroid precursors
- Elevated homocysteine level
- Normal MMA level

where MCV, mean corpuscular volume; MCHC, mean corpuscular hemoglobin concentration; mean corpuscular hemoglobin, MCH; RDW, red cell distribution width

In vitamin B12 deficiency, neurological abnormalities, elevated homocysteine, and elevated MMA levels are seen.

Solution to Question 24:

Zellweger syndrome is due to a defect in the biogenesis of peroxisomes.

Zellweger syndrome is also known as cerebrohepatorenal syndrome. It is caused by mutations in the PEX family of genes. This results in defective beta-oxidation of very long-chain fatty acids.

The child presents with

- Dysmorphic facial features.
- Weakness, hypotonia.
- Neonatal seizures
- Eye abnormalities.

Biochemical findings will show :

- Accumulation of very-long-chain fatty acids, phytanic acid
- Abnormalities in the synthesis of bile acids
- Marked reduction of plasmalogens

There is no appropriate treatment for Zellweger syndrome. It is very severe and infants rarely survive beyond a few months.

Solution to Question 25:

The correct statement about the action of enhancers is that they recruit chromatin and modify regulatory mechanisms.

A promoter is a sequence on DNA where RNA polymerase binds and acts as the start signal for transcription. They are present upstream or downstream but close to the gene that is undergoing transcription.

Enhancers or repressors are proteins that regulate the expression of a gene by recruiting chromatin(DNA bound with histones) and modifying regulatory mechanisms and making it more or less accessible to transcription by RNA polymerase. They may be located upstream or downstream and can be present at any distance from the promoter region. These are not specific to any single gene and interact with many other proteins.

Solution to Question 26:

Increased H^+ ions in the intermembrane space of mitochondria are due to decreased oxidative phosphorylation.

ATP is synthesized in mitochondria by a process called oxidative phosphorylation. The oxidation of substrates like carbohydrates, amino acids, or fats produces reducing equivalents (NADH, $FADH_2$). These equivalents enter the electron transport chain and transfer a proton (H^+) to the inter-membrane space. The inner mitochondrial membrane is not permeable to protons leading to the accumulation of H^+ in the intermembrane space. The flow of these protons through the F_o releases the energy that phosphorylates ADP to form ATP.

Drugs that inhibit oxidative phosphorylation like oligomycin inhibit the flow of protons through F_o causing an increase in H^+ ions in the intermembrane space.

Solution to Question 27:

Fructosamine is obtained by the glycation of albumin.

Fructosamine is a measure of non-enzymatic glycation of circulating proteins including albumin, globulins, and lipoproteins. It is now used as a reasonable alternative to HbA1c measurement when it is not reliable. Since albumin is the most abundant of the serum proteins, fructosamine is predominantly a measure of glycated albumin.

Fructosamine is used as a short-term marker of glucose control. It reflects the average glycemia over the preceding 2 to 3 weeks. HbA1c reflects glucose control over a period of the preceding 8 to 12 weeks.

Biochemistry NEET 2022

Question 1:

A patient from a Mediterranean country visits Africa, where he develops malaria. He is treated with primaquine and later develops hemolytic anemia. Deficiency of an enzyme involved in which of the following pathways could be the cause?

- a) Glycolysis
- b) Gluconeogenesis
- c) Hexose monophosphate (HMP) pathway
- d) Luebering–Rapoport pathway

Question 2:

Which of the following disorders follows autosomal recessive inheritance pattern?

- a) Huntington's disease
- b) Treacher Collins syndrome
- c) Cystic fibrosis
- d) Achondroplasia

Question 3:

Which of the following helps in the transport of fatty acids across the inner mitochondrial membrane?

- a) Acyl carrier protein
- b) Carnitine
- c) Lecithin-cholesterol acyltransferase
- d) Carnitine and albumin

Question 4:

A single mutation in a nucleotide base pair resulting in a termination codon is known as?

- a) Missense mutation

- b) Nonsense mutation
- c) Stop mutation
- d) Silent mutation

Question 5:

DNA packing is done by which of the following ?

- a) Histone
- b) Glycoprotein
- c) Nucleic acid
- d) Adenine

Question 6:

An adolescent male patient presents to you with exercise intolerance. He gives a history of developing cramps on exertion. Which of the following enzyme deficiencies could be the cause?

- a) Myophosphorylase
- b) Hexokinase
- c) Glucose-6-phosphatase
- d) Hepatic glycogen phosphorylase

Question 7:

Which of the following enzyme activities can be estimated in red blood cells to diagnose vitamin B2 deficiency?

- a) Transketolase
- b) Glutathione reductase
- c) Kynureninase
- d) Pyruvate dehydrogenase

Question 8:

The following is due to the deficiency of?



- a) Vitamin C
- b) Vitamin A
- c) Vitamin B
- d) Vitamin E

Question 9:

A child presents to you with anemia, thrombocytopenia, and bony pain. On examination, hepatosplenomegaly was noticed. If light microscopy of bone marrow aspirate reveals "crumpled tissue paper" appearance, what is the enzyme defect?

- a) Glucocerebrosidase
- b) Sphingomyelinase
- c) Hexosaminidase
- d) Glucose 6-phosphatase

Question 10:

The electron transport chain is a series of redox reactions that result in ATP synthesis. Which of the following is a cytochrome complex IV inhibitor?

- a) Cyanide
- b) Carbon dioxide
- c) Oligomycin

d) Ouabain

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	b
5	a
6	a
7	b
8	b
9	a
10	a

Detailed Explanations

Solution to Question 1:

The deficiency of glucose-6-phosphate dehydrogenase (G6PD) enzyme, which is involved in the hexose monophosphate (HMP) pathway, could be the cause behind this presentation.

G6PD is an enzyme that plays a critical role in redox reactions that occur in all aerobic cells. It is the only source of NADPH in erythrocytes, which helps in the formation of reduced glutathione. This reduced form protects erythrocytes against oxidative stress.

G6PD deficiency is an X-linked inheritance disorder. Most patients with this condition are asymptomatic. But when exposed to oxidative agents, they are prone to develop acute hemolytic anemia. The various oxidative triggers include:

- Fava beans
- Infections
- Drugs such as primaquine, dapsone, and nitrofurantoin

A hemolytic attack can present with weakness and abdominal/ lumbar pain. Later, the patient can develop jaundice, hemoglobinemia, and hemoglobinuria. Investigations will show elevated levels of lactate dehydrogenase (LDH) and low levels of plasma haptoglobin. Anisocytosis and spherocytes can be seen on peripheral smear. Acute renal failure is a complication seen in adults.

Other options:

Option A - Glycolysis, the main pathway of glucose, can occur either aerobically or anaerobically. A deficiency of the enzyme pyruvate kinase can present as hemolytic anemia. Fatigue can be the presentation when there is phosphofructokinase. Both these are enzymes of glycolysis.

Option B - Gluconeogenesis is the process by which non-carbohydrate precursors are used to synthesize glucose. Pyruvate carboxylase, phosphoenolpyruvate carboxykinase, and glucose-6-phosphatase are the enzymes in the pathway. Glucose-6-phosphatase deficiency leads to von Gierke disease, which is a glycogen storage disorder. Defective gluconeogenesis results in hypoglycemia.

Option D - Luebering–Rapoport pathway is a shunt mechanism that generates and dephosphorylates 2,3-bisphosphoglycerate (2,3-BPG). 2,3-BPG mutase is the enzyme involved. A deficiency of this enzyme can cause erythrocytosis.

Solution to Question 2:

Cystic fibrosis is a condition with an autosomal recessive mode of inheritance.

It results due to a defective mutation in the gene encoding for cystic fibrosis transmembrane conductance regulator (CFTR gene). It functions as an epithelial anion channel. It plays an important role in mucociliary transport along the respiratory mucosa. It also helps regulate the amount of exocrine gland secretion.

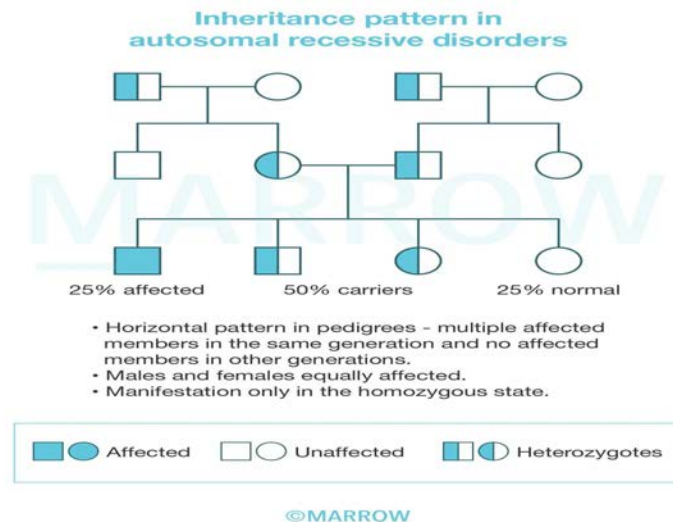
The defect in the ion channel results in the production of hyper viscous, adherent secretions which can obstruct the airways. Organisms such as *Pseudomonas aeruginosa* and *Staphylococcus aureus* can grow in the sputum. The exocrine pancreas can undergo fibrotic scarring, cyst formation, and acinar tissue loss. The viscid secretions can lead to pancreatic duct obstruction. The flow of digestive enzymes to the duodenum is hence impaired.

Symptoms include sinusitis, steatorrhea, and abdominal pain. Distal intestinal obstruction is a complication due to difficulty in excreting intestinal contents. In neonates, it can present as meconium ileus.

Detection of high levels of trypsinogen is used in newborn screening. Chloride level $>60\text{mEq/L}$ in the sweat of affected patients is diagnostic. Hyponatremic hypochloremic alkalosis and hypokalemia may also be seen.

Postural drainage, short-term antibiotics, and inhaled bronchodilators can be used for medical management. The definitive treatment is lung transplantation.

In autosomal recessive inheritance, multiple affected members are seen in the same generation. Other generations are not affected. The following pedigree chart depicts an autosomal recessive pattern of inheritance.



Other options:

Option A - Huntington's disease is a trinucleotide (CAG) repeat disorder with an autosomal dominant mode of inheritance. It is characterized by involuntary movements known as chorea.

Option B - Treacher Collins syndrome is an autosomal dominant disorder that presents as mandibular and malar hypoplasia.

Option D - Achondroplasia has an autosomal dominant inheritance pattern and presents with short limbs, saddle nose, and spinal deformity.

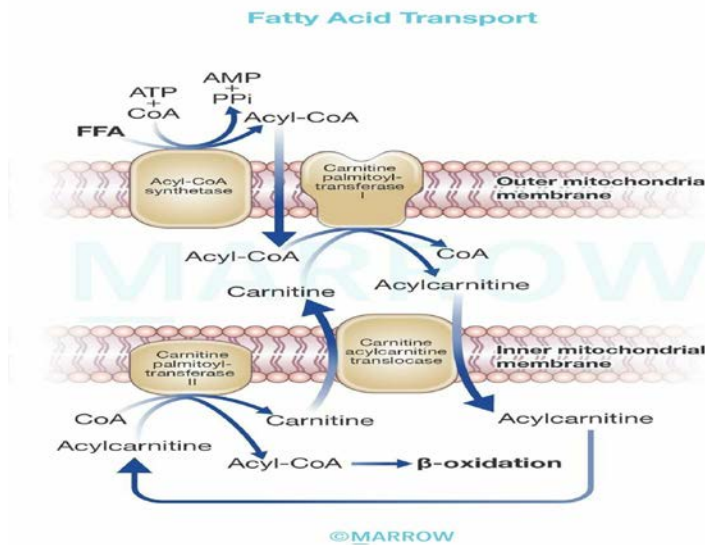
Solution to Question 3:

Carnitine is responsible for the transport of fatty acids or long-chain acyl-CoA across the inner mitochondrial membrane.

Fatty acids are oxidized in the mitochondria to produce acetyl-CoA to produce energy. They must first be converted to acyl-CoA (active fatty acid) before they can be catabolized.

Carnitine palmitoyltransferase - I (CPT- I) transports acyl-CoA across the inner mitochondrial membrane only in the presence of carnitine. The acyl group of acyl-CoA is transferred to carnitine. The acylcarnitine so formed can penetrate the inner mitochondrial membrane via the carnitine-acylcarnitine translocase. This transporter transfers the acyl group back to CoA, restoring acyl-CoA and liberating carnitine, which is moved back to the intermembrane space. Carnitine palmitoyltransferase -II catalyzes this reaction. The deficiency of this enzyme is a common cause of recurrent myoglobinuria.

Carnitine deficiency can lead to hypoglycemia and restrictive or dilated cardiomyopathy. It can also cause hepatic steatosis.



Other options:

Option A - Acyl carrier protein is a component of the fatty acid synthase complex involved in the biosynthesis of fatty acids. It has vitamin pantothenic acid (vitamin B5).

Option C - Lecithin-cholesterol acyltransferase (LCAT) is present in the plasma and plays an important role in the esterification of cholesterol. Deficiency leads to low levels of lysolecithin and cholesteryl esters.

Option D - The long-chain fatty acids are bound to albumin in the plasma. But albumin does not transport fatty acids across the inner mitochondrial membrane. Albumin is a negative acute-phase reactant synthesized by the liver. It is a carrier for acidic drugs.

Solution to Question 4:

A single mutation in a nucleotide base pair resulting in a termination codon is known as a nonsense mutation.

A codon is a sequence of three bases present on the mRNA (messenger RNA), and it codes for amino acids. Out of the 64 codons possible, only 61 code for the 20 amino acids. Three codons, UAA, UAG, and UGA do not code for any of the 20 amino acids. These are known as nonsense/punctator/termination/stop codons and function as translation termination signals. The UGA codon is traditionally considered a stop codon, but it codes for selenocysteine (21st amino acid).

The tRNA (transfer RNA) contains the anticodon which recognizes specific codons present on the mRNA. The sequence of the anticodon is complementary to the codon. The pairing between the codon and anticodon can wobble at the third base. For example, GGU, GGC, and GGA all pair with the anticodon CCI. The wobble phenomenon along with degeneracy reduces the effect of mutations.

A mutation involving a single nucleotide base is known as a point mutation. A nonsense mutation is a type of point mutation where a codon previously coding for an amino acid is changed to a stop

codon. This results in premature termination of translation and the formation of a truncated protein. mRNA containing premature stops are removed by the nonsense-mediated degradation pathway.

Other options:

Option A: A missense mutation is a type of point mutation where the new codon, formed by the changed base, codes for a different amino acid.

Option C: Stop mutation is a term that can be used interchangeably with nonsense mutation. Here, nonsense mutation is the best answer.

Option D: Silent mutations are point mutations where the new codon formed still codes for the same amino acid as the codon before the mutation.

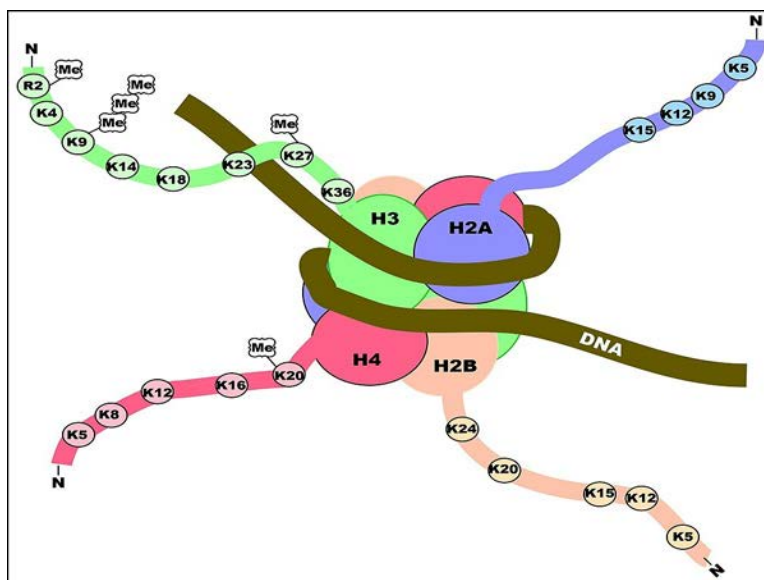
Solution to Question 5:

The proteins involved in DNA packing are histones.

Chromatin contains double-stranded DNA molecules and small basic proteins known as histones. In addition, nonhistone proteins are present that are involved in DNA replication and repair, and RNA synthesis. Histones play a role in condensing the DNA.

Nucleosomes are the organizational unit of chromatin and contain an octamer of histones around which 150 base pairs of DNA are packed. There are various types of histone proteins. H2A, H2B, H3, and H4 groups are components of nucleosomes and are known as core histones. They undergo post-translational modifications such as acetylation and methylation. This plays a critical part in chromatin structure and function.

The DNA in the nucleosome is arranged in a left-handed helical configuration over the histone octamer.



Other options:

Option B - Glycoproteins are proteins with oligosaccharide chains (glycans) bound to amino acids. Certain plasma proteins, peptide hormones, and cell membrane proteins are examples. Structural changes in the glycoproteins present in cancer cells may play a role in metastasis.

Option C - Nucleic acids can be either DNA or RNA. DNA is thought to be arranged in a double-stranded helix model. RNA on the other hand exists as a single strand. There are different types of RNA such as mRNA (messenger RNA), tRNA (transfer RNA), and rRNA (ribosomal RNA).

Option D - Adenine is a purine base that when linked to ribose forms adenosine, a nucleoside. When a nucleoside is esterified with a phosphate group a nucleotide is formed. Guanine is the other purine base. Cytosine is the pyrimidine base common to DNA and RNA. Thymine is present only in DNA and Uracil in RNA.

Solution to Question 6:

The clinical scenario points to a diagnosis of McArdle disease, which results due to a deficiency of the enzyme myophosphorylase (muscle glycogen phosphorylase).

McArdle disease, also known as type V glycogen storage disorder, is an autosomal recessive condition. It typically presents during late childhood or in the second decade of life. The muscle glycogen phosphorylase enzyme deficiency interferes with ATP generation by glycogenolysis in the muscles. This causes muscle glycogen accumulation.

Symptoms are exercise intolerance associated with muscle cramps. A second wind phenomenon, where the symptoms subside after a short period of rest, can also be seen. Myoglobinuria secondary to rhabdomyolysis leads to the passage of burgundy-colored urine in some patients.

Rarely, it can present during infancy as hypotonia, muscle weakness, and respiratory complications. It is generally fatal.

Muscle biopsy to measure glycogen content is the standard diagnosis. Avoiding strenuous exercise prevents symptoms. Glucose administration before exercise can improve tolerance. Vitamin B6 is also recommended.

Other options:

Option B - Hexokinase is a glycolytic enzyme. Its deficiency leads to hemolytic anemia. Splenomegaly may be another presenting feature.

Option C - Glucose-6-phosphatase is an enzyme involved in gluconeogenesis. Its deficiency leads to von Gierke disease also known as type I glycogen storage disease.

Option D - Hepatic glycogen phosphorylase deficiency results in type VI glycogen storage disease, also known as Hers disease.

Solution to Question 7:

Glutathione reductase activity in red blood cells can be estimated to diagnose vitamin B2 deficiency.

Vitamin B2, also known as riboflavin, is a water-soluble vitamin. Its active forms are FMN (flavin mononucleotide) and FAD (flavin adenine dinucleotide). Glutathione reductase is a FAD-containing enzyme (flavoenzyme). Hence, riboflavin deficiency can be detected by measuring erythrocyte glutathione reductase activity. Red blood cell and urinary concentrations of riboflavin can also be estimated to detect a deficiency.

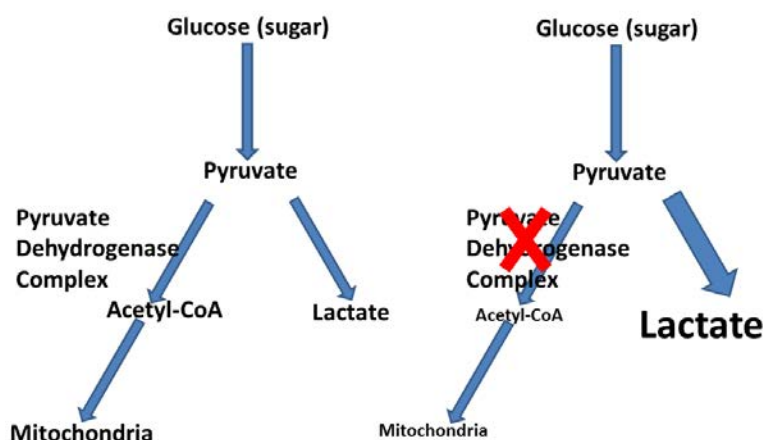
Deficiency of riboflavin can present as anemia, corneal vascularization, mucocutaneous lesions, and personality changes. Milk and other dairy products are important dietary sources.

Other options:

Option A - Transketolase contains thiamine diphosphate as a coenzyme. Estimation of erythrocyte apotransketolase (enzyme protein) after in vitro addition of thiamine diphosphate, can be used to assess thiamine nutritional status. Thiamine deficiency can result in peripheral neuritis, beriberi, and Wernicke encephalopathy with Korsakoff psychosis.

Option C - Kynureninase is an enzyme that requires pyridoxal phosphate as a coenzyme. It converts 3-hydroxykynurenine, a tryptophan metabolite, to 3-hydroxyanthranilate. In vitamin B6 deficiency, this is impaired and results in the formation and subsequent urinary excretion of xanthurenate. This can be used for diagnosis.

Option D - Pyruvate is decarboxylated by pyruvate dehydrogenase for which thiamine diphosphate acts as the cofactor. Therefore thiamine deficiency results in the accumulation of pyruvate. This leads to pyruvic and lactic acidosis, commonly seen in alcoholics.



Solution to Question 8:

The given image depicts Bitot's spots characteristically seen in deficiency of vitamin A. Bitot's spots are seen as white patches of keratinized epithelium on the sclera. It is more commonly bilateral and on the temporal side.

Vitamin A deficiency arises mostly due to the inadequate amounts of vitamin A in diet. It can present as night blindness, conjunctival xerosis, Bitot's spots, and keratomalacia. In later stages, it

can lead to xerophthalmia. It also increases the risk of developing measles, diarrhea, and dysentery. Skin manifestations include xerosis, scaling, and dermomalacia. Phrynoderma may also be seen as keratotic follicular papules affecting the thighs and upper arms. It is shown in the image below.



In patients above the age of 1 year, vitamin A deficiency is treated with 200,000 IU of vitamin A orally or 100,000 IU by intramuscular route. It should be repeated the next day and two weeks later.

Vitamin A is a fat-soluble vitamin that plays an essential role in vision, reproduction, and maintenance of epithelial integrity. The various forms of vitamin A include retinol (storage form), retinal, retinoic acid, and beta-carotene. It is stored in the liver perisinusoidal stellate (Ito) cells as retinyl ester (retinyl palmitate).

Other options:

Option A - Vitamin C is a water-soluble vitamin that aids in iron absorption and collagen maturation. It cannot be synthesized in humans because of the absence of L-gulonolactone oxidase. Vitamin C deficiency results in scurvy.

Option C - Vitamin B deficiency has clinical manifestations pertaining to each type.

Option D - Vitamin E deficiency can present as posterior column and spinocerebellar symptoms. Peripheral neuropathy, ophthalmoplegia, and pigmented retinopathy are other features seen.

Solution to Question 9:

The clinical scenario points to a diagnosis of Gaucher's disease, where there is a defect in the enzyme glucocerebrosidase.

Gaucher's disease is a lysosomal storage disease with autosomal recessive inheritance. The condition results due to the deficiency of beta-glucocerebrosidase. This results in the accumulation of glucosylceramide (glycolipid substrate) in the reticuloendothelial cells. There is progressive infiltration of the liver, spleen, and bone marrow.

There are three clinical types:

- Type 1: Adult (non-neuronopathic) form
- Type 2: Infantile (acute neuronopathic) form
- Type 3: Juvenile (subacute neuronopathic) form

Type 1 is the most common. It presents with a myriad of symptoms, such as bruising secondary to thrombocytopenia, fatigue due to anemia, hepatosplenomegaly, and bone pain. Pathologic fractures and lytic lesions of the femur, ribs, and pelvis may also be seen. A few patients may have pulmonary involvement. Growth retardation may be observed in cases presenting in the first decade.

Type 2 is rare, and presents with hypertonia, strabismus, and organomegaly. Neurodegeneration is the characteristic feature. Stridor and failure to thrive can also be the presenting features. Ultimately, death can occur due to respiratory arrest.

Type 3 presents with intermediate symptoms, as seen in the other two types. By the age of 10–15 years, the child succumbs to death.

The Gaucher cells, which are the reticuloendothelial cells with glucosylceramide, have a wrinkled-paper appearance due to intracytoplasmic substrate inclusions. These cells stain strongly with periodic acid-Schiff (PAS) stain. This is characteristic of Gaucher's disease.

Other options:

Option B - Acid sphingomyelinase deficiency results in Niemann-Pick disease types A and B. Both are autosomal recessive conditions. Type A has a neurodegenerative course, while type B is non-neuronopathic. The former presents with hepatosplenomegaly, lymphadenopathy, and psychomotor retardation. Death occurs by the age of 3 years. Type B is associated with hepatosplenomegaly and mild pulmonary involvement.

Option C - Hexosaminidase deficiency causes Sandhoff disease. It has an autosomal recessive pattern of inheritance. Clinical features include ataxia, dysarthria, and mental retardation.

Option D - Glucose-6-phosphatase deficiency leads to von Gierke disease, which is a glycogen storage disorder. It is an autosomal recessive condition. The affected children have characteristic doll-like facies. Hyperuricemia, lactic acidosis, and triglyceridemia can be seen. Hepatomegaly and renal enlargement is also associated.

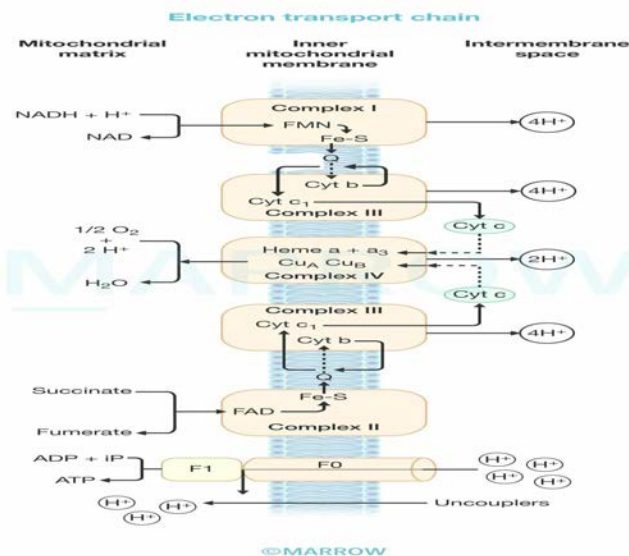
Solution to Question 10:

Cyanide is an inhibitor of cytochrome complex IV or cytochrome $a+a_3$ (cytochrome c oxidase). It is a component of the electron transport chain.

The electron transport chain (ETC) is a specialized set of electron carriers that can accept electrons. Energy-rich molecules like glucose are oxidized to produce water and carbon dioxide. The intermediate substrates in these reactions can pass on electrons to certain coenzymes such as nicotinamide adenine dinucleotide (NAD⁺) and flavin adenine dinucleotide (FADH₂). The reduced forms can then donate electrons to the ETC by which they lose most of their free energy which is used to move H⁺ through the inner mitochondrial membrane. The H⁺ gradient thus produced results in the production of ATP. ATP synthesis together with electron transport is

known as oxidative phosphorylation.

The reactions in oxidative phosphorylation and the arrangement of complexes that form the ETC are shown below.



These complexes can accept or donate electrons to coenzyme Q (CoQ) and cytochrome c, which are mobile electron carriers. Certain site-specific inhibitors can inhibit ATP synthesis.

Other options:

Option B - Carbon dioxide along with water is produced by the oxidation of energy-rich molecules in oxidative phosphorylation.

Option C - Oligomycin, an antibiotic, blocks the flow of protons through ATP synthase. This inhibits oxidative phosphorylation.

Option D - Ouabain is a cardiac drug that inhibits the $\text{Na}^+ - \text{K}^+$ ATPase of cell membranes by acting on the extracellular surface.

Complexes of ETC	Inhibitors
Complex I	Piericidin A, amobarbital, rotenone
Complex II	TTFA (thenoyltrifluoroacetone), carboxin, malonate
Complex III	BAL(British anti-lewisite), antimycin A
Complex IV	Hydrogen sulfide, carbon monoxide, cyanide

Biochemistry NEET 2023

Question 1:

Which of the following zymogens requires gamma-glutamyl carboxylase?

- a) Factors II, VIII, IX, X
- b) Factors II, VII, IX, and X
- c) Factors II, VII, IX, XI
- d) Factors II, VIII, X, XI

Question 2:

A patient presented with diarrhea, dermatitis, and dementia. Which of the following vitamin deficiencies would you suspect in this patient?

- a) Vitamin B3 deficiency
- b) Vitamin B1 deficiency
- c) Vitamin B6 deficiency
- d) Vitamin B12 deficiency

Question 3:

A young boy presents to the OPD with hypoglycemia and muscle cramps, on exertion or while playing. Then he becomes normal after resting for a while. These episodes are recurrent after a period of activity. He has decreased serum lactate and glucose levels. Which of the following diseases is he most likely to be suffering from?

- a) McArdle disease
- b) Hers disease
- c) Cori's disease
- d) Andersen disease

Question 4:

A family consumes only polished rice. Which of the following combination of vitamin deficiency and enzymatic defect will be present in this family?

- a) Riboflavin-glutathione reductase
- b) Thiamine-transketolase
- c) Thiamine-transaminase
- d) Riboflavin-transketolase

Question 5:

A chronic alcoholic patient presented to the casualty with confusion, ataxia, and painful eye movements. What could be the vitamin deficiency seen in this patient?

- a) Vitamin B deficiency
- b) Vitamin A deficiency
- c) Vitamin D deficiency
- d) Vitamin C deficiency

Question 6:

A 4-year-old child presents with easy fatigability. The mother also complains that the child has increased hunger between meals, which is relieved after food. Liver examination revealed no glycogen. The enzyme most likely deficient is _____.

- a) Glycogen phosphorylase
- b) Glycogen synthase
- c) Debranching enzyme
- d) Glucose 6 phosphatase

Question 7:

A child was brought with pedal edema and cheilosis. Cardiomegaly was present. What is the vitamin deficiency associated with this clinical presentation?

- a) Riboflavin
- b) Thiamine
- c) Pyridoxine
- d) Niacin

Question 8:

A chronic alcoholic patient is diagnosed with gouty arthritis. What biochemical changes will be seen in this patient?

- a) Decreased NADH/NAD⁺ ratio
- b) Increased urea and urate levels
- c) Increased urate and lactate levels
- d) Increased level of alkaline phosphatase

Question 9:

A child presents with erythematous scaly patches in the perioral region, mucosal ulcers, and impaired epithelial wound healing. The most likely mineral deficiency associated with this condition is _____.

- a) Iron deficiency
- b) Zinc deficiency
- c) Calcium deficiency
- d) Copper deficiency

Question 10:

A patient presenting with bleeding gums and easy bruisability was diagnosed with scurvy. This is due to _____.

- a) Low calcium
- b) Defective collagen formation
- c) Inhibition of clotting factors
- d) Increased keratinization of epithelium

Question 11:

Which of the following is seen in low insulin:glucagon ratio?

- a) Activation of lipoprotein lipase
- b) Activation of glycogen synthase

- c) Activation of phosphofructokinase 1
- d) Activation of hormone-sensitive lipase

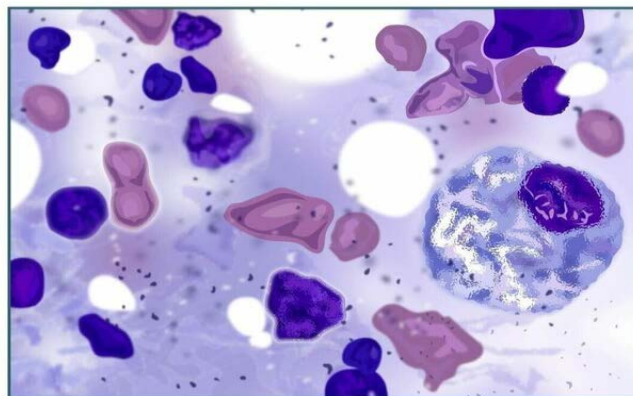
Question 12:

An infant is brought by his parents with complaints that his urine turns black on standing. Which of the following metabolic disorders is likely?

- a) Phenylketonuria
- b) Alkaptonuria
- c) Homocystinuria
- d) Maple syrup urine disease

Question 13:

A child presents with bone pain and hepatosplenomegaly. A trephine biopsy and aspirate show the following finding. Which of the following is the most likely enzyme deficient in this condition?



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- a) Hexosaminidase
- b) Glucocerebrosidase
- c) Sphingomyelinase
- d) Alpha 1,4-glucosidase

Question 14:

A child presents with thinning of hair and scaling of the scalp. On taking further history, it is found that she consumes raw eggs in her diet regularly. The given manifestations are due to the deficiency of which of the following vitamins?

- a) Niacin
- b) Pyridoxine
- c) Biotin
- d) Thiamine

Question 15:

A child presents with mental retardation, bone pain, and inability to walk. On funduscopy examination, a cherry red spot is seen. There is no organomegaly. What is the most likely diagnosis?

- a) Tay-Sachs disease
- b) Niemann-Pick disease
- c) Hurler syndrome
- d) Gaucher disease

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b
5	a
6	b
7	b
8	c
9	b
10	b
11	d

12	b
13	b
14	c
15	a

Detailed Explanations

Solution to Question 1:

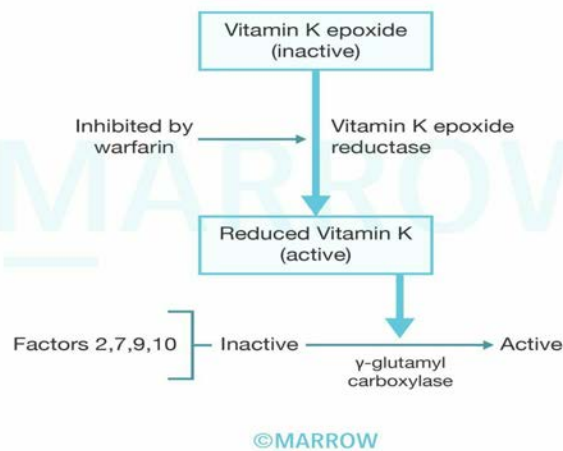
Clotting factors II, VII, IX, and X are zymogens of serine proteases that contain glutamic acid residues. These residues undergo vitamin-K-dependent carboxylation via gamma-glutamyl carboxylase to form gamma-carboxyglutamic acid residues. These residues chelate calcium ions, allowing the binding of clotting factors to the platelet surface.

Zymogens are the proprotein form of enzymes (proenzymes) that require proteolytic activation to carry out their function.

The reduced form of vitamin K acts as a cofactor in the post-synthetic modification of these zymogens and their activation. This reduced form of vitamin K is recycled back from its oxidized form by the enzyme vitamin K epoxide reductase, which is inhibited by warfarin.

The absence of vitamin K or the presence of warfarin inhibits carboxylation of the glutamate residues, leading to the release of abnormal precursors of clotting factors in the blood.

Mechanism of action of Warfarin



Solution to Question 2:

The given clinical scenario of a patient with diarrhea, dermatitis, and dementia is suggestive of niacin/vitamin B₃ deficiency.

Niacin deficiency causes pellagra, found mostly among people eating corn/maize-based diets, which are low in niacin and tryptophan. Excess leucine in jowar and maize disrupts the conversion of tryptophan to niacin.

Niacin can be synthesized in the body from tryptophan. Two compounds, nicotinic acid and nicotinamide, have the biological activity of niacin. The nicotinamide ring of the coenzymes nicotinamide adenine dinucleotide (NAD) and nicotinamide adenine dinucleotide phosphate (NADP) participate in oxidation/reduction reactions.

The initial symptoms of pellagra include loss of appetite, generalized weakness, irritability, abdominal pain, and vomiting. This is followed by bright red glossitis and a characteristic pigmented, scaly skin rash in sun-exposed areas. This rash is known as Casal's necklace when it forms a ring around the neck.

The 4Ds of pellagra:

- Diarrhea
- Dermatitis
- Dementia, leading to
- Death

It is diagnosed by measuring urinary levels of 2-methyl nicotinamide and 2-pyridone.

It is treated by oral supplementation of 100–200 mg of nicotinamide or nicotinic acid three times daily for 5 days.

Casal's necklace finding in niacin deficiency:



Other options:

Option B: Vitamin B1/Thiamine deficiency can manifest as wet beriberi (CVS features like heart failure and edema) or Shoshin beriberi, an acute and fulminant form of wet beriberi, dry beriberi (peripheral neuritis), or as Wernicke encephalopathy with Korsakoff psychosis (in alcoholics). Dietary deficiency of thiamine inhibits pyruvate dehydrogenase. Chronic alcoholics may develop potentially fatal pyruvic and lactic acidosis.

Option C: Vitamin B6/Pyridoxine deficiency is rare but has been observed in alcoholics, newborns, formulas low in vitamin B6, and women taking oral contraceptives. Isoniazid, an anti-TB drug, leads to pyridoxine deficiency resulting in peripheral neuropathy. Hence pyridoxine supplementation is included in ATT regimens.

Option D: Vitamin B12/Cobalamin deficiency causes megaloblastic anemia, dementia, spinal cord degeneration, homocysteinemia, etc.

Solution to Question 3:

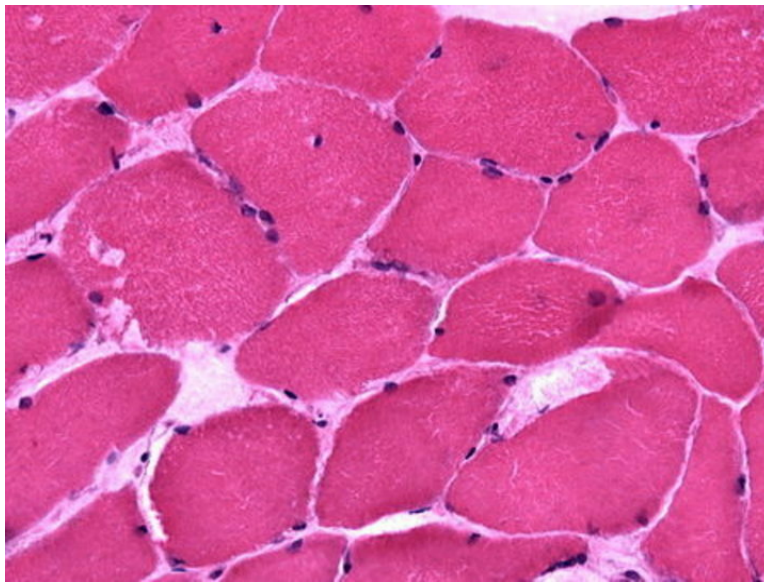
A clinical scenario of a patient with muscle cramps, pain, weakness with exertion, myoglobinuria, and the muscle biopsy specimen showing vacuolar myopathy are suggestive of McArdle's disease, which is due to the deficiency of muscle glycogen phosphorylase.

McArdle disease, also known as type V glycogen storage disorder, is an autosomal recessive condition. It typically presents during late childhood or in the second decade of life. The muscle glycogen phosphorylase enzyme deficiency interferes with ATP generation by glycogenolysis in the muscles. This causes muscle glycogen accumulation.

Symptoms are exercise intolerance associated with muscle cramps. Due to high-intensity exercise, sudden persistent muscle contraction occurs. This leads to severe muscle damage resulting in a massive release of muscle proteins and myoglobin in the blood. Myoglobinuria secondary to rhabdomyolysis leads to the passage of burgundy-colored urine in some patients.

A unique feature associated with McArdle is the improvement in symptoms after rest for a few minutes. This is termed as the second wind phenomenon. This occurs due to the increased blood flow to the muscles and increased utilization of free fatty acids (FFAs) for energy production after resting for a few minutes. This compensates for the impaired ATP generation from glycogenolysis. Patients with McArdle disease can benefit from engaging in regular aerobic exercises, such as walking or cycling, which can help increase their exercise tolerance and cardiovascular fitness. However, they should avoid high-intensity exercises or activities that can trigger muscle cramps and rhabdomyolysis. Glucose administration before exercise can improve tolerance. Vitamin B6 is also recommended.

Standard diagnosis includes muscle biopsy, which shows periodic acid Schiff (PAS)-positive vacuoles (Shown in the image below) of high glycogen content due to lack of myophosphorylase and genetic testing (PYGM gene sequencing).



Other options:

Option B: Hers disease (type VI glycogen storage disorder) - It is caused by the deficiency of liver glycogen phosphorylase. Patients with Hers disease exhibit growth retardation, hepatomegaly, and mild hypoglycemia. They do not typically present with muscle cramps and myoglobinuria.

Option C: Cori's disease (type III glycogen storage disorder) - It is caused by a deficiency of glycogen debranching enzyme. Patients with Cori's disease present with hepatomegaly, growth retardation, and hypoglycemia. Muscle weakness may be present but not muscle cramps or myoglobinuria.

Option D: Andersen disease (type IV glycogen storage disorder) - It is caused by the deficiency of glycogen branching enzyme. This disease presents with hepatomegaly, cirrhosis, and liver failure. It does not cause muscle cramps or myoglobinuria.

Solution to Question 4:

The consumption of polished rice leads to thiamine deficiency which leads to defective transketolase activity.

Polished rice is rice that has had its husk, bran, and germ removed, leaving only the endosperm. This processing method results in a loss of nutrients, including thiamine.

Thiamine pyrophosphate (Vit B1) is the cofactor of transketolase. This reaction requires two cofactors:

- Mg^{2+}
- Thiamine diphosphate (vitamin B1)

Transketolase and transaldolase are involved in the non-oxidative phase of the HMP pathway.

Thiamine deficiency can lead to beriberi, a condition characterized by peripheral neuropathy, weakness, and cardiovascular symptoms. In severe cases, it can also lead to Wernicke-Korsakoff syndrome, which is characterized by confusion, ataxia, and vision disturbances. Thiamine

deficiency can be prevented by consuming a balanced diet that includes unprocessed grains, legumes, and other sources of thiamine. Diagnosis of thiamine deficiency can be established by erythrocyte transketolase activity assessment.

Solution to Question 5:

The above clinical history is suggestive of Wernicke encephalopathy, which is caused by vitamin B1 (Thiamine) deficiency.

Vitamin B1/thiamine deficiency leads to following conditions:

- Wernicke encephalopathy:
 - An acute, neuropsychiatric syndrome characterized by nystagmus and ophthalmoplegia, mental status changes (confusion), and unsteadiness of stance and gait.
 - Wernicke's encephalopathy is a medical emergency, and thiamine should be initiated immediately, either intravenously or intramuscularly.
 - The current standard of treatment for patients with symptoms of Wernicke's encephalopathy is to give thiamine 100 mg intravenously before administering glucose-containing IV fluids.
- Korsakoff psychosis:
 - Korsakoff syndrome is characterized by a pronounced anterograde and retrograde amnesia along with potential impairment in visuospatial, abstract, and other types of learning.
 - Symptoms are similar to that of Wernicke encephalopathy, but also include memory loss and confabulation.
 - Permanent in at least a partial form in about 50% of the people affected.
- Beriberi: It can present with:
 - Dry beriberi: sensorimotor peripheral neuropathy, muscle weakness
 - Wet beriberi: tachycardia, cardiomegaly, high-output cardiac failure

Solution to Question 6:

The most likely enzyme deficiency in a child with fatigue, excessive hunger between meals, and absent liver glycogen is glycogen synthase deficiency.

Glycogen is a storage carbohydrate synthesized from glucose (glycogenesis) and is stored in the liver, muscles, and brain. Glycogen synthase is a regulating enzyme for glycogenesis.

Glycogen synthase exists in both phosphorylated (inactive) and dephosphorylated (active) forms. Phosphorylation is stimulated by epinephrine, norepinephrine, and glucagon via cAMP. Dephosphorylation is stimulated by insulin via phosphodiesterase.

A deficiency of glycogen synthase leads to glycogen storage disorder and presents in infancy with early morning drowsiness and fatigue, fasting hypoglycemia, pallor, emesis, and hypoglycemic convulsions. It leads to a marked decrease in liver glycogen stores. Hepatomegaly is absent.

Laboratory investigations reveal prolonged hyperglycemia, glycosuria, ketosis, lactic acidosis, low alanine levels, and normal insulin levels after glucose administration. A liver biopsy is diagnostic.

Treatment involves protein-rich frequent meals, and nighttime supplementation with cornstarch to prevent hypoglycemia and hyperketonemia.

Other options:

Option A: Glycogen phosphorylase deficiency, also known as McArdle's disease or glycogen storage disease type V, primarily affects skeletal muscle glycogen metabolism. Affected individuals experience muscle weakness, cramps, and exercise intolerance, but it doesn't typically cause the symptoms mentioned in the question.

Option C: Debranching enzyme deficiency, also known as Cori's disease or glycogen storage disease type III, presents with hepatomegaly, hypoglycemia, and growth retardation. The liver would still contain glycogen, albeit with an abnormal structure.

Option D: Glucose-6-phosphatase deficiency, known as von Gierke's disease or glycogen storage disease type I, is characterized by severe fasting hypoglycemia, lactic acidosis, hyperuricemia, and hepatomegaly. In this case, glycogen is present in the liver but cannot be broken down efficiently into glucose.

Solution to Question 7:

The given clinical presentation with pedal edema and cardiomegaly is suggestive of wet beriberi due to thiamine (vitamin B₁) deficiency.

Early symptoms of thiamine deficiency include peripheral neuropathy, irritability, loss of deep tendon reflexes and proprioception. As the disease progresses, increased intracranial pressure, meningismus and coma develop.

Wet beriberi results due to cardiac and renal dysfunction which leads to fluid accumulation. It is associated with tachycardia, cardiomegaly, cardiac failure, and edema. It is commonly seen in the refugee camps where people consume polished rice-based monotonous diets.

Dry beriberi is associated with neurological deficits such as peripheral neuropathy, optic atrophy, ataxia, and irritability. Complications include lactic acidosis and Wernicke encephalopathy in adults.

Diagnosis of thiamine deficiency can be established by erythrocyte transketolase activity assessment.

Other options:

Option A: Riboflavin deficiency can present with cheilosis, but it does not typically cause pedal edema and cardiomegaly.

Option C: Pyridoxine (vitamin B₆) deficiency can cause neurological symptoms, anemia, and dermatitis but is not associated with pedal edema, cardiomegaly, and cheilosis.

Option D: Niacin (vitamin B₃) deficiency leads to pellagra, characterized by the triad of dermatitis, diarrhea, and dementia. It is not associated with pedal edema, cardiomegaly, and cheilosis.

Solution to Question 8:

This patient with chronic alcoholism and gouty arthritis will most likely have elevated levels of lactate and urate.

The oxidation of ethanol to acetate by alcohol dehydrogenase is accompanied by the reduction of nicotinamide adenine dinucleotide (NAD⁺) to NADH. This results in a high concentration of NADH, leading to an increased NADH/NAD⁺ ratio (Option A).

Due to this, the following metabolic reactions that require NAD⁺ are inhibited:

- Glycolysis
- Citric acid cycle (ketogenesis favored)
- Pyruvate dehydrogenase
- Fatty acid oxidation

Decreased fatty acid oxidation leads to increased esterification and production of triacylglycerol, which causes fatty liver.

Also, there is an increase in two reactions:

- Pyruvate to lactate
- Oxaloacetate to malate

Increased lactate leads to hyper-lactic-acidemia, which leads to increased renal reabsorption of urate and decreased excretion of uric acid. This results in aggravation of gout. Alcoholics are deficient in thiamine which inhibits pyruvate dehydrogenase and leads to lactic acidosis.

A decrease in pyruvate and oxaloacetate levels, which are substrates for gluconeogenesis, leads to a decrease in blood glucose levels, causing symptoms of agitation, impaired judgment, and combativeness.

Other options:

Increased urea and urate levels are not specifically associated with alcoholism and gouty arthritis. While urate levels may be increased due to gout, urea levels are not directly related to the condition (Option B).

Increased levels of alkaline phosphatase can be seen in alcohol-induced hepatitis due to intrahepatic cholestasis but is not directly related to gouty arthritis in a chronic alcoholic patient. Therefore, this option is less relevant compared to increased urate and lactate levels (Option D).

Solution to Question 9:

The clinical features of periorificial dermatitis and stomatitis in a young child are suggestive of acrodermatitis enteropathica, seen in zinc deficiency.

Acrodermatitis enteropathica is a genetic disorder that results in the inability to absorb dietary zinc.

The clinical features develop after weaning and the infant presents with:

- Failure to thrive
- Skin manifestations such as symmetrical vesiculobullous dermatitis in the perioral, perianal, and acral regions
- Anorexia, diarrhea
- Recurrent infections due to T-cell immunodeficiency
- Alopecia with a reddish tint
- Impaired wound healing
- Male hypogonadism, growth retardation
- Neurological manifestations like tremors, nystagmus, photophobia, hypogeusia, irritability, and night blindness

Oral zinc supplementation is the treatment of choice.



Other options:

Option A- Iron deficiency is the most common micronutrient deficiency globally. It presents with iron-deficiency anemia and neurocognitive impairment. It presents clinically with pallor, fatigue, atrophic glossitis, and pica.

Option C- Calcium deficiency leads to rickets as a result of early weaning. It presents with typical symptoms of rickets, including failure to thrive; protruding abdomen; muscle weakness; fractures; craniotabes; delayed dentition; caries; enlargement of wrists and ankles; and hypocalcemic symptoms of tetany, stridor, and seizures.

Option D- Copper deficiency leads to anemia, growth retardation, defective keratinization and pigmentation of hair, hypothermia, degenerative changes in aortic elastin, osteopenia, and mental retardation.

Solution to Question 10:

Bleeding gums and easy bruisability seen in scurvy are due to defective collagen formation.

Scurvy results from the deficiency of vitamin C, which is required for the hydroxylation of prolyl and lysyl residues of collagen reactions. Hence, deficiency of vitamin C results in a defective prolyl hydroxylase activity, leading to defective collagen formation and resulting in generalized bleeding tendencies.

Hemorrhagic manifestations of scurvy include petechiae, purpura, ecchymoses at pressure points, epistaxis, gum bleeding, and characteristic perifollicular hemorrhages. Other manifestations are poor wound and fracture healing, hyperkeratosis of hair follicles, arthralgia, and muscle weakness. It also results in defective osteoid formation. Costo-chondral separation leads to the formation of tender nodules that are sharply angulated and are also known as the scorbutic rosary.

The cutaneous manifestations of scurvy include

- Phrynodema over posterolateral arms which can then generalize over the back and thighs and legs.
- Corkscrew hairs - fractured and coiled hairs due to impaired keratin crosslinks by disulfide bonds
- Perifollicular hemorrhage and purpura
- Lower limb edema
- Splinter hemorrhages

It is treated with vitamin C supplementation.

Other options:

Option A: Low calcium levels do not cause scurvy; they may contribute to other clinical conditions like osteoporosis or muscle cramps.

Option C: Inhibition of clotting factors is not responsible for scurvy. Vitamin K deficiency, however, can lead to the inhibition of clotting factors and cause bleeding tendencies.

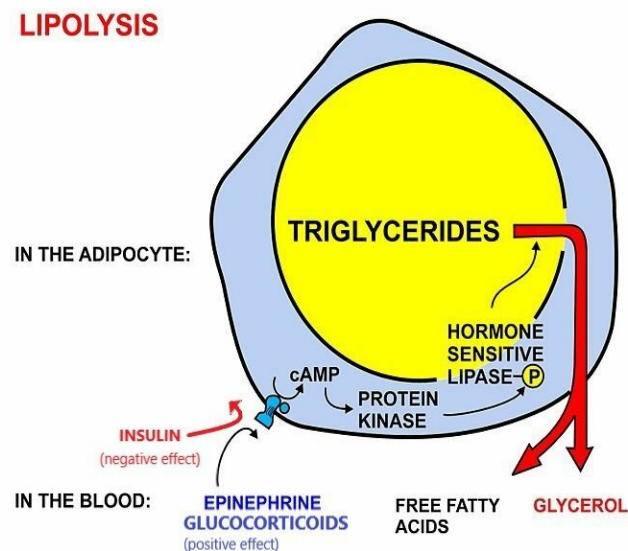
Option D: Increased keratinization of epithelium is not the cause of scurvy. Vitamin A deficiency, on the other hand, can result in hyperkeratoses and epithelial changes, such as xerosis and Bitot's spots.

Solution to Question 11:

Low insulin:glucagon ratio occurs in the fasting state. Since insulin has an inhibitory effect on hormone-sensitive lipase (lipolytic hormone), low insulin:glucagon ratio results in the activation of hormone-sensitive lipase.

In adipose tissues, decreased insulin results in the inhibition of lipogenesis and activation of intracellular hormone-sensitive lipase. The stored triglycerides are broken down (lipolysis) by hormone-sensitive lipase (HSL) into free fatty acids (FFAs) and glycerol. In the fasting state, glucose is spared for the brain and glycerol is used as a substrate for gluconeogenesis.

In diabetic patients, low insulin level corresponds to an increase in HSL activity, promoting the breakdown of lipids and the release of FFAs from adipose tissue. HSL activity is accelerated by hormones like epinephrine, norepinephrine, glucagon, ACTH, TSH, GH, ADH, and melanocyte-stimulating hormones (MSH).



Other options:

Option A: Activation of lipoprotein lipase (LPL) is promoted by insulin, not low insulin: glucagon ratio. LPL is involved in the breakdown of triglycerides and is active in the well-fed state. In low insulin: glucagon ratio, lipoprotein lipase activity is reduced, leading to decreased hydrolysis of triglycerides.

Option B: Activation of glycogen synthase is promoted by insulin. In low insulin: glucagon ratio, glycogen synthase activity is reduced, leading to decreased glycogenesis. Instead, glycogenolysis and gluconeogenesis are promoted to maintain blood glucose levels.

Option C: Activation of phosphofructokinase 1 (PFK1) is promoted by insulin. PFK1 is a rate-regulating enzyme in glycolysis. In low insulin: glucagon ratio, glycolysis is downregulated, and gluconeogenesis is upregulated to maintain blood glucose levels.

Well-fed state (Insulin)	During fasting (Glucagon)
Lipogenesis increased	Lipogenesis inhibited
Lipolysis inhibited	Lipolysis increased
Insulin inhibits HS-lipase	Glucagon activates HS-lipase
Lipoprotein lipase active	Lipoprotein lipase inactive

Solution to Question 12:

Blackening of urine on standing is characteristic of alkaptonuria.

Alkaptonuria is an autosomal recessive disorder of tyrosine metabolism, that develops due to the defect in homogentisate oxidase/homogentisic acid 1, 2 - dioxygenase, which oxidizes homogentisate to maleylacetoacetate. The defect in this enzyme leads to the accumulation of homogentisate, which is then excreted in the urine.

Alkaptonuria is characterized by homogentisic aciduria, ochronosis, and arthropathy (joint pain).

Homogentisic aciduria is the black discoloration of urine on prolonged exposure to air caused due to oxidation of the excreted homogentisate.

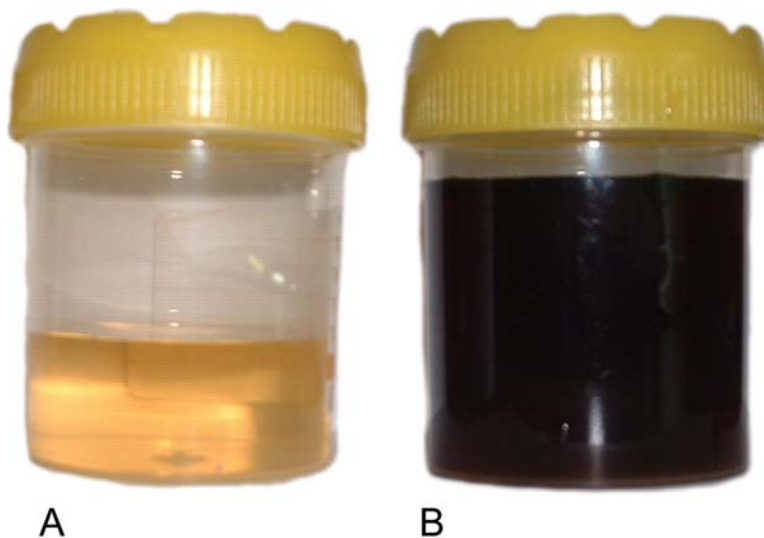
Ochronosis, which is bluish-black pigmentation in connective tissue, generally occurs after the third decade of life, and it occurs due to the accumulation of homogentisic acid and its oxidation products like benzoquinone acetic acid in connective tissue.

Arthritis generally begins in the third decade. Arthritis often begins in the spine and resembles ankylosing spondylitis in its large-joint distribution. Radiographs of the spine showing flattened and calcified intervertebral discs are pathognomonic. Other findings include degeneration of the intervertebral discs, followed by disc calcification and eventually fusion of the vertebral bodies. Osteophyte formation and calcification of the intervertebral ligaments also occur.

Diagnosis: Blackish discoloration of urine on standing is diagnostic of alkaptonuria (shown below). The ferric chloride test will be positive. The blackening of urine occurs due to the excretion of homogentisic acid in the urine.

Image A - Normal urine

Image B - Alkaptonuria



Other options:

Option A: Phenylketonuria (PKU) is a metabolic disorder resulting from a deficiency of the enzyme phenylalanine hydroxylase. It does not lead to black urine. Patients with PKU have a musty or mousy odor due to phenylacetic acid accumulation.

Option C: Homocystinuria is an inborn error of methionine metabolism caused by a deficiency of the enzyme cystathionine β -synthase. It is characterized by tall stature, long limbs, dislocation of

the lens of the eye, and predisposition to thromboembolism, but it does not lead to black urine.

Option D: Maple syrup urine disease (MSUD) is an inherited metabolic disorder caused by a deficiency of branched-chain α -ketoacid dehydrogenase. It does not cause black urine. Patients with MSUD have urine with a characteristic sweet odor, resembling maple syrup.

Solution to Question 13:

The given clinical scenario of a child with bone pain, hepatosplenomegaly, and biopsy showing Gaucher's cell with a characteristic crumpled tissue paper appearance is suggestive of Gaucher's disease. The deficient enzyme is glucocerebrosidase (also known as beta-glucosidase).

Gaucher's disease is an inborn error of metabolism caused due to a deficiency of beta-glucocerebrosidase. It is the most common lysosomal storage disorder and the mode of inheritance is autosomal recessive. There is an accumulation of lipid-laden macrophages in the spleen, liver, and other tissues.

There are three clinical subtypes - type I (chronic non-neuronopathic), type II (acute neuronopathic), and type III (intermediate). Clinical features include anemia, thrombocytopenia, hepatosplenomegaly, and neurological features depending on the subtype.

Gaucher's cells - These are formed as a result of the accumulation of massive amounts of glucocerebrosides within the phagocytes. The cells have a fibrillary cytoplasm and give a crumpled tissue paper appearance.

Option A: Deficiency of hexosaminidase A is seen in Tay-Sachs disease. The deficiency of beta-hexosaminidase A and B causes Sandhoff's disease. Clinical features are seizures, macrocephaly, and hyperacusis.

Option C: Deficiency of sphingomyelinase leads to Niemann-Pick disease. It is characterized by the presence of foam cells in the bone marrow. There may be pulmonary infiltrates leading to lung failure.

Option D: Pompe's disease (type 2 glycogen storage disease) occurs due to alpha 1,4-glucosidase (acid maltase) deficiency. It is the only glycogen storage disease that is a lysosomal storage disease and causes lysosomal glycogen accumulation in cardiac, skeletal, and smooth muscle cells. The clinical symptoms of Pompe disease are hypotonia (floppy infant), skeletal muscle weakness, hepatomegaly, hypertrophic cardiomyopathy, and macroglossia.

Solution to Question 14:

The given clinical scenario of scaling and alopecia and the history of raw egg consumption are suggestive of biotin deficiency.

Raw egg white contains a glycoprotein known as avidin, which tightly binds to biotin and renders it unabsorbable from the intestine. Hence, there can be a biotin deficiency state.

Biotin deficiency may cause:

- Dermatitis, scaling, seborrheic, and erythematous rash

- Mental status changes (depression, hallucinations)
- Hyperesthesia
- Alopecia
- Atrophic glossitis

In infants, biotin deficiency presents as hypotonia, lethargy, and apathy. In addition, infants may develop alopecia and a characteristic rash that includes the ears.

Solution to Question 15:

The given clinical scenario, along with the fundoscopy image showing a macular cherry-red spot, indicates that the most likely diagnosis is Tay-Sachs disease.

Tay-Sachs disease is a lysosomal storage disease caused by the deficiency of β -hexosaminidase. It can present during infancy with macrocephaly, increased startle reaction, gradual loss of motor skills, and a macular cherry-red spot. Affected children usually die by the age of 10–15 years.

It is of two types—juvenile-onset and adult-onset. The symptoms of the juvenile-onset form include ataxia and dementia. Dysarthria, lower motor neuron signs, and psychiatric disorders are common in the adult-onset type.

Other options:

Options B - Niemann-Pick disease is an autosomal recessive inborn error of lipid metabolism. It results due to the deficiency of the sphingomyelinase enzyme. This results in the accumulation of sphingomyelin in the brain, lungs, liver, and spleen. The clinical features include a combination of hepatosplenomegaly, developmental delay, failure to thrive, severe psychomotor regression, interstitial lung disease, and a cherry-red spot in the macula.

Option C - Hurler syndrome (mucopolysaccharidosis I) is an autosomal recessive condition that affects multiple organs. Presentation is usually at 6–24 months with corneal clouding, coarse facial features, joint stiffness, and skeletal dysplasia. Glaucoma, retinal degeneration, and nasal discharge are the other features. The condition is generally fatal.

Option D - Gaucher disease is a glycolipid storage disorder that can present with hepatosplenomegaly, mental retardation, and erosion of long bones. It is caused by the deficiency of β -glucosidase. The Gaucher cell has a wrinkled-paper appearance due to intracytoplasmic inclusions, which is a characteristic feature.

Biochemistry INI-CET 2023

Question 1:

Xanthurenic aciduria is seen in which of the following vitamin deficiency?

- a) Vitamin B2
- b) Vitamin B1
- c) Vitamin B12
- d) Vitamin B6

Question 2:

A 12-year-old boy with type 1 diabetes mellitus presents with unconsciousness. His urine is subjected to a test and the results are shown below. What is the test done?



- a) Seliwanoff's test- Fructose
- b) Benedict's test- Reducing sugars
- c) Fouchet's test- Bilirubin
- d) Rothera's test- Ketone bodies

Question 3:

The template DNA strand is 5'CATTCAATATCGATC3'. The resulting mRNA sequence after transcription is?

- a) 3'GAUCGAUAUUGAAUA5'
- b) 5'GATCGATATTGAATUT3'
- c) 5'GAUCGAUAUUGAAUG3'
- d) 3'GATCGATATTGAATU5'

Question 4:

Arrange the following steps of Karyotyping in the correct order :

- a) 1, 2, 3, 4, 5
- b) 1, 3, 2, 4, 5
- c) 1, 2, 3, 5, 4
- d) 2, 1, 3, 4, 5

Question 5:

A patient came with complaints of muscle weakness to the OPD. The doctor found that he has cardiomyopathy and diagnosed him to be a case of Barth Syndrome, where the patient has mitochondrial dysfunction. What is the defect in this patient?

- a) Cephalin
- b) Cardiolipin
- c) Lecithin
- d) Lysolecithin

Question 6:

In Pompe's disease, the enzyme deficient is?

- a) Lactic acid dehydrogenase
- b) Phosphofructokinase
- c) Muscle glycogen phosphorylase
- d) Lysosomal alpha glucosidase

Question 7:

The thermic effect is highest in _____.

- a) Carbohydrates
- b) Proteins
- c) Lipids
- d) Independent of components

Question 8:

Which of the following statements are false about collagen?

- a) Most common fibrous protein
- b) Prolsyl and lysyl hydroxylase are involved in post translational modification
- c) Phenylalanine is the most abundant AA in its structure
- d) Triple helix structure is important for its function

Question 9:

Ketone bodies are not utilized by the liver because it lacks:

- a) Thiokinase
- b) Thiophorase
- c) Thiolase
- d) Thiotransferase

Question 10:

Which of the following methods can detect the alterations in DNA sequence?

- a) 1 and 3
- b) 1, 2, and 3
- c) 2, 3, and 4
- d) 1, 2, and 4

Question 11:

Which anticoagulant present in the light blue vacutainer is used for plasma glucose estimation?

- a) Sodium fluoride
- b) Heparin
- c) Sodium Citrate
- d) Potassium oxalate

Question 12:

Which of the following is not involved in the repair of DNA following double-stranded breaks?

- a) ATM
- b) BRCA1
- c) MSH2
- d) Ku70

Question 13:

2-3 BPG binds to?

- a) Heme part of myoglobin
- b) Globin part of Hemoglobin
- c) Globin part of myoglobin
- d) Heme part of Hemoglobin

Question 14:

Which of the following is complementary to the active site of the enzyme?

- a) Product
- b) Substrate
- c) Allosteric activator
- d) Transition site

Question 15:

Cytochrome P450 monooxygenase utilises which of the following cofactors during the phase 1 reaction of detoxification of xenobiotics?

- a) FMN
- b) NADPH
- c) FAD
- d) NAD

Question 16:

Which biochemical cycle/process is used by monoclonal antibodies in hybridoma technology?

- a) TCA cycle
- b) Glycolytic pathway
- c) Pentose phosphate pathway
- d) Purine salvage pathway

Question 17:

Glycogen phosphorylase requires which of the following for its activation?

- a) Insulin
- b) Glucose 6 phosphate
- c) Ca^{2+}
- d) cAMP independent protein kinase

Answer Key

Question No.	Correct Option
1	d
2	d
3	c
4	a
5	b
6	d

7	b
8	c
9	b
10	a
11	c
12	c
13	b
14	b
15	b
16	d
17	c

Detailed Explanations

Solution to Question 1:

Xanthurenic aciduria is seen in deficiency of vitamin B6 (pyridoxal phosphate), a coenzyme required by kynureninase enzyme to convert 3-OH kynurenine to 3-OH anthranilic acid during tryptophan metabolism.

Other options-

Option A: Riboflavin or Vitamin B2 deficiency manifests as angular stomatitis, magenta tongue, seborrhea, corneal vascularization. It is seen in alcoholism, and individuals with a poor diet and decreased intake of dairy products.

Option B: A patient with thiamine or vitamin B1 deficiency presents either dry Beriberi with predominantly neurological symptoms like neuropathy, muscle weakness, and wasting or wet Beriberi with predominant cardiac symptoms like cardiomegaly, tachycardia, high output congestive cardiac failure.

Option C: Vitamin B12 or cobalamin deficiency presents as anemia, fatigue, and pallor symptoms. Other presenting complaints may include peripheral neuropathy, glossitis, diarrhea, headaches, neuropsychiatric disturbances like dementia, and subacute combined degeneration of the spinal cord (SACD) leading to ataxia. A peripheral smear shows macrocytic anemia and hypersegmented neutrophils.

Solution to Question 2:

The given clinical scenario points towards a diagnosis of diabetic ketoacidosis. Rothera's test as shown in the image is a specific test used for the determination of ketone bodies in the urine.

Type I diabetes mellitus is a low insulin state. There is an increased conversion of fatty acids to acetyl-CoA which in turn leads to increased ketone body production. The presence of

ketosis can be established by the detection of ketone bodies in urine by Rothera's test.

The test involves the saturation of 5 mL of urine with solid ammonium sulfate. This is followed by the addition of a few drops of freshly prepared sodium nitroprusside followed by 2 mL of liquor ammonia along the sides of the test tube. The development of a purple ring indicates the presence of ketone bodies in urine. Strip tests based on the same principle are also available.

Other options-

Option A: Seliwanoff's test will distinguish glucose and fructose. The reagent is resorcinol in HCl. 3 ml of the reagent and 0.5 ml of urine are heated for 30 seconds up to a boiling point. A red color indicates the presence of fructose.

Option B: Benedict's test is a test commonly used in the follow-up of diabetics for the detection of glucose in urine (glucosuria). It uses Benedict's reagent containing sodium carbonate, copper sulfate, and sodium citrate. Reducing sugars (containing a free aldehyde/ keto group) will reduce Benedict's reagent. Examples include glucose, fructose, and lactose. The test is a semi-quantitative test. The blue color indicates no reducing sugar.

Option C: Fouchet's test is used for detecting the presence of bile pigments in urine. Here, urine is heated with barium sulfate, which adsorbs bilirubin. Ferric chloride is added, which oxidizes the adsorbed bilirubin to biliverdin which produces a green color. Principle for all tests to detect bile pigments: Oxidizing agents readily oxidize bilirubin to biliverdin to form a green color.

Solution to Question 3:

5'GAUCGAUAUUGAAUG3' will be the sequence of mRNA produced from this fragment of template DNA.

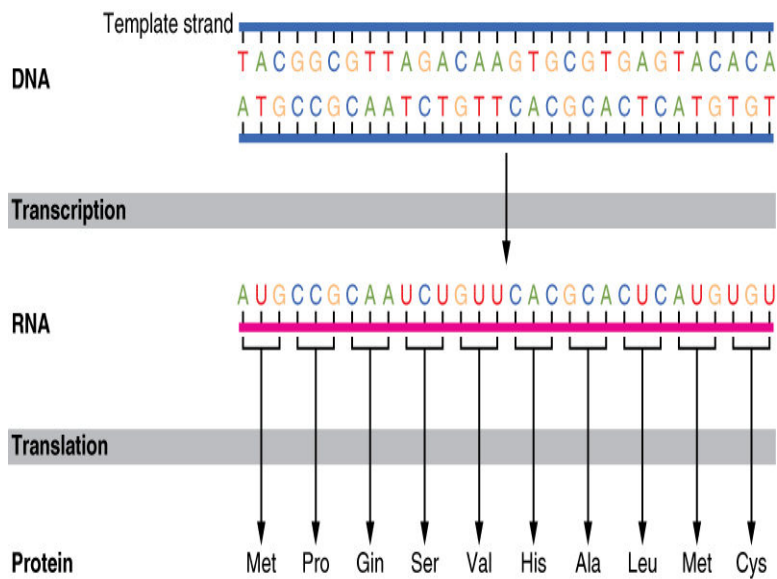
The non-template strand or coding strand or sense/plus strand in the DNA will have the same sequence of the RNA that is produced, except that the non-template or coding strand has T in place of U.

As all sequences are written in the 5' to 3' direction, unless otherwise specified, an RNA sequence of GAUCGAUAUUGAAUG would correspond to a DNA sequence, on the non-template strand, of CATTCAATATCGATC. Both the non-template strand and RNA produced will be a complementary sequence to the template strand.

Note:

Template strand - known as minus or antisense or non-coding strand.

Coding strand- known as plus or sense or non-template strand.



Solution to Question 4:

The correct order of steps of Karyotyping is as follows:

- Collect cells by peripheral vein using a heparin syringe.
- Culture in phytohaemoagglutinin.
- Incubation at 37 degrees C for 3 days.
- Harvest with colchicine.
- Routine staining with Giemsa

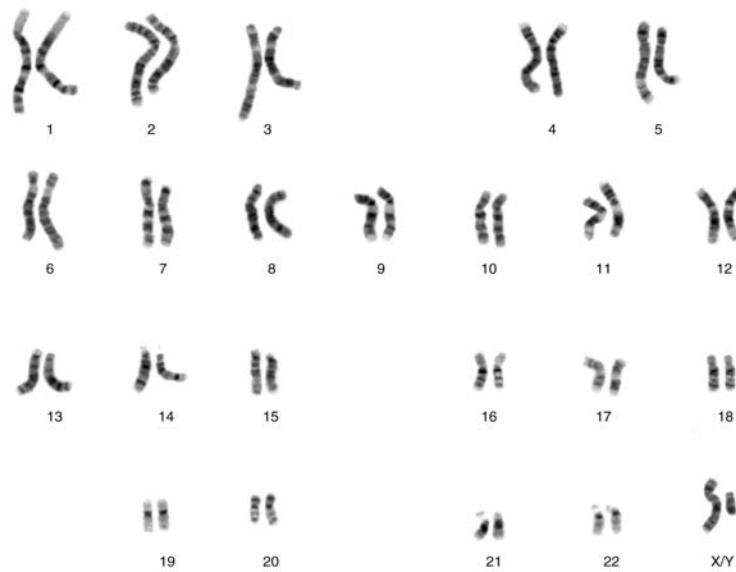
Karyotyping is the process by which chromosomes are isolated, stained, and examined to find chromosomal anomalies. It is used in prenatal diagnosis, investigation of repeat pregnancy loss, investigation of children and adults suspected of having a chromosomal disorder, and subtyping of lymphomas and leukemias for the purpose of exact diagnosis, giving a prognosis, and treatment.

Various techniques are used in karyotyping to study chromosomes. Staining with various compounds produces various results and helps us to understand the chromosomes better.

Common techniques used are:

- Q banding - fluorescence staining with quinacrine mustard
- G banding - Giemsa staining after partial denaturation of chromatin
- R banding - reverse Giemsa staining. In this, the light and dark areas are reversed
- C banding - staining of heterochromatin with silver salts
- T banding - staining the telomeres of chromosomes

The karyotyping of a normal human male is given below.



Solution to Question 5:

Decreased cardiolipin or alteration in its metabolism results in mitochondrial dysfunction seen in Barth syndrome (cardio skeletal myopathy) which is characterized by an enlarged and weakened heart (dilated cardiomyopathy), weakness in muscles used for movement (skeletal myopathy), recurrent infections due to small numbers of white blood cells (neutropenia), and short stature.

Diphosphatidyl glycerol, also called cardiolipin, is the phospholipid found only in the mitochondria. Cardiolipin, which is a phospholipid with antigenic properties, is responsible for the false-positive VDRL test.

It is an important component of the inner mitochondrial membrane, where it is essential for the following:

- Optimal function of numerous mitochondrial enzymes that are involved in mitochondrial energy metabolism.
- Maintenance of the structure of the mitochondrial membrane.

Other options-

Option A: Phosphatidylethanolamine, also called cephalin, is found in the cell membranes of nervous tissue, particularly in the brain.

Option C: Defect in the synthesis of dipalmitoyl lecithin leads to Respiratory distress syndrome (RDS), also known as hyaline membrane disease.

option D: Lysolecithin is obtained by removing one fatty acid from lecithin. Oxidized lysolecithin is associated with atherosclerosis.

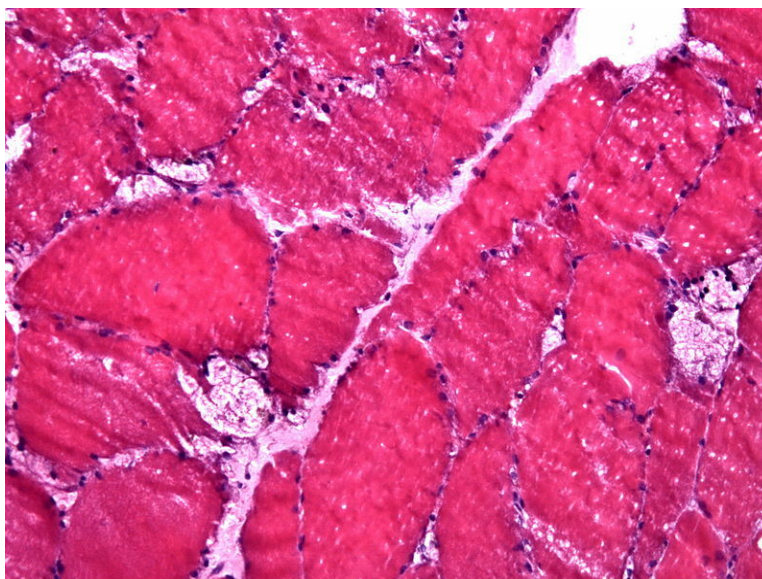
Solution to Question 6:

Pompe's disease results from a deficiency in the enzyme lysosomal alpha-glucosidase.

Pompe's disease is a type II glycogen storage disorder that results from a deficiency in the enzyme acid maltase/ lysosomal acid alpha-glucosidase. It is a rapidly progressive disorder that results in hypotonia and hepatosplenomegaly. Myocardial involvement helps to distinguish it from other similar glycogen storage disorders.

Hepatomegaly results from the ballooning of the lysosomes with glycogen. This creates a characteristic lacy cytoplasmic pattern. Cardiac and skeletal muscles will show enlargement with clear spaces in the cells, representing glycogen storage within the lysosomes. This disease classically has a poor prognosis, but enzyme replacement has helped to prolong survival in recent times.

A muscle biopsy with enlarged glycogen-containing vacuoles can be seen in the image below:



Other options:

Option A: Lactic acid dehydrogenase deficiency is not associated with any disorder.

Option B: Tarui's disease is characterized by a deficiency of muscle and erythrocyte phosphofructokinase 1. This results in abnormal glycogen metabolism and hence muscle cannot generate energy after exercise using glycogen reserves which leads to exercise intolerance.

Option C: Muscle glycogen phosphorylase deficiency is seen in McArdle's disease (type V glycogen storage disorder). Patients present with exercise-induced muscle cramping without elevated blood lactate levels which would normally be expected.

Solution to Question 7:

The thermic effect is highest in proteins.

Specific dynamic allowance (previously called the thermic effect of food): This refers to the increased heat production or increased metabolic rate following the intake of food (diet-induced

thermogenesis).

The values of SDA are

- 30% for proteins: This means that out of every 100 grams of proteins consumed, the energy available for doing useful work is 30% less than the calculated value - i.e 30% of the energy content in protein is used in its own metabolism.
- 15% for lipids, and
- 5% for carbohydrates.

Importance: It is a common experience that during hot weather following the consumption of a protein-rich meal, one feels hot and humid for a while. On the other hand, in cold weather, the same would provide a comfortable feeling.

The energy generated by the metabolism of the macronutrients is used for three energy-requiring processes that occur in the body:

- Resting metabolic rate
- Thermic effect of food (formerly termed specific dynamic action)
- Physical activity

Solution to Question 8:

Glycine is the most abundant amino acid(AA) in collagen and not phenylalanine.

Collagen is the most abundant protein in the animal world. It has a triple helical structure, forming a right-handed superhelix. A striking characteristic of collagen is the occurrence of glycine residues at every third position.

Proline and hydroxyproline confer rigidity on the collagen molecule. Hydroxyproline and hydroxylysine are formed by the post-translational hydroxylation of proline and lysine residues catalyzed by the enzyme prolyl and lysyl hydroxylase, whose cofactors are ascorbic acid (vitamin C) and α -ketoglutarate. Collagen is arranged in a quarter-staggered arrangement and further stabilized by covalent cross-links.

Solution to Question 9:

Ketone bodies are not utilized in the liver because it lacks the enzyme thiophorase.

Ketone bodies are synthesized in the liver mitochondria (by ketogenesis) and transported in the blood to the peripheral tissues. Here they are utilized by the following steps:

- β -hydroxybutyrate (the most predominant ketone body) is oxidized to acetoacetate by β -hydroxybutyrate dehydrogenase producing NADH.
- Acetoacetate is then activated to acetoacetyl CoA by transferring a CoA molecule from succinyl CoA by succinyl CoA-acetoacetyl CoA transferase (thiophorase).

- Acetoacetyl CoA is cleaved into two acetyl CoA molecules by thiolase and oxidized into CO₂ by the TCA cycle.

Thiokinase or Acyl CoA synthetase catalyzes the only energy-requiring step in fatty acid oxidation. In the presence of ATP and coenzyme A, thiokinase catalyzes the conversion of a fatty acid to acyl-CoA. Acyl-CoA is the active form of fatty acid. This step uses one high-energy phosphate and forms AMP and PPI.

When higher than normal ketone bodies are present in the blood or urine, it is called ketonemia or ketonuria, respectively. The overall condition is called ketosis. The metabolism of free fatty acids to produce ketone bodies is exaggerated in pathologic states such as starvation and diabetes mellitus, thus causing ketosis. The continued buffering and excretion of ketone bodies deplete the alkali reserve, causing ketoacidosis.

Solution to Question 10:

The alterations in DNA sequence can be detected by restriction fragment length analysis and pyrosequencing.

Restriction fragment length polymorphism (RFLP) is also called cleaved amplified polymorphic sequence (CAPS) assay. RFLP is a technique in which organisms may be differentiated by variations in their DNA sequence by restriction endonucleases. This technique involves cutting a particular region of DNA with known variability, with restriction enzymes, then separating the DNA fragments by agarose gel electrophoresis and determining the number of fragments and relative sizes.

Pyrosequencing is a synthesis type of approach of DNA sequencing. This technique employs four enzymatic reactions in a single tube to monitor DNA synthesis. Nucleotides are incorporated into the reaction and pyrophosphate (PPi) is released. PPi triggers a series of reactions resulting in the production of light, which is proportional to the amount of DNA and number of incorporated nucleotides. The generated light is detected and recorded by a detector system in the form of a peak signal, which reflects the activity of all four enzymes in the reaction.

The practical applications of recombinant DNA technology include:

- Gene mapping using techniques such as fluorescence in situ hybridization (FISH).
- Producing proteins such as:
 - Interferons, tPA (tissue plasminogen activator)
 - Insulin, growth hormone
 - Hepatitis B vaccine
- Molecular analysis of disease, to detect gene variations and mutations.
- Restriction Fragment Length polymorphism (RFLP) and single nucleotide polymorphism (SNP)
- Microsatellite DNA polymorphisms
- Gene therapy and bone marrow transplantation

Flow cytometry is a laser-based method that analyzes the cells and cell components based on the presence of certain cell surface molecules which are named as per cluster of differentiation (CD)

nomenclature. It is very effective for the study of the immune system and its response to infectious diseases and cancer. It cannot detect the alterations in DNA sequence.

Fluorescent in situ hybridization (FISH) is a technique where cells are stained with fluorescent stains. It uses DNA probes that recognize sequences specific to particular chromosomal regions. It cannot detect the alterations in DNA sequence. FISH can be performed on prenatal samples, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections. Uses include :

- Numerical abnormalities of chromosomes e.g., aneuploidy.
- Subtle microdeletions and rearrangements.
- Complex translocations that are not demonstrable by routine karyotyping.
- Gene amplification of HER2 in breast cancer or NMYC amplification in neuroblastomas.

Solution to Question 11:

Light blue vacutainers contain the anticoagulant sodium citrate.

Citrated plasma generated from whole blood is commonly utilized in laboratory testing for hemostasis. The recommended procedure involves the collection of specimens in tubes that are filled with a solution containing 3.2% sodium citrate (109 mM). The appropriate ratio for this collection is 9 parts blood to 1 part anticoagulant. The primary objective of citrate is to effectively eliminate calcium ions, which play a crucial role in the process of blood coagulation.

The following image shows color coding of vacutainers with the anticoagulant used and their practical uses:

Vacutainer	Additive	Uses
Light grey	Sodium fluoride and potassium oxalate	Blood glucose estimation
Light green	Lithium Heparin	Plasma testing
Light blue	Sodium Citrate	Coagulation tests (APTT and prothrombin time)

Color coding of vacutainers

					
EDTA For routine hematology	Sodium Citrate For coagulation studies	Potassium Oxalate or Sodium Fluoride Glucose determina- tion	No additive Collection of serum	Acid Citrate Dextrose Preserve RBC for blood banking and HLA typing	Heparin Inhibit thrombin activation

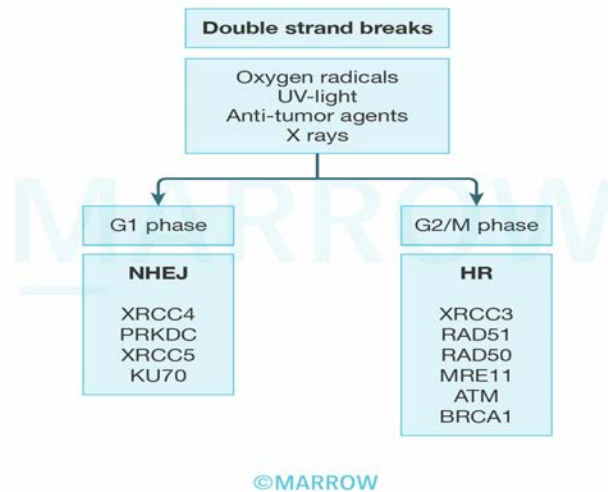
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Solution to Question 12:

MSH2 is involved in mismatch repair and is not involved in the repair of double-stranded DNA breaks as the repair involves the correction of a single mismatch base pair e.g. C to A, instead of T to A.

DNA mismatch repair proteins function as spell checkers during DNA replication by correcting the incorrect pairing. In HNPCC, the proofreading function of the mismatch repair systems (majorly MLH1 and MSH2) is lost, leading to the accumulation of errors throughout the genome. The genomes of these patients show microsatellite instability, which is characterized by changes in the length of short repeats.

Double-strand breaks (DSBs) in DNA pose a significant concern. Genetic recombination events can give rise to chromosomal translocation, chromosomal breakage, and ultimately cellular demise. Double-strand breaks (DSBs) have the potential to undergo repair through two distinct mechanisms: homologous recombination and non-homologous end joining. Homologous recombination is observed in yeasts, whereas in mammals, non-homologous end joining predominates.



Non-homologous end-joining (NHEJ) is a mechanism by which a group of proteins mediates the recognition, processing, and direct ligation of the ends of two DNA fragments where a double-strand break in DNA has occurred. This is in contrast to homologous repair (HR), another mechanism, where an undamaged homologous DNA template is required to replace an adjacent damaged one.

The type of repair depends on the phase of cell cycles:

Other options:

Option A: The protein known as ataxia telangiectasia mutated (ATM) serves as a crucial kinase that triggers a wide range of cellular stress responses through its signalling activity. The activation of ATM is induced by DNA double-strand breaks (DSBs) and oxidative stress, resulting in the phosphorylation of a wide range of target proteins. Recent advancements in molecular biology have brought to light the participation of the ataxia telangiectasia mutated (ATM) protein in a range of cellular mechanisms triggered by DNA damage. These mechanisms encompass the repair of double-strand breaks (DSBs), the halting of cell cycle progression through checkpoint activation, the initiation of programmed cell death (apoptosis), and the cessation of transcriptional activity. The compromised precision of repair resulting from the combinational failure of stress responses leads to heightened sensitivity to ionizing radiation (IR) in ataxia-telangiectasia (A-T) cells.

Option B: CtIP which is a DNA repair protein involved in Homologous Repair whose function is dependent on phosphorylation at serine 327 as well as the recruitment of BRCA 1.

Option D: The Ku heterodimer, consisting of Ku70 and Ku80 subunits, plays a pivotal role in the nonhomologous end-joining (NHEJ) pathway, which is responsible for the repair of DNA double-strand breaks (DSBs). Ku protein mediates the binding of the damaged DNA fragment and orchestrates the recruitment of additional proteins to enable the subsequent processing and ligation of the broken DNA end.

Phase of the cell cycle	Repair mechanism
G ₀ / G ₁ Phase	Nonhomologous End Joining Repair
S, G ₂ , M Phase	Homologous Repair

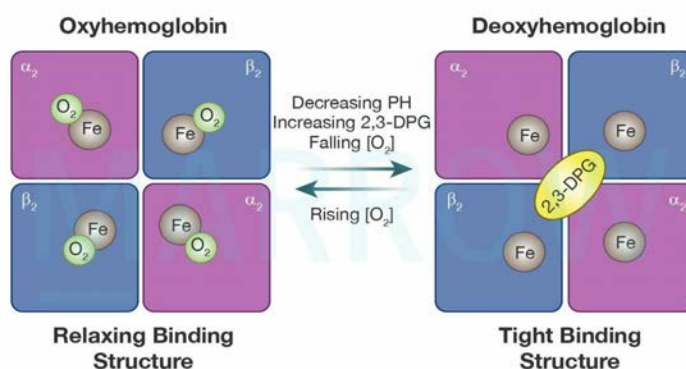
Solution to Question 13:

2,3 bisphosphoglycerate binds to the globin chains of the hemoglobin while oxygen binds to the heme part of the hemoglobin.

A single unit of 2,3-BPG forms a complex with one tetrameric unit of deoxyhemoglobin within the central cavity of its four subunits. The centre pocket of this structure contains two β -globin chains that are positively charged, such as histidine and lysine. Ionic bonds, also known as salt bridges, are established through the interaction between amino acids that carry a positive charge, specifically those found in globins, and phosphate groups that possess a negative charge, specifically those present in 2,3-BPG. The stabilization of deoxygenated haemoglobin (in its T-form) is achieved through the cross-linking of the α -chains by the binding of 2,3-BPG. Upon oxygenation of haemoglobin, the molecule 2,3-BPG is released from its binding site, resulting in the transformation of oxyhemoglobin into the R-form conformation.

A relaxed (R) configuration, exposes more O₂ binding sites. The quaternary structure of haemoglobin determines its affinity for O₂. In deoxyhemoglobin, the globin units are tightly bound in a tense (T) configuration, which reduces the affinity of the molecule for O₂. When O₂ is first bound, the bonds holding the globin units are released, producing a relaxed (R) configuration, which exposes more O₂ binding sites. The net result is a 500-fold increase in O₂. In tissues, these reactions are reversed, resulting in O₂ dissociation.

Oxygenated and deoxygenated hemoglobin



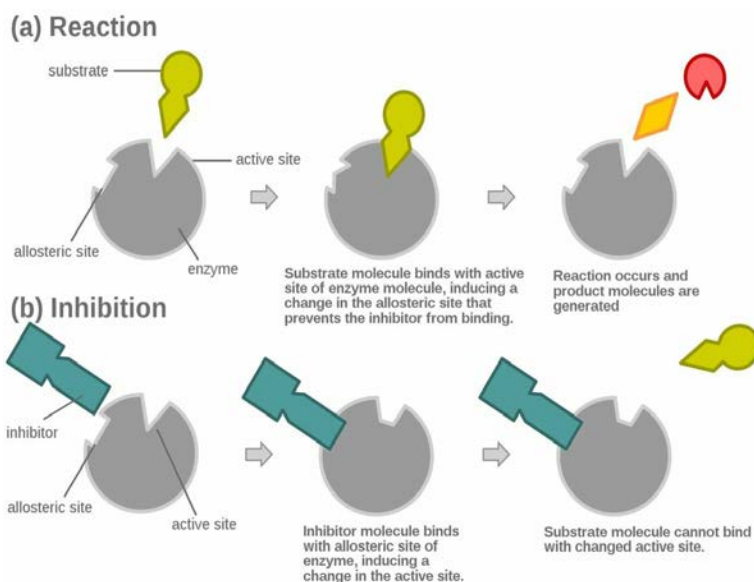
Solution to Question 14:

The substrate is complementary to the active site of the enzyme.

The Lock and Key concept was put forth by Emil Fischer. This model is the initial proposition aimed at elucidating the mechanism of an enzymatic catalytic process. Based on this concept, it can be inferred that the enzyme's structure or conformation exhibits a state of rigidity. The substrate effectively conforms to the binding site, which is now referred to as the active site, in a manner analogous to a key fitting precisely into its corresponding lock or a hand fitting snugly into a well-fitting glove. The active site of an enzyme can be described as a stiff and pre-configured template that selectively accommodates a certain substrate for binding.

The induced fit or Koshland theory states that the active site does not possess a firm and pre-determined conformation. The emerging active site exhibits the fundamental characteristics of the substrate binding site. The substrate's contact with the enzyme elicits a conformational change in the enzyme, leading to the establishment of a robust binding site for the substrate. Moreover, as a result of induced fit, the enzyme undergoes conformational changes that lead to the repositioning of the specific amino acids involved in the formation of the active site, hence facilitating the catalytic process. The induced fit model has garnered substantial empirical support through X-ray diffraction studies. Koshland's model additionally elucidates the mechanisms by which allosteric modulators and competitive inhibition exert their effects on enzymes.

The substrate strain theory states that the substrate experiences strain as a result of the conformational alteration generated by the enzyme. It is plausible that upon the binding of a substrate to the preexisting active site, the enzyme elicits a strain on the substrate. The presence of a stressed or strained substrate facilitates the generation of the desired product. The enzymatic action is believed to involve a combination of the induced fit model and substrate strain.



Other options:

Option A: The complementary active site of the enzyme binds to the substrate to form the product.

Option C: The active site of the enzyme is not complementary to the allosteric activator. Allosteric sites are separate and specific sites on the enzyme that are modulated with allosteric effect

modifiers and hence are complementary to them.

Option D: The active site of the enzyme binds to the substrate which form a transitional state which is unstable and later dissociates to form a product.

Solution to Question 15:

The microsomal cytochrome P450 system, located in the liver, facilitates the process of drug and foreign substance detoxification through hydroxylation processes that rely on the availability of NADPH.

Phase 1 reactions include oxidation, reduction, or hydrolytic reaction. CYPs- Cytochrome P family of enzymes play a major role in phase 1 reactions. Reactions carried out by phase 1 enzymes usually lead to the inactivation of a drug. The addition of functional groups does little to increase the water solubility of the drug but alters its biological properties.

The importance of NADPH is enumerated as follows:

- NADPH is needed for the reductive biosynthesis of fatty acids and steroids, hence HMP shunt is more active in lipogenesis tissues like adipose tissue and the liver.
- Along with Glutamate dehydrogenase helps in the synthesis of various amino acids.
- Maintaining reduced glutathione is critical for RBC membrane integrity. NADPH keeps haemoglobin's ferrous iron (Fe^{2+}) lowered to prevent methemoglobin (Fe^{3+}) buildup.
- High concentrations of NADPH are also required to maintain the transparency of the lens.

The cytochromes are a class of proteins that possess a conjugated heme group. The latter is comprised of a porphyrin ring containing an iron atom. The heme group present in cytochromes exhibits distinct characteristics when compared to the heme group observed in the structures of haemoglobin and myoglobin. The heme iron within cytochromes undergoes a cyclic process of oxidation (Fe^{3+}) and reduction (Fe^{2+}), playing a crucial role in facilitating electron movement throughout the electron transport chain (ETC). This stands in opposition to the heme iron found in haemoglobin and myoglobin, which remains in the ferrous (Fe^{2+}) form. The cytochromes were initially assigned the designations of a, b, and c based on the specific heme type they contained and their corresponding absorption spectra. Subsequent discoveries led to the identification of additional cytochromes, including c1, b1, b2, a3, among others.

Other options

Option A: Flavin adenine mononucleotide (FMN) is a phosphorylated form of riboflavin which functions as a cofactor for various enzymes. Some of these include as a cofactor for blue-light photoreceptors and oxidoreductases.

Option C: Flavin adenine dinucleotide (FAD) is nothing but phosphorylated riboflavin and is used in the electron transport chain and also is a cofactor for various enzymes. Some of these include oxidative phosphorylation, DNA repair, beta-oxidation of fatty acids as well as amino acid catabolism.

Option D: Nicotinamide adenine dinucleotide (NAD) consists of the vitamin B3 (niacin) and also serves as a coenzyme for various enzymatic reactions in the body. Some of these include oxidative phosphorylation (ATP production), DNA repair and intracellular calcium signalling.

Solution to Question 16:

It is the purine salvage pathway that is utilized by monoclonal antibodies in hybridoma technology.

Monoclonal antibodies refer to the antibodies produced by a group of lymphocytes that produce antibodies against a single antigen. These antibodies are produced by hybridoma technology.

In this technology, the B-lymphocytes of mice spleen are immunized with an antigen to produce a specific antibody. These specific antibody-producing B lymphocytes are then fused with myeloma cells that can multiply indefinitely.

The myeloma cells are deficient in the hypoxanthine-guanine-phosphoribosyltransferase (HGPRT) gene and do not produce immunoglobulins. Therefore, the resulting hybrid cell is producing antibodies against a single antigen indefinitely. These hybrid cells are cultured in a HAT medium (containing hypoxanthine, aminopterin, and thymidine) and the monoclonal antibodies are obtained. It is well-established that nucleotides present in DNA and RNA consist of purine and pyrimidine bases. The synthesis of these nucleotides can occur through either the De novo pathway or the salvage pathway. Consequently, the inclusion of aminopterin in the HAT medium effectively inhibits the De novo pathway, thereby rendering the synthesis of purine dependent on the salvage pathway. However, it should be noted that the salvage pathway exclusively takes place in the presence of the enzymes HGPRT and thymidine kinase. The hybridoma cell is uniquely capable of proliferating on the HAT medium due to its distinct composition. Specifically, the myeloma component of the hybridoma possesses the capacity for repetitive cellular division, while the plasma cell component contains the enzyme HGPRT. Consequently, the hybridoma cells can sustain growth and viability over an extended period when cultivated on the HAT medium.

Hybridoma technology is a method for producing large numbers of identical antibodies (monoclonal antibodies).

Hybridomas are somatic cell hybrids produced by fusing the antibody-forming spleen cells with myeloma cells. The resultant hybrid thus has both the antibody-producing ability of the B-cell and the ability to multiply indefinitely of the myeloma.

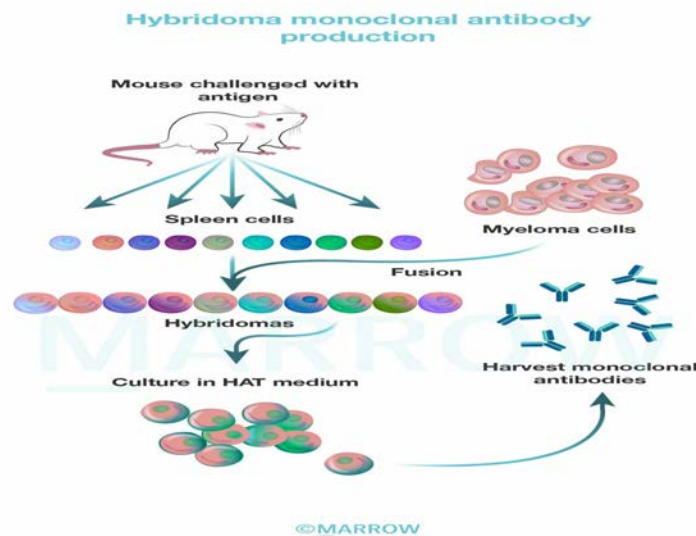
The following are the steps involved in hybridoma technology:

- Mice are immunized with an antigen.
- After several weeks of immunization, blood samples are obtained from mice for measurement of serum antibodies.
- When the antibody titer is high enough, mice are commonly boosted by injecting antigen without adjuvant, intraperitoneally or intravenously.
- Then the mice are euthanized and their spleens removed for in vitro hybridoma cell production.
- Fusing antibody-producing spleen cells, which have a limited life span, with myeloma cells (derived from an immortal tumour of lymphocytes) results in a hybridoma that is capable of unlimited growth.
- The cells are cultured in a medium containing hypoxanthine, aminopterin, and thymidine (HAT medium) which allows only the fused cells to survive in culture.

- Monoclonal antibodies are then harvested.

Applications of hybridoma technology

- Listing lymphocyte subpopulations
- Specific cell quantitative preparation
- As an immunosuppressant in therapeutic settings
- Blood component nephelometric assays
- ELISA testing
- Quantitative purification of antigens.
- Common therapeutic applications for monoclonal antibodies



Solution to Question 17:

Glycogen phosphorylase requires the presence of calcium to attain its fully activated state.

Glycogen phosphorylase is an enzyme required in glycogenolysis.

In the muscle: During muscle contraction → Ca^{++} is released → Ca^{++} binds to calmodulin forming Ca^{++} -Calmodulin complex → activates phosphorylase kinase from b form (inactive form) without phosphorylation → significant Ca^{++} is required to activate glycogen phosphorylase.

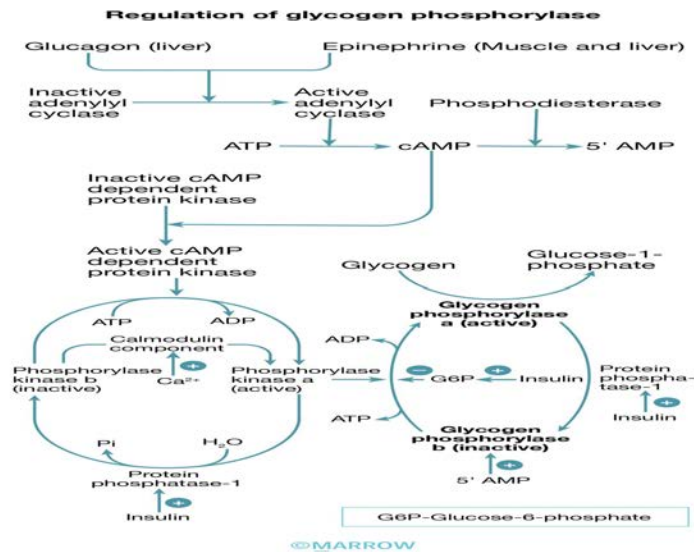
Hormonal regulation of glycogenolysis:

Glucagon (in liver) and epinephrine (in liver and muscle) acts via GPCR to activate adenylyl cyclase to catalyse synthesis of cAMP. cAMP dependent protein kinase activates phosphorylase kinase from inactive(b) to active(a) form, which in turn activates glycogen phosphorylase to active(a) form, hence leading to increased glycogenolysis. (Option D)

Option A: Insulin is a known inhibitor of glycogen phosphorylase which dephosphorylates the glycogen phosphorylase enzyme thus rendering it inactive.

Option B: Glucose 6 phosphate is also an inhibitor of glycogen phosphorylase which dephosphorylates it thus rendering it inactive.

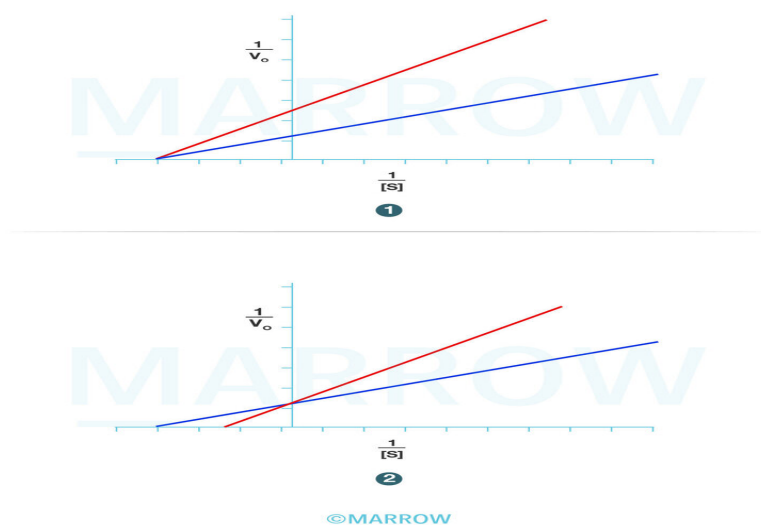
The regulation of glycogen phosphorylase is shown in the image below:



Biochemistry INI-CET 2024

Question 1:

The following Lineweaver-Burk plots depict enzymatic action in the presence of enzyme inhibitors. The enzyme action without inhibitor is represented by a blue line. What type of inhibitors are depicted in images 1 and 2?



- a) 1- Uncompetitive inhibition 2- Competitive inhibition
- b) 1- Noncompetitive inhibition 2- Competitive inhibition
- c) 1- Competitive inhibition 2- Uncompetitive inhibition
- d) 1- Competitive inhibition 2- Noncompetitive inhibition

Question 2:

FISH technique helps to identify all except?

- a) Gene amplification
- b) Deletion
- c) Point mutation
- d) Translocation

Question 3:

Which of the following micronutrients are associated with decreased sexual maturation and hypogonadism?

- a) Zinc
- b) Copper
- c) Selenium
- d) Iron

Question 4:

Which of the following is not a heme breakdown product?

- a) Carbon Monoxide
- b) Ferric / Fe 3^{+}
- c) ALA
- d) Biliverdin

Question 5:

Which type of mutation results in the reversal to wild type of phenotype when the mutant gene is suppressed?

- a) Deletion of mutated gene
- b) Frameshift of mutated gene
- c) Addition of another normal gene
- d) Mutations in tRNA

Question 6:

Which of the following is not a post-transcriptional modification?

- a) 5 prime capping
- b) Methylation
- c) Glycosylation
- d) Endonuclease cleavage

Question 7:

Match the following conditions :

- a) 1 - b, 2 -a, 3 -d, 4 -c
- b) 1 -c, 2 -d, 3 -b, 4 -a
- c) 1 -b, 2 -d, 3 -a, 4 -c
- d) 1 -c, 2 - b, 3 -a, 4 -d

Question 8:

All are true statements regarding folate deficiency except

- a) Increased levels of homocysteine and normal methylmalonic acid
- b) Folate supplementation on diagnosis of pregnancy prevents neural tube defect
- c) It is important to treat co-existing vitamin B12 deficiency before initiating folate therapy
- d) Folate deficiency is commonly seen in children with hemolytic anemia

Question 9:

Congenital lactic acidosis is seen due to a defect in which of the following enzymes?

- a) Pyruvate dehydrogenase
- b) Branched chain ketoacid dehydrogenase
- c) Transketolase
- d) Isocitrate dehydrogenase

Question 10:

Which of the following is not associated with G6PD deficiency?

- a) Equal incidence in males and females
- b) Defective NADPH synthesis
- c) Predisposes the patient to oxidative stress
- d) Patient presents with hemolysis

Question 11:

Which of the following is not a neurotransmitter derived from aminoacids ?

- a) Gamma amino butyric acid
- b) Dopamine
- c) 5 hydroxytryptamine
- d) Creatine

Question 12:

Which of the following methods is used to test the blood taken from a neonate for metabolic disease screening?

- a) Complete blood count
- b) Tandem Mass Spectrometry
- c) Next Generation Sequencing
- d) ELISA

Question 13:

Familial hypercholesterolemia is a result of a defect in which of the following?

- a) LDL receptor
- b) Apo E
- c) Apo CII
- d) Apo B48

Question 14:

In a patient with Diabetes Mellitus, the liver produces more TG and VLDL. What is the cause of the same?

- a) Decreased action of LPL with increased action of HSL
- b) Decreased action of LPL with Decreased action of HSL
- c) Increased action of LPL with increased action of HSL
- d) Increased action of LPL with decreased action of HSL

Question 15:

Nutritional deficiency of Copper, required by lysyl oxidase which is an important key step for formation for covalent cross linking in collagen synthesis, gives rise to which of the following condition?

- a) Menke's disease
- b) Scurvy
- c) Ehlers Danlos syndrome
- d) Scoliosis

Question 16:

Match the following with appropriate precursors

- a) i-b, ii-d, iii-a, iv-c
- b) i- c, ii-d, iii-a, iv-b
- c) i- a, ii-d, iii-b, iv-c
- d) i- b, ii-a, iii-d, iv-c

Question 17:

In the current scenario treatment for urea cycle disorder is available. Administration of amino acid phenylbutyrate can decrease ammonia formation. Which amino acid provides phenylbutyrate?

- a) Arginine
- b) Glutamate
- c) Aspartate
- d) Glycine

Question 18:

Which among the following are appropriate techniques for chromosomal aneuploidy?

- a) i,ii,iii,iv
- b) i,ii
- c) ii,iii
- d) iii,iv

Question 19:

Enzyme used in recombinant DNA technology are?

- a) 1,2,3,4
- b) 1,2,3
- c) 2,3,4
- d) 1,3,4

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	c
5	d
6	c
7	b
8	d
9	a
10	a
11	d
12	b
13	a
14	a
15	a
16	a
17	b
18	a
19	c

Detailed Explanations

Solution to Question 1:

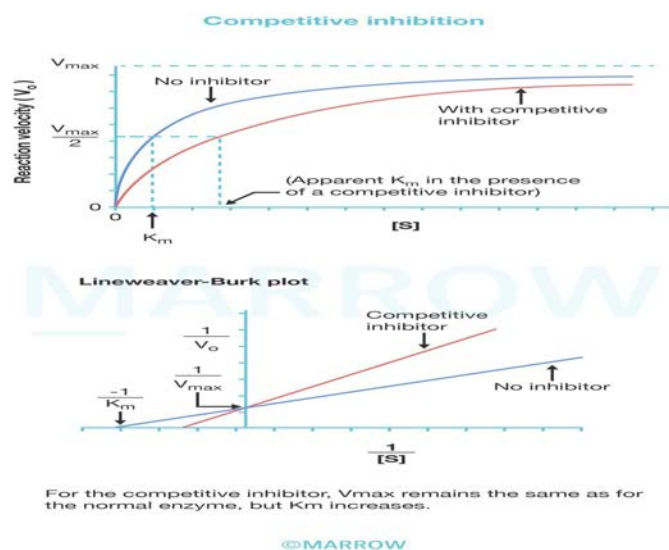
The Lineweaver-Burk plot in image 1 depicts non-competitive inhibition, and that in image 2 depicts competitive inhibition.

Important parameters in a reaction include:

- The affinity of the enzyme for its substrate, given by the Michaelis constant or K_m . A low K_m reflects a high affinity of the enzyme for substrate and vice versa. The value of K_m is given by the substrate concentration at which the reaction velocity is equal to one-half V_{max} . It does not change with changes in enzyme concentration.
- V_{max} is the maximum velocity of a reaction.

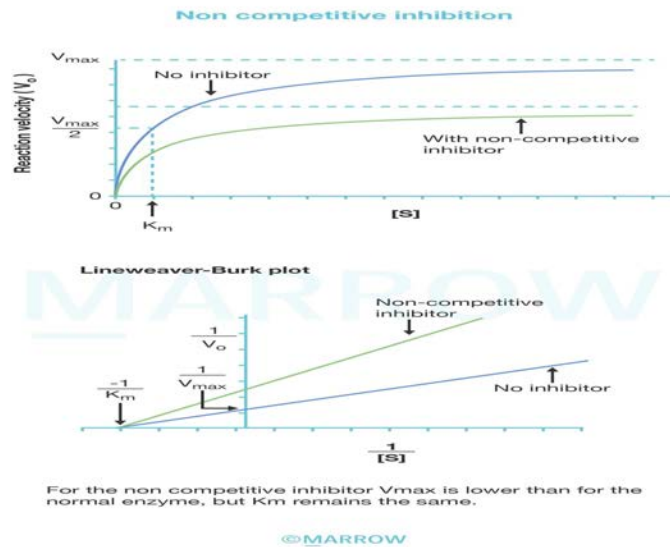
Competitive inhibition is when the inhibitor binds reversibly to the substrate-binding site and competes with the substrate for the active enzyme-binding site. This effect can be overcome by increasing the substrate concentration by which maximum velocity can still be achieved. Therefore, V_{max} is unaltered in this type of inhibition. However, the concentration of the substrate required to achieve half of the maximum velocity is increased. Therefore, K_m is increased in this condition.

In a Lineweaver-Burk plot, the addition of a competitive inhibitor causes plots with the same y-intercept (V_{max} is unaffected) but different x-intercepts (due to increased K_m).

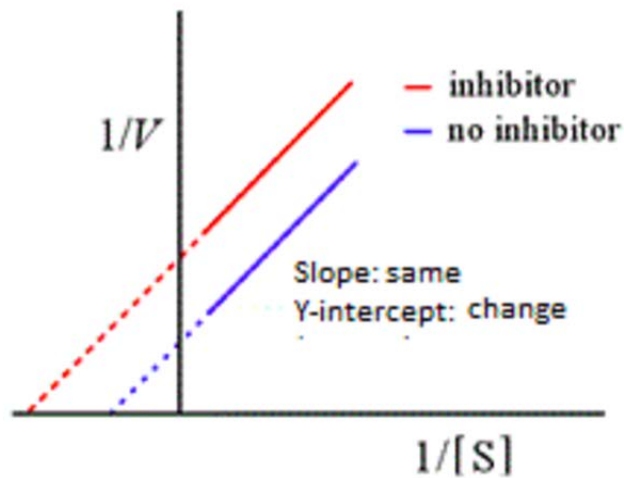


Noncompetitive inhibition is by binding of inhibitors to enzymes in locations remote from the active site. Increasing the concentration of the substrate does not overcome non-competitive inhibition. Thus, noncompetitive inhibitors decrease the V_{max} of the reaction. They do not interfere with the binding of the substrate to the enzyme. Thus, the enzyme shows the same K_m in the presence or absence of noncompetitive inhibitors.

In a Lineweaver-Burk plot, the addition of a non-competitive inhibitor causes plots with the same x-intercept (K_m is unaffected) but different slopes and y-intercepts (due to decreased V_{max}).



In uncompetitive inhibition, inhibitors bind with the enzyme-substrate intermediate complex and inhibit it. This leads to a decrease in both V_{max} and K_m .



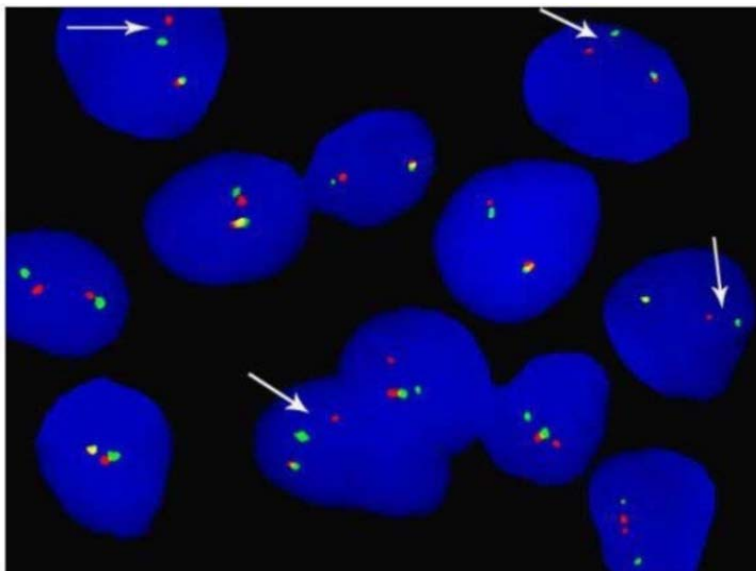
Solution to Question 2:

Fluorescent in situ hybridization (FISH) is not useful in identifying point mutation. Point mutations are identified by other means like sequencing studies and restriction fragment length polymorphism.

FISH is a method of molecular analysis of genetic lesions involving large deletions, duplications, or more complex rearrangements that are not readily amenable to detection using polymerase chain reaction or DNA sequencing approaches.

FISH uses DNA probes that recognize sequences specific to particular chromosomal regions. To perform FISH, large fragments of cloned genomic DNA spanning up to 200 kb are labeled with fluorescent dyes and applied to metaphase chromosome preparations or interphase nuclei. The probe hybridizes to its homologous genomic sequence and thus labels a specific chromosomal region that can be visualized under a fluorescent microscope.

FISH can be performed on prenatal samples, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections. It is used to detect numeric abnormalities of chromosomes (aneuploidy), subtle microdeletions and complex translocations that are not demonstrable by routine karyotyping, and gene amplification (e.g., HER2 in breast cancer or NMYC amplification in neuroblastomas).



Solution to Question 3:

Zinc deficiency is associated with decreased sexual maturation and hypogonadism.

Zinc is the second most common trace element in the body after iron. As a component of zinc finger proteins, zinc regulates gene transcription and participates in nucleic acid metabolism, protein synthesis and thereby cell growth. Zinc is a major antioxidant being part of the enzyme superoxide dismutase.

Zinc is derived mainly from animal protein. Diets based on cereals, plants and legumes are associated with zinc deficiency due to dietary phytic acid decreasing the bioavailability of zinc.

Zinc is absorbed throughout the small intestine by a process of facilitated diffusion. Almost 90% of the total body zinc is localized in bone and skeletal muscle.

Deficiency -

- Poor physical growth is an important feature of zinc depletion in preschool and school-age children. Delayed sexual maturation and hypogonadism are prominent features of zinc deficiency in adolescents. Other features include anemia, anorexia, diarrhea, alopecia, impaired immune function and poor wound healing.

- Acrodermatitis enteropathica is an autosomal recessive disorder of severe zinc deficiency, caused by impaired intestinal absorption due to a defect in intestinal zinc transporter protein. It presents in early infancy with vesiculobullous, dry, scaly, or eczematous skin lesions chiefly in the periorificial and acral areas.

Normal requirements for children range between 3.5 and 5.0 mg/day. Acquired zinc deficiency states can be treated with 0.5 to 1.0 mg/kg/day for several weeks. Malnourished children have much higher requirements of 2-4mg/kg/day.

Other options :

Option B - Copper deficiency results in microcytic, hypochromic anemia unresponsive to iron therapy, depigmentation of hair, neutropenia, neurological problems and osteoporosis. Disruption in copper transport is linked to Wilson's disease and Menke's disease.

Option C - Selenium deficiency is a cause of Keshan disease, a form of cardiomyopathy in young children seen in some regions of China. In combination with iodine deficiency, lack of selenium can result in myxedematous endemic cretinism.

Option D - Iron deficiency causes anemia, irritability, anorexia, pica, angular stomatitis, leg cramps and congestive heart failure.

Solution to Question 4:

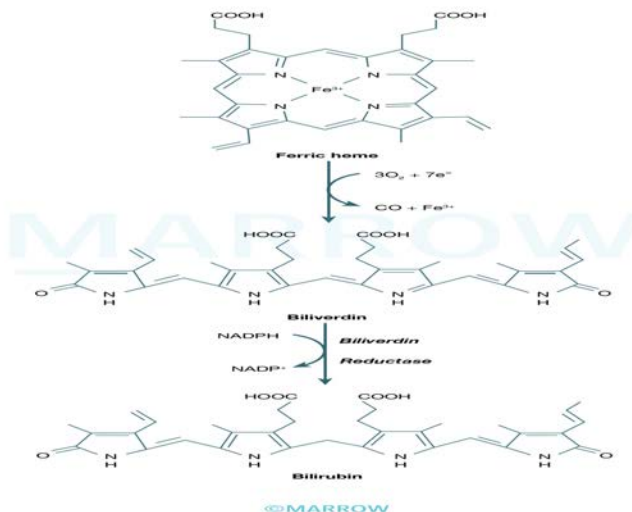
Aminolevulinic acid (ALA) is not a product of heme breakdown; rather it is an intermediate in the synthesis of heme molecules.

The process of heme breakdown occurs over a series of reactions. In the reticuloendothelial system of the liver, spleen and bone marrow, heme is degraded by microsomal heme oxygenase. It requires NADPH and cytochrome c. In this process, the alpha methenyl bridge of the heme molecule is liberated as carbon monoxide. The Fe⁺ liberated is oxidized to Fe³⁺⁺ and taken up by transferrin.

The linear tetrapyrrole formed is biliverdin (green pigment), which is further reduced to bilirubin (red-yellow pigment) by an NADPH-dependent biliverdin reductase. The bilirubin is transported in the plasma bound to albumin and is taken up by the liver, where it is conjugated with glucuronic acid to make it water-soluble, which is finally excreted in the bile.

The intestinal bacteria deconjugate the bilirubin, which is reduced to urobilinogen. It is further reduced to stercobilinogen, which is excreted through feces.

Steps in the breakdown of heme



Solution to Question 5:

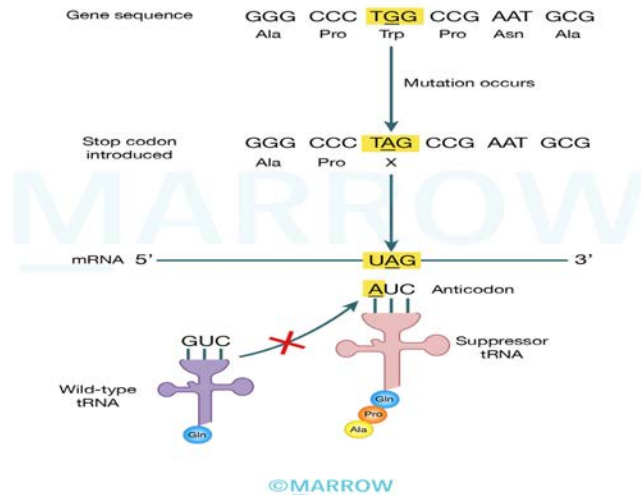
Mutations in tRNA are known to suppress the mutant gene and cause reversal to wild phenotype. Such mutations are called suppressor mutations.

Suppressor mutations are secondary mutations that alter the phenotypes caused by an existing mutation. This is not to be confused with reversion mutations in which the original mutation has been lost. Instead, suppressor mutations return the phenotype caused by the original mutation to a more wild-type phenotype despite the presence of the original mutation. The suppressor mutation's effect on the phenotype can vary from partial restoration to full restoration of the wild-type phenotype.

In prokaryotic and lower eukaryotic organisms, abnormally functioning tRNA molecules have been discovered that are themselves the results of mutations. Some of these abnormal tRNA molecules are capable of binding to and decoding altered codons, thereby suppressing the effects of mutations in the structural genes.

These suppressor tRNA molecules, usually formed as a result of alterations in their anticodon regions, are capable of suppressing certain missense mutations, nonsense mutations, and frameshift mutations. An example of suppressor mutation is illustrated below.

A primary mutation occurs changing the original codon sequence from UGG (Trp) to UAG (Stop). A secondary mutation occurs in the gene for a glutamine tRNA leading to the anticodon switch from a GUC to AUC. This mutant tRNA can recognize the UAG stop codon and insert glutamine in place of the stop, allowing translation to continue past the stop and suppressing the original mutation.



In the laboratory context, such suppressor tRNAs, coupled with mutated variants of aminoacyl-tRNA synthetases, can be utilized to incorporate unnatural amino acids into defined locations within altered genes that carry engineered nonsense mutations. The resulting labeled proteins can be used for in vivo and in vitro cross-linking and biophysical studies.

Solution to Question 6:

Glycosylation is not a post-transcriptional modification; it is a post-translational modification.

A primary transcript is the initial, linear, RNA copy of a transcription unit. The collection of all the primary transcripts synthesized in the nucleus by RNA polymerase II is known as heterogeneous nuclear RNA (hnRNA). The pre-mRNA components of hnRNA undergo extensive co- and posttranscriptional modification in the nucleus and become mature mRNA. These modifications usually include the following :

- Addition of a 5' cap - This is the first of the processing reactions for pre-mRNA. The cap is a 7-methylguanosine attached to the 5' terminal end of the mRNA and is resistant to most nucleases. Creation of the cap requires the removal of the γ -phosphoryl group from the 5' triphosphate of the pre-mRNA, followed by the addition of guanosine monophosphate. Methylation of this terminal G occurs in the cytosol and is catalyzed by guanine-7-methyltransferase. Additional methylation steps may occur (options A and B).
- Addition of a 3'-poly-A tail - Most eukaryotic mRNA have a chain of 40 to 250 adenylates (adenosine monophosphates) attached to the 3' end. This poly-A tail is added by the nuclear enzyme, polyadenylate polymerase. The poly-A tail terminates eukaryotic transcription. In addition, it helps stabilize the mRNA, facilitates its exit from the nucleus, and aids in translation.
- Splicing - Maturation of eukaryotic mRNA usually involves removal from the primary transcript of RNA sequences (introns) that do not code for protein (endonuclease cleavage). The remaining coding (expressed) sequences, the exons, are joined together to form the mature mRNA. The process of removing introns and joining exons is called splicing. The molecular complex that accomplishes these tasks is known as the spliceosome (option D).

Solution to Question 7:

The correct answer is option B: 1 -c, 2 -d, 3 -b, 4 -a

Von Gierke disease (type I glycogen storage disorder) occurs due to the deficiency of glucose-6-phosphatase enzyme in the liver, kidney, and intestinal mucosa. Patients present commonly by 3-4 months of age with hypoglycemia (leading to seizures), lactic acidosis, and hepatomegaly. Affected children often have a doll-like face with fat cheeks, relatively thin extremities, short stature, and a protuberant abdomen due to massive hepatomegaly. The kidneys are also enlarged, whereas spleen and heart are not involved. Laboratory findings include hypoglycemia, lactic acidosis, hyperuricemia, and hyperlipidemia.

Pompe's disease (type II glycogen storage disorder) occurs due to a mutation in alpha-1,4-glucosidase (also known as acid maltase), a lysosomal enzyme. A defect in this enzyme leads to the accumulation of glycogen in the lysosomes of peripheral, voluntary muscles and progresses to involve the cardiac muscle and diaphragm. Clinical features include hypotonia (floppy baby syndrome), cardiomyopathy, hepatomegaly, and failure to thrive. Myozyme can prevent worsening or can reverse cardiac and skeletal abnormalities.

Cori disease (type III glycogen storage disorder) occurs due to a deficiency of the glycogen debranching enzyme. The features of Cori disease include hypoglycemia, hepatomegaly, and hyperlipidemia but with normal blood lactate and uric acid levels. Muscle weakness and atrophy can also occur in these patients. Liver biopsy shows the accumulation of glycogen in the form of limit-dextrins (branched structures).

McArdle disease (type V glycogen storage disorder) typically presents during late childhood or in the second decade of life. The muscle glycogen phosphorylase enzyme deficiency interferes with ATP generation by glycogenolysis in the muscles. This causes muscle glycogen accumulation. Symptoms are exercise intolerance associated with muscle cramps. This leads to severe muscle damage resulting in a massive release of myoglobin in the blood.

1) Glycogen accumulation in liver and renal cells and lactic acidemia, associated with hypoglycemia	c) Von Gierke disease
2) Glycogen accumulation in lysosomes	d) Pompe disease
3) Exercise intolerance	b) McArdle disease
4) Accumulation of branched structures	a) Cori disease

Solution to Question 8:

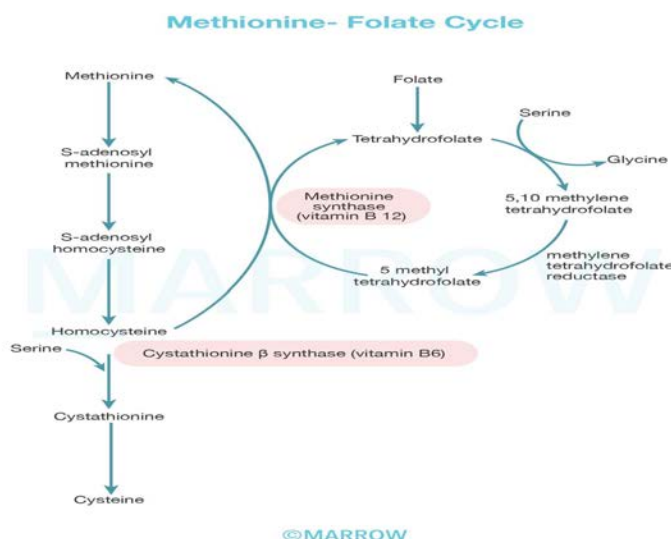
Folate deficiency is seen in children with macrocytic anemia and not with hemolytic anemia.

Folic acid (vitamin B9) is the synthetic form of folate used in supplements and fortification of foods. Folic acid plays a key role in one-carbon metabolism, essential for the biosynthesis of several compounds. Tetrahydrofolate (THF), the reduced, coenzyme form of folate, receives one-carbon fragments from donors such as serine, glycine, and histidine and transfers them to intermediates in the synthesis of amino acids, purine nucleotides, and thymidine monophosphate (TMP), a pyrimidine nucleotide incorporated into DNA.

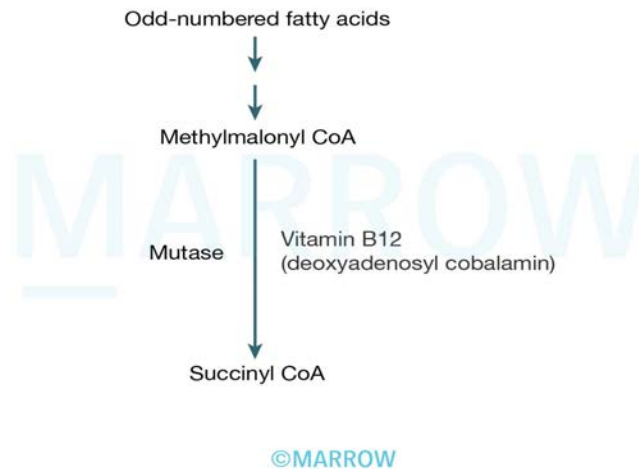
Leafy vegetables such as spinach, turnip, lettuce and fortified cereal products are rich sources.

Folate deficiency is caused by increased demand (e.g., pregnancy and lactation), poor absorption caused by pathology of the small intestine, alcoholism, or treatment with drugs (e.g., methotrexate) that are dihydrofolate reductase inhibitors. Children with hemolytic anemia have high folate requirements due to increased erythropoiesis predisposing them to deficiency (option B). Its manifestations are :

- Anemia - A primary result of folic acid deficiency is megaloblastic anemia, caused by diminished synthesis of purine nucleotides and TMP, which leads to an inability of cells to make DNA and, therefore, an inability to divide.
- Neural tube defects - Maternal deficiency of folate, during pregnancy, is implicated in neural tube defects. Periconceptional folate supplements, begun 1 month before conception and continued for 3 months after, are recommended to reduce the risk.
- Hyperhomocysteinemia - Folate, along with vitamin B12, is involved in converting homocysteine to methionine, as depicted below. Hence, folate deficiency results in increased levels of homocysteine.



However, levels of methylmalonic acid are normal. Methylmalonic acid levels are elevated in deficiency of vitamin B12, which is required as a coenzyme in the conversion of methylmalonyl CoA to succinyl CoA (option A).



Treatment: In patients with megaloblastic anemia, it is imperative to exclude and treat vitamin B12 deficiency before treating with folate. Otherwise the neurological signs of vitamin B12 deficiency may develop and progress irreversibly (option C). Deficiency is corrected using folic acid at a dose of 0.5-1mg/day orally for 3-4 weeks.

Solution to Question 9:

Congenital lactic acidosis is seen due to a defect in pyruvate dehydrogenase.

The major source of acetyl CoA for the tricarboxylic acid (TCA) cycle is the oxidative decarboxylation of pyruvate by the multienzyme pyruvate dehydrogenase complex (PDHC). The PDHC is a protein aggregate of multiple copies of three enzymes, pyruvate decarboxylase (E1), dihydrolipoyl transacetylase (E2), and dihydrolipoyl dehydrogenase (E3). Each catalyzes a part of the overall reaction.

A deficiency of the E1 component of the PDHC is the most common biochemical cause of congenital lactic acidosis. The deficiency results in a decreased ability to convert pyruvate to acetyl CoA, causing pyruvate to be shunted to lactate via lactate dehydrogenase. This creates particular problems for the brain, which relies on the TCA cycle for most of its energy and is particularly sensitive to acidosis. Symptoms include neurodegeneration, muscle spasticity, and, in the neonatal-onset form, early death. The inheritance pattern is X-linked dominant, with both males and females affected. Although there is no proven treatment for PDHC deficiency, dietary restriction of carbohydrates and supplementation with thiamine may reduce symptoms in select patients.

Other options :

Option B - Branched-chain ketoacid dehydrogenase deficiency results in maple syrup urine disease. There is defective oxidative decarboxylation of leucine, isoleucine and valine. The disease is characterized by feeding problems, ketoacidosis, changes in muscle tone, and coma.

Option C - Defective transketolase is a rare disorder of pentose phosphate metabolism with symptoms apparent in infancy including developmental delay and intellectual disability, delayed or absent speech, short stature, and congenital heart defects.

Option D - Isocitrate dehydrogenase is an enzyme of the TCA cycle which converts isocitrate to alpha-ketoglutarate. Defects in this enzyme are linked to carcinogenesis.

Solution to Question 10:

G6PD Deficiency does not have an equal incidence in males and females. It is an X-linked recessive (XLR) disorder, hence males are more affected than females.

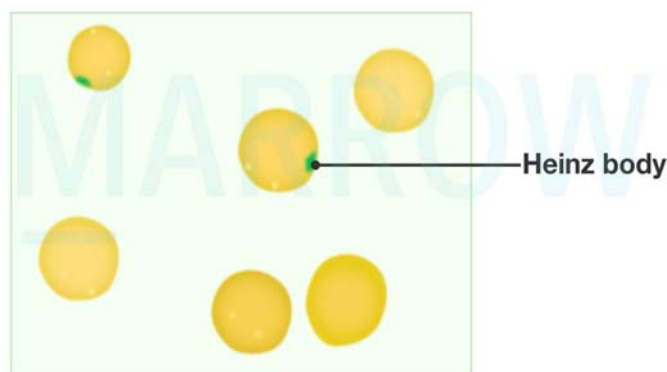
Glucose-6-phosphate dehydrogenase (G6PD) deficiency causes congenital hemolytic anemia due to enzyme deficiency. G6PD is required to generate NADPH while oxidizing glucose-6-phosphate in the hexose monophosphate pathway (HMP). G6PD deficiency leads to reduced NADPH levels (option B) which then reduces the level of reduced glutathione. Reduced glutathione is required to protect the cells against oxidative stress to RBCs and hemolysis.

Clinically, most patients are asymptomatic. When exposed to triggers like fava beans, infections, drugs such as primaquine, chloroquine, sulfonamides/sulfones, dapsone, and nitrofurantoin patient develops:

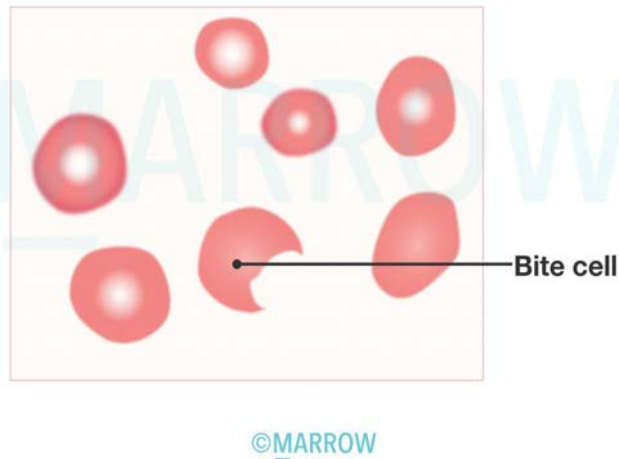
- Episodic hemolysis, both intravascular and extravascular
- Hemolytic anaemia
- Hemoglobinemia
- Hemoglobinuria

Increased oxidative damage to RBCs leads to the formation of denatured precipitates known as Heinz bodies. These bodies can damage the RBC membrane leading to intravascular hemolysis. Additionally, when such RBCs pass through the splenic cords, macrophages pluck out these precipitates, giving them the characteristic appearance of bite cells.

The images below show the presence of Heinz bodies and bite cells.



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Solution to Question 11:

Creatine is not a neurotransmitter derived from amino acids.

Creatine is an intermediate product in the process of formation of creatinine. The steps involved in the formation are as follows:

- The amidino group of arginine is transferred to glycine to form guanidoacetic acid. This step occurs in the kidneys and pancreas.
- Guanido acetic acid is methylated by S-adenosyl methionine to form creatine. This step occurs in the liver.

Creatine is phosphorylated to creatine phosphate by the enzyme creatine kinase, which is present in the muscle, brain and liver. The reaction needs hydrolysis of ATP. The stored creatine phosphate in the muscle serves as an immediate store of energy in the muscle. During muscle contraction, the energy is first derived from ATP hydrolysis. Thereafter, the ATP is regenerated by the hydrolysis of creatine phosphate. This is called the Lohmann's reaction.

The creatine phosphate may be converted to its anhydride, creatinine, which is excreted in the urine. Creatinine level in the blood is an indicator of renal function.

Other options :

Option A - Gamma amino butyric acid (GABA) functions in brain tissue as an inhibitory neurotransmitter. It is formed by decarboxylation of glutamate by L-glutamate decarboxylase.

Option B - Dopa is a precursor in the formation of the neurotransmitters dopamine, norepinephrine and epinephrine. It is formed by the hydroxylation of tyrosine by tyrosine hydroxylase.

Option C - 5 Hydroxytryptamine is an important neurotransmitter in the brain, where it functions as an antidepressant. It is derived from the amino acid tryptophan, in a process of hydroxylation followed by decarboxylation.

Solution to Question 12:

Tandem mass spectrometry is used to screen a blood sample for metabolic diseases.

Mass spectrometry is a very sensitive quantitative analytical technique used to measure a wide range of clinically relevant analytes. It identifies molecules based on their mass or molecular size. A sample is vaporized such that a positive charge is produced on the molecule. An electric field is applied so that particles move. At the same time, a magnetic field is applied, so the cations are deflected and hit on the detector. For molecules of identical charge, the time required for the molecule to reach the detector is inversely proportional to the mass of the molecule.

In tandem mass spectrometry, two mass spectrometers (MS) are arranged sequentially. The first MS separates big peptides, based on their mass. From this observation, the parent ion can be directed into the second MS. Here the ions collide with argon gas molecules and are broken into smaller ions. These daughter ions are detected and quantified by their mass spectrum in the second instrument.

Tandem MS is used to screen blood samples from newborns for the presence and concentrations of amino acids, fatty acids and other metabolites. Abnormalities in metabolite levels can serve as diagnostic indicators for a variety of genetic disorders like phenylketonuria.

Solution to Question 13:

Familial hypercholesterolemia occurs due to a defect in the LDL (low-density lipoprotein) receptor.

The primary function of LDL particles is to provide cholesterol to the peripheral tissues (or return it to the liver). They do so by binding to plasma membrane LDL receptors that recognize apo B-100.

After binding, the LDL–receptor complex is endocytosed. The pH of the endosome falls, which allows the separation of the LDL from its receptor. The receptors can be recycled, whereas the lipoprotein remnants in the vesicle are transferred to lysosomes and degraded by lysosomal acid hydrolases, releasing free cholesterol, amino acids, fatty acids, and phospholipids. These compounds can be reutilized by the cell.

Defects in the synthesis of functional LDL receptors cause a significant elevation in plasma LDL-C. Patients with such deficiencies have type IIa hyperlipidemia /familial hypercholesterolemia (FH). This autosomal dominant condition predisposes individuals to premature atherosclerosis and coronary artery disease.

Patients with homozygous FH must be treated aggressively to delay the onset and progression of CVD. Two drugs that reduce the hepatic production of VLDL and thus LDL, a small-molecule inhibitor of the microsomal TG transfer protein (MTP) and an antisense oligonucleotide to apoB, and an antibody that inhibits ANGPT3 are approved and should be considered in patients who have insufficient response to statins and PCSK9 inhibitors. LDL apheresis should be considered in patients who have persistently elevated LDL-C levels despite drug therapy.

Other options :

Option B - Defects in apo-E lead to type III hyperlipoproteinemia / familial dysbetalipoproteinemia. It is characterized by defective clearance of chylomicron remnants and intermediate-density lipoproteins by the liver. This leads to hypercholesterolemia, xanthomas, and atherosclerosis.

Option C - Defective apo-C II results in type I hyperlipoproteinemia/ familial lipoprotein lipase deficiency. The resulting inactive lipoprotein lipase leads to slow clearance of chylomicrons and VLDL, and low levels of LDL and HDL. Patients present with eruptive xanthomas and acute pancreatitis.

Option D - Apo-B 48 is involved in the assembly of chylomicrons. Defects in this protein are not associated with familial hypercholesterolemia.

Solution to Question 14:

In a patient with Diabetes Mellitus, the liver produces more TG and VLDL, due to decreased action of LPL with increased action of HSL.

- Lipoprotein lipase (LPL): Enzyme responsible for hydrolyzing triglycerides (TG) in VLDL and chylomicrons into free fatty acids (FFAs) for storage or energy use in tissues.
- Hormone-sensitive lipase (HSL): Enzyme that breaks down stored triglycerides in adipocytes into free fatty acids and glycerol.

Insulin has an inhibitory effect on hormone-sensitive lipase (HSL). In diabetes, low insulin corresponds to an increase in HSL activity. This promotes the breakdown of lipids and a release of TG and free fatty acids from the adipose tissue.

Insulin induces the activity of lipoprotein lipase (LPL) which is present on the capillary walls. LPL promotes hydrolysis of TG and the degradation of the transport proteins, VLDL and chylomicrons. In diabetes, low insulin leads to decreased LPL activity which corresponds to an increase in VLDL and TG.

This results in increased production of triglycerides and VLDL in the liver.

Solution to Question 15:

Nutritional deficiency of copper, required by lysyl oxidase for formation for covalent cross linking in collagen synthesis, gives rise to Menke's disease.

Menke's disease is an X-linked recessive condition caused due to a mutation in the ATP7A gene on chromosome X. This causes impaired transport of dietary copper from the small intestine to the liver leading to low levels of copper-containing enzymes.

The final step of collagen synthesis involves the formation of covalent cross-links. This is done by the enzyme lysyl oxidase, a copper-containing protein. This is essential for the strength and rigidity of collagen. As a result of deficiency of copper in Menkes disease, this step of collagen synthesis is affected.

The patients presents with depigmented kinky hair, failure to thrive, neurodegenerative symptoms (loss of developmental milestones, truncal hypotonia, epilepsy), multiple fractures, anaemia due to ceruloplasmin deficiency, neutropenia, and recurrent infections.

Other Options:

Option B: Scurvy: Caused by vitamin C deficiency, leading to defective collagen synthesis due to impaired hydroxylation of proline and lysine.

Option C: Ehlers-Danlos syndrome: A group of genetic disorders affecting collagen structure or production.

Option D: Scoliosis: A structural deformity of the spine.

Solution to Question 16:

The correct matched sequence is : i-b, ii-d, iii-a, iv-c.

i. Melatonin – b. Tryptophan

Tryptophan is an essential amino acid and the precursor for the synthesis of melatonin, produced by the pineal gland. The process begins with tryptophan being converted into serotonin, and then serotonin is further converted into melatonin.

ii. Histamine – d. Histidine

Histamine is derived from the amino acid histidine. The conversion of histidine into histamine is catalyzed by the enzyme histidine decarboxylase. Histamine is involved in local immune responses, acts as a neurotransmitter, and plays a role in regulating gastric acid secretion.

iii. Norepinephrine – a. Tyrosine

Norepinephrine (noradrenaline) is synthesized from tyrosine. Tyrosine is first converted to L-DOPA by the enzyme tyrosine hydroxylase, and then L-DOPA is converted into dopamine. Finally, dopamine is hydroxylated to form norepinephrine. Norepinephrine is a neurotransmitter involved in the sympathetic nervous system's fight-or-flight response.

iv. Glutathione – c. Cystine

Glutathione is a tripeptide composed of glutamate, cystine, and glycine. The cystine component is crucial because it contains a thiol (-SH) group, which is responsible for the antioxidant properties of glutathione.

i. Melatonin	b. Tryptophan
ii. Histamine	d. Histidine
iii. Norepinephrine	a. Tyrosine
iv. Glutathione	c. Cystine

Solution to Question 17:

The amino acid that provides phenylbutyrate in the treatment of urea cycle disorders is glutamate.

Ammonia to Urea cycle: Ammonia gets converted to glutamate (non toxic form) through transamination. Glutamate is transported to the liver as glutamine (by glutamine synthase). In the liver, ammonia is released from glutamine, converting it to alpha keto glutamate. The released ammonia enters the urea cycle.

In acetylation therapy, Phenylbutyrate (a prodrug) which is metabolized to phenylacetate. Phenylacetate combines with glutamine (a nitrogen-rich amino acid) to form phenylacetylglutamine. Phenylacetylglutamine is water-soluble and is excreted in the urine. This process helps trap excess nitrogen, bypassing the urea cycle.

Other drugs that can be used to achieve the same outcome (ammonia scavengers) include:

- Phenylacetate
- Benzoate

Other Options:

Option A and C: Arginine and Aspartate do not play a role in this specific reaction.

Option D: Glycine: The combination of benzoate and glycine forms hippuric acid, a non-toxic compound. When Benzoate is administered to the body, it acts as a nitrogen-scavenging agent. Hippuric acid is water-soluble and is excreted in the urine, effectively removing one nitrogen molecule from the body per molecule of hippuric acid formed. This mechanism is similar to how phenylacetate acts as a scavenger for glutamate.

Solution to Question 18:

In situ hybridization, Conventional cytogenetics, Sanger sequencing and Polymerase chain reaction are appropriate techniques for chromosomal aneuploidy.

Techniques to detect aneuploidy:

- In situ hybridization: Fluorescent In Situ Hybridisation (FISH) helps in the detection of specific genetic information using fluorescent probes. FISH is used to detect chromosomal aneuploidy, subtle microdeletions and rearrangements, complex translocations, and gene amplification.
- Conventional cytogenetics: This technique involves the study of chromosomes under a microscope, typically using methods like karyotyping, which arranges chromosomes by size, shape, and number. The analysis allows the identification of large chromosomal abnormalities, such as aneuploidy or structural changes.
- Sanger sequencing: DNA sequencing that uses chain-termination to identify the exact sequence of nucleotides in a DNA molecule. Simple Sanger sequencing can detect aneuploidy only with massive parallel sequencing.
- Polymerase chain reaction: Polymerase chain reaction (PCR) is a rapid automated method for the amplification of specific DNA sequences (or genes). It can accurately quantify the target nucleic acid sequence within a sample. It is useful for determining somatic mutation with low allele frequency. Quantitative Fluorescence (QF) PCR is a method of rapid aneuploidy detection. It

is less expensive and allows simultaneous processing of large number of samples of FISH.

- Array Comparative Genomic Hybridization: microarray based technique which can reveal imbalances across the genome.
- Multiplex Ligation Probe Amplification: to detect gene dosage.

Solution to Question 19:

Phosphatase, Terminal transferase and Crispr Cas9 are used in recombinant DNA technology.

Recombinant DNA technology is a technique where DNA from different sources is combined to create new genetic material. Applications of recombinant DNA technology include producing insulin, growth hormones, vaccines, gene therapy, and studying gene functions and disease mechanisms.

Phosphatase dephosphorylates 5' ends of RNA and DNA. It is primarily used for the removal of phosphate groups prior to kinase labelling and it is also used to prevent self ligation.

Terminal transferase catalyzes the addition of nucleotides to the 3' terminus of a DNA molecule. It is used in homopolymer tailing.

CRISPR-Cas9 acts as an RNA-directed nuclease. It is a DNA editing/gene regulatory system. It involves two essential components: Cas9, an endonuclease that causes a double-stranded DNA break, allowing modifications to the genomic DNA to be made; and a synthetic single-guide RNA, or sgRNA, responsible for guiding Cas9 to its target.

Isomerase is a class of enzymes that catalyze the structural rearrangement of molecules, converting one isomer into another. Isomerase is not commonly used directly in recombinant DNA technology, it has broader applications. They are used in metabolic pathways and as biocatalysts in industrial processes.

Biochemistry NEET 2024

Question 1:

A patient underwent gastrectomy for gastric adenocarcinoma. Which of the following vitamins needs to be supplemented in this patient?

- a) Vitamin B12
- b) Vitamin B1
- c) Vitamin A
- d) Vitamin D

Question 2:

A 10-year-old child from a southern Indian state, hailing from a poor family, presents with rough and scaly skin. The child's diet lacks vegetables and fruits. After treatment with linseed oil, Vitamin A, and Vitamin B, the symptoms subsided. What is the most likely diagnosis?

- a) Lichen planus
- b) Erythroderma
- c) Phrynoderma
- d) Keratosis pilaris

Question 3:

A patient presents to the emergency department with severe abdominal pain, nausea, and vomiting. The laboratory results show serum triglyceride levels > 2000 mg/dl. Which deficiency could be contributing to the patient's condition?

- a) Apolipoprotein C II
- b) Apolipoprotein B48
- c) Apolipoprotein B100
- d) Apolipoprotein A I

Question 4:

After the age of 12, the boy had an intellectual disability. On examination, fingernail abrasions, excessive biting of lips, and increased levels of serum uric acid were observed. What is the likely diagnosis?

- a) Hurler syndrome
- b) Lesch-Nyhan disease
- c) Hunter syndrome
- d) Von Gierke syndrome

Question 5:

A neonate was brought with features of lethargy, and poor oral intake. On examination, a cataract was seen. Urine Benedict's test was positive but negative for Glucose. What will be the enzyme deficiency in this condition?

- a) Hexokinase
- b) Galactokinase
- c) Galactose-1-phosphate uridyl transferase
- d) Phosphofructokinase

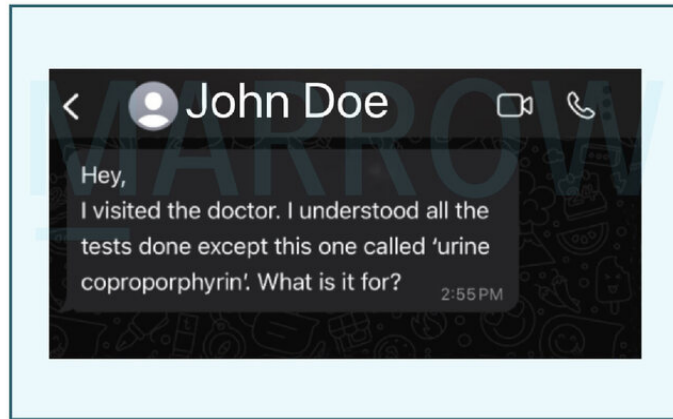
Question 6:

A patient diagnosed with a rare homozygous recessive condition presented with ectopia lentis. In the peripheral smear, there was no megaloblastic anemia. Later, he was found to have elevated homocysteine levels in the blood. Which of the following supplementations will help the patient?

- a) Thiamine
- b) Pyridoxine
- c) Biotin
- d) Riboflavin

Question 7:

You received a text message from your friend JOHN DOE about occupational exposure after he visited his doctor as shown below. What is the test used to diagnose?



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- a) Anthracosis
- b) Lead poisoning
- c) Silicosis
- d) Asbestosis

Question 8:

A 3-year-old child is brought to the OPD with complaints of persistent delayed developmental milestones, and hyperpigmented macular lesions on the body which are more pronounced on exposure to sunlight. What is the error in DNA multiplication?

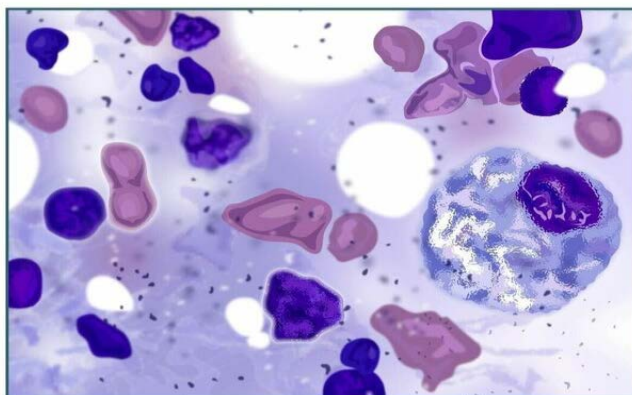
- a) Base excision repair
- b) Nucleotide excision repair
- c) DNA mismatch repair
- d) Homologous recombination

Question 9:

A patient with a family history of coronary heart disease presented with tendon xanthomas. His lipid profile showed serum LDL to be 280 mg/dL and triglyceride levels was 398 mg/dl. What is the likely defect in this patient?

- a) LDL receptor defect
- b) LDL metabolism
- c) VLDL metabolism

A child presents to you with bone pain, hepatosplenomegaly, and pancytopenia. A trephine biopsy aspirate shows the following finding. Which of the following is the most likely enzyme deficient in this patient?



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- a) Galactocerebrosidase
- b) Glucocerebrosidase
- c) Sphingomyelinase
- d) Alpha 1,4-glucosidase

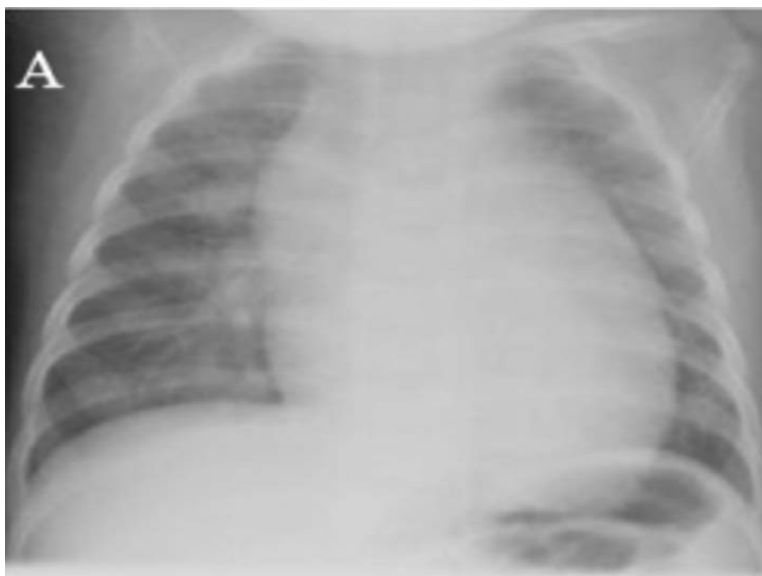
Question 13:

A 32-year-old lady comes with complaints of erythematous scaly rash, loose stools, and cognitive dysfunction. The rash involved her forearm, neck, and face. She seems to have difficulty remembering things from the recent past. Her electrolytes are within normal range. She admits to having shifted to a corn-based diet due to economic reasons. Which of the following would improve her symptoms?

- a) Oral Nicotinamide supplementation
- b) High dose of IV thiamine
- c) High Dose Vitamin C
- d) Stop corn diet

Question 14:

A 3-month-old child was brought to the clinic with a history of difficulty in feeding and failure to thrive. On examination, the child is floppy, 48 cm in length and weighs 2.8 kg. There is generalized hypotonia, macroglossia and hepatomegaly. The chest x-ray of the child is given below. What is the most likely diagnosis?



- a) Ebstein's anomaly
- b) Pompe's disease
- c) Transposition of great arteries
- d) Von- Gierke's disease

Question 15:

A patient presented with multiple yellowish raised plaques on eyelids, xanthomas on the achilles tendon. What is the likely cause for the patient's presentation?

- a) Familial hypercholesterolemia
- b) Tangier's disease
- c) Xanthomatosis
- d) Fabry's disease

Question 16:

A child presented to the OPD with complaints of fatigue and muscle weakness while exercising. Laboratory investigations found that LCFA oxidation was affected. A muscle biopsy was done which revealed the presence of fat vacuoles. What is the probable diagnosis?

- a) LDL defect
- b) LPL defect
- c) Fatty acid synthase defect
- d) Carnitine deficiency

Question 17:

A mother came back home to see her child suffering from difficulty breathing. The child had contact with rotenone while trying to bathe their pet. The child was taken to the emergency department and she later passed away. Inhibition of which of the following enzymes was the likely cause of her death?

- a) Cytochrome c reductase
- b) Succinate dehydrogenase
- c) NADH-Q oxidoreductase
- d) Cytochrome oxidase

Question 18:

A child suffering from phenylketonuria was started on necessary dietary modifications but the neurological symptoms persisted. Phenylalanine hydroxylase levels in the serum are normal. Which of the following enzymes is likely to be deficient in this case?

- a) Tyrosine hydroxylase
- b) Homogentisate oxidase
- c) Tetrahydrobiopterin synthase
- d) Dopa decarboxylase

Question 19:

Which micronutrient is most likely deficient in a patient with the radiographic findings given below along with anemia, neutropenia, depigmented hair and skin?



- a) Iron
- b) Zinc
- c) Copper
- d) Fluoride

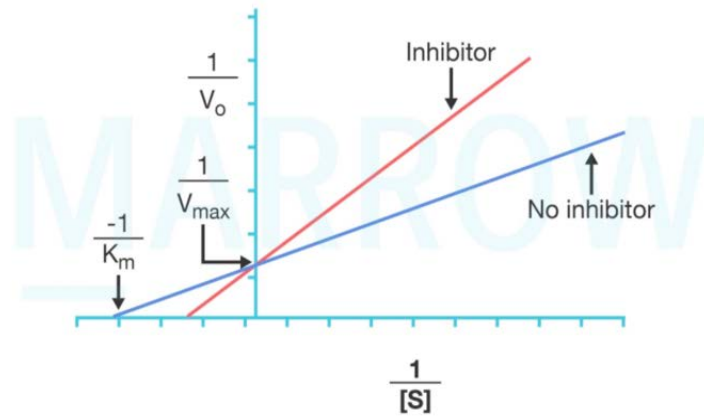
Question 20:

A child presented with recurrent abdominal pain, vomiting, and diarrhea following ingestion of milk-based products. Which enzyme is likely to be deficient?

- a) Lipase
- b) Maltase
- c) Amylase
- d) Lactase

Question 21:

In the given Lineweaver-Burk plot, the blue line represents the action of an enzyme, and the red line represents the action of the same enzyme in the presence of an inhibitor. What is the type of inhibition shown here?



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- a) Competitive inhibition
- b) Non-competitive inhibition
- c) Uncompetitive inhibition
- d) Allosteric inhibition

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	b
5	c
6	b
7	b
8	b
9	a
10	d
11	c
12	b
13	a
14	b

15	a
16	d
17	c
18	c
19	c
20	d
21	a

Detailed Explanations

Solution to Question 1:

A patient who has undergone gastrectomy for gastric adenocarcinoma needs to be supplemented with vitamin B12.

Vitamin B12 absorption is a complex process that relies on several factors. Intrinsic factor, a glycoprotein secreted by the parietal cells in the stomach, is essential for this process. Gastric acid and pepsin release vitamin B12 from dietary proteins, allowing it to bind with cobalophilin in the saliva. In the jejunum, cobalophilin is broken down, freeing vitamin B12 to bind with intrinsic factor. This complex is then absorbed in the distal ileum through specific receptors. Following gastrectomy, the production of intrinsic factor is impaired due to the removal of gastric tissue, making vitamin B12 absorption difficult. Therefore, vitamin B12 supplementation is necessary after gastrectomy to prevent deficiency.

Other complications of gastrectomy include:

- Nutritional deficiencies- Iron deficiency anemia (most common), Vitamin B12 deficiency, and osteoporosis
- Dumping syndrome: early and late
- Diarrhea
- Duodenal stump leak/blow out
- Bile reflux gastritis and esophagitis
- Gall stones
- Roux syndrome

Solution to Question 2:

The cutaneous lesions with rough and scaly skin, with dietary history stating lack of vegetables and fruits is suggestive of phrynoderma.

Phrynoderma are keratotic follicular papules commonly present over the elbows and knees. They can also extend to the shoulders, neck, back, thighs, buttocks, and abdomen. It is also known

as toad skin, due to the rough cone-shaped elevated papules. It can be caused by deficiency of vitamins A, B complex, C, and E and essential fatty acids.

When it occurs due to vitamin A deficiency, it is associated with ophthalmic symptoms like xerophthalmia, night blindness, and Bitot's spots and cutaneous symptoms like xerosis, scaling, and acne.

Management includes nutritional improvement through diet, multivitamin supplementation, and treatment of the underlying cause.

Note: Linseed oil is rich in alpha-linolenic acid, which is an essential fatty acid.

The toad skin appearance of a child with phrynoderma is given below:



Other options:

Option A: Lichen planus is a papulosquamous lesion presenting as pruritic, purple, polygonal, planar, papules and plaques. These papules are often highlighted by white dots or lines called Wickham striae, which are created by areas of hypergranulosis. The characteristic lesions in lichen planus are shown below:



Option B: Erythroderma is a serious skin condition characterized by widespread redness (erythema) and scaling affecting more than 90% of the body's surface area. It is also known as exfoliative dermatitis and often results from underlying skin disorders, systemic diseases, or drug reactions.

Option D: Keratosis pilaris is an inherited abnormality of keratinization. It affects the follicular orifices leading to keratotic follicular plugging, perifollicular erythema, and follicular atrophy. It usually occurs during puberty. They present as small, keratotic papules on the extensor aspects of the limbs, particularly the arms which can lead to ingrowing of the hair. Treatment is purely reserved for cosmetic purposes.

Keratosis pilaris



Solution to Question 3:

The elevated triglyceride levels and symptoms of acute pancreatitis are likely related to a deficiency in apolipoprotein C II.

Apolipoprotein C2 is a lipoprotein lipase cofactor that catalyzes cleavage of chylomicrons. Defective apo C II results in type I hyperlipoproteinemia/ familial lipoprotein lipase deficiency. The resulting inactive lipoprotein lipase leads to slow clearance of chylomicrons and VLDL, and low levels of LDL and HDL. Patients present with eruptive xanthomas and acute pancreatitis.

Lipoprotein lipase is an extracellular enzyme that is present in the capillary walls of most tissues. It is chiefly synthesized by the adipose tissue, and cardiac and skeletal muscles. A deficiency of lipoprotein lipase or apo C-II leads to type 1 hyperlipoproteinemia or familial chylomicronemia which leads to an increased risk of acute pancreatitis.

Other options:

Option B: Apo B48 is involved in the assembly of chylomicrons. Defects in this protein are not associated with familial hypercholesterolemia.

Option C: Apo B100 binds low-density lipoprotein receptors. For the uptake of LDL, apo B100 acts as a ligand for the low-density lipoprotein receptor.

Option D: Apo A1 activates lecithin cholesterol acyltransferase (LCAT). LCAT is the enzyme present in HDL, which is activated by apo A1. LCAT converts free cholesterol (amphipathic) in HDL to cholesterol ester(hydrophobic). The initial discoidal shape now becomes spherical. This facilitates reverse cholesterol transport.

Solution to Question 4:

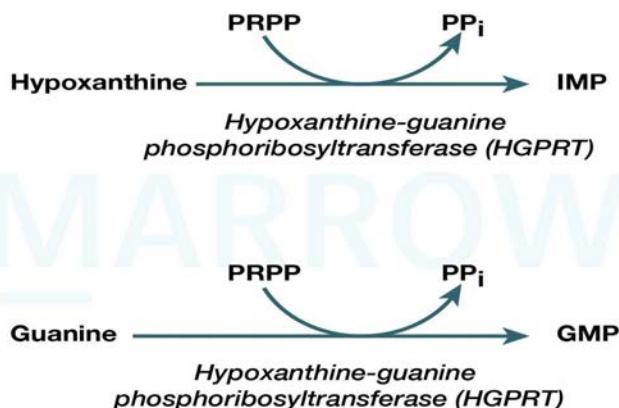
The above clinical scenario with findings showing self-mutilation and laboratory values of hyperuricemia is most likely suggestive of Lesch-Nyhan disease.

Lesch-Nyhan syndrome is an X-linked recessive disorder due to a complete deficiency of the hypoxanthine-guanine phosphoribosyltransferase (HGPRT). This deficiency results in an inability to salvage hypoxanthine or guanine leading to increased levels of phospho-ribosyl pyrophosphate (PRPP). Increased PRPP leads to increased purine synthesis, which is later degraded to uric acid. Increased hypoxanthine is also converted to uric acid.

The patients have intellectual disability, behaviors of self-injury like lip and finger biting, developmental delay, muscle dystonia, and extrapyramidal symptoms.

Diagnosis is made by the elevated levels of uric acid and by HGPRT enzyme assay.

The image given below shows the metabolism of hypoxanthine and guanine by HGPRT.



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Orange diaper syndrome can be an early indicator of Lesch-Nyhan syndrome. At birth, individuals with Lesch-Nyhan syndrome typically appear asymptomatic, with normal prenatal growth and development. The condition is often first detected through the presence of orange-coloured crystals in the diaper, resulting from hyperuricemia. Since hyperuricemia is a common feature across all variants of HGPRT deficiency, which causes Lesch-Nyhan syndrome, predicting associated neurodevelopmental abnormalities requires enzyme activity testing.

Solution to Question 5:

The given clinical scenario of a neonate presenting with lethargy, and poor oral intake, with examination revealing a cataract and urine benedict's test was positive but negative for glucose, is suggestive of classical galactosemia and enzyme deficient is galactose-1-phosphate uridyl transferase.

Galactose-1-phosphate uridyl transferase deficiency leads to classical galactosemia. It is an autosomal recessive disorder. Normally, newborns receive up to 40% of their intake as lactose (glucose + galactose). The deficiency of this enzyme leads to an inability to convert galactose-1-phosphate to uridine diphosphate galactose. This leads to the accumulation of galactose 1-phosphate and causes injury to the kidney, liver, and brain.

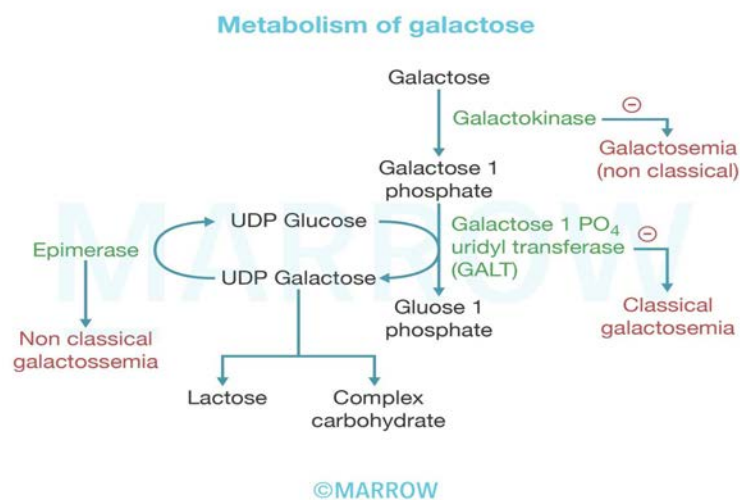
Clinical features:

- Accumulation of galactose-1-phosphate in the liver causes liver damage & leads to hepatomegaly, vomiting, and convulsions.
- Accumulation of galactose-1-phosphate depletes the liver of inorganic phosphate leading to diminished ATP synthesis which affects gluconeogenesis resulting in hypoglycemia.
- Accumulation of galactose-1-phosphate in the nerves causes intellectual disability.
- Accumulation of galactitol in the lens causes oil-drop cataracts.
- Patients with galactosemia are at an increased risk of developing Escherichia coli neonatal sepsis.

In uridyl transferase deficiency, the epimerase is present in adequate amounts, so that the galactosemic individual can still form UDPGal from glucose. This explains how normal growth and development of affected children can occur despite the galactose-free diets.

Urinary reducing sugar (Benedict's test) and mucic acid test are positive. Direct enzyme assay using erythrocytes establishes the diagnosis.

Treatment is a galactose-free diet.



Other options:

Option A: Hexokinase is a glycolytic enzyme, involved in maintaining intracellular glucose concentration by catalysing the first step in glucose metabolism, i.e., phosphorylation of intracellular glucose. Its deficiency leads to hemolytic anaemia. Splenomegaly may be another presenting feature.

Option B: Galactokinase is the enzyme responsible for the conversion of galactose to galactose 1-phosphate. Only cataract is seen in the deficiency of this enzyme because galactose-1-phosphate is not formed. Hence no toxic effects of this compound are manifested.

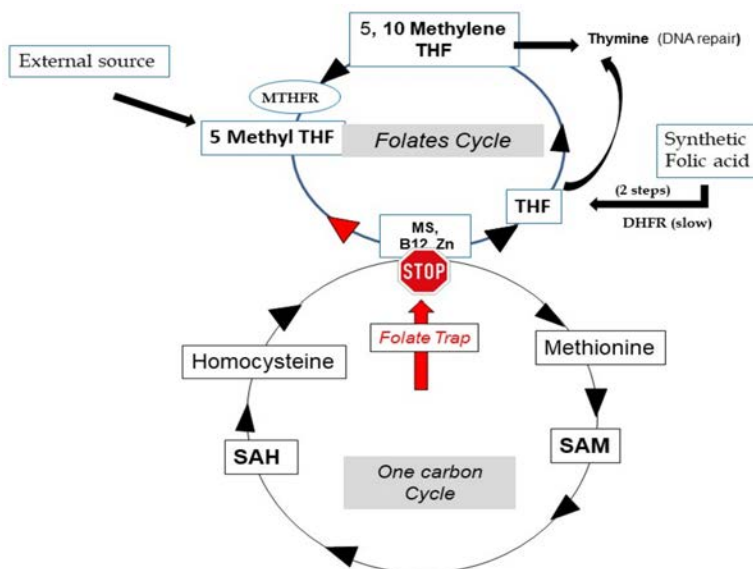
Option D: Phosphofructokinase is a key regulatory enzyme of glycolysis and is necessary for the ATP-dependent conversion of fructose-6-phosphate to fructose-1,6-diphosphate. Tauri's disease is type VII glycogen storage disorder that results due to the deficiency of phosphofructokinase.

Solution to Question 6:

The vitamin that can be used as supplementation in this patient with homocystinuria is pyridoxine (vitamin B6).

Classical homocystinuria is an autosomal recessive disease most commonly caused by a deficiency of the cystathionine β -synthase. The deficiency of this enzyme leads to an accumulation of homocysteine in the blood and a clinical syndrome of intellectual disability, subluxated ocular lenses, long and slender extremities, arachnodactyly, and arterial and venous thrombi. The skeletal features of homocystinuria often mimic Marfan syndrome. Folic acid and vitamin B12

deficiency can also lead to homocystinuria with megaloblastic anemia.



Demonstration of elevated total homocysteine in the blood confirms the diagnosis. 50% of the cases of homocystinuria are termed pyridoxine-responsive because large doses of pyridoxine can enhance the activity of deficient enzymes in these patients. Administration of large doses of pyridoxine (vitamin B6) restores the activity of the deficient enzyme in half of all cases of homocystinuria.

Folate supplementation is also added to overcome any folate deficiency that might occur due to increased conversion of homocysteine to methionine. For the other 50% of the patients, the treatment consists of methionine restriction and cysteine supplementation. In addition, betaine (methyl group donor), folate and vitamin B12 may be added.

Solution to Question 7:

The test for coproporphyrinogen levels is used as a screening test to diagnose lead poisoning.

Lead inhibits the enzymes delta-aminolevulinic acid (ALA) dehydratase and ferrochelatase, which are involved in heme synthesis. ALA is converted to porphobilinogen by ALA dehydratase, inhibited by lead. So, serum porphobilinogen levels are decreased in lead poisoning.

Laboratory findings include the following:

- An elevated serum lead level
- An elevated level of protoporphyrin in blood and elevated ALA and coproporphyrinogen in urine
- Microcytic anemia with basophilic stippling in peripheral smear due to ineffective heme synthesis

Clinical features of chronic lead poisoning include facial pallor, anemia, basophilic stippling of red cells, blue lines in the gums, and retinal stippling. In the later phase, the findings are colic abdominal pain, constipation, nerve palsy, encephalopathy, and disturbance of the genito-urinary and cardiovascular systems.

Diagnosis of lead poisoning:

Laboratory parameter	Value	Remarks
Coproporphyrin in urine	>150 mcg/L	Screening test- indicates exposure
Aminolevulinic acid (ALA) in urine	>5 mg/L	Indicates lead absorption
Lead in blood	>70 mcg/dL	Clinical symptoms appear
Lead in urine	0.8 mg/L	Indicates lead absorption
Basophilic stippling of RBCs		Sensitive indicator

Solution to Question 8:

The history points to a diagnosis of xeroderma pigmentosum. It is characterized by defective nucleotide excision repair.

UV radiation leads to pyrimidine dimerization in DNA strands, preventing further replication by DNA polymerase. The nucleotide excision repair mechanism corrects the dimer formation in DNA. Xeroderma pigmentosum (XP) is an autosomal recessive genetic disorder characterized by an inability to repair DNA pyrimidine dimers caused by UV exposure. This leads to clinical and cellular hypersensitivity to ultraviolet (UV) radiation.

In XP the skin is normal at birth and has a three-stage progression:

- Diffuse erythema, scaling, and freckle-like areas over the sun-exposed regions of the body at around 6 months of age.
- The second stage is characterized by poikiloderma. This results in skin atrophy, telangiectasias with mottled hyperpigmentation, and hyperpigmentation.
- In the third stage appearance of cutaneous malignancies like squamous cell carcinomas, malignant melanoma, basal cell carcinoma, and fibrosarcoma.

Most of the cases with XP have ocular manifestations. Initially, it appears as photophobia, conjunctivitis, ectropion, and symblepharon with ulceration. Accumulation of pyrimidine dimer causes corneal opacification and neovascularization. Ocular malignancies are common in the lids, conjunctiva, and cornea as they are exposed to as much as facial skin. XP is ultimately fatal, and early detection of the skin lesions and their management with prevention of UV ray exposures prolong life.

Solution to Question 9:

The clinical presentation with tendon xanthomas, elevated serum LDL and elevated serum triglycerides points to a diagnosis of familial hypercholesterolemia and is due to a defect in LDL

receptors.

Familial hypercholesterolemia (type IIa hyperlipoproteinemia) is an inherited disorder due to defects in LDL receptors. This defect causes a decrease in the uptake of LDL and cholesterol. As a result, the blood levels of both LDL and cholesterol are elevated. Such patients are at increased risk of developing atherosclerosis and coronary artery disease.

Features of FH:

- Painful polyarthritis and tendinitis are common.
- Physical findings seen are corneal arcus and tendon xanthomas over the Achilles tendon and the dorsum of the hands. Xanthomas are tumorous masses produced by foamy cell clusters (lipid-laden macrophages) found in the connective tissue of the skin and in tendons.

Diagnosis is done clinically based on hypercholesterolemia with

- LDL $>$ 190 mg/dl,
- Family history of hypercholesterolemia and/or
- Premature coronary artery disease.

Treatment includes lifestyle changes, restriction of a fatty diet, and pharmacotherapy with statins in heterozygous FH patients.

Recent drugs for specific treatment of homozygous FH are:

- Lomitapide, a microsomal triglyceride transfer protein (MTP) inhibitor, and
- Mipomersen, an antisense oligonucleotide against apoB.

The image below shows Achilles tendon xanthoma and a plain lateral radiograph of both ankles showing bilateral symmetrical swellings (arrows) superior to the calcaneum respectively.





Note: Lipoprotein lipase (LPL) deficiency causes type I hyperlipoproteinemia due to defects in LPL or apo C-II. Eruptive xanthomas are seen. Chylomicrons and VLDL are elevated. LDL and HDL are low. There is no risk for coronary disease.

Solution to Question 10:

Among the various food items given on the plate, the primary source of thiamine (vitamin B₁) in the Indian diet is option 4 i.e. chapati and rice.

The thiamine content in various foods in decreasing order are as follows:

- Gingelly seeds
- Groundnut
- Bengal gram
- Whole wheat
- Rice

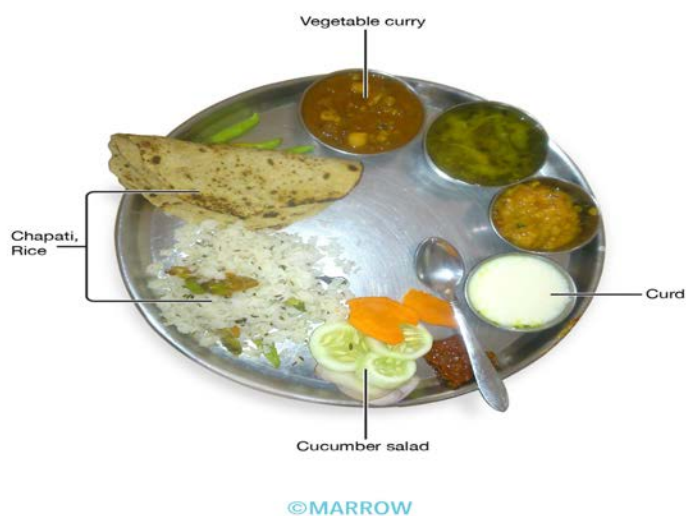
Gingelly (sesame) seeds have the highest content of thiamine.

Thiamine is lost from rice during the process of milling. As it is a water-soluble vitamin, washing, and cooking rice cause further losses. Hence, it is advisable to avoid highly polished rice and eat parboiled or under-milled rice.

Thiamine is essential for energy metabolism, particularly in carbohydrate processing. Its active form, thiamine diphosphate, acts as a coenzyme for three critical enzyme complexes: pyruvate dehydrogenase, α -ketoglutarate dehydrogenase, and branched-chain keto acid dehydrogenase. This coenzyme facilitates oxidative decarboxylation by forming a reactive carbanion that attaches to carbonyl groups like those in pyruvate, leading to the release of CO₂. Thiamine diphosphate is also vital for transketolase in the pentose phosphate pathway.

Thiamine triphosphate, another form of thiamin, helps regulate nerve function by activating chloride channels in nerve membranes.

A thiamine deficiency can cause several conditions such as beriberi, characterized by peripheral neuritis, which can also involve heart failure and edema. Additionally, thiamine deficiency can lead to Wernicke's encephalopathy and Korsakoff's psychosis, commonly associated with alcohol or drug abuse. The lack of thiamine impairs the conversion of pyruvate to acetyl-CoA, leading to increased levels of lactate and pyruvate in the blood, which can result in dangerous lactic acidosis, especially in those consuming high-carbohydrate diets.



Solution to Question 11:

Phenobarbital is a barbiturate that inhibits complex I to Q of the electron transport chain.

Complex I of the electron transport chain is NADH-Q oxidoreductase. It oxidizes NADH and reduces coenzyme Q. Phenobarbital inhibits complex I to Q so that the electrons will not be able to be transferred from NADH to coenzyme Q, which is the first step in the electron transport chain. As a result, the rate of oxygen consumption will decrease.

The electron transport chain (ETC) represents the final pathway through which electrons, originating from the oxidation of carbohydrates, fatty acids, and amino acids, are channeled toward a final reaction with oxygen to produce water and ATP. The ETC complexes are embedded in the inner mitochondrial membrane.

The oxidation of compounds by ETC complexes leads to the release of free energy. This is used to pump H^+ to the intermembrane space. This results in the formation of a proton motive force that is used for the phosphorylation of ADP to ATP. Thus, oxidation is coupled with phosphorylation to generate ATP. NADH generates 2.5 ATP and $FADH_2$ generates 1.5 ATP.

The table below summarizes the inhibitors of the electron transport chain.

Complexes of ETC	Inhibitors
Complex I:NADH-Q oxidoreductase	Piericidin A, phenobarbital, rotenone
Complex II:Succinate-Q reductase	TTFA (thenoyltrifluoroacetone), carboxin, malonate
Complex III:Q- Cyt c oxidoreductase	BAL (British anti-lewisite), antimycin A
Complex IV:Cyt c oxidase	Hydrogen sulfide, carbon monoxide, cyanide
Complex V:ATP Synthase	Atractyloside, oligomycin

Solution to Question 12:

The given clinical scenario of a child with bone pain, hepatosplenomegaly, and trephine biopsy showing Gaucher's cell with a characteristic crumpled tissue paper appearance is suggestive of Gaucher's disease. The deficient enzyme is glucocerebrosidase (also known as beta-glucosidase).

Gaucher's disease is an inborn error of metabolism caused due to a deficiency of beta-glucocerebrosidase. It is the most common lysosomal storage disorder and the mode of inheritance is autosomal recessive. There is an accumulation of lipid-laden macrophages in the spleen, liver, and other tissues.

There are three clinical subtypes - type I (chronic non-neuronopathic), type II (acute neuronopathic), and type III (intermediate). Clinical features include anemia, thrombocytopenia, hepatosplenomegaly, and neurological features depending on the subtype.

Gaucher's cells are formed as a result of the accumulation of massive amounts of glucocerebrosides within the phagocytes. The cells have a fibrillary cytoplasm and give a crumpled tissue paper appearance.

Solution to Question 13:

The given clinical features suggest the diagnosis of pellagra, due to her corn-based diet. Oral nicotinamide supplements can improve her symptoms.

Niacin can be synthesized in the body from tryptophan. Two compounds, nicotinic acid and nicotinamide, have the biological activity of niacin. The nicotinamide ring of the coenzymes nicotinamide adenine dinucleotide (NAD) and nicotinamide adenine dinucleotide phosphate (NADP) participate in oxidation/reduction reactions.

The initial symptoms of pellagra include loss of appetite, generalized weakness, irritability, abdominal pain, and vomiting. This is followed by bright red glossitis and a characteristic pigmented, scaly skin rash in sun-exposed areas. This rash is known as Casal's necklace when it forms a ring around the neck.

The 4Ds of pellagra:

- Diarrhea
- Dermatitis
- Dementia, leading to
- Death

It is diagnosed by measuring urinary levels of 2-methyl nicotinamide and 2-pyridone.

Other options:

Option B: Thiamine (Vitamin B1) deficiency leads to conditions like beriberi and Wernicke-Korsakoff syndrome, which are characterized by symptoms such as muscle weakness, peripheral neuropathy, cardiovascular issues, and confusion. While cognitive dysfunction can result from thiamine deficiency, rash and loose stools do not fit with thiamine deficiency. Therefore, thiamine would not address the full spectrum of this patient's symptoms.

Option C: Vitamin C deficiency causes scurvy, which is associated with symptoms like bleeding gums, bruising, joint pain, and fatigue. Scurvy is not associated with a rash like the one described, nor does it typically cause cognitive dysfunction or diarrhea. Hence, high-dose vitamin C would not improve this patient's symptoms.

Option D: Corn-based diets are low in niacin and tryptophan, leading to pellagra over time, especially if the diet is not balanced with other niacin-rich foods. While stopping the corn-based diet could prevent further niacin deficiency, it will not immediately address the existing deficiency and symptoms like rash, diarrhea, and cognitive dysfunction. The patient would still need niacin supplementation to resolve the current symptoms.

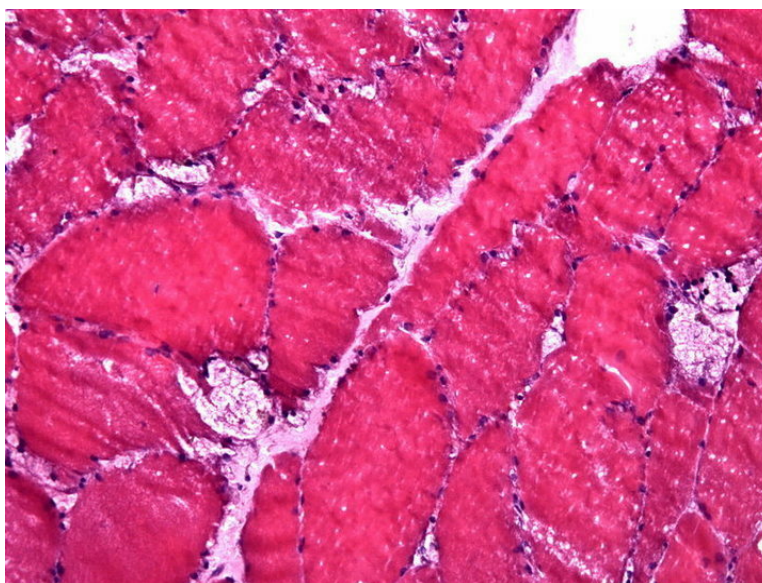
Solution to Question 14:

The child with a clinical history of difficulty feeding and failure to thrive with generalised hypotonia, macroglossia, hepatomegaly and chest x-ray findings of massive cardiomegaly suggest a diagnosis of Pompe's disease.

Pompe's disease is a glycogen storage disorder(type II) caused by a deficiency of lysosomal acid maltase (alpha-glucosidase) which leads to the accumulation of glycogen in the lysosomes of the liver (hepatomegaly), skeletal muscles in the form of large vacuoles (hypotonia and weakness), cardiac muscles (hypertrophic cardiomyopathy). The child generally dies by the age of 2 years due to cardiac failure. The adult form presents with muscle dystrophies.

Laboratory findings show elevated levels of serum creatine kinase (CK), aspartate transaminase (AST), alanine transaminase (ALT), and lactate dehydrogenase (LDH). Urine glucose tetrasaccharide, a glycogen breakdown metabolite, is a reliable biomarker for disease severity and treatment response. Treatment includes enzyme replacement therapy with recombinant enzyme α -glucosidase alfa (Myozyme).

Given below is the image of muscle biopsy showing glycogen-positive vacuoles.



Other options:

Option A- Ebstein's anomaly is the teratogenic effect of maternal lithium consumption which causes a cyanotic heart disease in the newborn with atrialization of the right ventricle and conduction defects. The chest X-ray will show a box-shaped enlargement of the right atrium and pulmonary oligemia.

Option C- Transposition of great vessels is cyanotic congenital heart disease where there is a switch in the position of two main arteries i.e. pulmonary artery and aorta. X-ray shows of egg-on-string appearance with pulmonary plethora.

Option D- Von Gierke disease occurs due to a deficiency of glucose-6-phosphatase. Patients present commonly in 3-4 months of age with hypoglycemia, lactic acidosis, hepatomegaly, growth retardation, hyperuricemia, and hyperlipidemia.

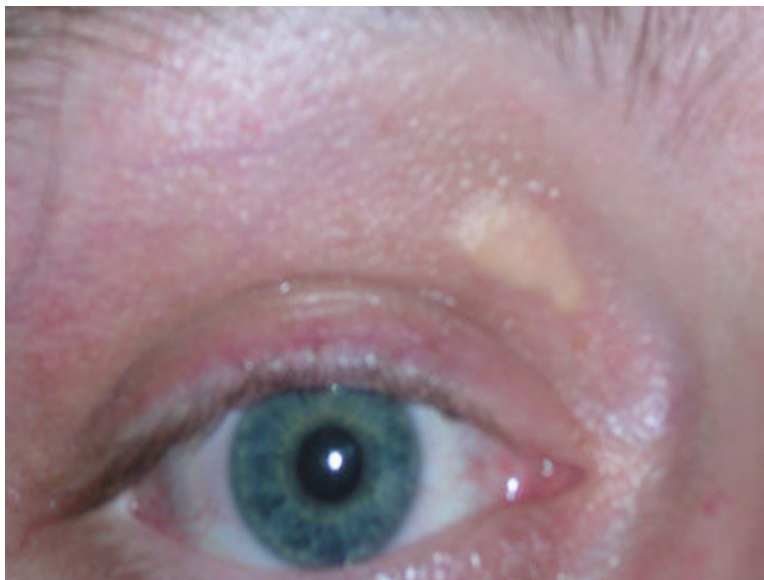
Solution to Question 15:

The clinical presentation with yellowish, raised plaques on eyelids, and tendon xanthomas points to a diagnosis of familial hypercholesterolemia (FH).

Familial hypercholesterolemia (type IIa hyperlipoproteinemia) is an inherited disorder due to defects in LDL receptors and Apo B100. This decreases the uptake of Low-density lipoprotein and cholesterol leading to their increased levels in the blood. The increased LDL accumulates in the blood vessels causing atherosclerosis (premature coronary artery disease) and in the Achilles tendon causing tendon xanthomas. Xanthomas are tumorous masses produced by foamy cell clusters (lipid-laden macrophages) found in the connective tissue of the skin and tendons. Patients can also have yellowish plaques around the eyes (Xanthelasma) and corneal arcus.

Treatment includes lifestyle changes, restriction of a fatty diet, and pharmacotherapy with statins in heterozygous FH patients.

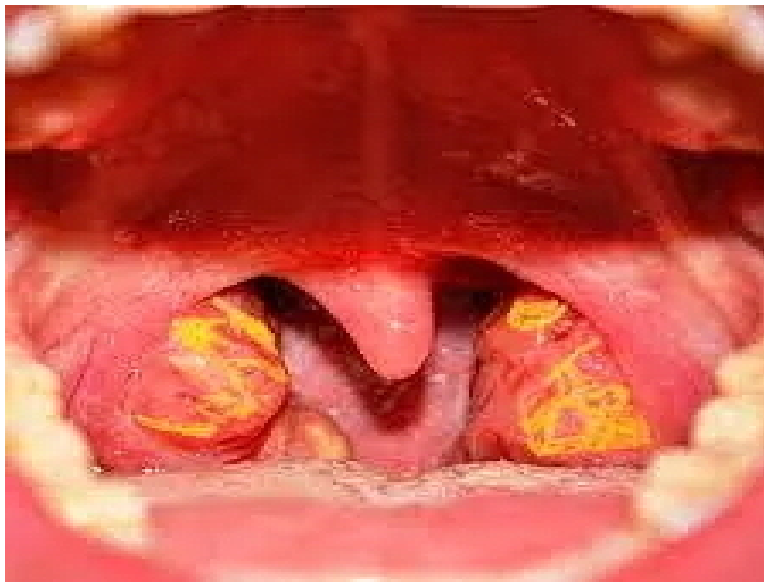
The images below show Achilles tendon xanthoma and xanthelasma respectively:



Other options:

Option B: Tangier's disease is a hypolipoproteinemia caused by a mutation of the ABCA1 gene. Enlarged orange tonsils are pathognomonic. Low HDL levels are seen. There is an increased risk for atherosclerosis.

The image below shows enlarged yellow-orange coloured tonsils in Tangier's disease.



Option C: Xanthomatosis is the deposition of lipids in various parts of the body like tendon xanthomas, xanthelasma, eruptive and tuberous xanthomas. This is a sign of underlying lipid metabolism disorders.

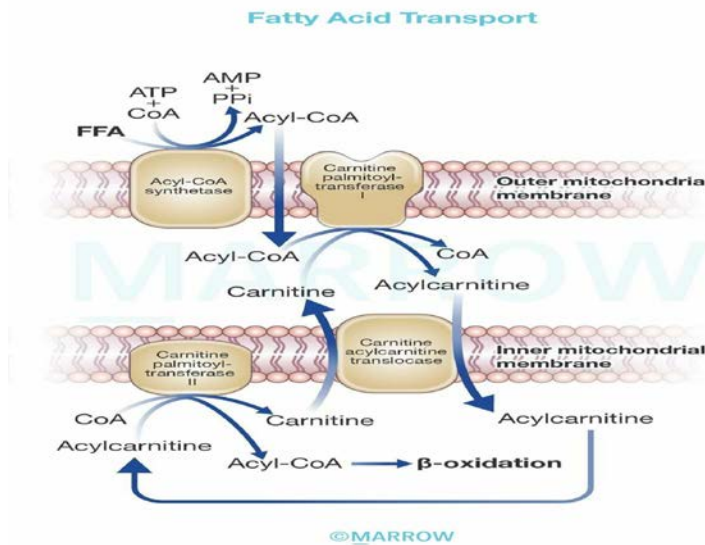
Option D: Fabry's disease is a lysosomal storage disorder with a defect in alpha-galactosidase. It causes angiokeratomas, hypohidrosis and characteristic Maltese cross deposits in urine.

Solution to Question 16:

The clinical scenario of a child with fatigue and muscle weakness on exercise with laboratory findings of LCFA oxidation deficiency and fat vacuoles on muscle biopsy suggests a diagnosis of carnitine deficiency.

Carnitine deficiency is a defect in fatty acid oxidation that leads to hypoketotic hypoglycemia and hyperammonemia. Carnitine transports activated LCFA (long chain fatty acyl CoA) from the cytoplasm to the mitochondria for beta-oxidation (as they cannot cross the mitochondrial membrane), carnitine deficiency impairs this transport causing accumulation of lipids (skeletal muscle weakness). This excess fatty acid inhibits the urea cycle causing hyperammonemia. Since beta-oxidation cannot occur, energy production is reduced, and there is hypoglycemia without a corresponding increase in ketone bodies (hypoketotic) during times of glucose depletion such as exercise.

The role of carnitine in the metabolism of fatty acids is demonstrated in the image below:



Clinical features of carnitine deficiency begin in childhood and include poor feeding, lethargy, liver dysfunction and hepatomegaly, hypotonia, and dilated cardiomyopathy. Treatment involves oral supplementation of carnitine.

Other options:

Option A: An increase in LDL can cause hyperlipoproteinemia, and reduced LDL can cause defective absorption of fat-soluble vitamins.

Option B: LPL is involved in hydrolysis of lipoproteins in the blood to free fatty acid and glycerol which are then taken up by the tissues. A defective LPL can cause type 1 familial chylomicronemia syndrome presenting as pancreatitis.

Option C: Fatty acid synthase is a multienzyme complex involved in fatty acid synthesis in the cytosol which when defective may impair energy production, essential fatty acids, and affect metabolic pathways. There is no accumulation of fat vacuoles since fatty acids are not synthesised.

Solution to Question 17:

In the given clinical vignette, inhibition of NADH-Q oxidoreductase by rotenone was the likely cause of breathlessness, followed by death in this child.

Rotenone inhibits complex I, thereby inhibiting the respiratory chain, so electrons cannot be transferred to oxygen. This halts ATP production, leading to an energy deficit in the cells, eventually causing cell death. The ETC constitutes four enzyme complexes:

- Complex I: NADH-Q oxidoreductase
- Complex II: Succinate-Q reductase
- Complex III: Q-cytochrome c oxidoreductase
- Complex IV: cytochrome c oxidase

Components of complex I are FMN and Fe-S complex. NADH-Q oxidoreductase catalyses the transfer of electrons from NADH to coenzyme Q, along with the transfer of four H⁺ across the membrane.

Solution to Question 18:

This child with phenylketonuria with neurological symptoms not improving despite dietary modifications and normal serum phenylalanine hydroxylase level is suggestive of tetrahydrobiopterin (BH₄) synthase deficiency.

A diet low in phenylalanine (eg: tapioca) is the mainstay of treatment for classic phenylketonuria but may not be sufficient in BH₄ deficiency where supplementation with BH₄ and deficient neurotransmitter precursors (L-dopa) is required to overcome the block.

Phenylketonuria (PKU) is an inborn error of metabolism seen due to a defect in the enzyme phenylalanine hydroxylase (classic phenylketonuria) and less commonly, due to a defect in BH₄ synthase (non-classical). These defects result in the impaired conversion of phenylalanine to tyrosine. The patient develops clinical features due to elevated levels of phenylalanine metabolites and decreased levels of tyrosine. The excess phenylalanine in the blood is metabolized to phenylpyruvate (positive ferric chloride) and phenylacetate (mousy odour) which are excreted in the urine. Infants appear normal at birth and gradually develop symptoms of hyperreflexia, seizures, hypopigmentation, low IQ, hyperactivity, and tremors.

Tandem mass spectroscopy is the investigation of choice. Guthrie test is a screening test that shows with absence of growth of *Bacillus subtilis*. The ferric chloride test is positive with the development of a transient blue-green colour.

Other options -

Option A: Tyrosine hydroxylase is the rate-limiting enzyme in the synthesis of epinephrine, converts tyrosine to DOPA, its deficiency does not cause phenylketonuria

Option B: Homogentisic acid oxidase converts homogentisic acid to maleylacetoacetic acid in the tyrosine degradation pathway and its deficiency causes alkaptonuria with a dark colour of urine on standing, bluish-black pigmentation in connective tissue (ochronosis) and arthritis.

Option D: Dopa decarboxylase is another enzyme involved in the synthesis of epinephrine. It converts DOPA to dopamine in the presence of vitamin B₆. While enzymes involved in epinephrine synthesis cause neurological dysfunction, they do not cause phenylketonuria as such.

Solution to Question 19:

The given clinical features of anaemia and neutropenia, along with depigmented hair and skin and the radiological findings suggestive of pelkan spurs, trummerfeld zone, Frenkel's line, and Wimberger's ring sign point to the diagnosis of Menke's disease. Patients with this condition have low serum levels of copper.

Menke's disease or Menke-Kinky hair syndrome is an X-linked recessive condition caused due to a mutation in the ATP7A gene on chromosome X. This causes impaired transport of dietary copper

from the small intestine to the liver leading to low levels of copper-containing enzymes.

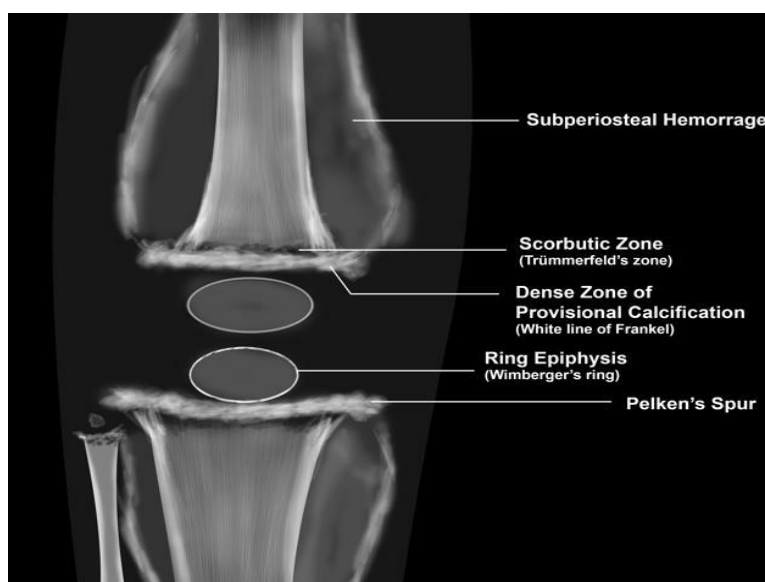
The final step of collagen synthesis involves the formation of covalent cross-links. This is done by the enzyme lysyl oxidase, a copper-containing protein. This is essential for the strength and rigidity of collagen. As a result of deficiency of copper in Menkes disease, this step of collagen synthesis is affected.

The patient presents with depigmented kinky hair, failure to thrive, neurodegenerative symptoms (loss of developmental milestones, truncal hypotonia, epilepsy), multiple fractures, anaemia due to ceruloplasmin and hephaestin deficiency (converts ferrous to ferric to be bound to transferrin), neutropenia, and recurrent infections.

Lab investigations show low serum copper ($<65 \mu\text{g/dL}$) and low ceruloplasmin ($<20 \text{ mg/dL}$). The disease is fatal and has no curative treatment.

A variety of distinct anomalies that are representative of scurvy and acquired copper insufficiency can be seen on radiographs of people with Menke's disease.

These include



Early administration of L-histidine and copper chloride injections may be beneficial.

Other options:

Option A: Iron deficiency causes iron deficiency anemia, and hair loss (telogen effluvium). It does not cause neutropenia, depigmentation of hair and skin, or bony abnormalities.

Option B: Zinc deficiency causes acrodermatitis enteropathica characterized by periorificial dermatitis, diarrhea, and alopecia.

Option D: Fluoride is essential for bone and teeth mineralization and its deficiency causes decreased bone-mineral density, osteoporosis, and tooth decay.

Solution to Question 20:

The given scenario indicates intolerance of milk-based products in this child, hence, the enzyme likely to be deficient is lactase.

Lactase is the enzyme that is responsible for breaking down lactose, a disaccharide found in milk and dairy products. When this enzyme is deficient, it can lead to the accumulation of lactose in the intestinal lumen where it is fermented by intestinal bacteria to lactate which produces symptoms of lactose intolerance, including recurrent abdominal pain, vomiting, and diarrhea after consuming lactose-rich food items. Lactase and maltase are present in the intestinal brush border.

Other options:

Option A- Lipases are involved in the hydrolysis of dietary triglycerides to free fatty acids and glycerol. It is deficient in exocrine pancreatic deficiency states such as chronic pancreatitis, cystic fibrosis, and Crohn's. Although it might also present with abdominal pain and diarrhea, the stools are foul-smelling(due to gut bacteria) and greasy due to the presence of fat (steatorrhea) and are also associated with fat-soluble vitamin deficiency.

Option B- Maltase breaks down maltose which is a disaccharide intermediate of starch breakdown.

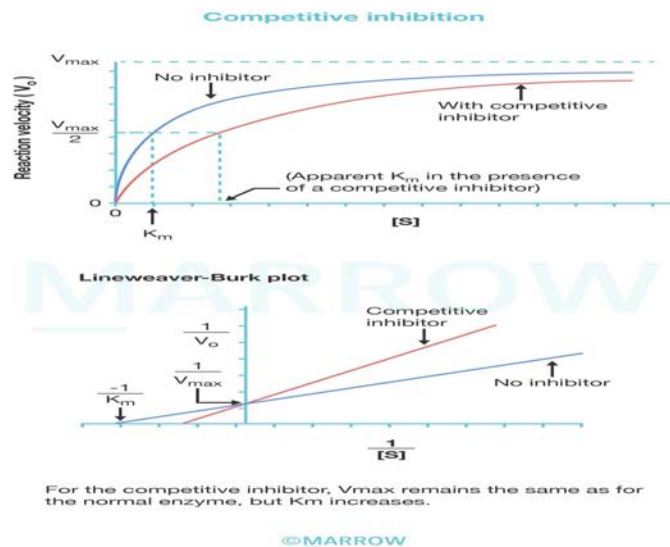
Option C- Amylase hydrolyses starch which is a glucose homopolymer and is present in germinating cereals, potatoes, legumes, and malt.

Solution to Question 21:

The given graph shows competitive inhibition, in which the value of V_{max} remains constant and the K_m value increases. The intercept on the y-axis remains unchanged while the intercept on the x-axis shifts close to zero forming a steeper slope.

Competitive inhibition is when the inhibitor binds reversibly to the substrate-binding site and competes with the substrate for the active enzyme-binding site. This effect of competitive inhibition can be overcome by increasing the substrate concentration by which maximum velocity can still be achieved. Therefore, V_{max} (maximum velocity of the reaction) is unaltered in this type of inhibition. However, the concentration of the substrate required to achieve half of the maximum velocity is increased. Therefore, K_m is increased in this condition.

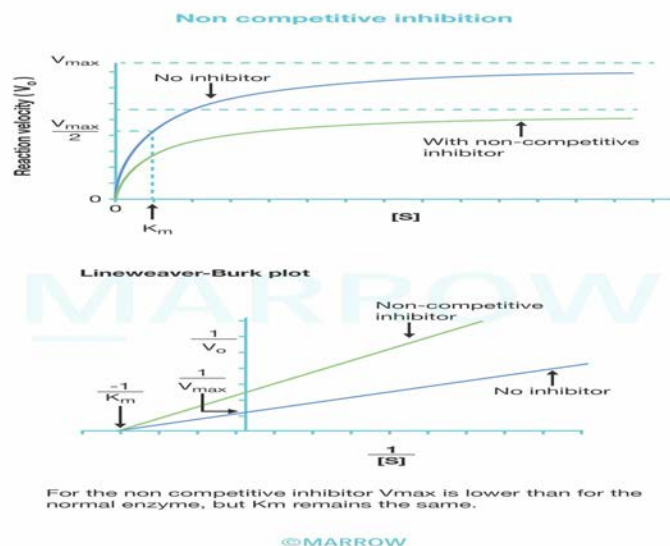
The graph below shows the changes in V_{max} and K_m in competitive inhibition.



Other options:

Option B: Non-competitive inhibition is by binding inhibitors to enzymes in locations remote from the active site. Increasing the concentration of the substrate does not overcome non-competitive inhibition. Thus, noncompetitive inhibitors decrease the V_{max} of the reaction. They do not interfere with the binding of the substrate to the enzyme. Thus, the enzyme shows the same K_m in the presence or absence of noncompetitive inhibitors.

The graph shows the changes in V_{max} and K_m in non-competitive inhibition.



Option C: In uncompetitive inhibition, inhibitors bind with the enzyme-substrate intermediate complex, and inhibit it. This leads to a decrease in both V_{max} and K_m . An example is the inhibition of placental alkaline phosphatase (ALP) by phenylalanine.

Option D: In Allosteric inhibition, inhibitors bind to a site other than the active site i.e. allosteric site, and induce a conformational change (change in the shape of the active site unlike in non-competitive inhibition) thus, inhibiting the enzyme. ATP is an allosteric inhibitor of

Phosphofructokinase in glycolysis.